DIGITAL COMBAT SIMULATOR



OH-58D
KIOWA WARRIOR

USER MANUAL

EAGLE DYNAMICS

BELL

DCS

Contents

Introduction **12**

1.1 About this document 12

AIRCRAFT **13**

Aircraft and Module Overview **14**

2.1 Aircraft History 14

2.2 External Overview 16

2.3 Module Options 17

2.3.1 Control Display Location *17*

2.3.2 Cyclic Trim Type *18*

2.3.3 Pedal Trim Type *18*

2.3.4 Throttle Travel Range *19*

2.3.5 ODA Eye Display *19*

2.3.6 Cockpit Vibration *19*

2.3.7 FFB Force Strength *19*

2.4 Aircraft Additional Properties 20

2.4.1 Aircraft Control Priority *20*

2.4.2 Remove Doors *20*

2.4.3 Install Pilot Display Unit *20*

2.4.4 Equip Personal Weapons *21*

2.4.5 Remove Mast Mounted Sight *21*

2.4.6 Rapid Deployment Gear (MPLH) *21*

2.4.7 Install AN/ALQ-144 IRCM *21*

2.4.8 Import Editor Drawings *22*

2.4.9 IDM Net *22*

Quick Start Guide **23**

3.1 Control Bindings 23

3.1.1 Single- and Multicrew Experience *23*

3.1.2 Suggested Control Bindings *23*

3.2 Start-up 28

3.2.1 Before Engine Start *28*
3.2.2 Engine Start *28*
3.2.3 Systems *29*
3.2.4 NAV Alignment *29*
3.2.5 Further Systems *29*
Cockpit Familiarization **30**
4.1 Pilot Instruments 31
4.2 Copilot Instruments 32
4.3 Center Console 33
4.4 Overhead Console 34
4.5 Center Post Circuit Breaker Panel 35
SYSTEMS **36**
Flight Control System **37**
5.1 Pilot Cyclic Grip 37
5.2 Pilot Collective Grip 38
5.3 Copilot Cyclic Grip 39
5.4 Force Trim System 40
5.5 Stability and Control Augmentation System 40
5.5.1 SCAS Control Panel *40*
5.5.2 SCAS Heading Hold and Trim Hat *41*
Full Authority Digital Engine Control **42**
6.1 Overview 42
6.2 FADEC Power and Control 42
6.2.1 FADEC Power Source *42*
6.2.2 FADEC Control *42*
6.3 FADEC Modes 44
6.3.1 Automatic *44*
6.3.2 Manual *44*
Engine **45**
7.1 Overview 45

POLYCHOP SIMULATIONS

7.2 Controls and Switches 45
7.2.1 Throttle 45
7.2.2 Ignition Keylock Switch 46
7.2.3 Start Switch 46
7.2.4 Ignition Circuit Breaker Switch 47
7.2.5 Fuel Boost Switch 47
7.2.6 Engine Anti-Ice Switch 47
7.2.7 Engine Oil Bypass Switch 47
Electrical System 48
8.1 Overview 48
8.2 Controls and Switches 48
8.2.1 Battery 1 Switch 48
8.2.2 Battery 2 Switch 48
8.2.3 AC Generator Switch 48
8.2.4 DC Generator Switch 48
8.2.5 Essential Bus Switch 48
Multiparameter Display and Vertical Scale Instruments 49
9.1 Multiparameter Display 49
9.2 Vertical Scale Instruments 50
Standby Instruments 51
10.1 Standby Airspeed Indicator 51
10.2 Barometric Altimeter 51
10.3 Standby Magnetic Compass 51
10.4 Chronometer 52
10.5 Free Air Temperature Gauge 52
Control Display Subsystem 53
11.1 Introduction 53
11.2 Improved Master Controller Processor Unit 53
11.3 Multifunction Display 54
11.4 Multifunction Keyboard 56
11.5 Optical Display Assembly 57

11.5.1 ODA Symbology *58*
11.5.2 ODA Declutter Levels *59*
11.5.3 ODA Symbology – Gun *60*
11.5.4 ODA Symbology - Rockets *60*
11.5.5 ODA Symbology - Hellfire *61*
11.5.6 ODA Symbology - ATAS *62*
11.6 Pilot Display Unit 63
11.7 Personal Computer Data Transfer System 67
11.8 MFD Auxiliary Control Panels 67
11.8.1 Pilot MFD Auxiliary Control Panel *67*
11.8.2 Copilot MFD Auxiliary Control Panel *68*
11.9 MFD Interaction and Pages 69
11.9.1 Common Elements *69*
11.10 INITIAL Pages 74
11.10.1 Initial 1 ..... *74*
11.10.2 Initial 2 ..... *75*
11.10.3 NAV Align *76*
11.10.4 NAV Mode Select *77*
11.10.5 Power Assurance *78*
11.10.6 Performance Setup *79*
11.10.7 Hover Performance *80*
11.10.8 Cruise Performance *81*
11.10.9 Data Loader *82*
11.10.10 Load Mission *83*
11.10.11 Store Mission *84*
11.10.12 Notebook *85*
11.10.13 Ground Setup *86*
11.10.14 Fuel Management *87*
11.10.15 Fault Detection/Location Menu (?) *87*
11.10.16 Clock Timer *88*
11.11 VSD Pages 89

11.11.1 VSD *89*
11.11.2 HVR *93*
11.12 HSD Pages 95
11.12.1 HSD- *95*
11.12.2 RMS *97*
11.12.3 RMS Setup *101*
11.12.4 RMS Feature Select *102*
11.12.5 NAV Setup *103*
11.12.6 Route Setup *104*
11.12.7 Waypoint List *106*
11.12.8 New Waypoint *107*
11.12.9 Control Point List *108*
11.12.10 New Control Point *109*
11.12.11 Target Point List *110*
11.12.12 New Target Point *111*
11.12.13 Direct Point *112*
11.12.14 Battlefield Graphics *113*
11.13 COMM Pages 116
11.13.1 COMM *116*
11.13.2 IFF 1 ..... *117*
11.13.3 IFF 2 ..... *118*
11.13.4 IFF 3 ..... *119*
11.14 Weapons Pages 120
11.14.1 Weapons *120*
11.14.2 Weapons BIT/Setup *124*
11.14.3 Pilot Sparse Weapons Page (Gun) *129*
11.14.4 Pilot Sparse Weapons Page (Rockets) *130*
11.14.5 Pilot Sparse Weapons Page (Hellfire) *131*
11.14.6 Pilot Sparse Weapons Page (ATAS) *132*
11.14.7 Copilot Sparse Weapons Page (Gun) *133*
11.14.8 Copilot Sparse Weapons Page (Rockets) *134*

11.14.9 Copilot Sparse Weapons Page (Hellfire) *135*
11.14.10 Copilot Sparse Weapons Page (ATAS) *136*
11.15 Pages 137
11.15.1 ASE Setup *137*
Armament Control Panel 138
12.1 Overview 138
Mast Mounted Sight 139
13.1 Overview 139
13.2 Mast Mounted Sight System Control Panel 141
13.3 Pages 142
13.3.1 MMS Page *142*
13.3.2 Laser Codes Page *144*
13.4 Modes 145
13.4.1 Stow *145*
13.4.2 Preflight *146*
13.4.3 Pre-Point *147*
13.4.4 Forward *148*
13.4.5 Search *149*
13.4.6 Manual *151*
13.4.7 Uncommanded *152*
13.4.8 Point Track *153*
13.4.9 Area Track *154*
13.4.10 Acquire On The Move *155*
13.4.11 Offset Acquire *156*
13.4.12 Offset Track *157*
13.4.13 Target Locate Mode *158*
Radios 159
14.1 Overview 159
14.2 Communications System Control Panel 160
14.3 Radio Select Switch 161
14.4 Channel Select Switch 162

14.5 Remote Frequency Display 163
Aircraft Survivability Equipment 164
15.1 Introduction 164
15.2 Flare Dispenser 164
15.3 AN/APR-39 Radar Detection Set 164
15.3.1 Radar Detector Indicator Symbology 165
15.3.2 Radar Detector Annunciator Messages 166
15.4 AN/AVR-2 Laser Detection Set 167
15.5 AN/ALQ-144 Infra-red Missile Countermeasures Device 167
15.5.1 IR Jammer Circuit Breaker Switch 167
15.6 AN/AAR-57 Common Missile Warning System 168
15.6.1 CMWS Circuit Breaker Switch 168
15.6.2 CMWS Control Panel 169
15.6.3 CMWS Control Indicator 170
Lighting 171
16.1 Exterior Lighting 171
16.1.1 Anti-Collision Lights 171
16.1.2 Position Lights 172
16.1.3 Searchlight 173
16.1.4 Formation Lights 174
16.2 Interior Lighting 175
16.2.1 Instrument Lights 175
16.2.2 Console Lights 176
16.2.3 Flood Light 177
16.2.4 Flashlight 178
Level 2 Manned/Unmanned System (L2MUMS) 179
17.1 Overview 179
17.2 Pages 180
17.2.1 Platform Reception Selection 180
17.2.2 Platform Transmission Selection 181
17.2.3 Video Display 182

17.2.4 Map Display *184*

PROCEDURES **186**

Aviation **187**

18.1 Startup 187

18.1.1 Before Engine Start *187*

18.1.2 Engine Start *187*

18.1.3 ..... *188*

18.1.4 ENGINE RUNUP — CPO/PILOT. *188*

18.2 Shutdown 189

18.2.1 ENGINE SHUTDOWN. *189*

18.3 FADEC & Engine Management 190

18.3.1 Switching to FADEC Manual mode *190*

18.3.2 MPD Bit Operation *190*

18.4 Warnings, Cautions, Advisories 190

18.4.1 Acknowledgement and Recall of Cautions and Advisories *190*

Navigation **192**

19.1 Overview 192

19.2 Data Transfer System 192

19.2.1 Saving Mission Data *192*

19.2.2 Loading Mission Data *193*

19.3 Waypoints 194

19.3.1 Creating a Waypoint *194*

19.3.2 Editing a Waypoint *195*

19.3.3 Deleting a Waypoint *196*

19.3.4 Deleting all Waypoints *197*

19.4 Control Points 198

19.4.1 Creating a Control Point *198*

19.4.2 Editing a Control Point *199*

19.4.3 Deleting a Control Point *199*

19.4.4 Deleting all Control Points *200*

POLYCHOP SIMULATIONS

19.5 Target Points 200
19.5.1 Creating a Target Point 200
19.5.2 Creating a Target Point using the MMS 'Store' function 201
19.5.3 Creating a Target Point using the MMS 'Store More' function 201
19.5.4 Editing a Target Point 202
19.5.5 Deleting a Target Point 202
19.6 Direct To Mode 203
19.6.1 Creating a Direct To Point 203
19.7 Routes 203
19.7.1 Creating a Route 203
19.7.2 Editing a Route 203
19.7.3 Selecting a Route 203
19.8 Battlefield Graphics 204
19.8.1 Creating a Line Graphic 204
19.8.2 Editing a Line Graphic 204
19.8.3 Deleting a Line Graphic 204
19.8.4 Creating an Area Graphic 205
19.8.5 Editing an Area Graphic 205
19.8.6 Deleting an Area Graphic 205
Communication 206
20.1 Radios 206
20.1.1 Selecting a Radio 206
20.1.2 Transmitting on a Radio 206
20.1.3 Adding or Editing a Channel 206
20.1.4 Deleting a Channel 206
20.1.5 Editing Frequencies from the Remote Frequency Display 206
20.2 Improved Data Modem (IDM) 207
20.2.1 Overview 207
20.2.2 Special Controls 207
20.2.3 Changing Callsign ID 207
20.2.4 Sending a Position Report (POSREP) 208

20.2.5 Receiving a Message *208*
20.2.6 Sending a Free Text Message *209*
20.2.7 Sending a Predetermined Free Text Message *210*
20.2.8 Sending a Spot Report (SPOTREP) *211*
20.2.9 Storing a Target Point from a Received Message *212*
Sensors 213
21.1 Mast Mounted Sensor (MMS) 213
21.1.1 Starting the MMS *213*
21.1.2 Shutting down the MMS *213*
21.1.3 Controlling the MMS in Manual Mode *213*
21.1.4 Using the Laser Rangefinder *213*
21.1.5 Using the Laser Designator *213*
21.1.6 Slaving to Forward Mode *214*
21.1.7 Pre-Point Mode *215*
21.1.8 Search Mode – Point *216*
21.1.9 Search Mode – Sweep *217*
21.1.10 Area Track *218*
21.1.11 Point Track *219*
21.1.12 Acquire On The Move *220*
21.1.13 Offset Acquire *221*
21.1.14 Offset Track *222*
Weapons 223
22.1 M3P Machine Gun 223
22.1.1 Firing *223*
22.1.2 Infrared Zoom Laser Illuminator Designator *224*
22.2 Rockets 225
22.2.1 Rocket Cue Setup *225*
22.2.2 Unguided *226*
22.2.3 APKWS *226*
22.3 Hellfire 227
22.3.1 LOBL *227*

22.3.2 LOAL *227*
22.3.3 Hellfire parameters *229*
22.4 ATAS 230
22.4.1 Unslaved *230*
22.4.2 Slaved *230*
22.5 Copilot's Personal Weapon (M4) 231
22.5.1 Using VR Hand Controllers *231*
22.5.2 Using Mouse Input *231*
L2MUMS 232
23.1 Receiving from an Unmanned Platform 232
23.1.1 Prerequisites *232*
23.1.2 Steps *232*
23.2 Receiving from a Manned Platform 233
23.2.1 Prerequisites *233*
23.2.2 Steps *233*
23.3 Transmitting 234
23.3.1 Prerequisites *234*
23.3.2 Steps *234*
Mission Editor 235
24.1 Creating the Default Route 235
24.2 Creating Non-Route Waypoints 237
24.3 Creating Control Points 238
24.4 Creating Target Points 239
24.4.1 Target Subtypes *241*
24.5 Creating Routes with Drawings 242
24.6 Creating Battlefield Graphics with Drawings 243
24.6.1 Creating Line Drawings *244*
24.6.2 Creating Area Drawings *244*
APPENDIX 245
Glossary 245

POLYCHOP SIMULATIONS

Warnings Advisories & Cautions..... 246
26.1 WARNING messages..... 246
26.2 CAUTION messages..... 247
26.3 ADVISORY messages..... 248

Introduction

Version: 1.0

1.1 About this document

- This User Guide is a work in progress and does not yet fully document all the features available on the aircraft. The manual will be updated as further improvements to the module release.

- The DCS OH-58D Kiowa Warrior User Guide is split into three main sections.

    - The Aircraft section introduces the Kiowa Warrior and the DCS module, and focuses on setting up the module and control preferences and basic familiarization. It also includes a Quick Start guide.

    - The Systems section covers in detail the various onboard mechanical and electrical systems of the aircraft.

    - The Procedures section provides a set of checklists for completing various tasks in the aircraft. Each procedure contains links referring to items in the Systems section that can provide more detail.

- This document uses internal links to allow navigation to another relevant area of the document. Whenever text is presented in the **link format** you can click on the text to navigate to the target page.

- There are multiple terms used for the aircrew positions in the OH-58D Kiowa Warrior. For example, the left seat can variously be referred to as 'CP/G', 'CP/O', 'Copilot' or just 'Left seater'. For the sake of consistency, within this document we use the term 'Pilot' for the right seat and 'Copilot' for the left seat positions. This also extends to any duplicated controls and interfaces, e.g. "Copilot Cyclic Grip".

Section 1

AIRCRAFT



Aircraft and Module Overview

2.1 Aircraft History

The Bell OH-58 Kiowa is a remarkable piece of aviation history. This single-engine, single-rotor military helicopter has served numerous roles, from observation and utility to direct fire support.

The Bell OH-58 Kiowa was initially developed during the early 1960s as the D-250 for the Light Observation Helicopter (LOH) program. Despite facing stiff competition from rival Hughes OH-6 Cayuse, Bell refined its design and successfully submitted a variant to the reopened LOH competition. The initial model, designated as the OH-58A, was introduced in May 1969. Successive models followed, featuring uprated engines, enhanced protection systems, and other improvements, culminating in the OH-58F.

In the 1970s, the US Army expressed interest in an advanced scout helicopter. The OH-58 Kiowa was further developed, evaluated, and ultimately procured as the OH-58D Kiowa Warrior. This variant was equipped to perform armed reconnaissance missions and provide fire support to friendly ground forces. It featured a distinctive Mast Mounted Sight (MMS) containing various sensors for target acquisition and laser designation.

The OH-58D Kiowa Warrior introduced perhaps the most distinctive feature of the Kiowa family, the Mast Mounted Sight (MMS). This gyro-stabilized platform contained a television system, a thermal imaging system, and a laser range finder/designator. These features gave the OH-58D the additional mission capability of target acquisition and laser designation in day or night, limited-visibility, and adverse weather.

The Bell OH-58 Kiowa has a rich operational history, having been deployed in numerous war zones over the decades. Only two months after its entry into service, it was first deployed into the Vietnam War. The US Army made extensive use of various OH-58 models, seeing active combat during the Gulf War, the Invasion of Panama, and the War in Afghanistan among others.

Despite its proven capabilities, the OH-58 Kiowa was retired from active service in the US Army in 2017. However, it continues to serve in several other countries' armed forces, including Greece, Taiwan, and Saudi Arabia. The Kiowa's retirement was part of a broader restructuring of the US Army's aviation assets, which saw a shift towards the use of unmanned aerial vehicles (UAVs) for reconnaissance roles.

The Bell OH-58 Kiowa is a testament to the ingenuity and adaptability of military aviation design. From its early days as a light observation helicopter to its evolution into a versatile combat platform, the Kiowa has proven its worth time and again. Its legacy continues to influence the design of future military helicopters, ensuring that the spirit of the Kiowa lives on.

2.2 External Overview

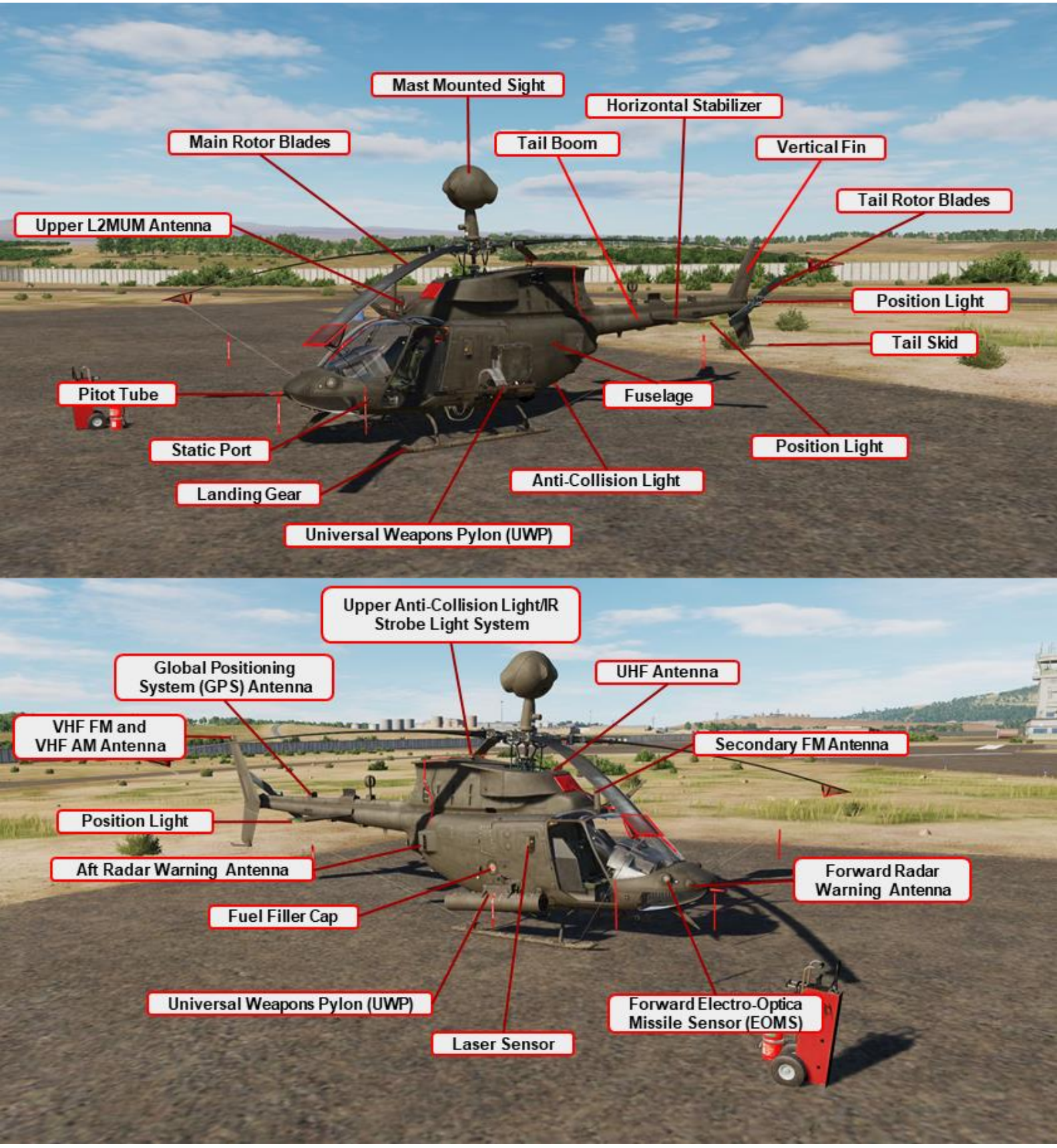


Figure 1 - External Overview 1

POLYCHOP SIMULATIONS

2.3 Module Options

The Options for the DCS OH-58D Kiowa Warrior can be found by accessing 'Options' from the Main Menu, then selecting 'Special' and navigating to 'OH-58D(R)' on the list of installed modules.

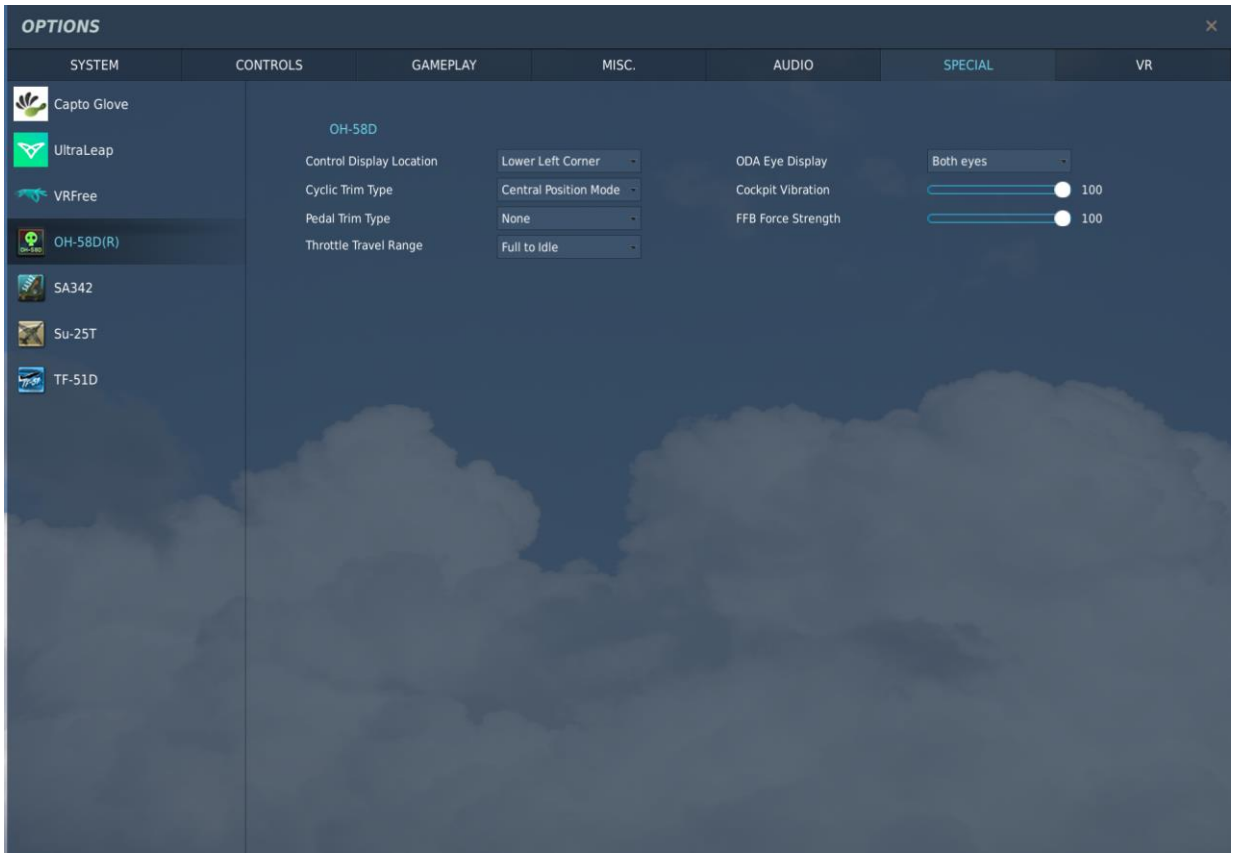


Figure 2 - Module Options

2.3.1 Control Display Location

The Control Display Location option selects the position at which the Controls Indicator is displayed on the screen

2.3.2 Cyclic Trim Type

There are four options for Cyclic Trim Type:

- None – pressing and releasing the Force Trim button has no effect on the cyclic
- Instant – pressing and releasing the Force Trim button instantly sets the center position of the joystick to the currently held cyclic position
- Fade In/Fade Out – pressing and releasing the Force Trim button moves the center position of the joystick to the currently held cyclic position over a fraction of a second
- Central Position Mode - pressing and releasing the Force Trim button instantly sets the center position of the joystick to the currently held cyclic position, but no joystick input is then registered until the joystick is centered

WARNING:

Cyclic Trim may not function correctly if the 'Force Feedback' checkbox in OPTIONS > MISC is not correctly set.

2.3.3 Pedal Trim Type

There are four options for Pedal Trim Type:

- *None* – pressing and releasing the Force Trim button has no effect on the pedals
- *Instant* – pressing and releasing the Force Trim button instantly sets the center position of the pedals to the currently held input position
- *Fade In/Fade Out* – pressing and releasing the Force Trim button moves the center position of the pedals to the currently held input position over a fraction of a second
- *Central Position Mode* - pressing and releasing the Force Trim button instantly sets the center position of the pedals to the currently held input position, but no further movement is then registered until the input is centered

2.3.4 Throttle Travel Range

The OH-58D Kiowa Warrior has a rotary throttle grip on the collective that governs fuel flow to the engine. In normal flight the fuel flow is controlled by the Full Authority Digital Engine Control (FADEC) and the throttle has no authority when above 68% when the FADEC switch is in the AUTO position. In the event that the FADEC fails or is disabled, fuel flow must be controlled using the throttle after placing the FADEC switch in the MAN position.

The throttle axis has a stop that prevents travel below 50% (the IDLE position) to 0% (CLOSED position). Throttle at IDLE correlates to an engine speed of 63% to 65% Ng. A button on the Pilot Collective Grip must be depressed to allow for travel below the IDLE position.

Since the lower range is not required outside of shutdown procedures, we have provided an option to dedicate the entire throttle axis to the upper half of the throttle range (IDLE to FULL) which allows much finer control of fuel flow and no risk of passing the IDLE stop when there is no hardware detent or stop.

For most players, we advise using this option by selecting the 'Full to Idle' range.

Players who have the hardware capable to do so can allow the full range of throttle motion by selecting the 'Full to Closed' range.

2.3.5 ODA Eye Display

The Optical Display Assembly (ODA) is a helmet mounted display that attaches to the player's night vision goggles. The ODA Eye Display option determines in which eye the ODA is displayed for VR users.

2.3.6 Cockpit Vibration

Adjusts the level of background vibration in the aircraft. Vibration due to effects such as ETL or overspeed are not affected.

2.3.7 FFB Force Strength

Allows adjustment of the strength of force exerted by a Force Feedback Stick.

2.4 Aircraft Additional Properties

There are a variety of options available in the DCS Mission Editor that allow a degree of customization of the DCS OH-58D Kiowa Warrior. Many of these options reflect real world changes to the airframe over its 33-year service history, allowing mission creators to modify the helicopter to represent a specific time period or mission capability.

These options can be found by placing an OH-58D(R) and selecting the 'Aircraft Additional Properties' tab.

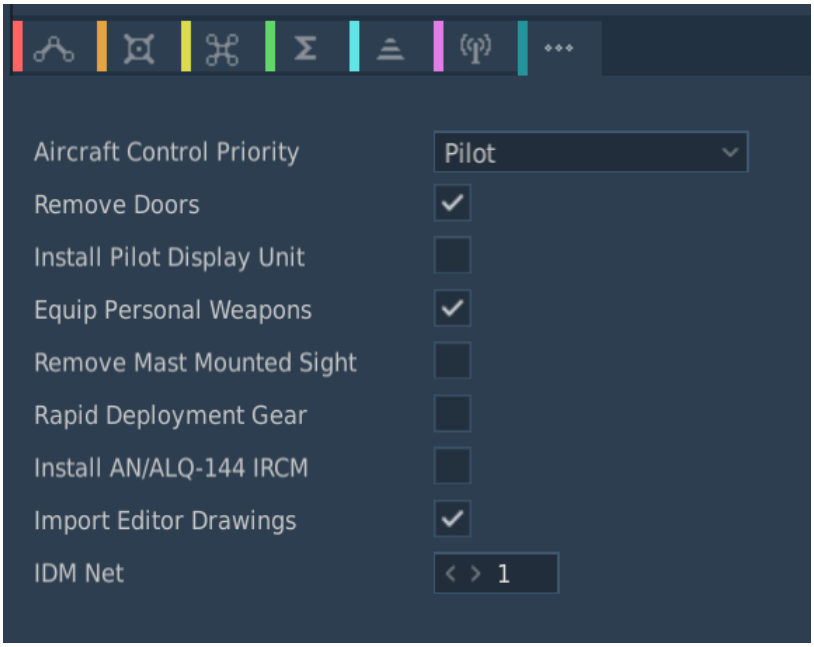


Figure 3 - Aircraft Additional Properties

2.4.1 Aircraft Control Priority

Determines which player has priority control of the aircraft in multicrew.

2.4.2 Remove Doors

Enable to remove doors from the aircraft. Doors can also be removed or added via the Ground Crew radio menu.

2.4.3 Install Pilot Display Unit

The Pilot Display Unit (PDU) is a roof mounted Heads Up Display system that was largely removed from aircraft due to safety issues with the night vision goggles used by pilots.

For more information on the PDU, see **Pilot Display Unit.**

2.4.4 Equip Personal Weapons

Crew deployed in combat would often be equipped with personal weapons, primarily used for defense in case the aircraft needs to be evacuated in hostile territory. In the case of the Kiowa Warrior, these weapons were sometimes used offensively. In the DCS OH-58D Kiowa Warrior module we have included these weapons by default, and the Copilot may employ their weapon offensively.

More information on how to employ this weapon can be found in **Copilot's Personal Weapon (M4)**.

2.4.5 Remove Mast Mounted Sight

The **Mast Mounted Sight** (MMS) is the large optical sensor mounted on top of the rotor mast. While the Kiowa Warrior almost always flew with the MMS attached, it could be removed for transportation and there are cases of the aircraft flying without the sensor attached both in deployment and in training.

INFO:

Removing the MMS will disable all relevant functionality such as the use of the TV and IR cameras, Laser Rangefinder and Laser Designator systems

2.4.6 Rapid Deployment Gear (MPLH)

Multi-Purpose Light Helicopter Rapid Deployment Gear was fitted to aircraft being transported as it allowed the aircraft to be lowered when stowed.

2.4.7 Install AN/ALQ-144 IRCM

The AN/ALQ-144 IRCM is a passive infrared countermeasures device that disrupts the seekers of missiles targeting the aircraft.

The AN/ALQ-144 was replaced by the **AN/AAR-57 Common Missile Warning System** around 2009.

More information on this system can be found in **AN/ALQ-144 Infrared Countermeasures Device.**

INFO:

Installing the AN/ALQ-144 will remove the Common Missile Warning System and the cockpit Control Indicator from the aircraft.

Installing the AN/ALQ-144 will remove L2MUM capability.

2.4.8 Import Editor Drawings

The OH-58D Kiowa Warrior navigation system allows for the creation of multiple flight routes, as well as the drawing of Battlefield Graphics that are used to display locations on the map.

To facilitate the creation of these extra routes and graphics, we have implemented the ability to import Polygon and Line Drawings from the DCS Mission Editor as Routes and Graphics.

By default, these drawings will be imported (subject to use of the correct naming convention). Disabling this option will prevent the drawings from being loaded into the OH-58D Kiowa Warrior's mission computers.

For more information on the use of Mission Editor Drawings to create Routes and Battlefield Graphics, see **Creating Routes with Drawings** and **Creating Battlefield Graphics with Drawings.**

2.4.9 IDM Net

The IDM Net value indicates which network the Improved Data Modem will connect to. The IDM will only be able to communicate with aircraft on the same network.

Quick Start Guide

3.1 Control Bindings

3.1.1 Single- and Multicrew Experience

The OH-58D Kiowa Warrior is a two-seater aircraft where the workload is shared between Pilot and Copilot.

In order to separate and manage the inputs available to each position, we have created two separate input profiles, one for each seat. The profiles can be selected from the drop-down list of aircraft in the Controls page and are named 'OH-58D Pilot' and 'OH-58D Copilot'.

In order to facilitate solo flight in DCS World, we have set up the control system in such a fashion that a solo player may control both the Pilot and Copilot inputs from a single seat.

When in a solo mission, or in a multiplayer environment where there is no second player in the aircraft, the player may switch seats at will, using the 'Switch to Pilot Seat' or 'Switch to Copilot Seat' commands.

From either seat the player has access to all inputs of both Pilot and Copilot.

In the event of a control binding conflict, the priority is given to the currently occupied seat.

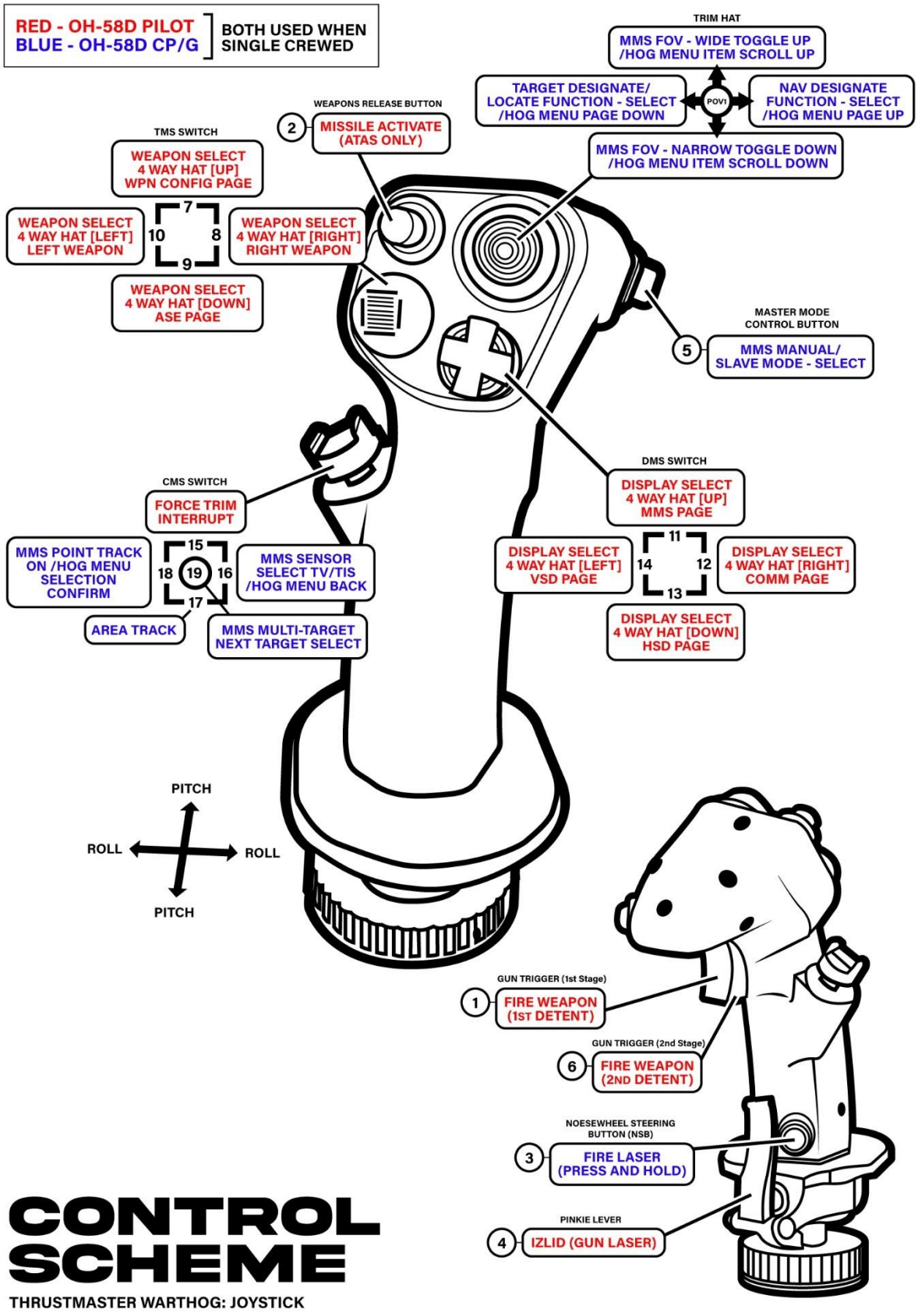
For example, if the player binds the 'OH-58D Pilot' input 'Fire Weapon [1st Detent]' to the 'Enter' key, and also binds the 'OH-58D Copilot' input 'Fire Laser' to the 'Enter' key, then when the player sits in the Pilot seat, activation of the 'Enter' key will fire the weapons, but when sitting in the Copilot seat, it will fire the laser.

When a second player joins the aircraft in a multiplayer environment, the inputs are restricted to the occupied position. For example, a player in the Pilot position will no longer be able to use the 'Fire Laser' input.

3.1.2 Suggested Control Bindings

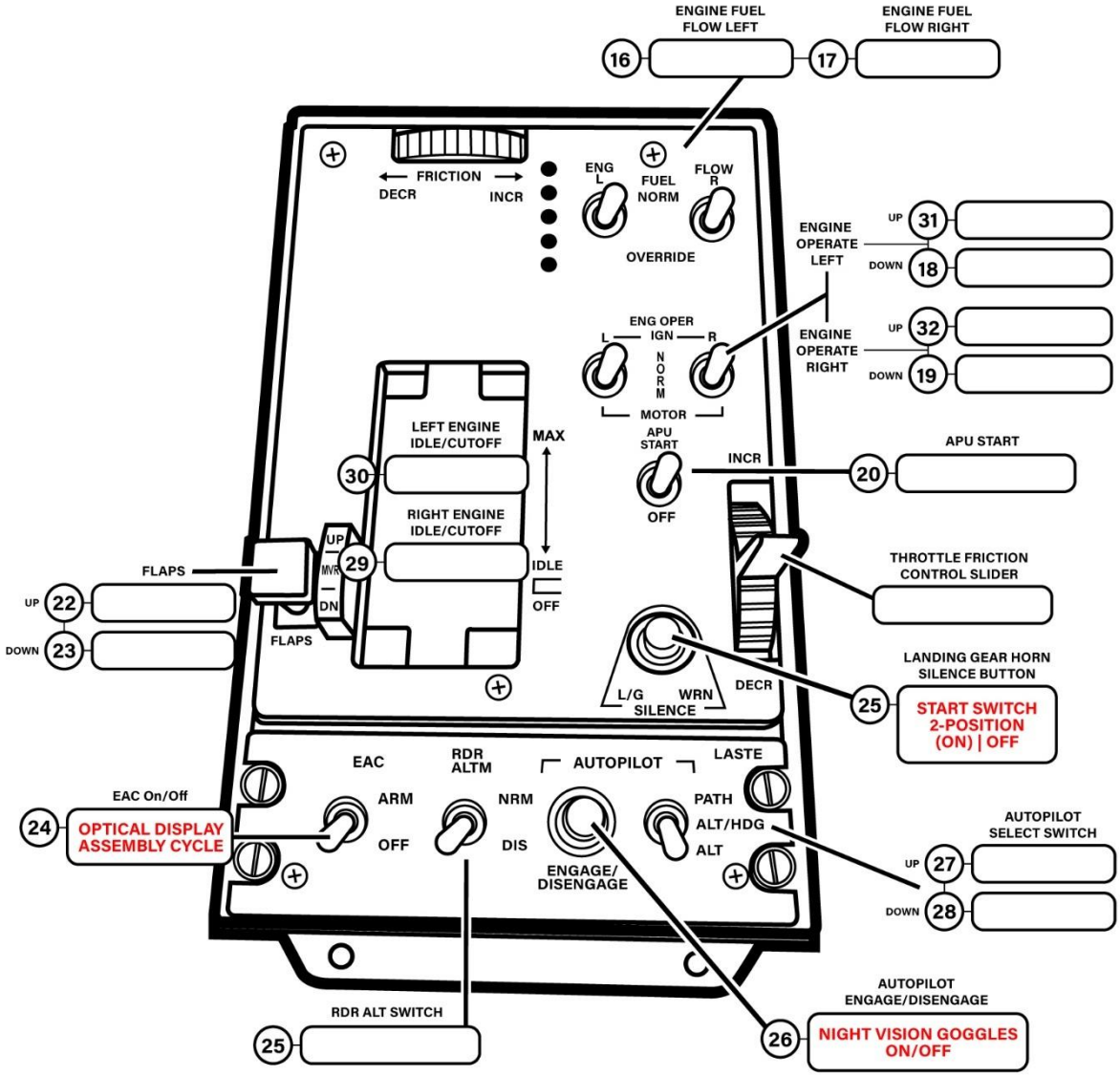
The following diagrams show suggested bindings for the DCS OH-58D Kiowa Warrior. These are based on the most commonly used inputs in DCS missions, and are based on the use of standard HOTAS hardware.

3.1.2.1 Joystick



3.1.2.2 Throttle

RED - OH-58D PILOT
BLUE - OH-58D CP/G
BOTH USED WHEN SINGLE CREWED

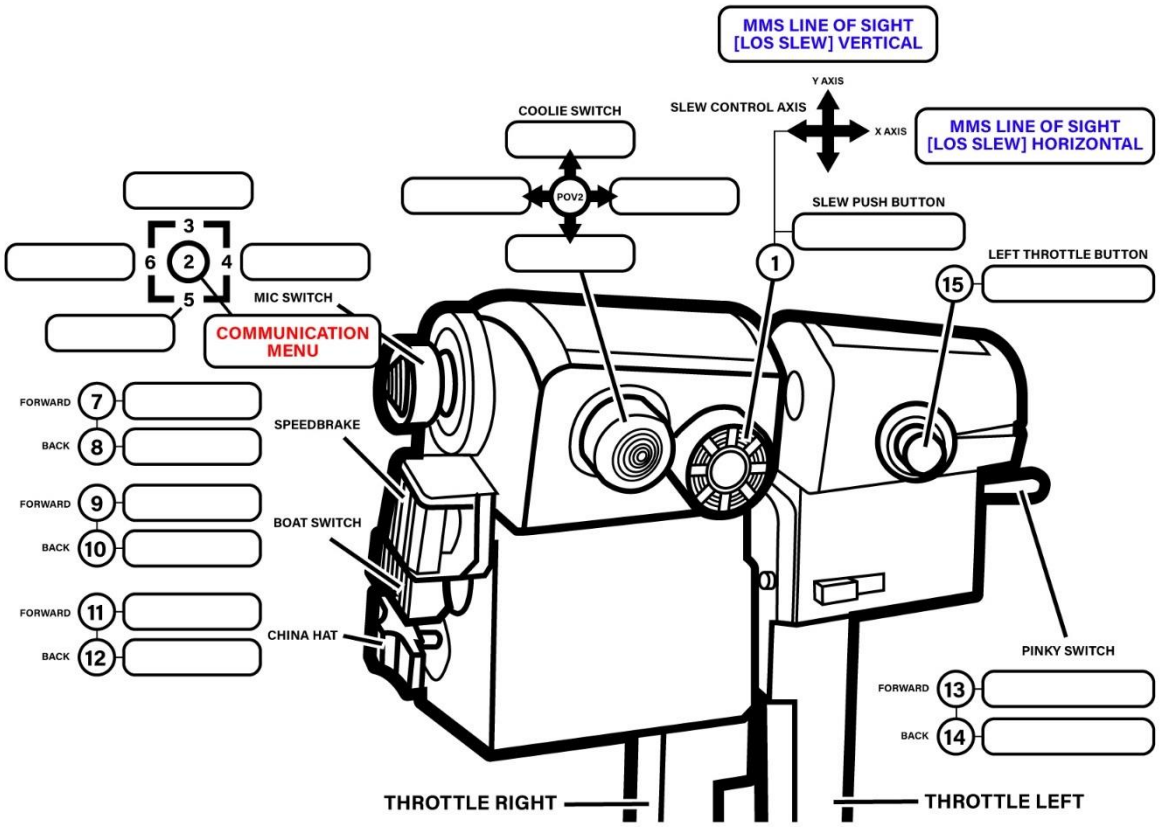


CONTROL SCHEME

THRUSTMASTER WARTHOG: THROTTLE BASE

POLYCHOP SIMULATIONS

RED - OH-58D PILOT
BLUE - OH-58D CP/G
BOTH USED WHEN SINGLE CREWED

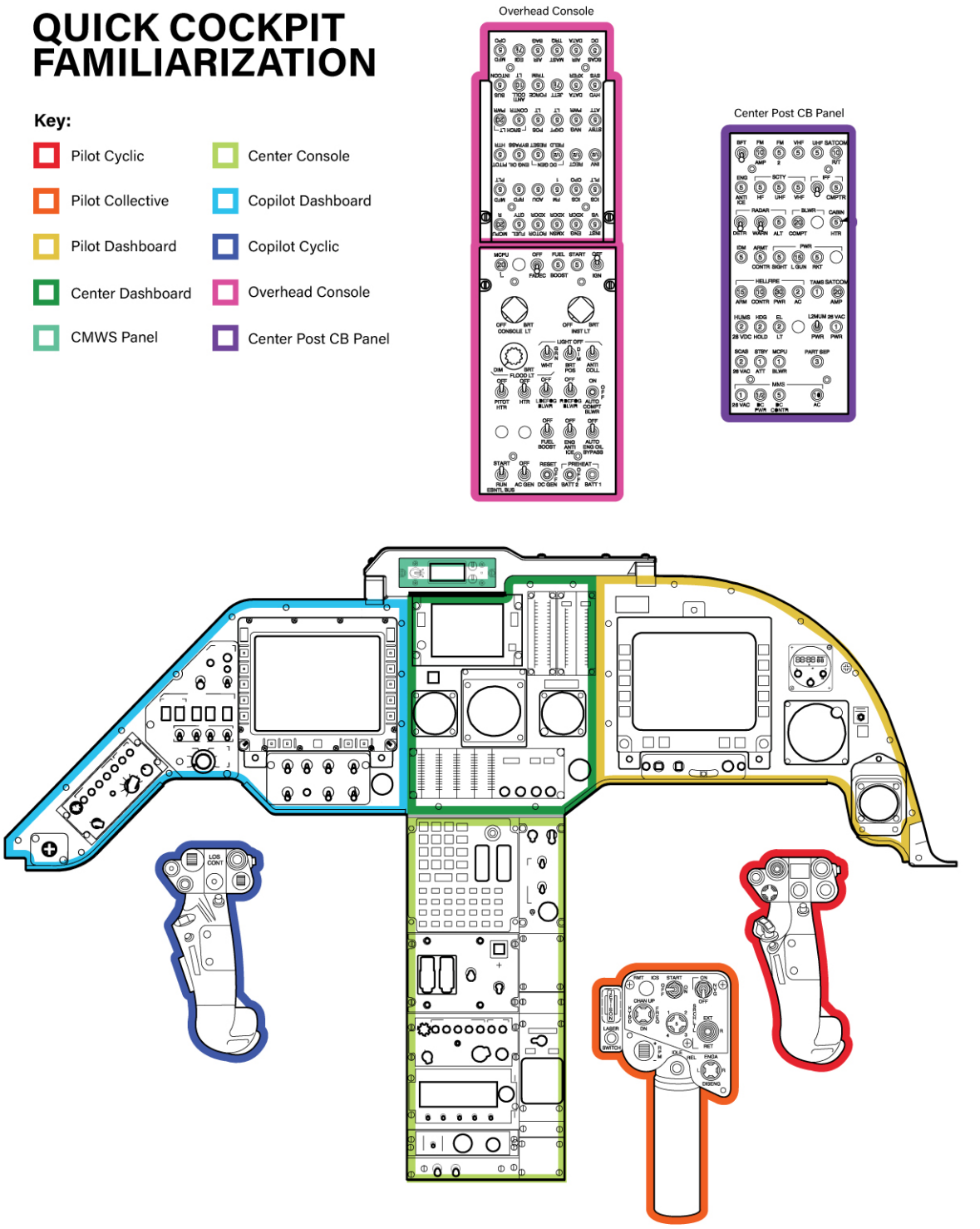


CONTROL SCHEME

THRUSTMASTER WARTHOG: THROTTLE GRIP

POLYCHOP SIMULATIONS

3.1.2.3 Quick Cockpit Familiarization



POLYCHOP SIMULATIONS

3.2 Start-up

3.2.1 Before Engine Start

1. Set **Essential Bus Switch** to START
2. Set **FADEC Circuit Breaker Switch** to ON
3. Set **Ignition Circuit Breaker Switch** to ON
4. Turn **Ignition Keylock Switch** to ON position
5. Set **Battery 1 Switch** to ON
    a. Pilot MFD, RFD, VSI and MPD will switch on
    b. Acknowledge Warning Cautions Advisories (WCA's) as required by pressing the REC/ACK switch DOWN.
6. Verify **FADEC Auto/Manual Switch** (on dash) is in AUTO

3.2.2 Engine Start

- Advance **Throttle** to IDLE position
    - The Start Switch must be activated within 60 seconds of advancing the throttle or the engine will not start. To reset, set Throttle to OFF and toggle the FADEC Circuit Breaker Switch OFF, then ON. Then restart the procedure.
- Hold **Start Switch** for two seconds then release
    - Pilot MFD may flicker as voltage drops during start process
    - Acknowledge cautions as necessary
- Wait for NG to stabilize between 63-65%
- **DC Generator Switch** – ON
- **AC Generator Switch** – ON
- **Essential Bus Switch** to RUN
- **Fuel Boost Switch** - ON
- Particle Separator Circuit Breaker – IN
- Advance Throttle to OPEN
    - Acknowledge cautions as necessary
    - Wait for TGT and RPM to stabilize
- Set **MPD Selector Switch** to display NR/NP
- Increase NP/NR to 100% using the **RPM Trim Switch**
    - Increase until one amber chiclet is visible on the NR **VSI**

3.2.3 Systems

- **SCAS Power Switch** – ON
- Set **SCAS Pitch/Roll Engage Switch** to ON
- Set **SCAS Yaw Engage Switch** to ON
- Set **MPD Selector Switch** to display FUEL QTY/ENG TRQ
- Switch Pilot MFD to **VSD** page

3.2.4 NAV Alignment

EGI alignment will start automatically in most cases. It will take a maximum of four minutes to align (and often much less time), during which time the VSD and HVR pages will display reduced information.

3.2.5 Further Systems

At this point the aircraft is ready for flight. The following systems can be enabled at the player's discretion.

- IFF Circuit Breaker Switch – ON
- Radar Detector Circuit Breaker Switch – ON
- Radar Warning Circuit Breaker Switch – ON
    - AN/APR-39 and AN/AVR-2 powerup and self-tests will begin automatically
- L2MUM Circuit Breaker Switch – ON
- CMWS CI **OFF/ON/TEST Knob** – ON
    - AN/AAR-57 powerup and self-test begins automatically
- Copilot MFD – Set to **MMS** page
- **MMS Mode Selector** – FWD
- MMS **Manual/Slave Switch** – MANUAL, then SLAVED
- Select LASER CODE by repeatedly pressing L1 bezel on Copilot MFD until the desired code is highlighted
- MMS **Laser Switch** – As desired
- **Master Arm Switch** - STANDBY

Cockpit Familiarization

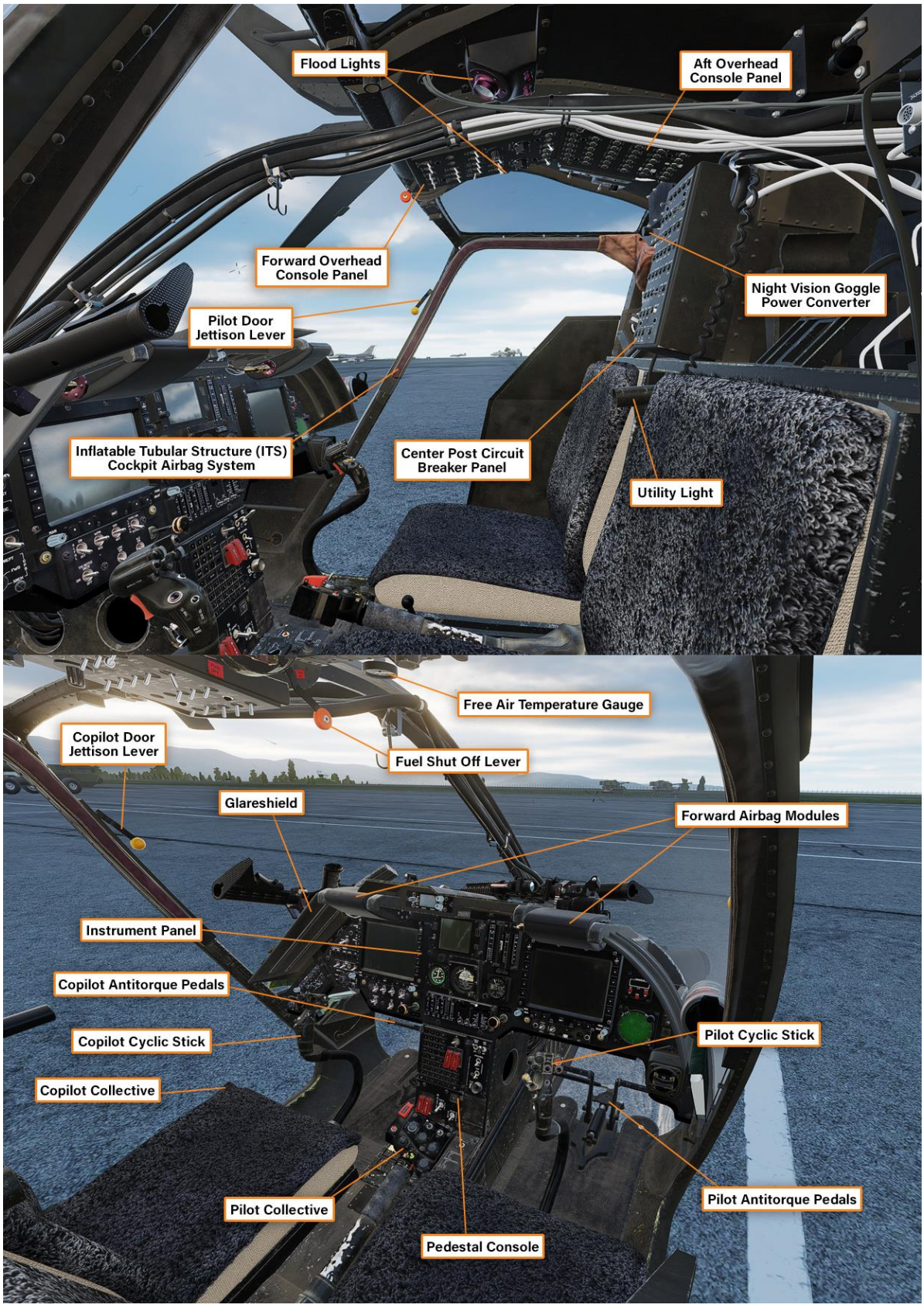


Figure 4 - Cockpit Overview 2

POLYCHOP SIMULATIONS

4.1 Pilot Instruments



Figure 5 - Pilot Instruments

POLYCHOP SIMULATIONS

4.2 Copilot Instruments

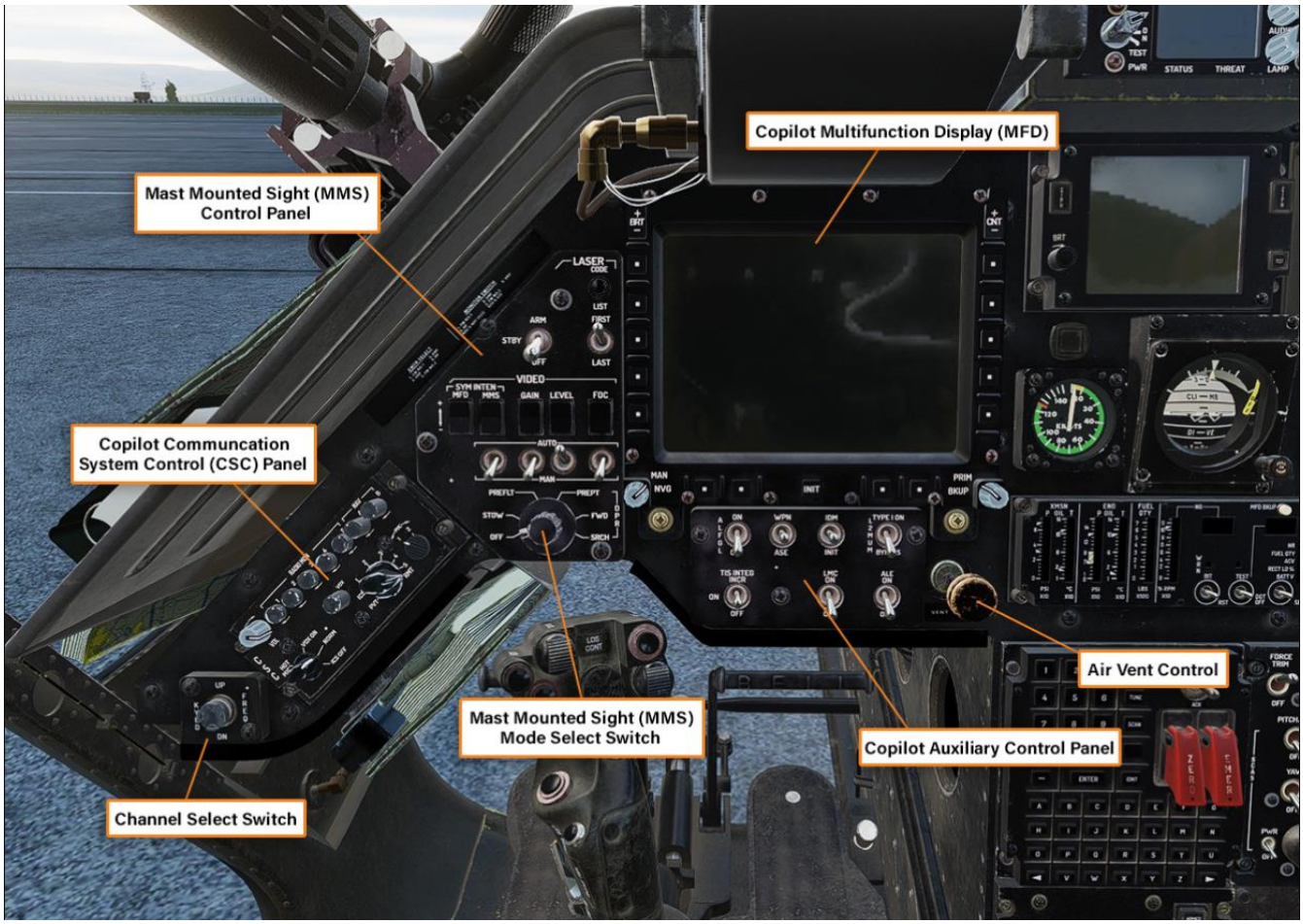


Figure 6 - Copilot Instruments

POLYCHOP SIMULATIONS

4.3 Center Console



Figure 7 - Center Console Overview

POLYCHOP SIMULATIONS

4.4 Overhead Console

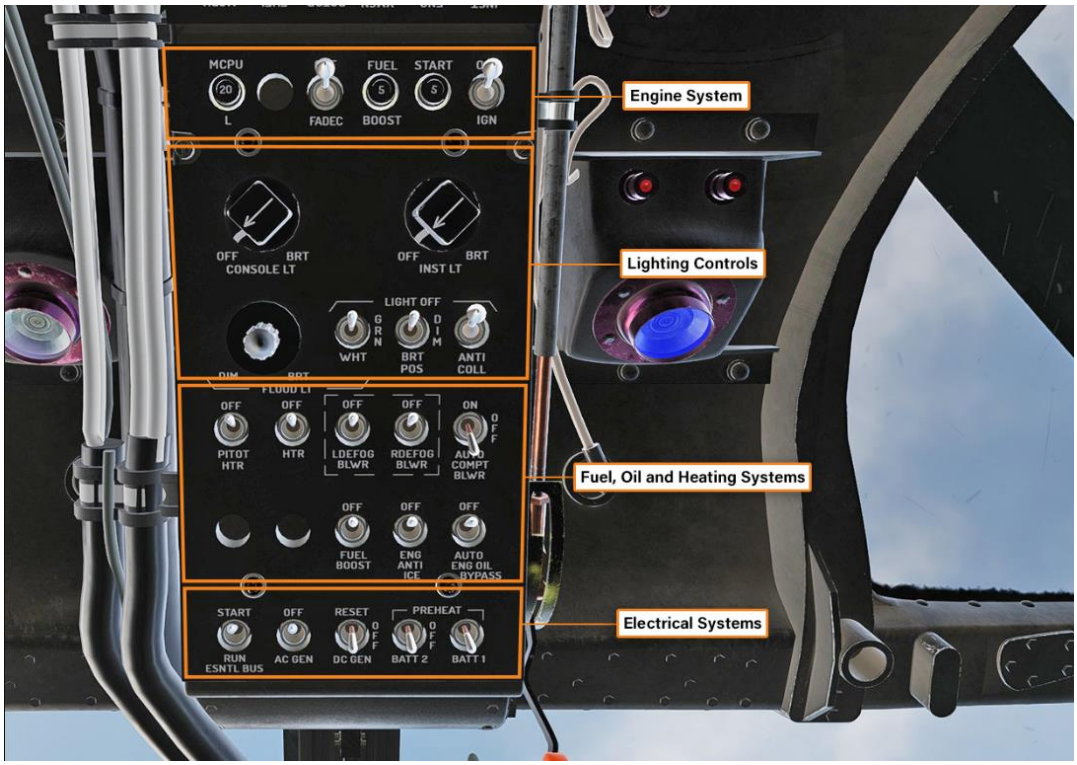


Figure 8 - Overhead Console Overview 1

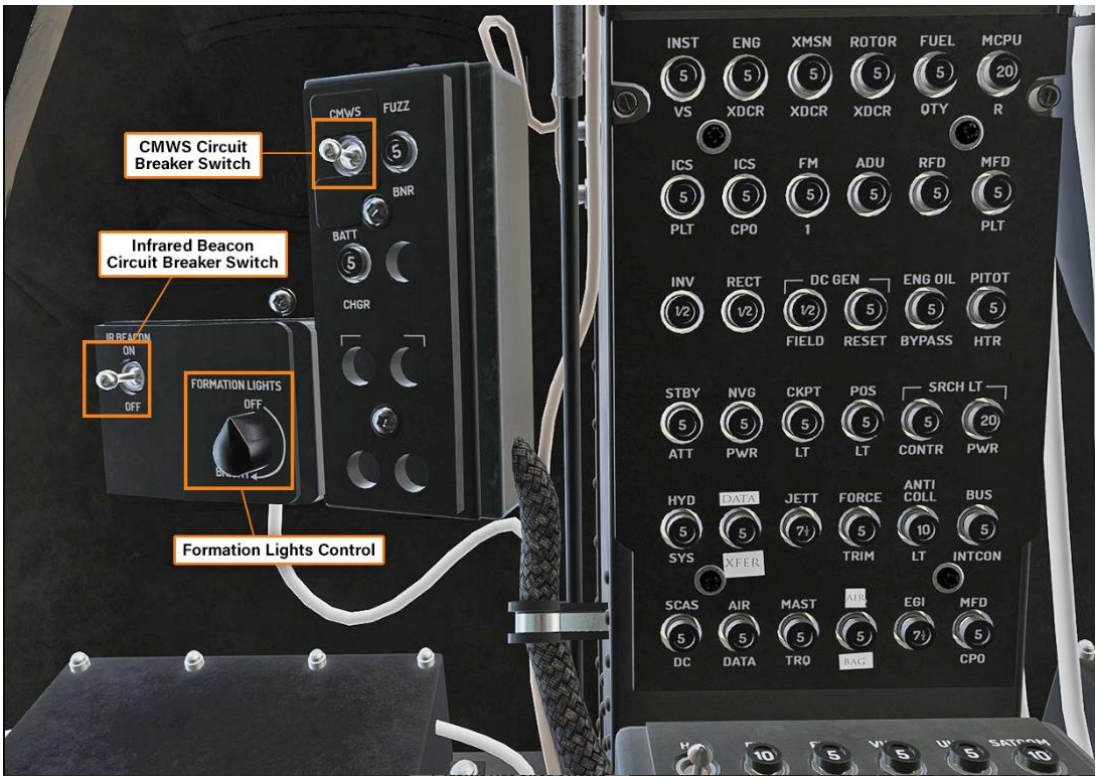


Figure 9 - Overhead Console Overview 2

POLYCHOP SIMULATIONS

4.5 Center Post Circuit Breaker Panel

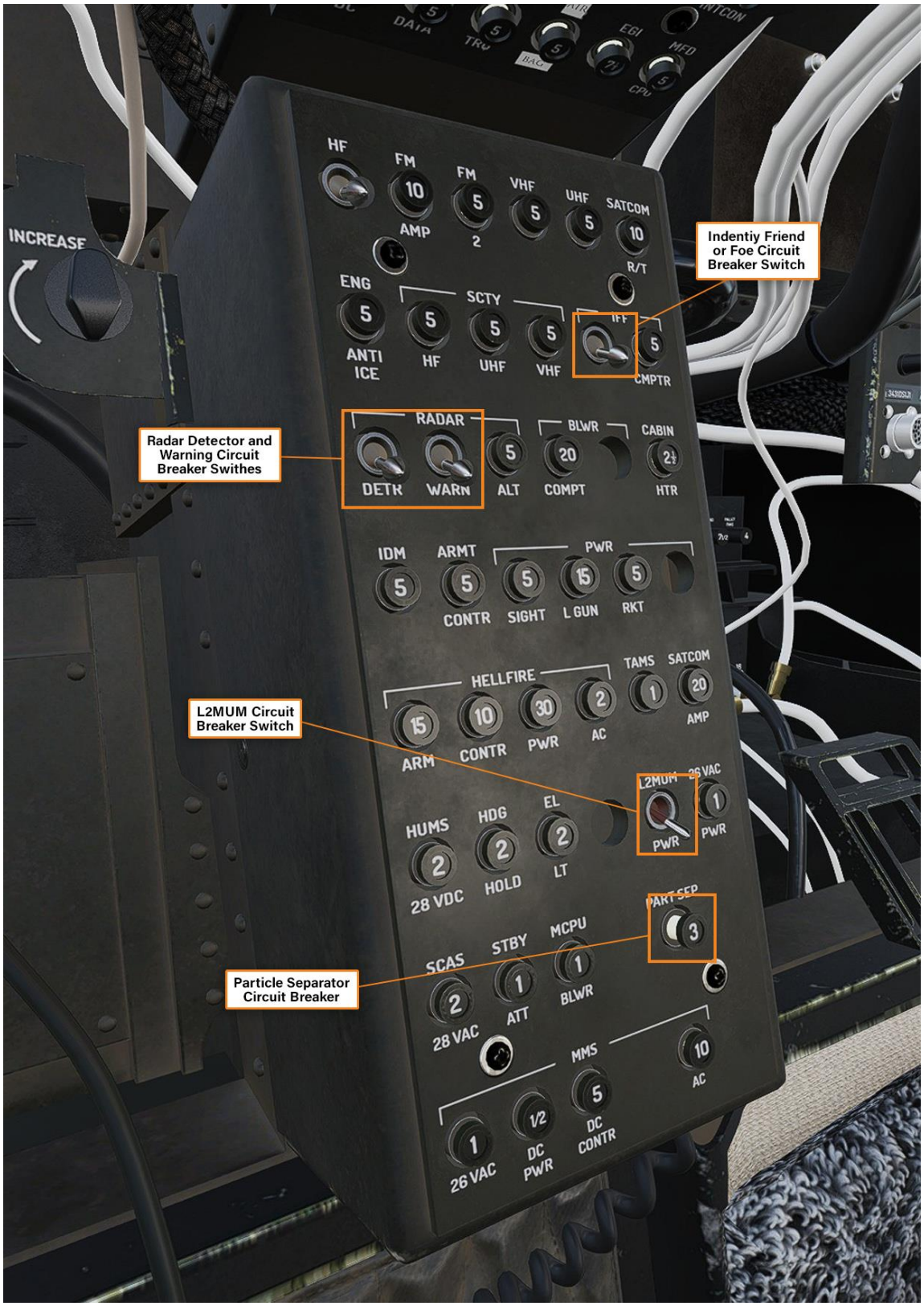


Figure 10 - Center Post Overview

POLYCHOP SIMULATIONS



Section 2

SYSTEMS

Flight Control System

5.1 Pilot Cyclic Grip

The Pilot Cyclic is primarily used for controlling the pitch and roll of the helicopter. The Cyclic Grip contains inputs for actioning weapons, making trim inputs and selecting MFD pages.

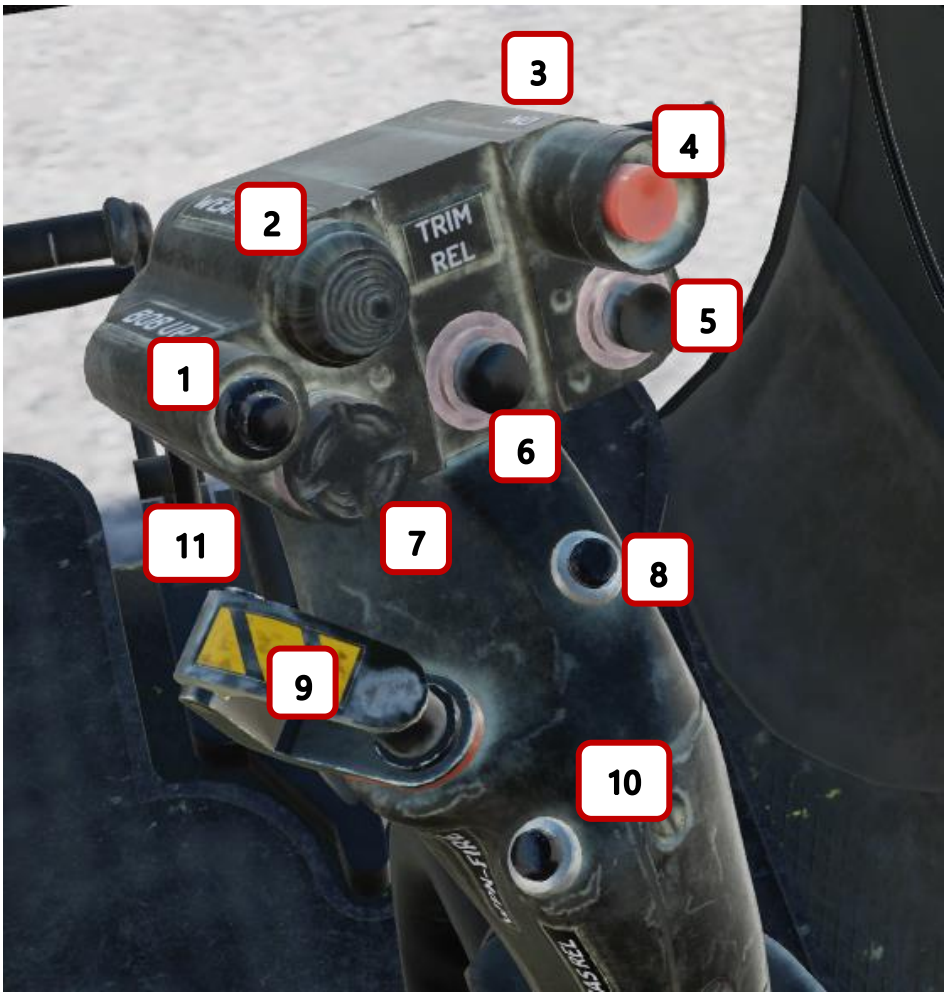


Figure 11 - Pilot Cyclic Grip

1. Bob Up/Drop Waypoint Switch
2. Weapon Select Switch
3. (Behind Grip) Optical Display Assembly Switch / ODA
4. Flare Dispense Switch
5. Missile Step Switch
6. Bob Up/Drop Waypoint Switch
7. Weapon Select Switch
8. (Behind Grip) Optical Display Assembly Switch / ODA
9. Flare Dispense Switch
10. Missile Step Switch
11. Trim Release Switch
12. Display Select Switch
13. Missile Activate Switch
14. Weapon Fire Switch
15. SCAS Release Switch
16. ICS/Radio Switch (Trigger)

POLYCHOP SIMULATIONS

5.2 Pilot Collective Grip

The Pilot Collective Grip contains a variety of inputs including Searchlight control, Radio/ICS controls and SCAS heading controls.

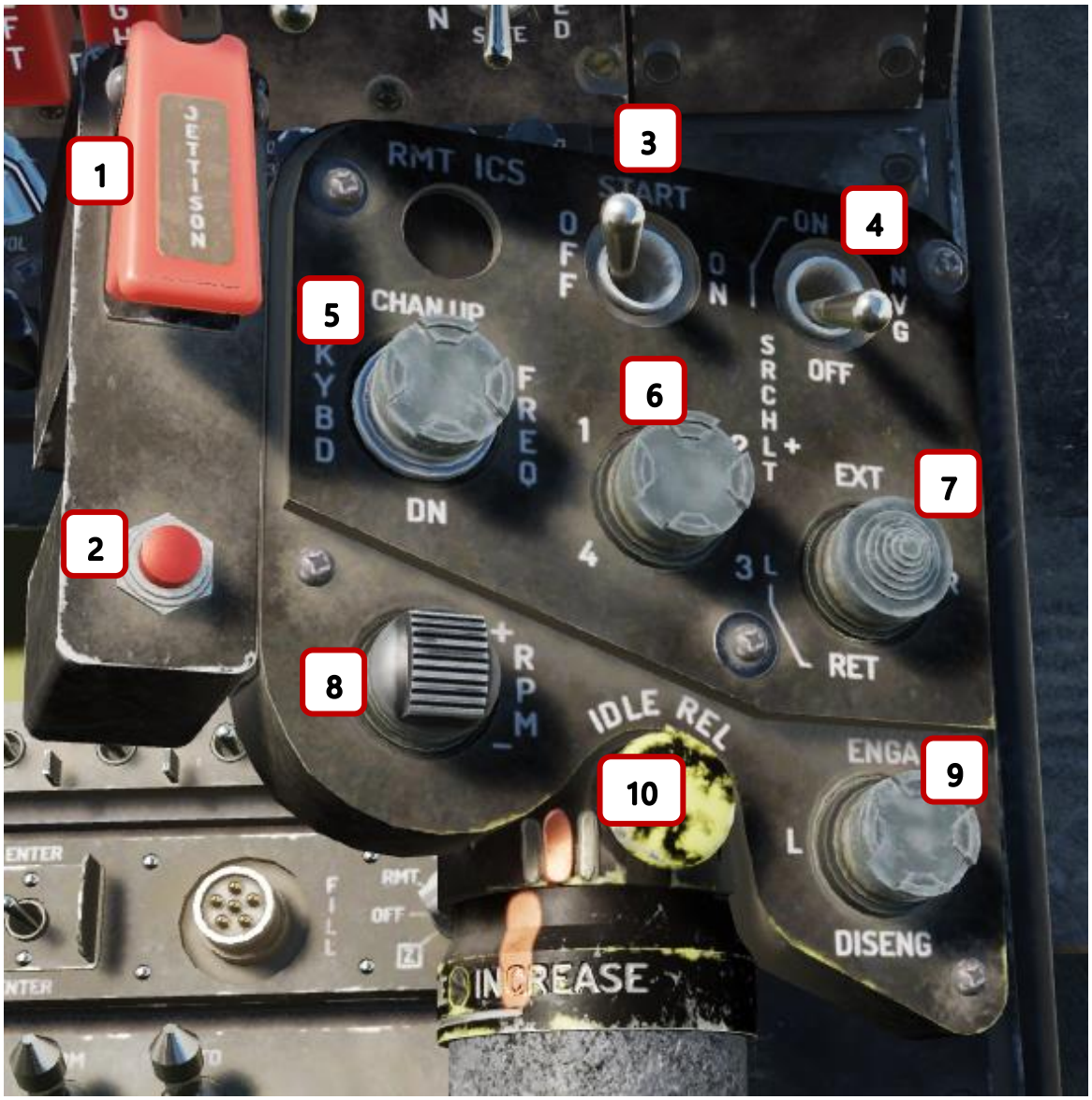


Figure 12 - Pilot Collective Grip

1. Flare Jettison Switch and Cover
2. IZLID Switch
3. Engine Start Switch
4. Searchlight ON/OFF Switch
5. Channel Select Switch
6. Radio Select Switch
7. Searchlight Control Switch
8. Engine RPM Trim Switch
9. SCAS Heading Hold/Trim Switch
10. Throttle Idle Release Switch

POLYCHOP SIMULATIONS

5.3 Copilot Cyclic Grip

The Copilot Cyclic Grip is primarily used for control of the Mast-Mounted Sight.

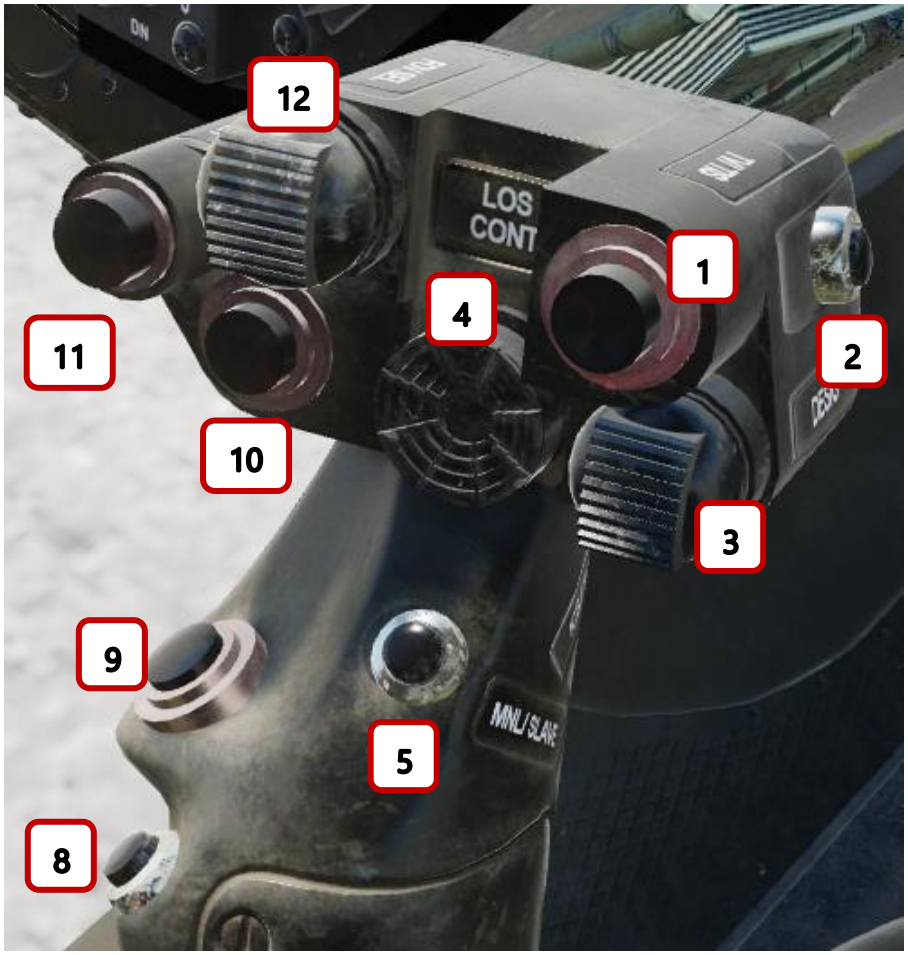


Figure 13 - Copilot Cyclic Grip

1. TV/TIS select/HOG BACK
2. MMS Frame Freeze/Target Select / HOG ON/OFF
3. NAV/Target Designate / HOG SCROLL BACKWARDS/ SCROLL FORWARD
4. LOS Slew Control
5. Manual/Slave
6. ICS/Radio Transmit (Trigger)
7. MMS Point track/ HOG Enter-Confirm (Pinky Switch)
8. SCAS Release
9. Trim Release
10. Area Track
11. Laser Fire
12. MMS FOV Select/ HOG LIST SELECT

POLYCHOP SIMULATIONS

5.4 Force Trim System

The Force Trim System is a magnetic brake system which holds the cyclic in position when activated. Force Trim does not affect the pedals or collective.

Force Trim can be activated by the Force Trim Switch on the SCAS Control Panel, and can be interrupted using the Force Trim Interrupt Switch on the Pilot or Copilot Cyclic Grips to provide a new reference point.

5.5 Stability and Control Augmentation System

The Stability and Control Augmentation System (SCAS) is a flight control augmentation system that works in three channels – pitch, roll and yaw. SCAS improves flight handling qualities and reduces pilot workload by assisting desired response characteristics and damping external helicopter inputs. The system also includes a Heading Hold Mode (HHM).

5.5.1 SCAS Control Panel

The SCAS Control Panel contains controls for the SCAS and Force Trim System.

1. Force Trim ON/OFF
2. Hydraulic System ON/OFF
3. SCAS Pitch/Roll Engage Switch
4. SCAS Yaw Engage Switch
5. SCAS Power Switch
6. SCAS Test Button



Figure 14 - SCAS Control Panel

POLYCHOP SIMULATIONS

5.5.2 SCAS Heading Hold and Trim Hat

The SCAS Heading Hold and Trim Hat controls the Heading Hold Mode (HHM). The Heading Hold Mode provides a limited heading hold and heading trim functionality.

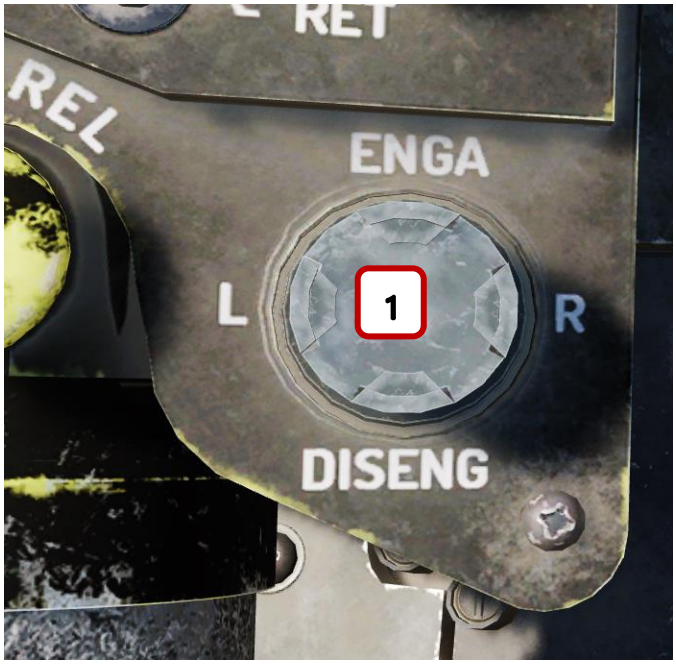


Figure 15 - SCAS Heading Hold and Trim Hat

1. SCAS Heading Hold and Trim Hat.
    I. UP: Engage Heading Hold Mode
    II. DOWN: Disengage Heading Hold Mode
    III. LEFT: Trim Helicopter Left
    IV. RIGHT: Trim Helicopter Right

POLYCHOP SIMULATIONS

Full Authority Digital Engine Control

6.1 Overview

The Full Authority Digital Engine Control (FADEC) is an electrical system that governs rotor speed, limits engine torque and provides automatic start sequencing and surge protection functions. The FADEC has a hydromechanical backup mode (Manual mode) that only provides NP overspeed protection.

6.2 FADEC Power and Control

6.2.1 FADEC Power Source

The FADEC is powered from two sources; the Battery Emergency Bus or a Permanent Magnet Alternator (PMA) located in the engine compartment. The PMA is capable of providing power to the FADEC when NP is greater than 85%. Below 85% NP power is supplied from the aircraft via the Battery Emergency Bus and the FADEC Circuit Breaker Switch.

6.2.2 FADEC Control

6.2.2.1 FADEC Circuit Breaker Switch

The FADEC Circuit Breaker Switch is located in the Overhead Console Panel and must be placed into the ON position to provide aircraft power to the FADEC during start-up and shutdown.

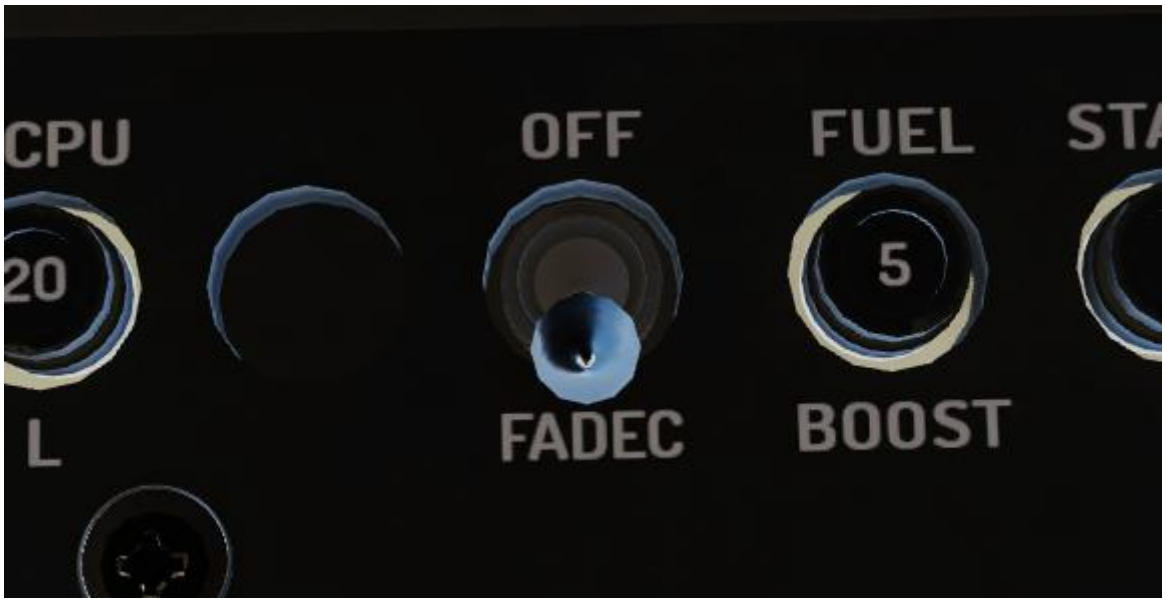


Figure 16 - FADEC Circuit Breaker Switch

POLYCHOP SIMULATIONS

6.2.2.2 FADEC Auto/Manual Switch

The FADEC can be switched between Automatic and Manual (Backup) mode using the FADEC Auto/Manual Switch located on the dashboard above the Standby Airspeed Indicator.



Figure 17 - FADEC Auto/Manual Switch

6.2.2.3 RPM Trim Switch

With FADEC in the AUTO mode, the RPM Trim Switch is located on the Pilot Collective Grip and allows adjustment of engine speed (NP) between 95%-105%.



Figure 18 - RPM Trim Switch

POLYCHOP SIMULATIONS

6.3 FADEC Modes

6.3.1 Automatic

In Automatic mode the Hydromechanical Unit (HMU) regulating fuel flow to the engine is controlled from the Electronic Control Unit (ECU). The ECU monitors engine and rotor speeds, throttle and collective positions and engine temperatures in order to maintain the set rotor RPM.

When the FADEC is in Automatic mode, the ECU will govern fuel flow when the throttle is above 65%. If the FADEC is placed in Automatic mode and the throttle is below 65%, the 'Check Throttle' advisory will be displayed on the MFD.

During start-up the ECU will govern fuel flow, starter and ignition logic to control NG acceleration and limit engine start temperatures.

The FADEC provides an overspeed protection system that will be activated at 124.5% NP and resets at 118% NP. When the overspeed protection system is activated fuel flow will be rapidly reduced until the system resets.

6.3.2 Manual

When placed into Manual mode, the HMU is controlled by the throttle. Only the overspeed protection system remains active in Manual.

WARNING:

In the event of a FADEC failure or switch from Automatic to Manual mode, the throttle should immediately be placed to ~76% to prevent rapid engine temperature and speed increases. *The pilot will then need to manually adjust throttle position to match power and fuel demand to maintain rotor RPM at 100%.*

Engine

7.1 Overview

The OH-58D Kiowa Warrior is equipped with a single Rolls-Royce Allison turboshaft engine that powers the rotors through the transmission system.

The engine is comprised of a two-stage gas producer turbine and a two-stage power turbine assembly. It is air-cooled and provides power to the aircraft via the AC and DC generators and the PMA.

7.2 Controls and Switches

7.2.1 Throttle

A twist grip throttle is available on the Pilot and Copilot Collectives.

The throttle regulates fuel flow to the engine via the Hydromechanical Unit (HMU) when the FADEC is in Manual mode or in Automatic mode and the throttle is below 65%.

The throttle includes a stop at the IDLE position to prevent accidental moving of the throttle to the OFF position.

The Idle Release Button allows the throttle to be rotated below the IDLE position.



Figure 19 - Throttle

POLYCHOP SIMULATIONS

7.2.2 Ignition Keylock Switch

The Ignition Keylock Switch is a security key switch which controls power to the ignition system. The Ignition Keylock Switch can be found under the Collective.



Figure 20 - Ignition Keylock Switch

7.2.3 Start Switch

The Start Switch is found on the Pilot Collective Grip and is a momentary switch that actuates the engine starter system.



Figure 21 - Start Switch

INFO:

The Start Switch includes a safety feature that requires the switch to be activated within 60 seconds of advancing the throttle from the OFF position or the engine will not start.

To clear this safety feature, the throttle must be placed into the OFF position and the **FADEC Circuit Breaker Switch** cycled OFF then ON.

7.2.4 Ignition Circuit Breaker Switch

The Ignition Circuit Breaker Switch is found on the Overhead Console Panel and provides electrical power to the ignition system.



Figure 22 - Ignition Circuit Breaker Switch

7.2.5 Fuel Boost Switch

The Fuel Boost Switch activates the fuel boost pump, assisting in providing fuel to the engine during critical operating periods.



Figure 23 - Fuel Boost Switch

7.2.6 Engine Anti-Ice Switch

The Engine Anti-Ice Switch controls the engine inlet anti-icing system, causing hot air to flow over the engine inlet and prevent build-up of ice.



Figure 24 - Engine Anti-Ice Switch

7.2.7 Engine Oil Bypass Switch

The Engine Oil Bypass Switch is not functional.

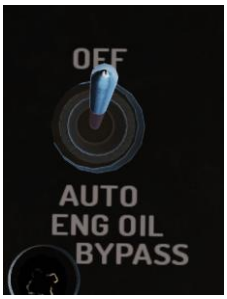


Figure 25 - Engine Oil Bypass Switch

Electrical System

8.1 Overview

8.2 Controls and Switches

The main row of switches for the Electrical System can be found on the forward portion of the Overhead Console Panel.

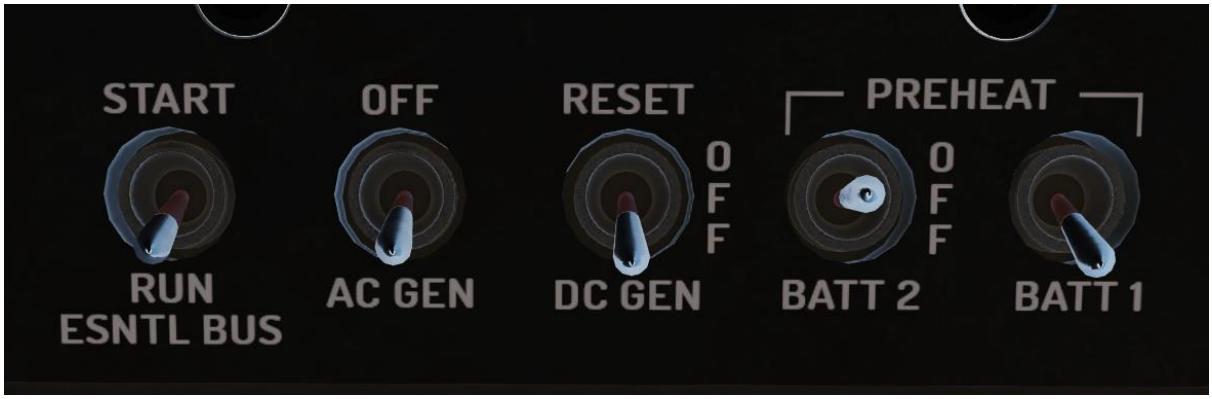


Figure 26- Electrical System Switches

8.2.1 Battery 1 Switch

The OH-58D Kiowa Warrior has a single 24 volt battery for supplying starting and emergency power. The Battery 1 switch enables battery power to be supplied to the aircraft.

8.2.2 Battery 2 Switch

The Battery 2 Switch has no function in this aircraft.

8.2.3 AC Generator Switch

The AC Generator Switch allows power to be supplied to the aircraft from the AC Generator.

8.2.4 DC Generator Switch

The DC Generator Switch allows power to be supplied from the DC Generator. The DC Generator Switch must be placed in the OFF position when connected to Ground Power.

8.2.5 Essential Bus Switch

The Essential Bus Switch allows the battery to supply power to the battery emergency bus when in the START position.

POLYCHOP SIMULATIONS

Multiparameter Display and Vertical Scale Instruments

9.1 Multiparameter Display

The MPD is mainly divided by two parts. The left part where the vertical scales display the different values. From left to right the values are: transmission pressure, transmission temperature, engine oil pressure, engine oil temperature, fuel quantity and NG or gas generator speed.

On the right side, there is boxed values, starting from the left there is again the NG value. The two other boxes represent the values selected by a green colored light in the vertical scale below, that can be changed with the MPD Selector Switch.

There are also two separate lights and the MFD Backup button. The last displays at the top part of the MFD various labels with basic information of the aircraft status.

On the other side, the light labeled by WRN will turn on every startup, if after resetting it by the Reset Switch just below it doesn't turn off, the aircraft is due a revision, also a message on the boxes at the right side will notify the nature of the error. The light of the right side is nonfunctional.

The test switch is used to test all the lights under this section of the manual (MPD and Vertical Scale).

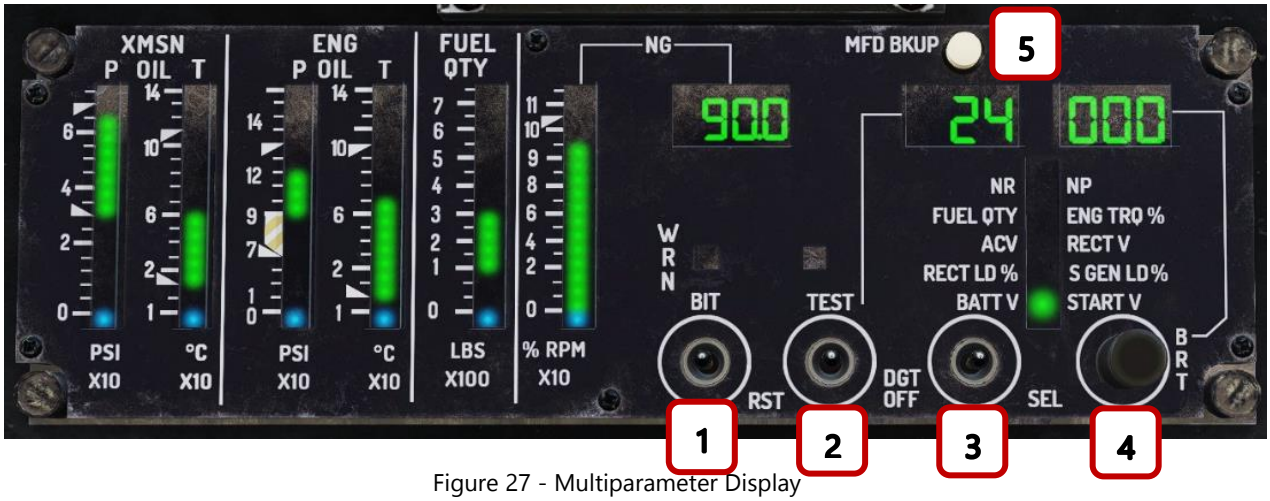


Figure 27 - Multiparameter Display

1. Built In Test / Reset Switch
2. Test Switch
3. MPD Selector Switch
4. Brightness rotator
5. MFD Backup Button

POLYCHOP SIMULATIONS

9.2 Vertical Scale Instruments

The Vertical Scale Instruments (which includes the TGT/TRQ Dual Indicator) are lighted displays found on the Pilot Instruments side of the main dashboard. From left to right they display NR (Relative Rotor RPM), NP (Relative Engine Power), Turbine Gas Temperature and Torque. Two digital displays show the Turbine Gas Temperature and Torque values.

The scales have blue, green, yellow and red lit segments.

The blue segment will display if the instrument is operational.

The remaining segments will be lit as the value of each scale is passed, for example when NR passes 100% the segment between 100% and 101% will be lit.

Yellow and red segments represent precautionary areas and maximum operating limits respectively.



Figure 28 - Vertical Scale Instruments

POLYCHOP SIMULATIONS

Standby Instruments

10.1 Standby Airspeed Indicator

The image below depicts the standby airspeed indicator.

It is operated by the differential between impact and static air pressure and requires no electrical power.



Figure 29 - Standby Airspeed Indicator

10.2 Barometric Altimeter

The following image shows the barometric altimeter, that will give the pilot a reference of altitude MSL and uses the static air pressure for measuring it.

It is positioned to the right of the standby attitude.

There is input of electricity from the power assured bus. Adjusting the pressure with the knob will also adjust it in the VSD altimeter.

1. Pressure adjustment knob.



Figure 30 - Barometric Altimeter

10.3 Standby Magnetic Compass

Provides the magnetic heading, and it is positioned at the right and below the RWR.

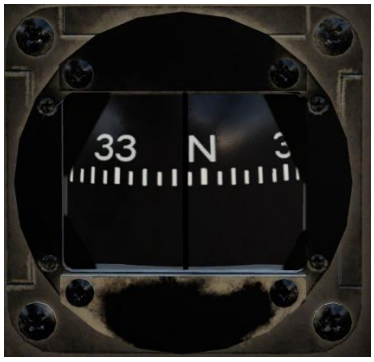


Figure 31 - Standby Magnetic Compass

POLYCHOP SIMULATIONS

10.4 Chronometer

This indicator is located at the right of the Right Screen and above the RWR indicator.

It has 3 modes:
LT (Local Time),
UTC (Coordinated Universal Time) and SW (stopwatch).

You can use the Start/Stop and Reset buttons when using the SW mode.

1. Reset/Set button.
2. Mode button.
3. Start/Stop button.



Figure 32 - Chronometer

10.5 Free Air Temperature Gauge

This gauge represents the current external temperature, and it is in front of the fuel lever on the top of the windshield.



Figure 33 - Free Air Temperature Gauge

POLYCHOP SIMULATIONS

Control Display Subsystem

11.1 Introduction

The Control Display Subsystem (CDS) is the primary interface for both Pilot and Copilot to display and control flight and navigation data, communications, sensors and weapons.

The CDS is comprised of two Improved Master Controller Processor Units (IMCPUs), two Multifunction Displays (MFDs), one Multifunction Keyboard (MFK), one Remote Frequency Display (RFD), two Optical Display Assemblies (ODAs) and one Personal Computer Data Transfer System (PC-DTS).

Other systems linked to the CDS include the Mast Mounted Sensor (MMS), AN/APR-39 Radar Detector Set, AN/AVR-2 Laser Detection Set, AN/ALQ-144 Infra-red Countermeasures Device and the AN/AAR-57 Common Missile Warning System.

11.2 Improved Master Controller Processor Unit

The Left and Right Improved Master Controller Processor Units handle all inputs to the CDS and processes and displays data to the various output interfaces.

POLYCHOP SIMULATIONS

11.3 Multifunction Display

Two Multifunction Displays housed in the instrument display function as the main output device for the CDS. The MFDs operate independently allowing different functions to be performed simultaneously by the Pilot and Copilot.



Figure 34 - Multifunction Display

Each MFD includes fifteen bezel buttons for mode and function selection, a Brightness (BRT) and Contrast (CNT) rocker for adjusting the display picture.

A Manual/NVG Mode selector allows for changing the display image between a daylight setting and an NVG compatible mode.

The Primary/Backup selector allows for selection of the alternate IMCPU as the input source in the event that the primary source fails.

POLYCHOP SIMULATIONS



Figure 35 - MFD in Manual mode



Figure 36 - MFD in NVG mode

POLYCHOP SIMULATIONS

11.4 Multifunction Keyboard

The Multifunction Keyboard (MFK) is located on the center console and is available to both crewmembers. Data entry on the MFK is shared between both MFDs and so only a single entry mode may be active at any time.

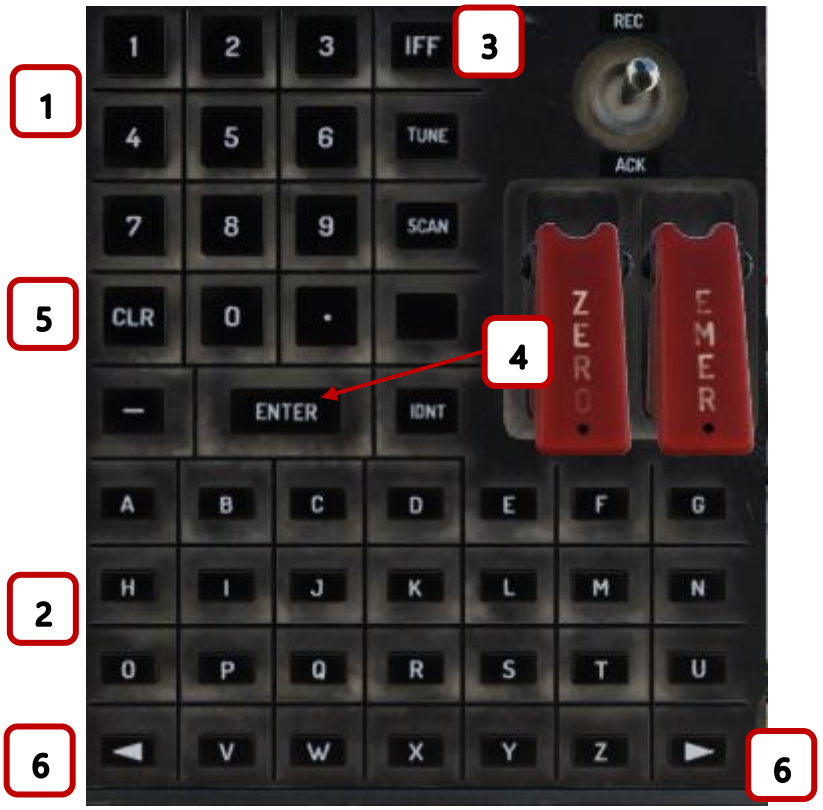


Figure 37 - Multifunction Keyboard

1. *Numeric keys.* Used to enter numerals. A decimal point and minus key are also included
2. *Alpha keys.* Used to enter characters
3. *Labelled keys*
    - IFF. Displays IFF page 1 on the Copilot MFD
    - TUNE. No function
    - SCAN. No function
    - BLANK. Inserts a blank space
    - IDNT. Broadcasts IFF identification
4. *ENTER* key. Confirms entry of input data. Exact function depends on data entry mode
5. *CLEAR* key. Clears all input
6. *CURSOR LEFT/RIGHT* keys. Advances/Retreats the cursor. Left key will delete any preceding input. Right key will add a space if at end of input

POLYCHOP SIMULATIONS

11.5 Optical Display Assembly

The Optical Display Assembly (ODA), also referenced as the ANVIS Display Symbology Subsystem (ADSS) is a helmet mounted display system that is attached to the Night Vision Goggles and provides primary flight data similar to the **VSD** and **Sparse VSD** displays.

The Pilot and Copilot have independent ODAs that can be configured from the **Initial 2** page on their respective MFDs. The display mode and data are determined by the respective MFD page and weapon selections.



Figure 38 - View through the Optical Display Assembly

POLYCHOP SIMULATIONS

11.5.1 ODA Symbology

The ODA shares many symbology components with the **VSD** and **Sparse Weapons** pages on the MFDs. The symbology displayed varies depending on the selected weapon and weapon modes. The ODA also provides declutter options.

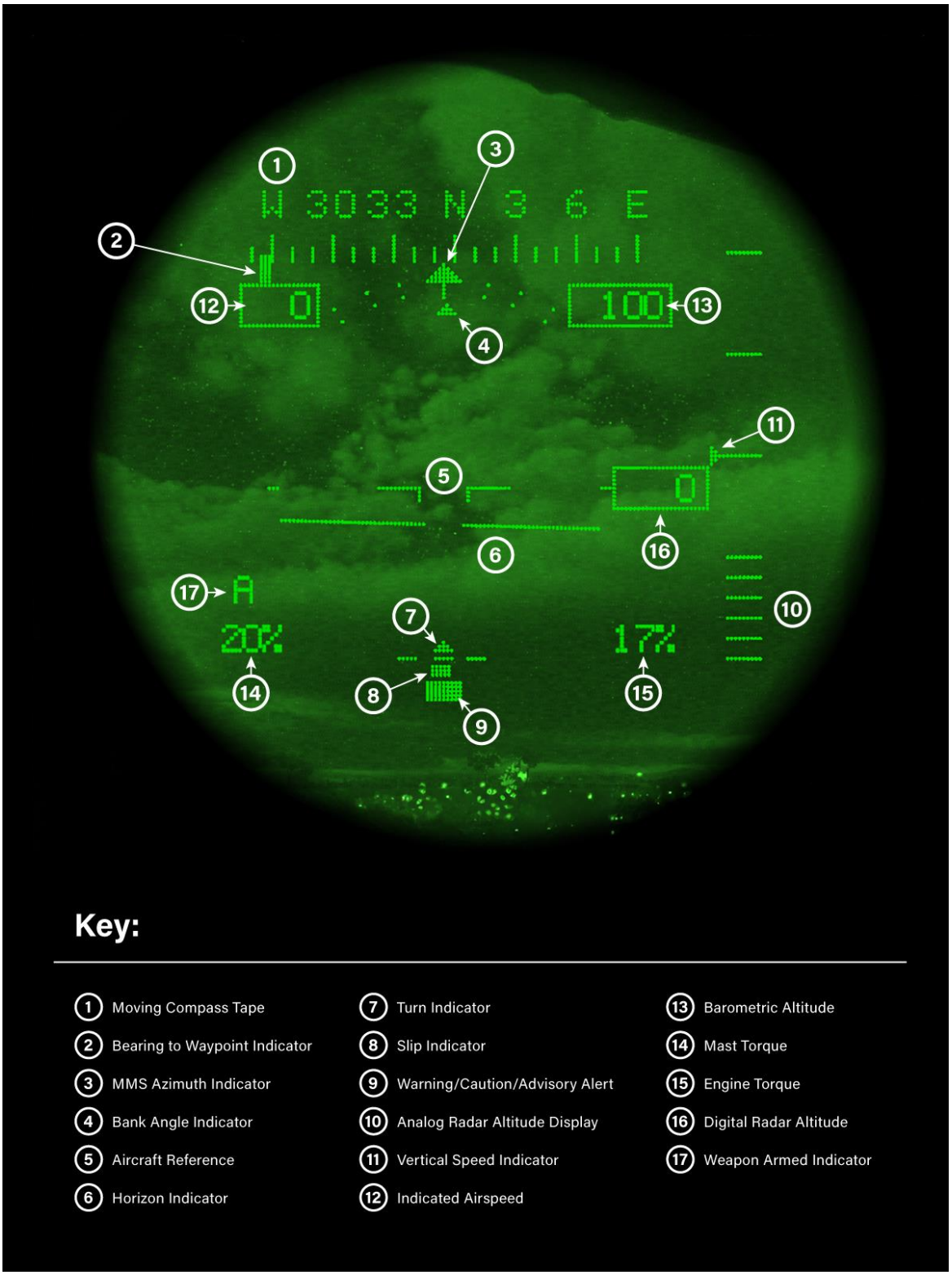


Figure 39 - ODA Symbology

11.5.2 ODA Declutter Levels

The ODA Declutter Levels can be selected from the MFD **Initial 2** page.

When Level 1 is selected, the Vertical Speed Indicator and Analog Radar Altitude display are hidden.



Figure 40 - ODA Symbology, Declutter Level 1

In Declutter Level 2, the Vertical Speed Indicator, Analog Radar Altitude Display, Indicated Airspeed, Barometric and Radar Altitude displays, Engine and Mast Torque indicators, Weapon Armament Indicator and all weapon symbologies are hidden.



Figure 41 - ODA Symbology, Declutter Level 2

POLYCHOP SIMULATIONS

11.5.3 ODA Symbology – Gun

When the Gun is selected from the Pilot's Weapon Selector switch, the symbology includes the following displays:



Figure 42 - ODA Symbology, Gun

1. MMS Line of Sight (LOS) Indicator
2. Gun Reference Cue

11.5.4 ODA Symbology - Rockets



Figure 43 - ODA Symbology, Rockets

POLYCHOP SIMULATIONS

1. Rocket Cue

11.5.5 ODA Symbology - Hellfire

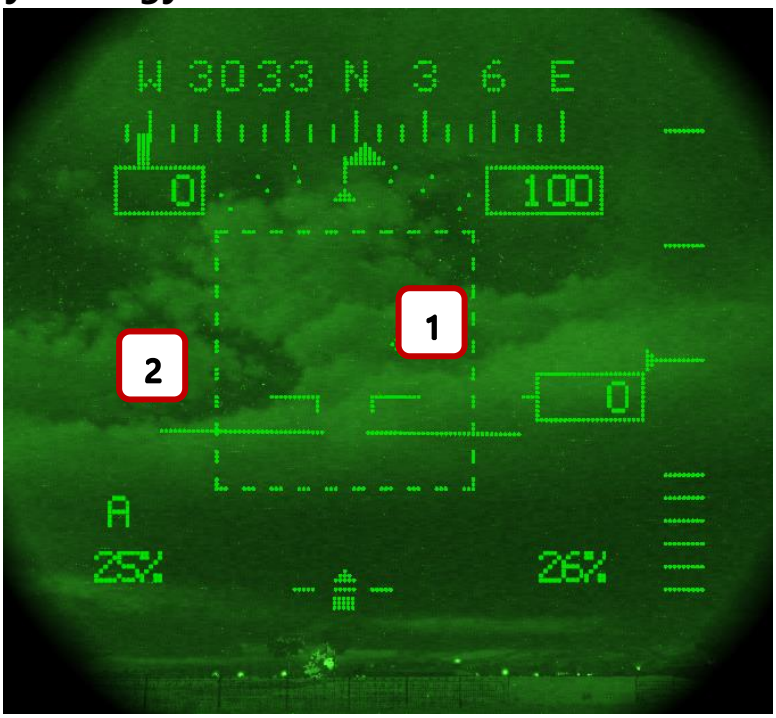


Figure 44 - ODA Symbology, Hellfire (LOBL)

1. MMS Line of Sight (LOS) Indicator
2. Hellfire Constraints Indicator (LOBL)

POLYCHOP SIMULATIONS



Figure 45 - ODA Symbology, Hellfire (LOBL, tracking within constraints)

11.5.6 ODA Symbology - ATAS

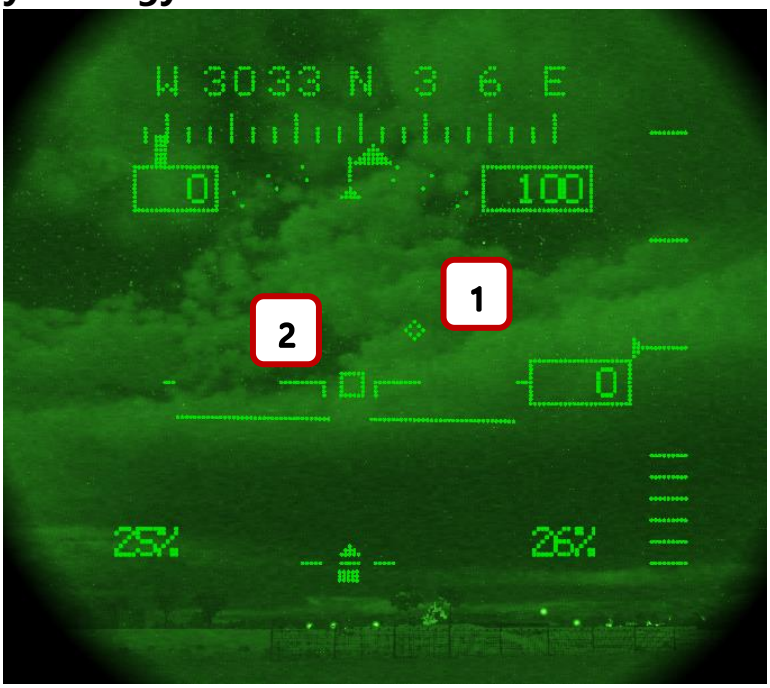


Figure 46 - ODA Symbology, ATAS (Caged)

1. MMS Line of Sight (LOS) Indicator
2. ATAS Seeker Cue

POLYCHOP SIMULATIONS



Figure 47 - ODA Symbology, ATAS (Uncaged, Tracking)

3. Super elevation Bar

11.6 Pilot Display Unit

The Pilot Display Unit (PDU) is mounted to the helicopter airframe above the pilot's windshield.

The PDU can be adjusted vertically for differing pilot eye levels. The PDU control panel has a BRT knob which allows pilot to adjust display brightness and a TEST button which initiates the built-in test (BIT) of the PDU and Electronics Unit (EU).

The PDU provides symbology and weapon system status for ATAS and symbology only for 2.75-inch rocket and .50 caliber machine gun weapon systems.

POLYCHOP SIMULATIONS

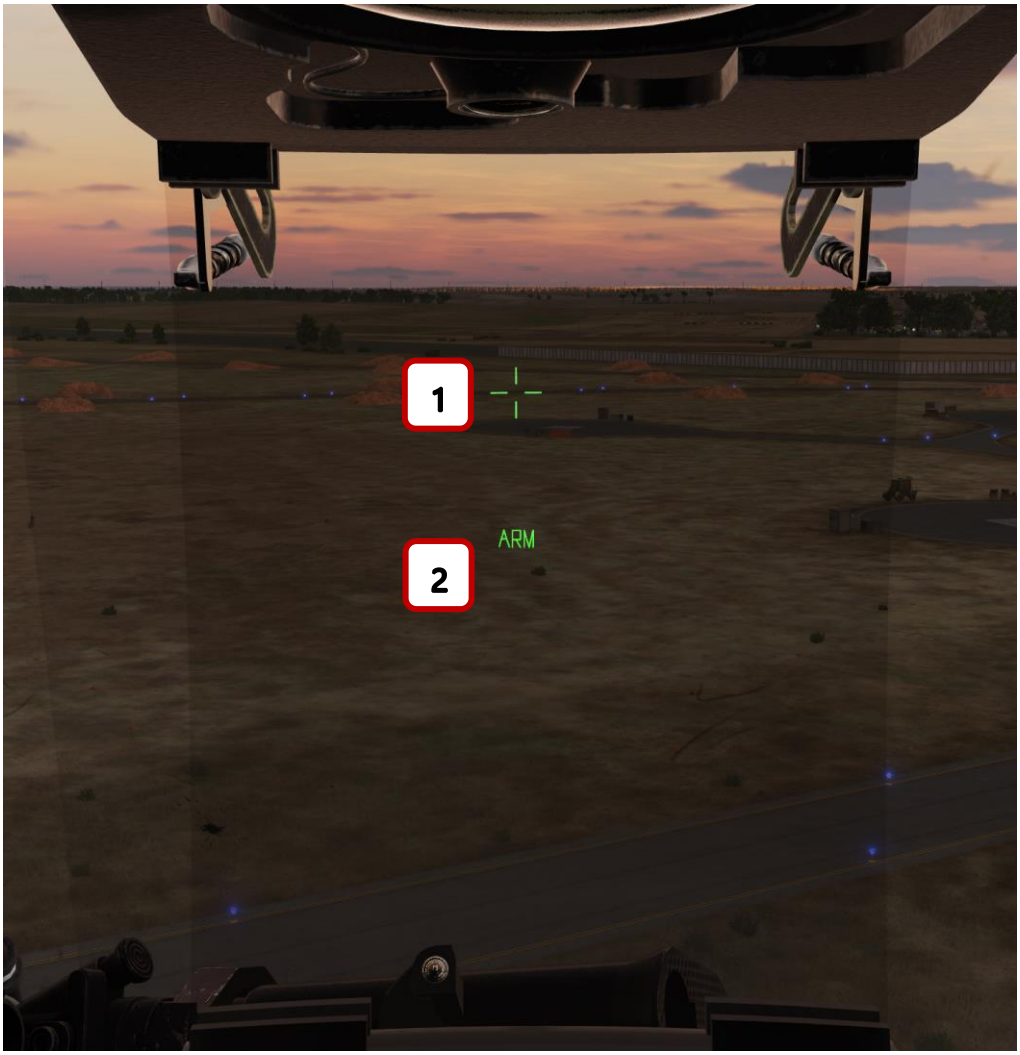


Figure 48 - PDU Gun Reticule

1. Gun crosshair
2. Master Arm status

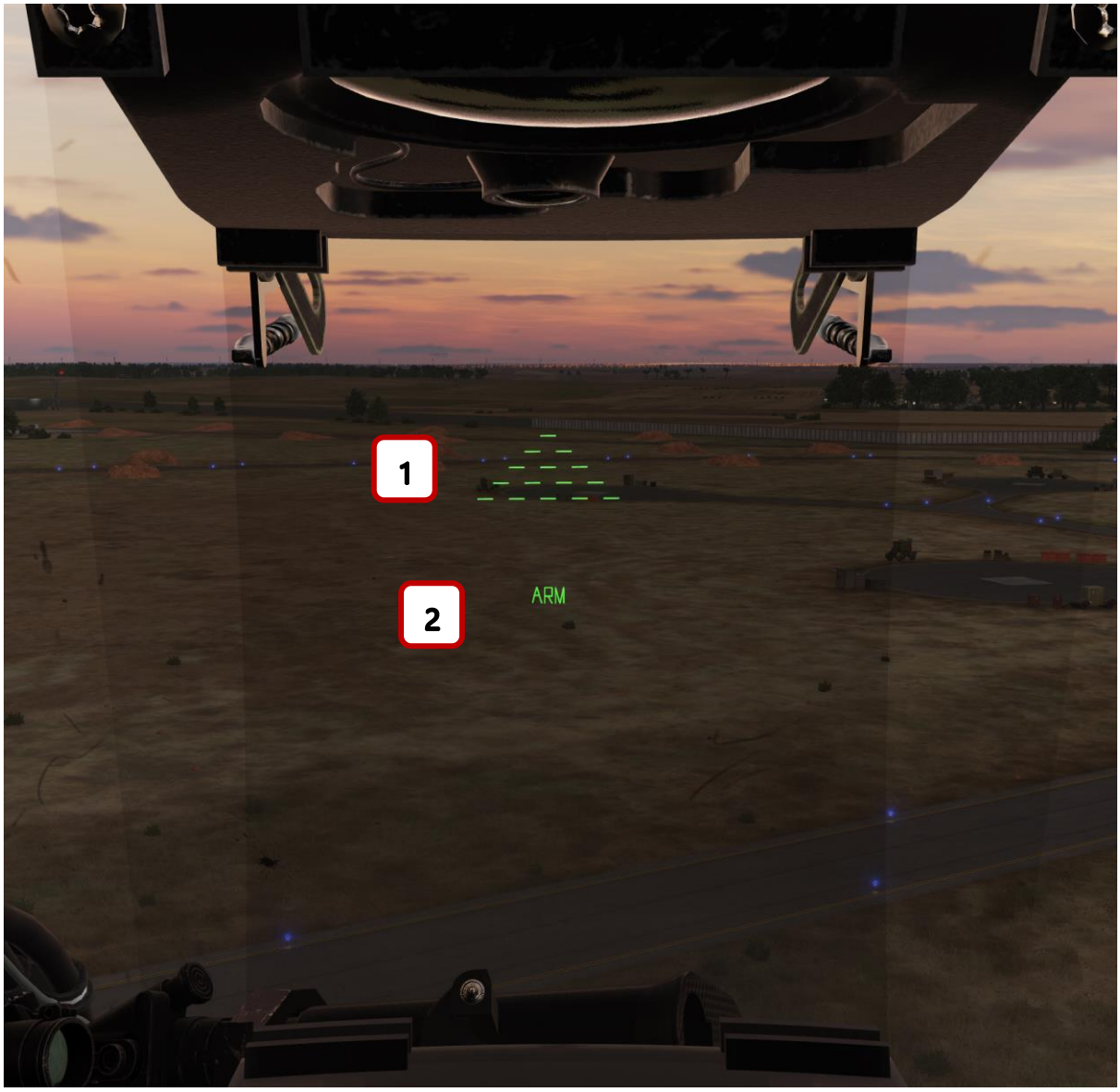


Figure 49- PDU Rockets Reticule

3. Projected rocket impact area / beaten zone
4. Master Arm status

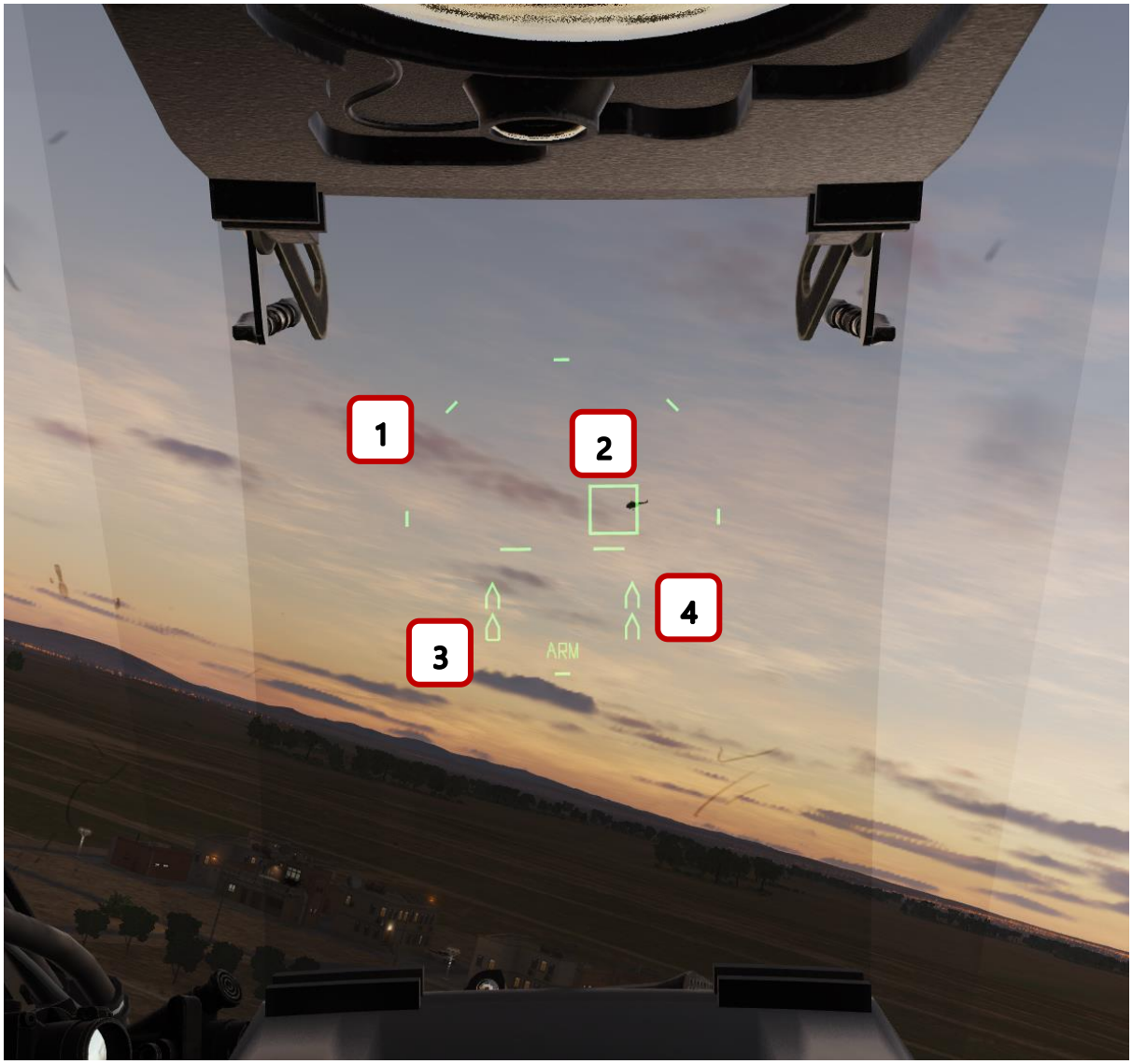


Figure 50- PDU ATAS Reticule

1. ATAS 20-degree Field of View outline
2. ATAS Tracking Reticule
3. Missile Selected (closed icon at bottom left)
4. Missile Present (open icons)

11.7 Personal Computer Data Transfer System

The Personal Computer Data Transfer System (PC-DTS) is a data storage system that allows for various navigation, weapons, radio communications and mission data to be stored to and loaded from a memory card.

The DCS OH-58D Kiowa Warrior module emulates this feature by allowing players to save and load this same data to and from JSON files that can be edited and shared offline.

For more information on using the PC-DTS, refer to the **Data Transfer System** process documentation, and the **Data Loader**, **Load Mission** and **Store Mission** MFD pages.

11.8 MFD Auxiliary Control Panels

Both Pilot and Copilot have MFD Auxiliary Control Panels that include extra inputs for MFD control.

11.8.1 Pilot MFD Auxiliary Control Panel

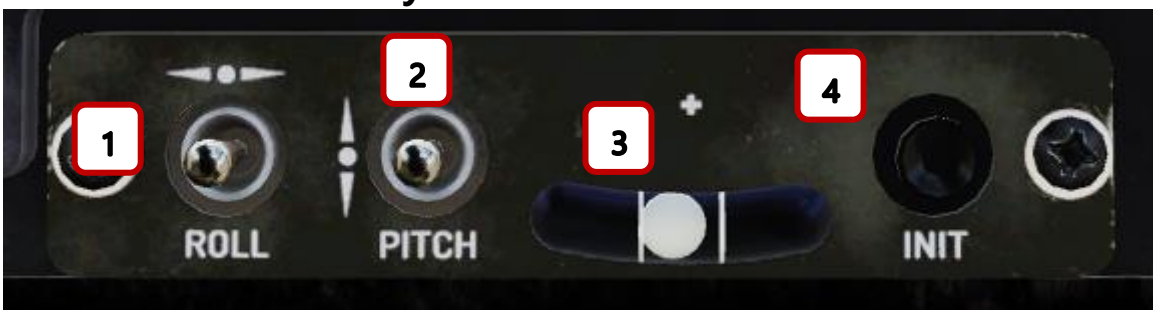


Figure 51 - Pilot MFD Auxiliary Control Panel

1. VSD Roll Correction – used to adjust the roll angle as displayed in the **VSD**. Pressing and holding the switch adjusts the roll angle in the direction of the switch. The roll angle adjustment is limited to +/- five degrees
2. VSD Pitch Correction – adjusts the pitch angle as displayed in the **VSD**. Pressing and holding the switch adjusts the pitch angle in the direction of the switch. The pitch angle adjustment is limited to +/- ten degrees
3. Standby Slip Indicator
4. INIT button – enters the **Initial 1** page on the Pilot's MFD

POLYCHOP SIMULATIONS

11.8.2 Copilot MFD Auxiliary Control Panel

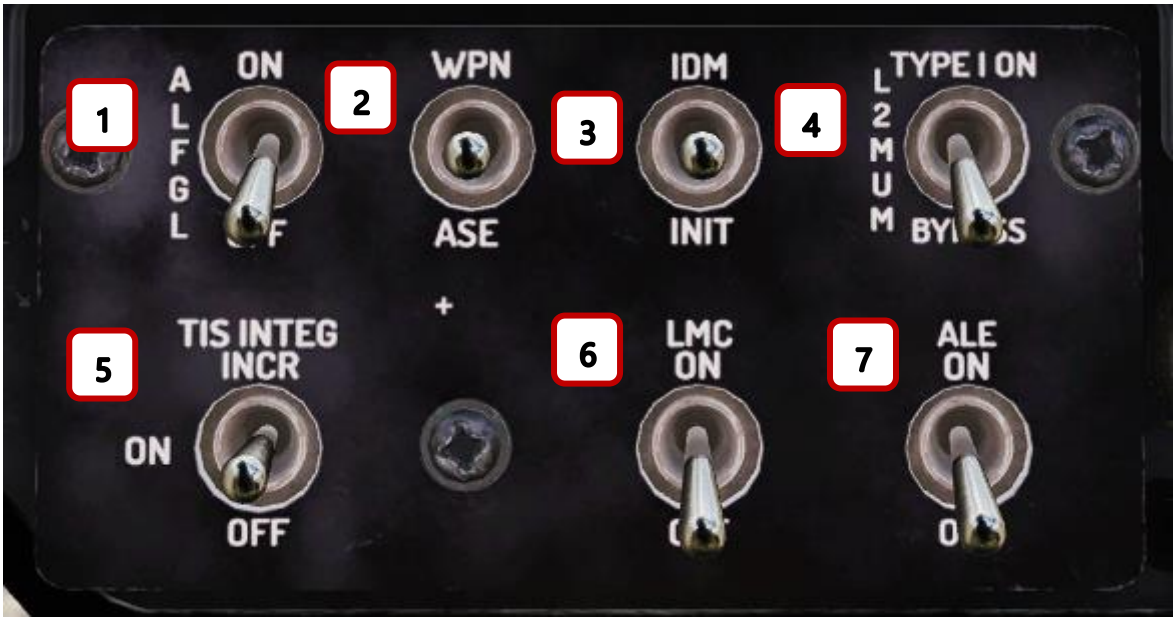


Figure 52 - Copilot MFD Auxiliary Control Panel

1. AUTOMATIC LOW FREQUENCY GAIN LIMITING – no function in DCS
2. WPN/ASE – Enters the **Weapons** page or the **ASE Setup** page on the Copilot's MFD. Successive presses of WPN will further cycle the Copilot's MFD through the available **Copilot Sparse Weapons** pages if weapons are installed.
3. IDM/INIT – IDM has no function in DCS. INIT will enter the **Initial 1** page on the Copilot's MFD
4. L2MUM TYPE 1 ON/BYPASS – no function in DCS
5. TIS INTEGRATION INCREMENT/ON/OFF – no function in DCS
6. LINEAR MOTION COMPENSATION ON/OFF – toggles Linear Motion Compensation when MMS is in Manual mode
7. AUTOMATIC LEVELING EQUALIZATION – no function in DCS

11.9 MFD Interaction and Pages

11.9.1 Common Elements

This section covers MFD elements that are present on most or all pages.

11.9.1.1 Bezel Interaction

Each MFD is surrounded by fifteen input keys or 'bezels'. There are five input bezels on the left and right side of the display, referred to as L1-L5 and R1-R5 from top to bottom. Along the bottom are four input bezels, referred to as B1-B4 from left to right, and an 'INIT' bezel in the center.

The INIT bezel on the Pilot's MFD has no function.

The INIT bezel on the Copilot's MFD is used to toggle between the display of the CDS and L2MUMS.

INFO:

The INIT bezel should not be confused with the nearby INIT button on the Pilot's MFD Auxiliary Control Panel, nor the INIT switch on the Copilot's MFD Auxiliary Control Panel.

POLYCHOP SIMULATIONS

11.9.1.2 Mode Keys (Bezels B1-B4)

The MFD Mode Keys are the primary method to access the top-level pages of the VSD, HSD, MMS and COMM pages. Each is accessed using the relevant bezel. The indicated top level pages may be accessed from any other page. In some cases, the bezel text is not displayed on the page, but the bezels remain functional.



Figure 53 - MFD Mode Keys

11.9.1.3 Data Entry Mode

Data Entry Mode is activated when the user is required to enter data into the CDS. The type of data required depends on the selected page and bezel. For information on valid data see the corresponding MFD page documentation.

Upon activation of Data Entry Mode, a cursor will appear next to the corresponding bezel and inputs from the **Multifunction Keyboard** will be accepted and displayed.



Figure 54 - Data Entry Mode Example with Cursor

For specific information on the behavior of each input, see **Multifunction Keyboard.**

Upon successful validation of the input data, data entry mode is cancelled.

If the data is invalid, the cursor will remain and the blinking text 'ERR' will be displayed. The data can be corrected by clearing and re-inputting the entry.

Selection of another bezel will cancel Data Entry Mode and the input data will be cleared.

Activation of Data Entry Mode on the other MFD will cancel Data Entry Mode on the primary MFD and input data will be cleared. The **Multifunction Keyboard** can only be used to enter data on the MFD that is currently in Data Entry Mode.

POLYCHOP SIMULATIONS

11.9.1.4 Top Level Display

The top display is present on all pages. It contains the following:

- Weapon Status
- Heading Hold/SCAS Airspeed Invalid Alert
- Throttle Position Indicator
- Backup Instruments indicating Engine RPM, Engine Torque and Mast Torque

The visibility of the Backup Instruments can be controlled by the MFD BACKUP switch on the MPD.

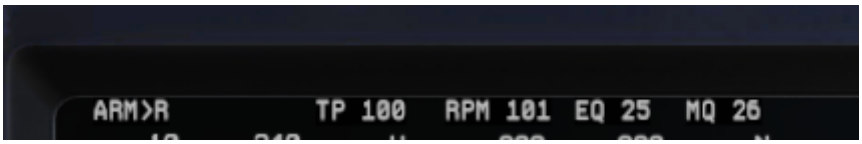


Figure 55 - MFD Backup Instruments

11.9.1.5 Heading Tape

The Heading Tape is present on the VSD, HVR, MMS and RMS pages as well as the various Sparse Weapons pages.

The Heading Tape is comprised of a linear compass scale, a lubber line indicating the aircraft heading, a chevron indicating the bearing to selected navigation point, and the MMS Azimuth Indicator.



Figure 56 - MFD Heading Tape

11.9.1.6 MMS Azimuth & Elevation Indicator

The MMS Azimuth & Elevation Indicator shows the azimuth and elevation of the MMS relative to the aircraft in degrees. An arrow is displayed next to the digits when they are greater or less than zero in order to indicate direction.

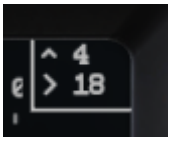


Figure 57 - MMS Azimuth & Elevation Indicator

POLYCHOP SIMULATIONS

11.9.1.7 Warnings, Cautions, Advisories

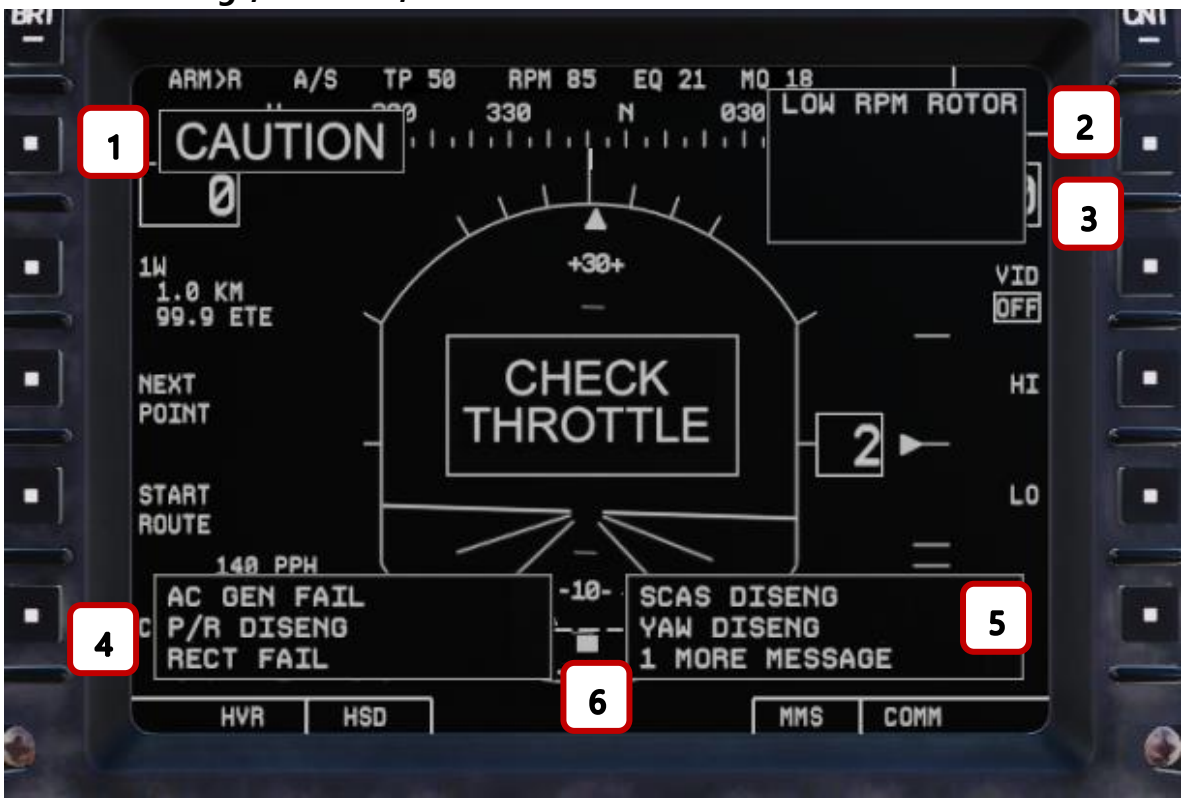


Figure 58 - MFD Warnings, Cautions, Advisories

The warning, caution, and advisory system visually and audibly advises the crew of specific faults in the helicopter systems.

1. CAUTION Annunciator
2. WARNING Annunciator
3. WARNING Messages
4. CAUTION Messages
5. ADVISORY Messages
6. Total of messages left to be displayed/read

Pressing the REC/ACK switch to the ACK position will delete the message from the screen. The total number of active advisories is displayed to the right of the letter A at the bottom of the screen.

INFO:

When a warning condition occurs, the warning is displayed, accompanied by a warning audio. The FADEC FAIL warning consists of a high/low tone.

The LOW RPM ROTOR warning consists of a continuous whoop.

The audio signal for all other warnings is a continuous three “bongs” per second. The audio signal can be silenced by moving the REC/ACK (recall/acknowledge) switch on the MFK to the ACK position

The Full list of **Warnings Cautions & Advisories** is included in the **Appendix**

11.9.1.8 Hands-On General User Interface (HOG)

The Hands-On General User Interface (HOG) is only available on the left MFD and can only be controlled using the Copilot Cyclic Grip. It provides an interface to various data-sharing and target management features. More information on the HOG can be found in the **Improved Data Modem** procedures.

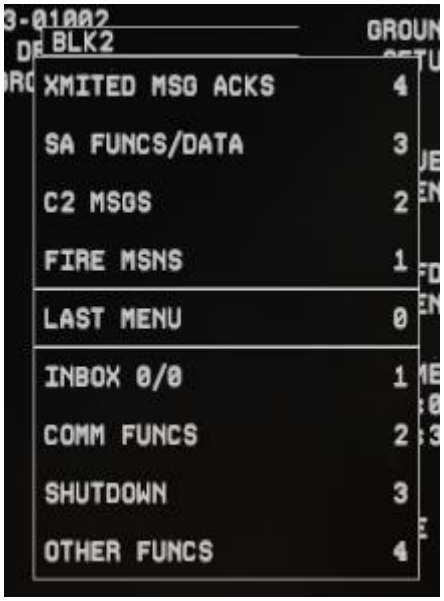


Figure 59 - MFD Hands-On General User Interface

POLYCHOP SIMULATIONS

11.10 INITIAL Pages

11.10.1 Initial 1

The Initial page is the page loaded when the CDU receives power. It is used to access performance, setup and management pages for various CDU systems. The Initial page can also be accessed at any time on either MFD by pressing the INIT button on the respective **MFD Auxiliary Control Panel**.

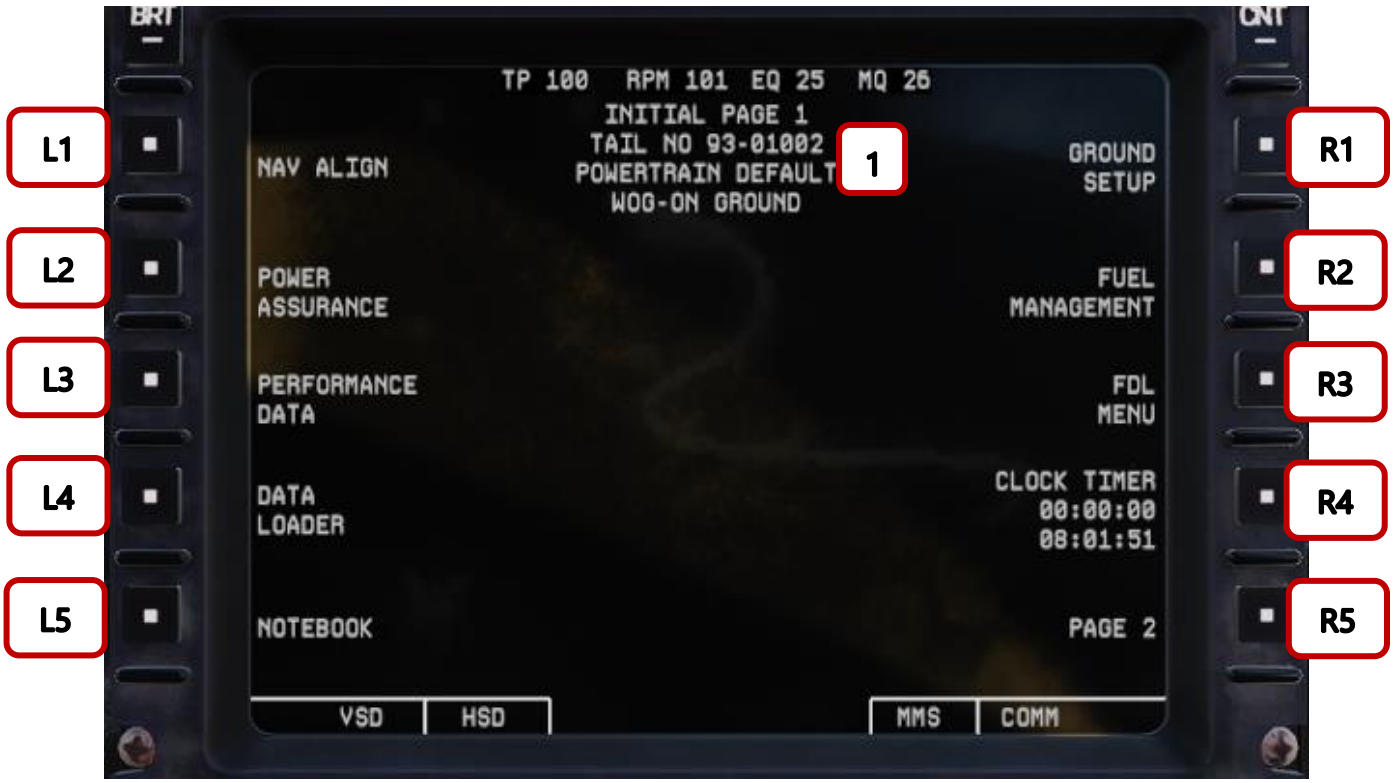


Figure 60 - Initial page 1

1. Aircraft Tail Number, Powertrain setting and Weight on Ground status

L1. *NAV ALIGN*. Enters the **NAV Align** page

L2. *POWER ASSURANCE*. Enters the Power Assurance page

L3. *PERFORMANCE DATA*. Enters the Performance Data page

L4. *DATA LOADER*. Selecting Data Loader will activate Data Entry mode in order to enter the password to access **PC-DTS** functions. Validation of the

R1. *GROUND SETUP*. Enters the **Ground Setup** page

R2. *FUEL MANAGEMENT*. Enters the **Fuel Management** page

R3. *FDL MENU*. Enters the **Fault Detection/Location Menu** page

R4. *CLOCK TIMER*. Displays the Mission Elapsed Time and the Local or Zulu Time (depending on selection). Activation of the bezel enters the **Clock Timer** page

POLYCHOP SIMULATIONS

correct password (**58D**) will access the **Data Loader** page. The password only needs to be entered upon the first time selection of the Data Loader bezel

L5. *NOTEBOOK*. Enters the **Notebook** page

R5. *PAGE 2*. Enters the **Initial 2** page

11.10.2 Initial 2

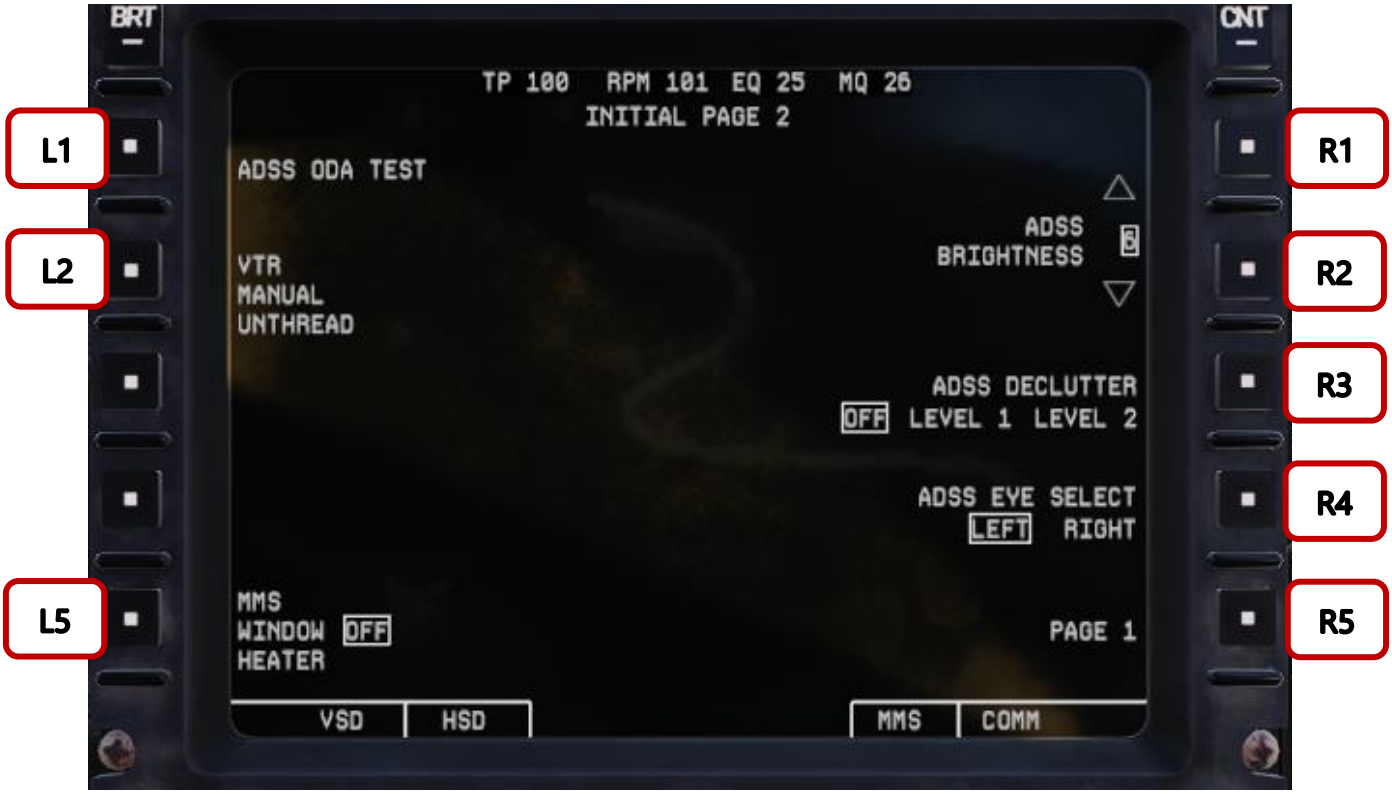


Figure 61 - Initial page 2

L1. *ADSS ODA TEST*. Toggles the TEST page on the **ODA**

L2. *VTR MANUAL UNTHREAD*. No function

L5. *MMS WINDOW HEATER ON/OFF*. No function

R1. *ADSS BRIGHTNESS INCREASE*. Increases the brightness of the **ODA**

R2. *ADSS BRIGHTNESS DECREASE*. Decreases the brightness of the **ODA**

R3. *ADSS DECLUTTER*. Activation of the bezel toggles **ODA** declutter between OFF, LEVEL 1 and LEVEL 2

R4. *ADSS EYE SELECT*. Toggles **ODA** between LEFT and RIGHT eye display mode

R5. *PAGE 1*. Enters the **Initial 1** page

POLYCHOP SIMULATIONS

11.10.3 NAV Align



Figure 62 - NAV Align page

L1 + L2. Allows entry of a position using either Latitude/Longitude or UTM dependent on selection at R5. A point identifier such as a waypoint, target point or control point may also be used

L3. *MAG HDG*. No function

L4. *ELEV*. Allows entry of an elevation in feet or meters

L5. *DATUM/SPHEROID*. No function

R1. *AUTO/MANUAL*. Allows selection of AUTO or MANUAL alignment mode. If boxed, indicates that alignment in the boxed mode is currently in progress

R2. *FAST*. Selects FAST alignment mode. If boxed, indicates that FAST alignment is currently in progress

R3. *NAV MODE*. Enters **Nav Mode Select** page

R4. *SHIP ALGN*. Selects SHIP ALIGN alignment mode. If boxed, indicates that SHIP ALIGN is currently in progress

R5. *UTM/LAT/LONG*. Toggles coordinate mode between UTM and Lat/Long

POLYCHOP SIMULATIONS

11.10.4 NAV Mode Select

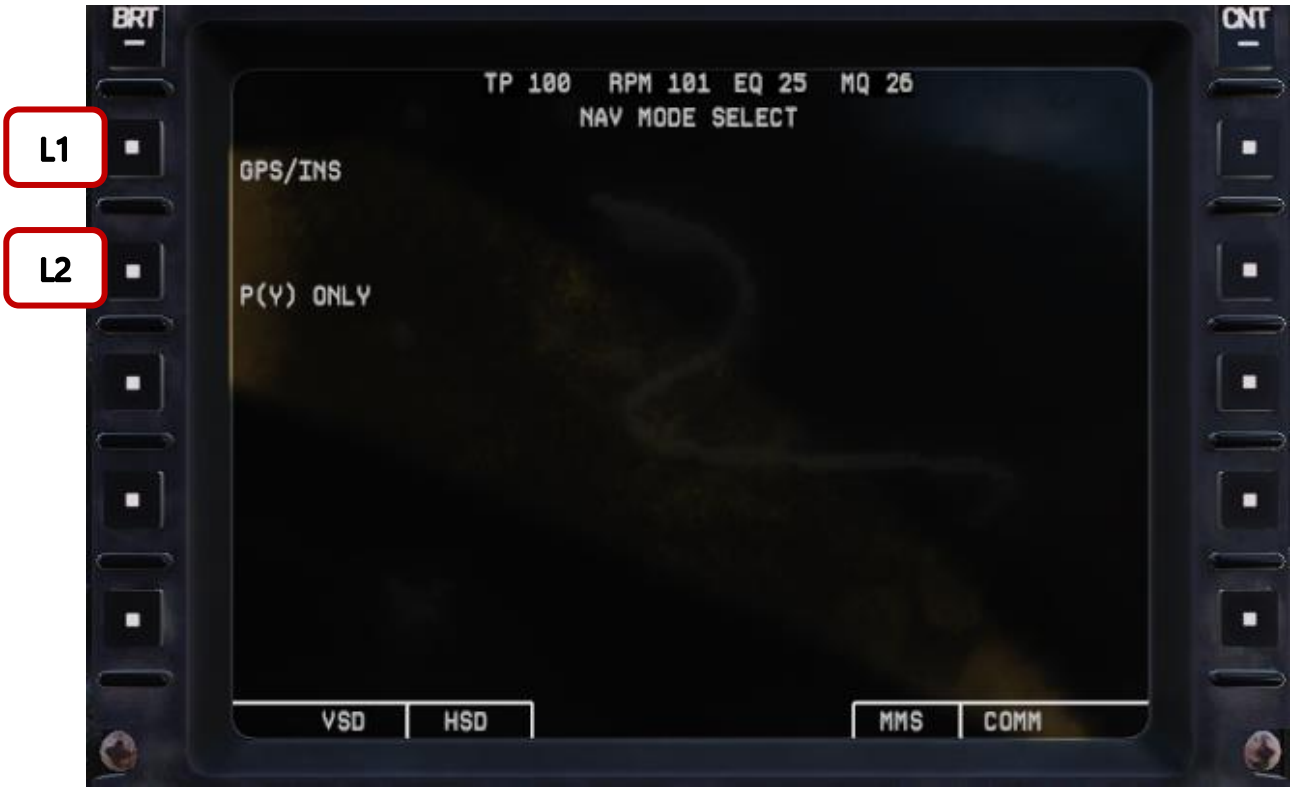


Figure 63 - NAV Mode Select page

L1. Toggles EGI mode between GPS/INS, GPS and INS

L2. No function

POLYCHOP SIMULATIONS

11.10.5 Power Assurance

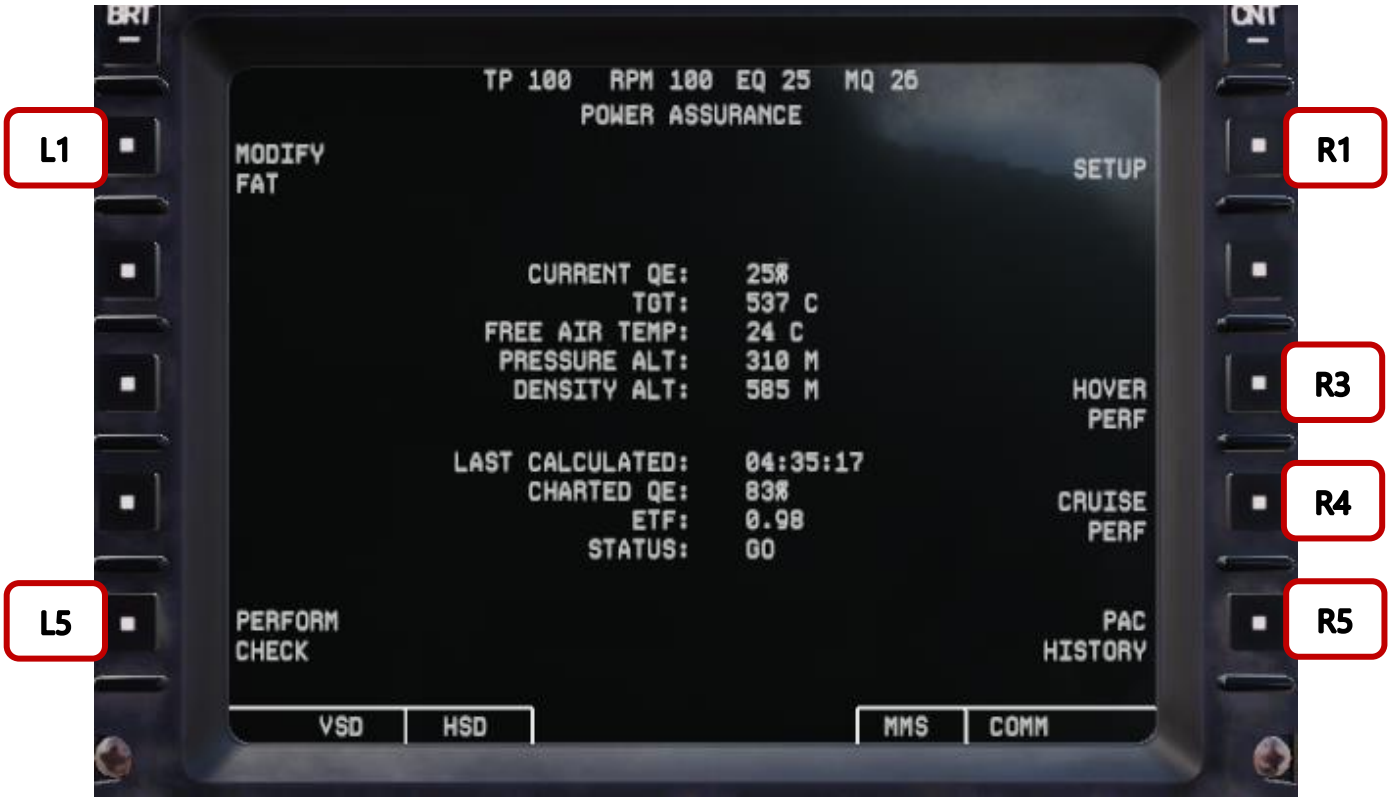


Figure 64 - Power Assurance page

L1. *MODIFY FAT*. Enables data entry mode. Allows input of a Free Air Temperature for power calculations

L5. *CALC REQD/PERFORM CHECK*. Activation of the bezel commands the system to calculate the performance data based on the current parameter setup

R1. *SETUP*. Enters the **Performance Setup** page

R3. *HOVER PERF*. Enters the **Hover Performance** page

R4. *CRUISE PERF*. Enters the **Cruise Performance** page

R5. *PAC HISTORY*. No function

POLYCHOP SIMULATIONS

11.10.6 Performance Setup

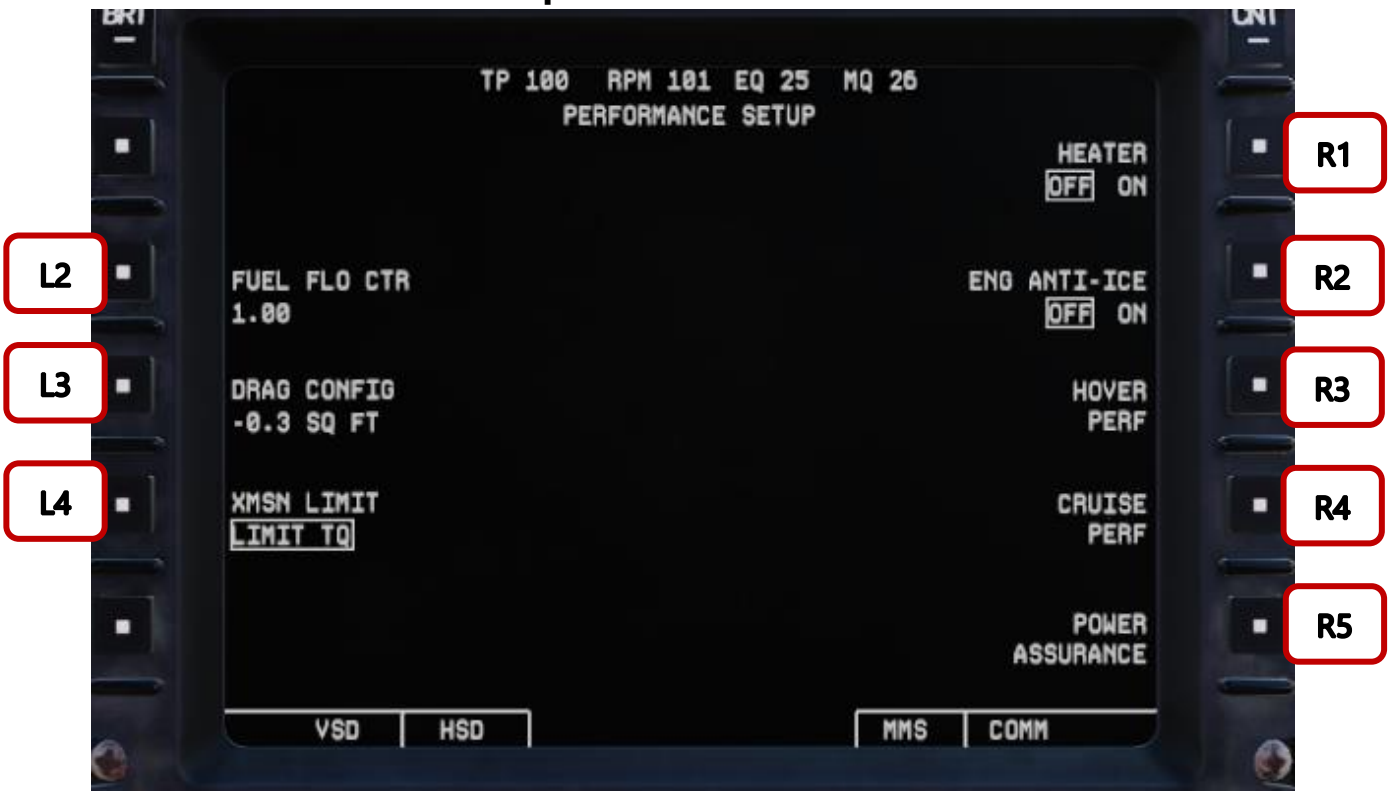


Figure 65 - Performance Setup page

L2. *FUEL FLO CTR*. No function

L3. *DRAG CONFIG*. Activates data entry mode. Allows overriding of the Drag Configuration value

L4. *XMSN LIMIT*. Toggles transmission limiting for the purpose of performance calculations

R1. *HEATER ON/OFF*. Toggles use of the Heater performance modifier

R2. *ENG ANTI-ICE*. Toggles use of the Engine Anti-Ice performance modifier

R3. *HOVER PERF*. Enters the **Hover Performance** page

R4. *CRUISE PERF*. Enters the **Cruise Performance** page

R5. *POWER ASSURANCE*. Enters the **Power Assurance** page

POLYCHOP SIMULATIONS

11.10.7 Hover Performance

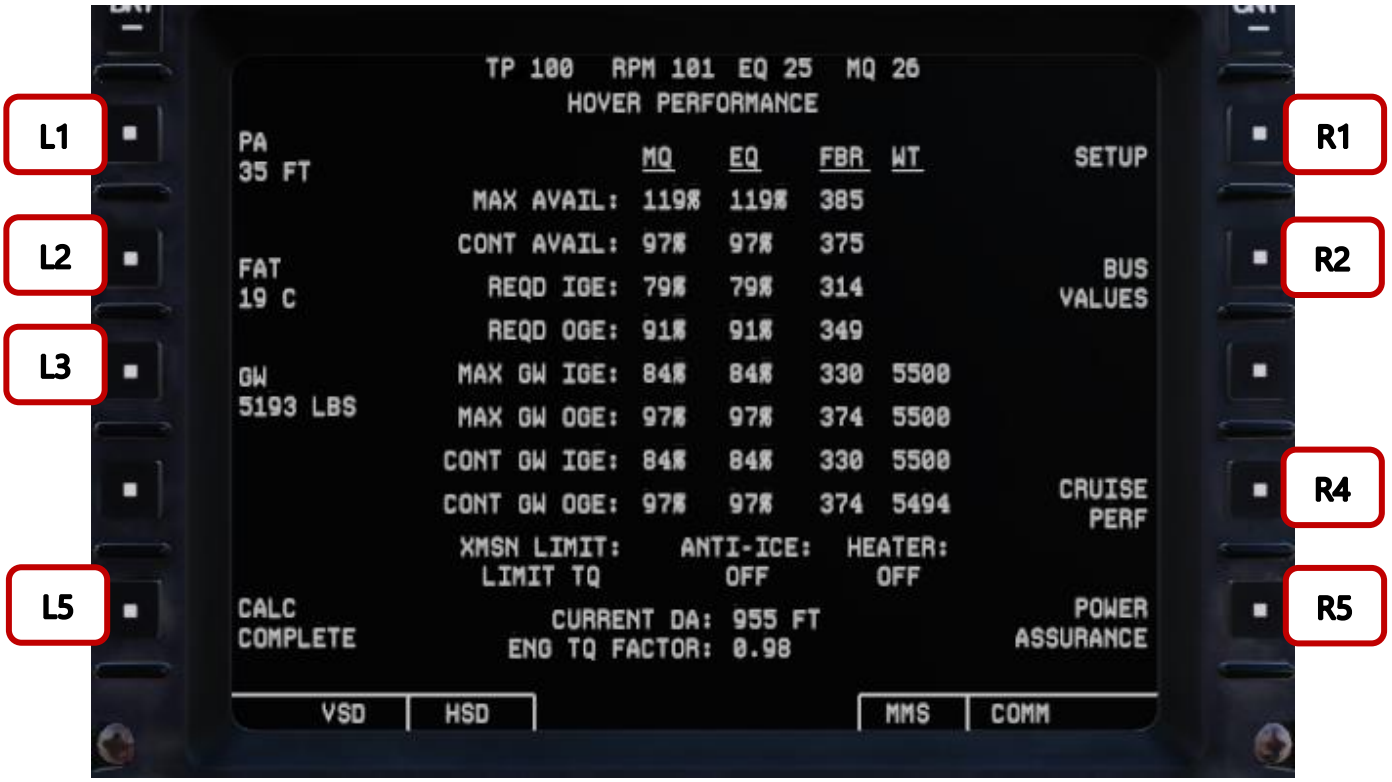


Figure 66 - Hover Performance page

L1. *PA* key. Activates data entry mode. Enter a number between −1,000 and 20,000

L2. *FAT* key. Activates data entry mode. Enter a number between −50 and +60 °C

L5. *CALC COMPLETE* key. Press key to initiate hover performance check.

R1. *SETUP* key. Provides access to the Performance Setup page.

R2. *BUS VALUES* key. Press to reset hover PA and hover FAT to current bus values

R4. *CRUISE PERF* key. Provides access to the Cruise Performance page.

R5. *POWER ASSURANCE* key. Provides access to the Power Assurance page.

POLYCHOP SIMULATIONS

11.10.8 Cruise Performance

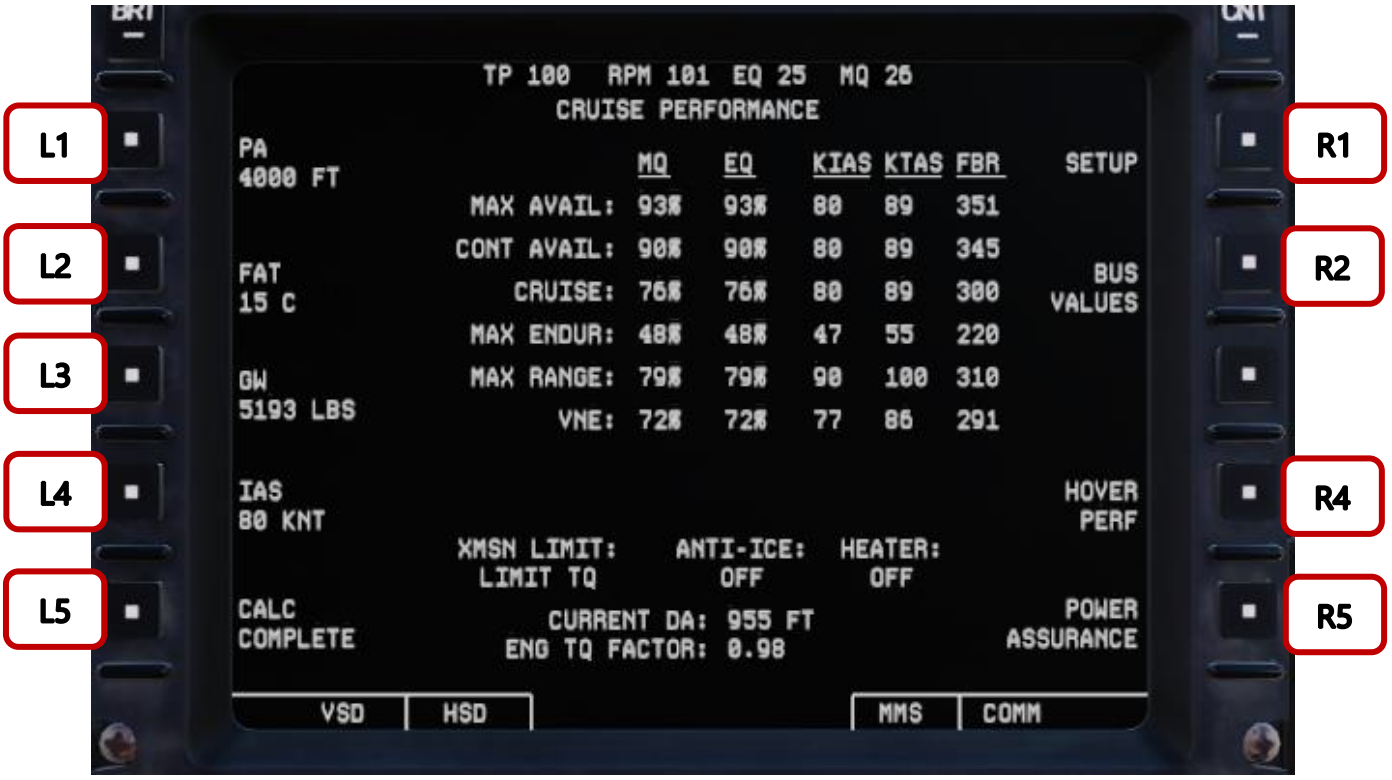


Figure 67 - Cruise Performance page

L1. *PA* key. Activates data entry mode. Enter a number between −1,000 and 20,000

L2. *FAT* key. Activates data entry mode. Enter a number between −50 and +60 °C

L3. *GW* key. Enter a number between 0 and 5500 lbs.

L4. *IAS* key. Enter a number between 0 and 133

L5. *CALC COMPLETE* key. Press key to initiate hover performance check.

R1. *SETUP* key. Provides access to the Performance Setup page.

R2. *BUS VALUES* key. Press to reset hover PA and hover FAT to current bus values

R4. *HOVER PERF* key. Provides access to the Hover Performance page.

R5. *POWER ASSURANCE* key. Provides access to the Power Assurance page.

POLYCHOP SIMULATIONS

11.10.9 Data Loader



Figure 68 - Data Loader page

L1. *LOAD MISSION* key. Calls up LOAD MISSION x page.

L3. *LOAD MISSION* key. Calls up LOAD MISSION x page.

L5. *DECLASSIFY SYSTEM* key. Displays when system detects classified data in NVM. Otherwise, legend is blank.

R1. *STORE DATA IN MISSION 1* key. Calls up STORE MISSION x page.

R2. *STORE DATA IN MISSION 2* key. Calls up STORE MISSION x page.

R3. *STORE DATA IN MISSION 3* key. Calls up STORE MISSION x page.

R4. *INIT PAGE 1* key. Calls up INITIAL PAGE 1.

R5. *INIT PAGE 2* key. Calls up INITIAL PAGE 2.

POLYCHOP SIMULATIONS

11.10.10 Load Mission

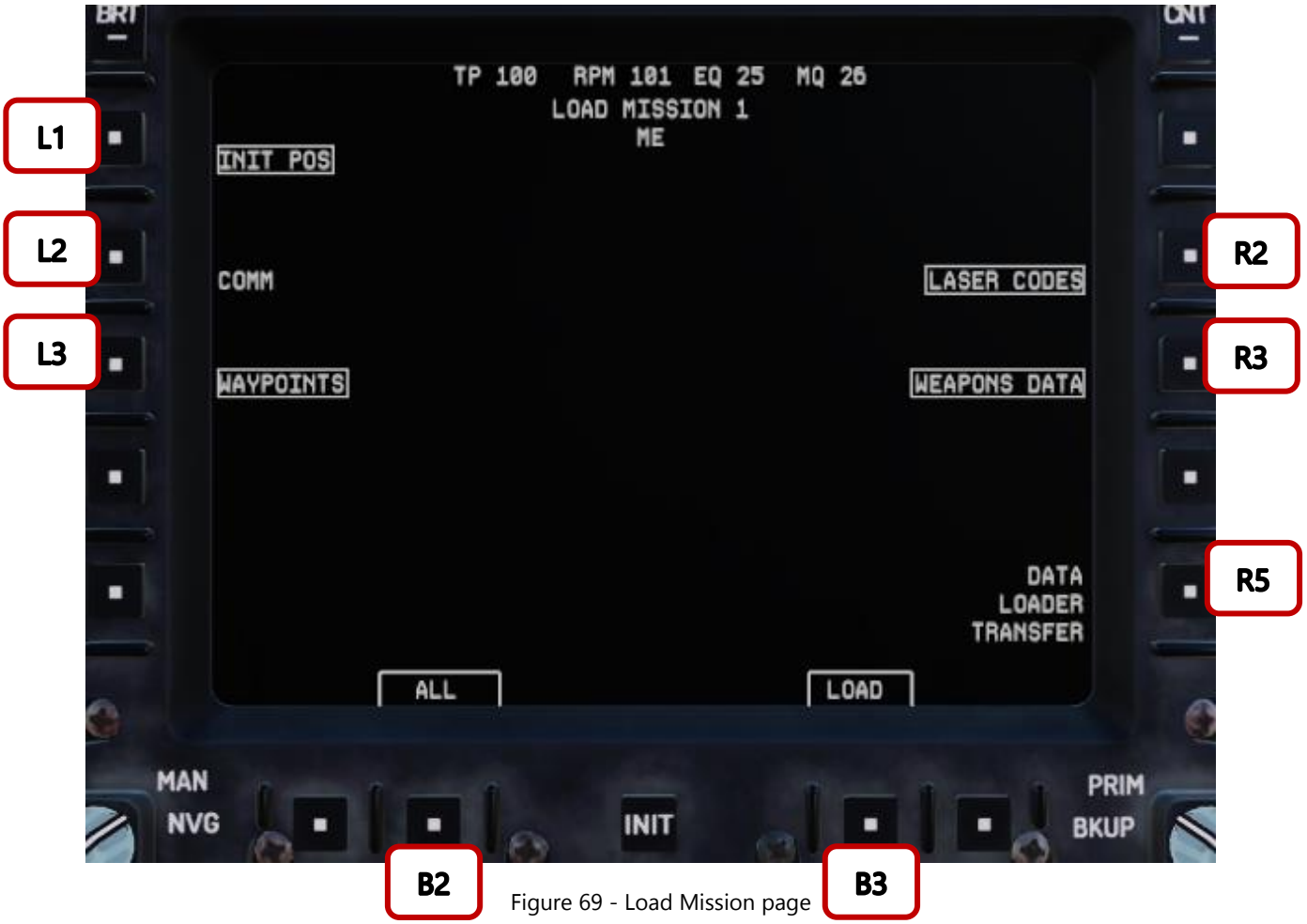


Figure 69 - Load Mission page

L1. *INIT POS* key. Selects/deselects initial position coordinates for loading.

L2. *COMM* key. Selects/deselects communication data for loading.

L3. *WAYPOINTS* key. Selects/deselects WPT and TGT data for loading.

R2. *LASER CODES* key. Selects/deselects laser data for loading.

R3. *WEAPONS DATA* key. Selects/deselects WEAPONS data for loading.

R5. *DATA LOADER TRANSFER* key. Calls up DATA LOADER TRANSFER page.

B2. *ALL* key. Selects all parameters for loading.

B3. *LOAD* key. Initiates loading of all selected parameters.

POLYCHOP SIMULATIONS

11.10.11 Store Mission

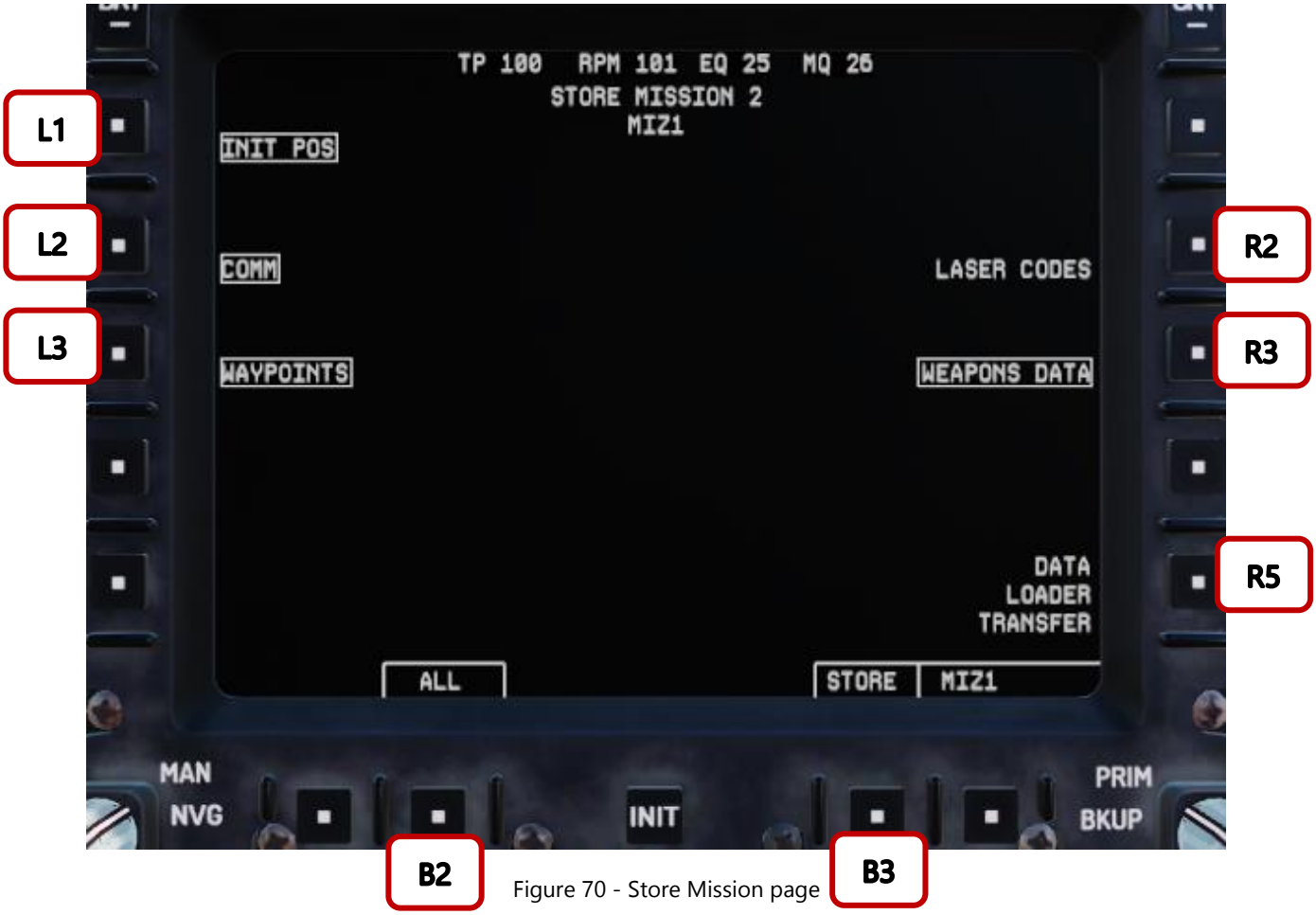


Figure 70 - Store Mission page

L1. *INIT POS* key. Selects/deselects initial position coordinates for storage.

L2. *COMM* key. Selects/deselects communication data for storage.

L3. *WAYPOINTS* key. Selects/deselects WPT and TGT data for storage.

R2. *LASER CODES* key. Selects/deselects laser data for storage.

R3. *WEAPONS DATA* key. Selects/deselects WEAPONS data for storage.

R5. *DATA LOADER TRANSFER* key. Calls up DATA LOADER TRANSFER page.

B2. *ALL* key. Selects all parameters for storage.

B3. *STORE* key. Initiates storage of selected data

B4. *Mission Identifier*. When STORE key pressed, a cursor appears allowing the operator to enter an identifier for data to be stored.

POLYCHOP SIMULATIONS

11.10.12 Notebook

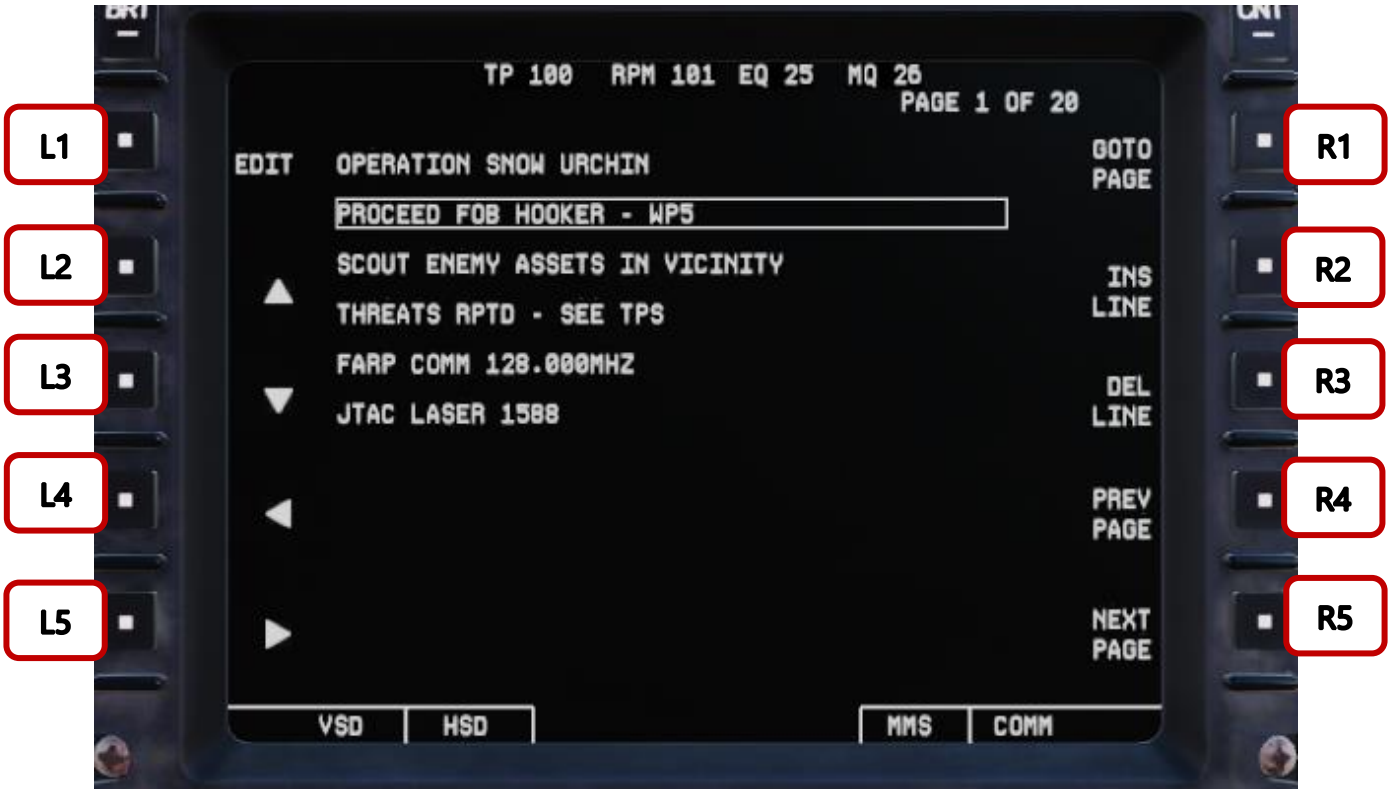


Figure 71 - Notebook page

L1. *EDIT*. Pressing L1 places the selected line into the edit mode

L2. *UP Arrow*. Pressing L2 moves the selected line up the page to the previous line

L3. *DOWN Arrow*. Pressing L3 moves the selected line down the page to the next line

L4/ L5. *LEFT/RIGHT Arrow*. Pressing L4 and L5 moves the cursor to next left or right character position respectively

R1. *GO TO PAGE*. The functionality of this bezel is determined by whether the edit mode (L1) is active or inactive.

R2. *INS LINE*. Pressing R2 inserts a blank line on the currently selected page at the location of the cursor box

R3. *DEL LINE*. Pressing R3 deletes the line currently selected by the cursor box

R4. *PREV PAGE*. Pressing R4 moves to the previous page, if more, then 1 page is present.

R5. *NEXT PAGE*. Pressing R5 moves to the next page, if more, then 1 page is present.

POLYCHOP SIMULATIONS

11.10.13 Ground Setup

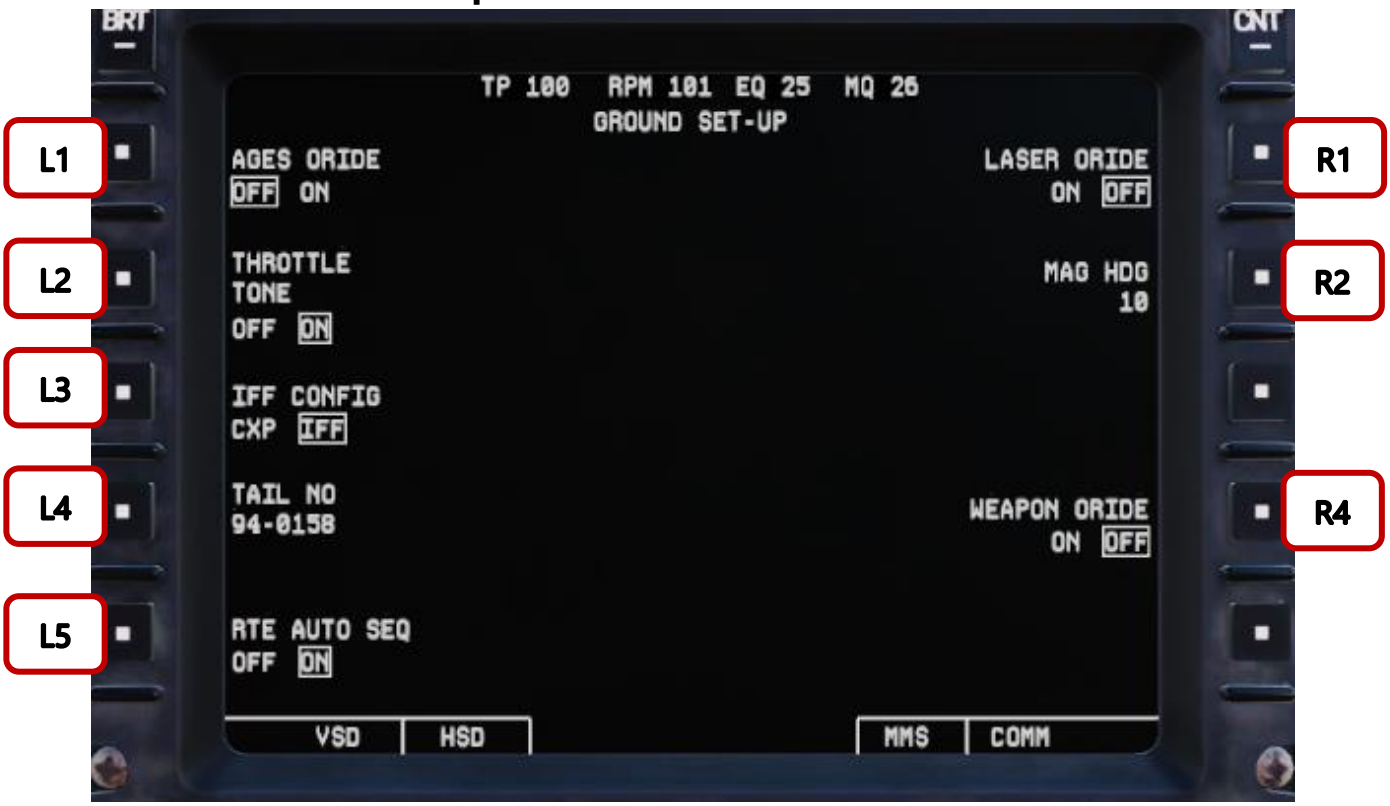


Figure 72 - Ground Setup page

L1. *AGES ORIDE*. IN OP

L2. *THROTTLE TONE*. IN OP

L3. *IFF CONFIG*. IN OP

L4. *TAIL NO*. IN OP

L5. *RTE AUTO SEQ*. Pilot or copilot can set the auto next point feature

R1. *LASER ORIDE*. This function allows the operator to override the "on ground" laser safety interlock

R2. *MAG HDG*. IN OP

R4. *WEAPON ORIDE*. This function allows the operator to override the "on ground" weapons safety interlock

POLYCHOP SIMULATIONS

11.10.14 Fuel Management

L3. *START FUEL CK*. Bezel position L3 allows the user to initiate a fuel check

L4. *FUEL CK LENGTH*. Bezel position L4 allows the user the ability to change the time intervals between the fuel check: 20, 30, or 60 minutes

R1. *RESET AVERAGE*. Bezel position R1 provides a double line legend that allows the user to restart averaging calculations.

R2. *VSD/HSD DISPLAY*. Bezel position R2 provides a triple line legend that allows for selection of either average (AVG) or instantaneous (INST) fuel burn rate and fuel time remaining data on the VSD and HSD pages. INST is the default.

R4. *BINGO ALARM*. Bezel position R4 provides a triple line legend that allows for the setting of the BINGO ALARM mode ON or OFF

R5. *SET BINGO*. Bezel position R5 provides a double line legend that allows the user to set the BINGO timer.

11.10.15 Fault Detection/Location Menu (?)

L1. *BIT*. IN OP

L2. *CW HISTORY*. IN OP

L3. *ENGINE HISTORY*. IN OP

L4. *FADEC MONITOR*. IN OP

L5. *SW VERSION CHECKS*. IN OP

R1. *MAINTAINERS CHECK*. IN OP

R2. *SCAS MONITOR*. IN OP

R4. *RAD ALT TEST*. Pressing and holding R4 on the FDL Menu Page initiates the Radar Altimeter Test.

POLYCHOP SIMULATIONS

11.10.16 Clock Timer

L1. *ZULU*. The EGI provides Zulu Time (based upon GPS time) to the CDS on a periodic basis for display at L1

L2. *LOCAL*. Local time is displayed at L2

L3. *TIMER* 1. A countdown timer function with a user-defined label is provided at L3

L4. / L5. *SELECT*. The up-arrow at L4 and the down-arrow at L5 are used to select which time (Zulu Time or Local Time) is to be displayed at R4 on Initial Page 1

R1. *JULIAN DATE*. The EGI provides the Julian Date for display at R1

R2. *ALARM* 1. An alarm function with a user-defined label is provided at R2

R3. *MISSION*. Mission time is displayed at R3 in HH:MM:SS format

R5. *NAV ALIGN*. Pressing R5 on the Clock Timer page displays the NAV Align page

POLYCHOP SIMULATIONS

11.11 VSD Pages

11.11.1 VSD

The Vertical Situation Display (VSD) page shows attitude and navigation data. The VSD is a top-level page that can be accessed from any other page by pressing the B1 bezel. The VSD page can also be accessed on the RMFD using the Display Select switch on the pilot's cyclic. This page acts as an access point to the **HVR** page.

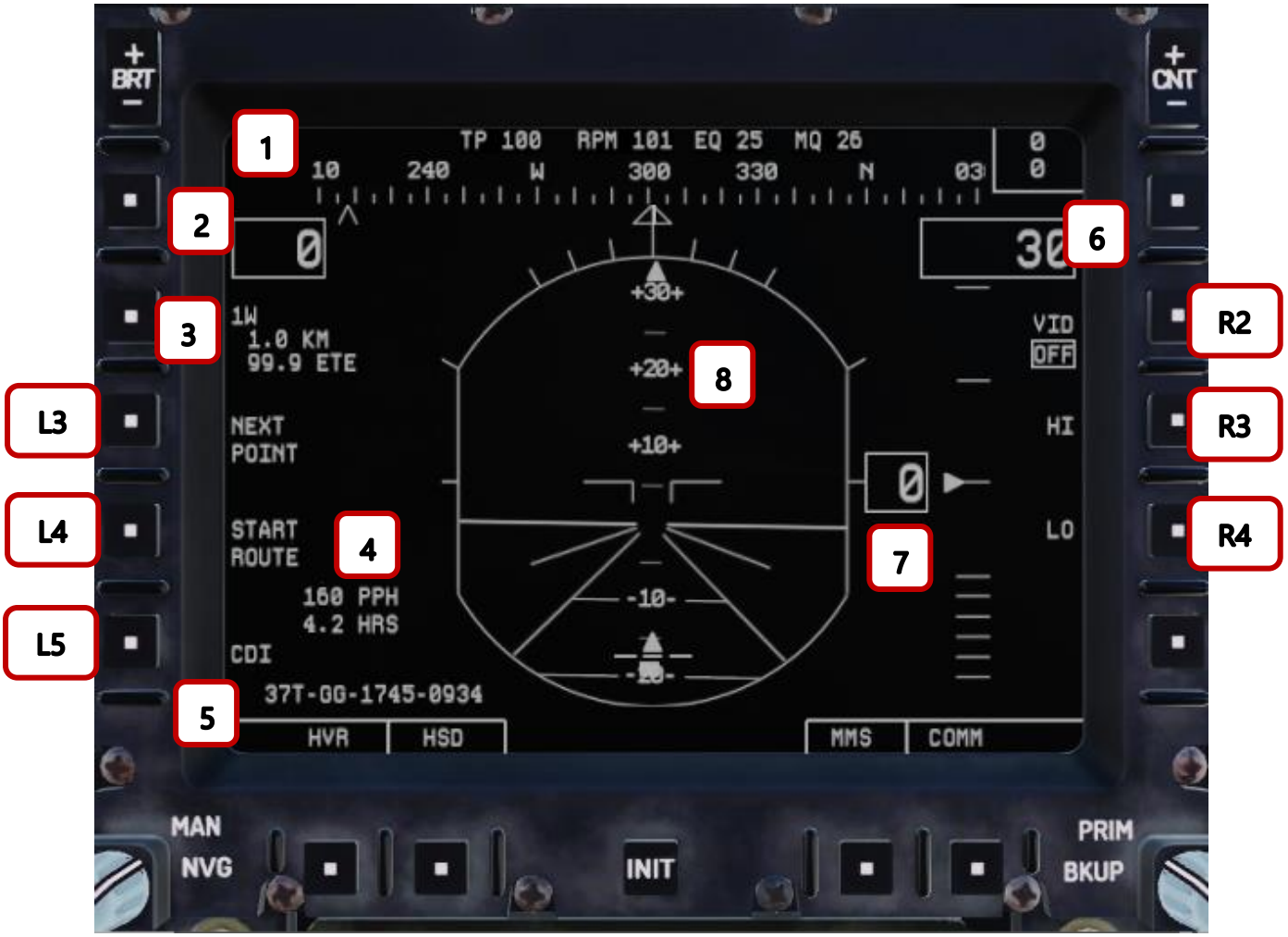


Figure 73 - VSD page

1. Compass Tape and MMS Indicator
2. Indicated airspeed in knots. Blanked above 130 knots
3. Selected navigation point, distance and estimated time enroute. Distance is displayed in KM (UTM) or NM (LL) depending on current coordinates mode
4. Fuel flow and estimated flight time remaining
5. Present position (UTM) or (LL)
6. Barometric altitude in feet
7. Radar Altimeter Digital & Analog Altitude Indicators and Rate of Climb Indicator. Digital Altitude Display is blanked above 1425 feet. Analog Altitude Indicator is blanked above 200 feet
8. Attitude Window

POLYCHOP SIMULATIONS

L3. *NEXT POINT*. Selects next navigation point in current route. If last navigation point is already selected, will display END ROUTE

L4. *PREV POINT*. Selects previous navigation point in current route. If first navigation in the route is already selected, will display START ROUTE

L5. *CDI*. Toggles display of the Course Deviation Indicator (CDI)

R2. *VID OFF/TV/TIS*. Cycle background MMS display mode between OFF, TV and TIS (Pilot) or between OFF and ON (Copilot)

R3. *HI*. Enables data entry of HIGH radar alert altitude

R4. *LO*. Enables data entry of LOW radar alert altitude

POLYCHOP SIMULATIONS

11.11.1.1 Attitude Window

The Attitude Window comprises of a Pitch Ladder, Roll Reference Indicator, Aircraft Symbol, Skid/Slip Indicator and Turn Rate Indicator.

Displays pitch and roll attitude references.

It is a circle truncated by vertical lines at the ±60 degrees marks from the vertical center line.

Lubber line is zero roll reference line.

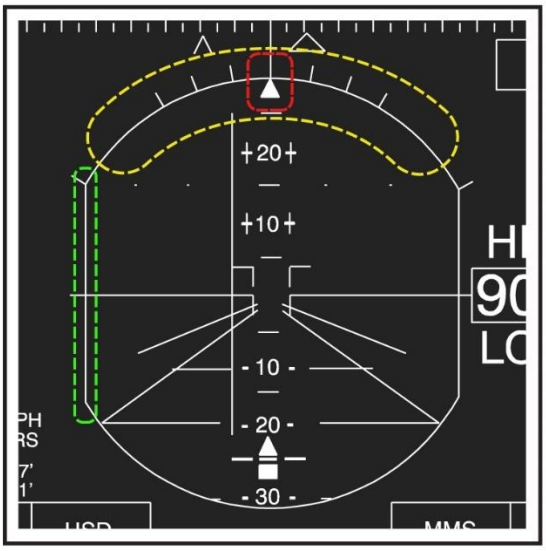


Figure 74. Attitude Window

POLYCHOP SIMULATIONS

11.11.1.2 Course Deviation Indicator

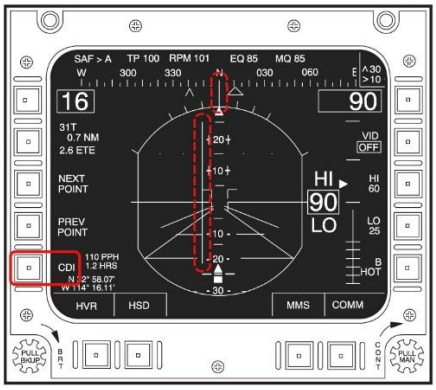
The Course Deviation Indicator displays a course deviation of up to 10 degrees left or right.

Located with respect to the lubber line and when aircraft is on course, aligns with lubber line. Indicates a course deviation of up to 10 degrees left, or right, with relation to the linear scale. Each dot on the two-dot linear scale represents 5 degrees.

MFD VSD page

Pressing L5 *CDI* toggles the course deviation indicator line (vertical line to left of center screen, plus horizontal line of dots) ON and OFF.

The CDI indicates the course deviation up to 10° left or right with relation to the two-dot scale.

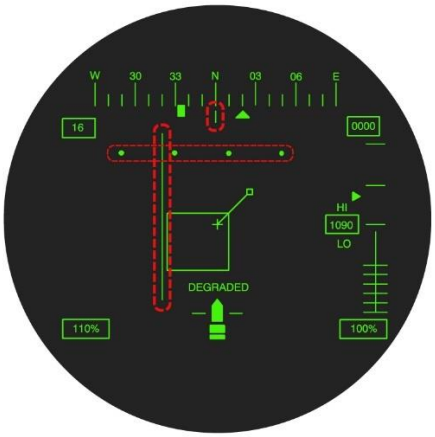


Pilot ODA

Pressing L-5 toggles between on and off. Located with respect to the lubber line and when aircraft is on course, aligns with lubber line.

Indicates a course deviation of up to 10 degrees left, or right, with relation to the linear scale.

Each dot on the two-dot linear scale represents 5 degrees.



The pilot and CPO can independently set whether the CDI is displayed on their respective MFDs.

INFO:

CDI is reliable only if the aircraft present position is within the bounds of the particular route leg and there is a previous point selected. For example;

If the aircraft present position is between the two WPs and START ROUTE or PREV POINT is selected, the CDI is reliable.

If aircraft position is outside the leg route structure, the CDI indication is not reliable.

POLYCHOP SIMULATIONS

11.11.2 HVR

The Hover (HVR) page is a sub-mode of the VSD page. It can be accessed by selecting the B1 bezel while in the VSD page. The HVR page shares many elements of the **VSD** page, and so only the elements unique to the HVR page are detailed here.

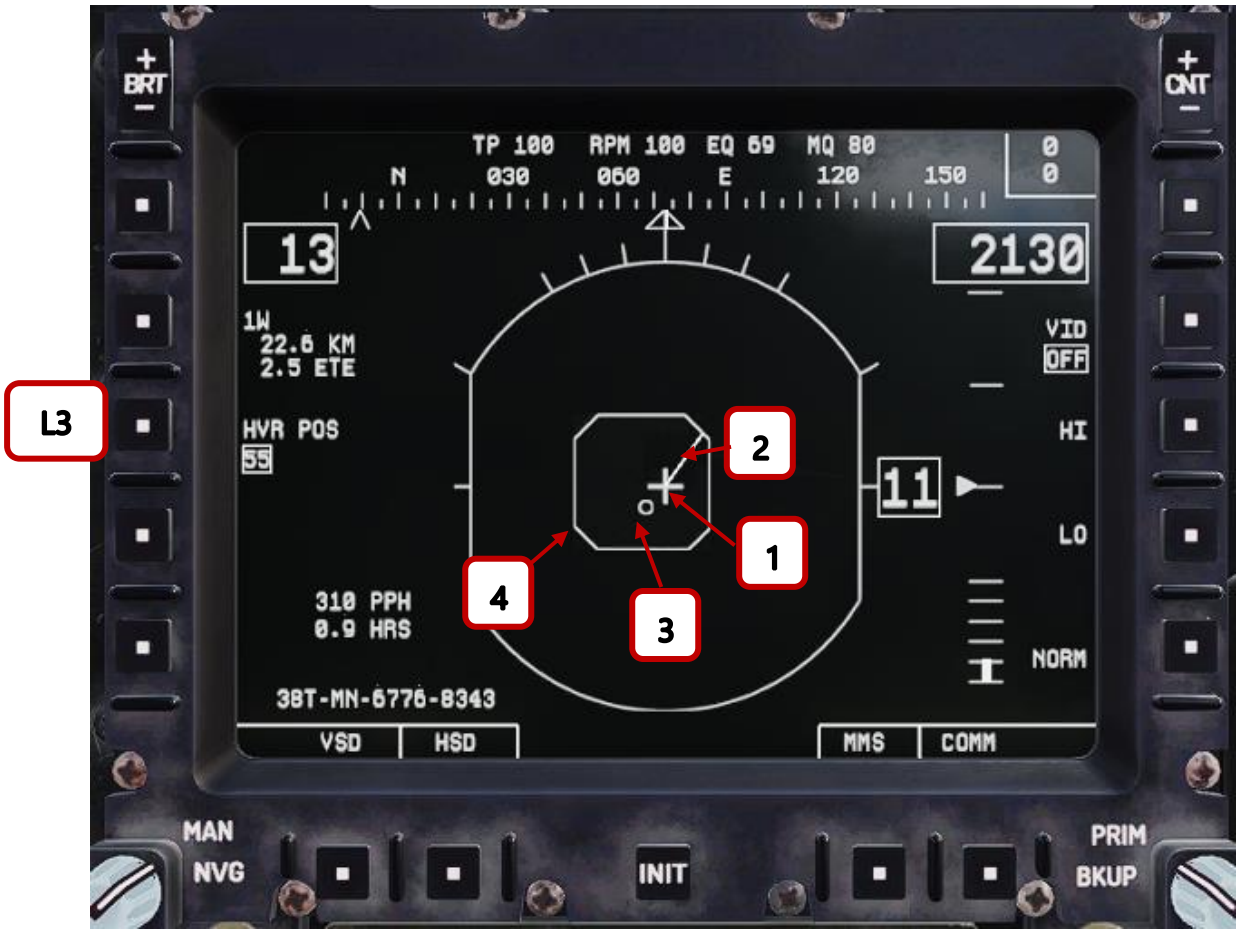


Figure 75 - HVR page

1. Hover Aircraft Symbol
2. Velocity Vector
3. Acceleration Cue
4. Bob Up box

L3. *HVR POS*. Allows for data entry of Hover Drift Limit, limited from 0 to 150 in ft. When the aircraft has drifted beyond the set limit, a HVR DRIFT advisory is displayed

POLYCHOP SIMULATIONS

11.11.2.1 Bob Up box

The Bob Up box is an element inside the HVR page that gives the pilot a reference from an initial point that is selected from the position of the aircraft at the time the switch was pressed. Half scale deflection of the position box (i.e., from the center of the MFD to the boundary of the attitude window) is equal to approximately 88 feet.

To enter this mode the pilot needs to press **Bob Up/Drop Waypoint Switch** for 1.5 seconds. To remove the symbology from the screen, repeat the action.

11.12 HSD Pages

11.12.1 HSD-

The Horizontal Situation Display (HSD) page can be displayed as either HSD- or HSD+. The primary difference between these two modes is the display of the RMS map in the background of the HSD+ page. Selection of either mode is performed via the B2 bezel when in either page.

The HSD is the primary navigation page and serves as an entry point for a variety of navigation subpages for managing Waypoints, Control Points, Target Points, Routes, Battlefield Graphics and the Rotorcraft Map System.

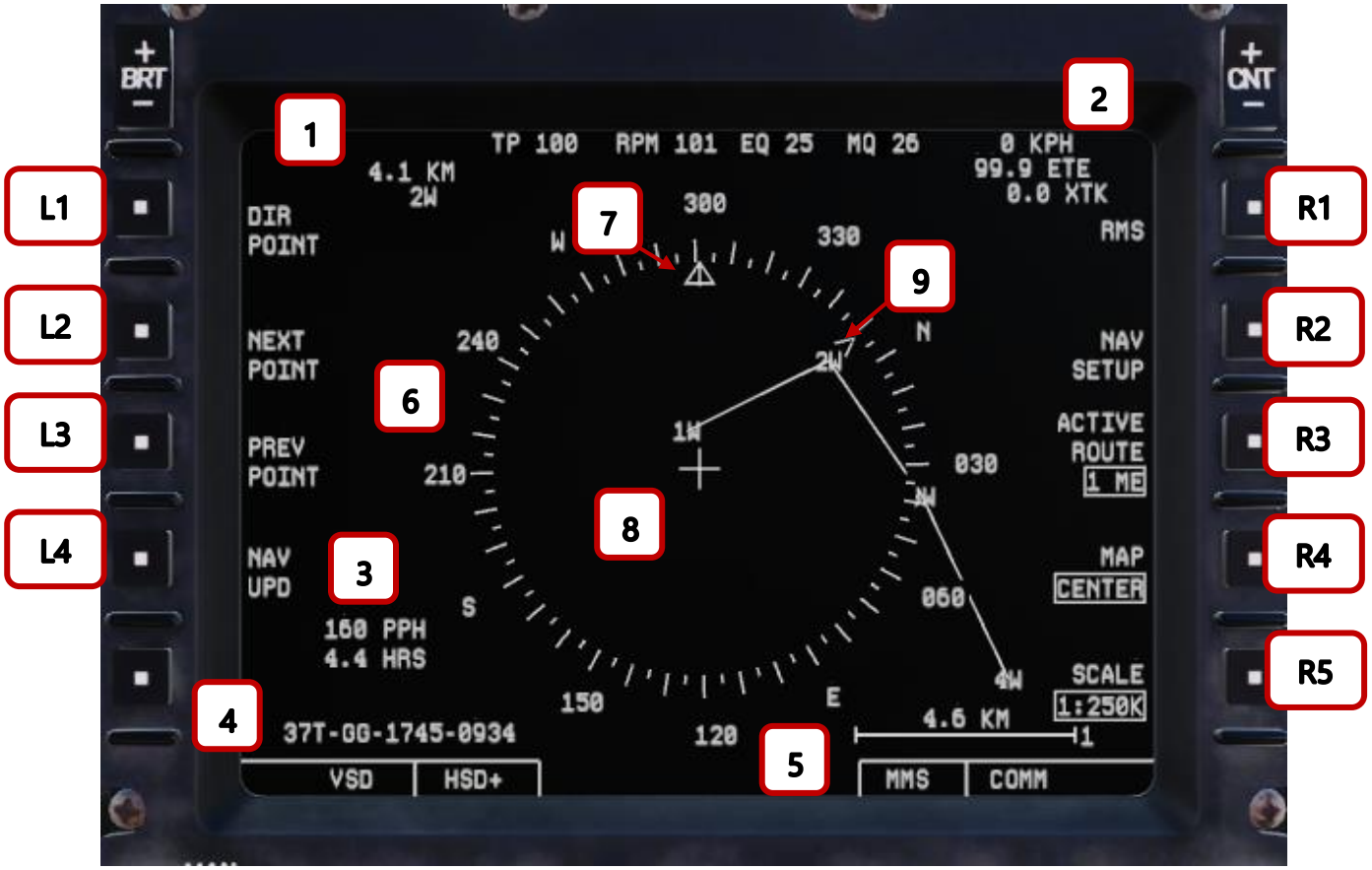


Figure 76 - HSD- page

1. ID and distance to selected Waypoint, Control Point or Target Point. Distance is displayed in KM (UTM) or NM (LL)
2. Groundspeed in KPH (UTM) or KTS (LL), Estimated Time Enroute (minutes) and Cross Track KM (UTM) or NM (LL)
3. Fuel Flow and Estimated Fuel Time Remaining
4. Present position in UTM or LL Scale indicator in KM (UTM) or NM (LL)
5. Compass rose
6. Lubber Line and Bearing to MMS LOS Indicator
7. Aircraft Symbol
8. Bearing to Waypoint Indicator

POLYCHOP SIMULATIONS

L1. *DIR POINT*. Enters the **Direct Point** page

L2. *NEXT POINT*. Cycles to the next point in the route. The steering cues and waypoint data are updated automatically. Displays END ROUTE if there are no more points in the current route

L3. *PREV POINT*. Selects the previous point in the route. Displays START ROUTE if there are no previous points in the route

L4. NAV UPD. Enables selection of the Nav Update functions

R1. *RMS*. Opens the **Rotorcraft Map System** page

R2. *NAVSETUP*. Opens the **NAV Setup** page

R3. *ACTIVE ROUTE*. Selection will cycle between the three available routes. The currently active route name is displayed

R4. *MAP CENTER/OFFSET*. Toggles between Offset and Center map display

R5. *SCALE*. Cycles between the available map scales. Scales are 1:50K, 1:100K, 1:250K and 1:500K



Figure 77 - HSD+ page

POLYCHOP SIMULATIONS

11.12.2 RMS

The Rotorcraft Map System (RMS) page provides display customization and navigation features for the Rotorcraft Map System.

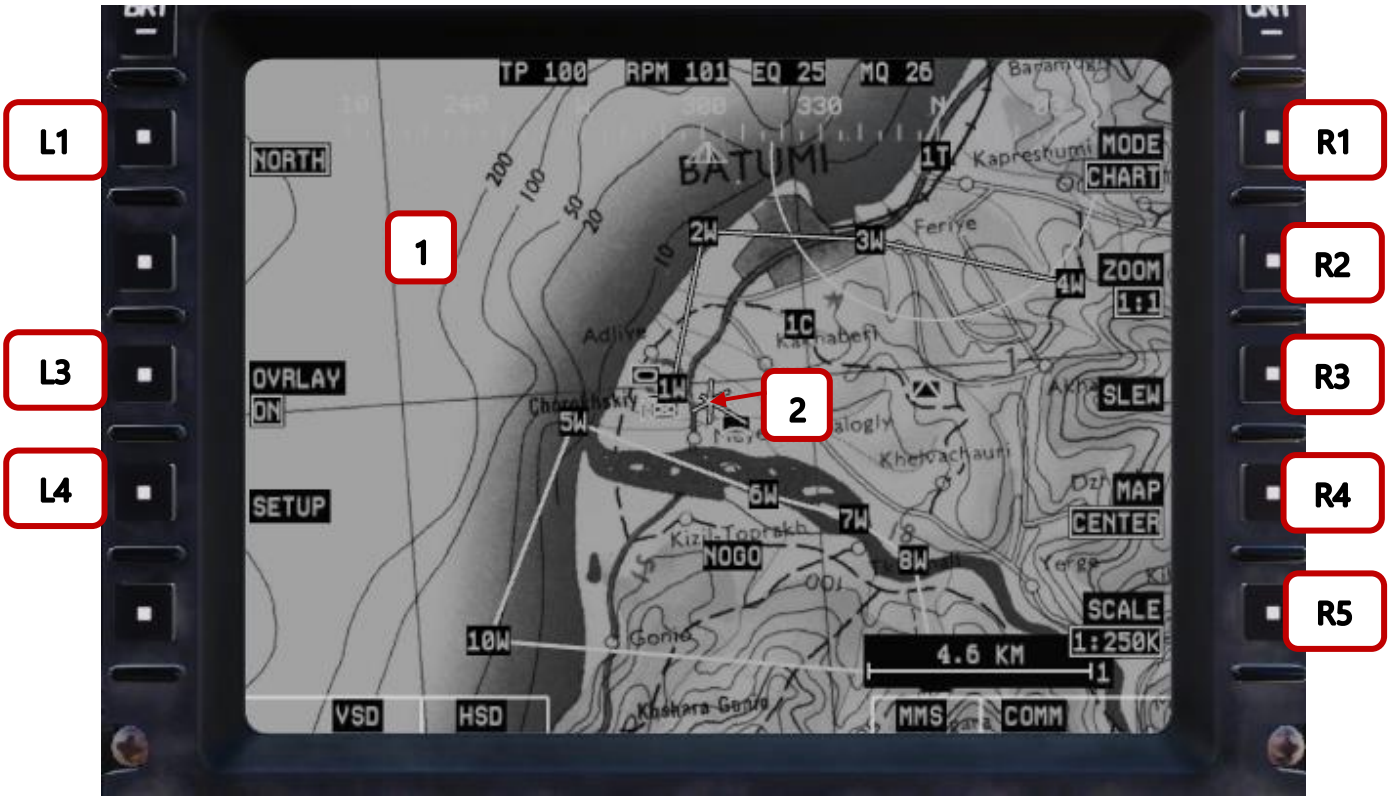


Figure 78 - RMS page (Chart display)

INFO:

Any changes made to the map display while in the RMS page will also be applied upon returning to the HSD pages

1. Map display
2. Aircraft Symbol

L1. *HDG/NORTH*. Pressing L1 allows the map to toggle between North Up or Heading Up orientation

L3. *OVRLAY*. Determines whether overlays are displayed on the map. Overlays are currently limited to JVMF symbology and cannot be configured

R1. *MODE*. Cycles between the available map display modes. Currently only CHART (Raster Chart) and DTED (Digital Terrain Elevation Display) display imagery. Selection of DATA or NONE will result in a blank map display. Bezel options are limited by display mode

L4. *SETUP*. Enters the **RMS Setup** page

R2. *ZOOM*. Allows selection of zoom level. Available zoom levels are 1:1, 1.25:1, 1.5:1, 1.75:1 or 2:1

R3. *SLEW*. Enters RMS Slew Mode. Functionality in RMS Slew Mode is covered in more detail below

R4. *MAP CENTER/OFFSET*. Cycles between display of the map in Centered or Offset mode

R5. *SCALE*. Cycles between the available map scales. Scales are 1:50K, 1:100K, 1:250K and 1:500K

POLYCHOP SIMULATIONS

11.12.2.1 Digital Terrain Elevation Display

Selecting the Digital Terrain Elevation Display (DTED) mode presents a digital image of the surrounding terrain with a configurable elevation band. In this mode a new option at the L5 bezel is made available to toggle visibility of the elevation band. The elevation can be configured in the **RMS Setup** page.

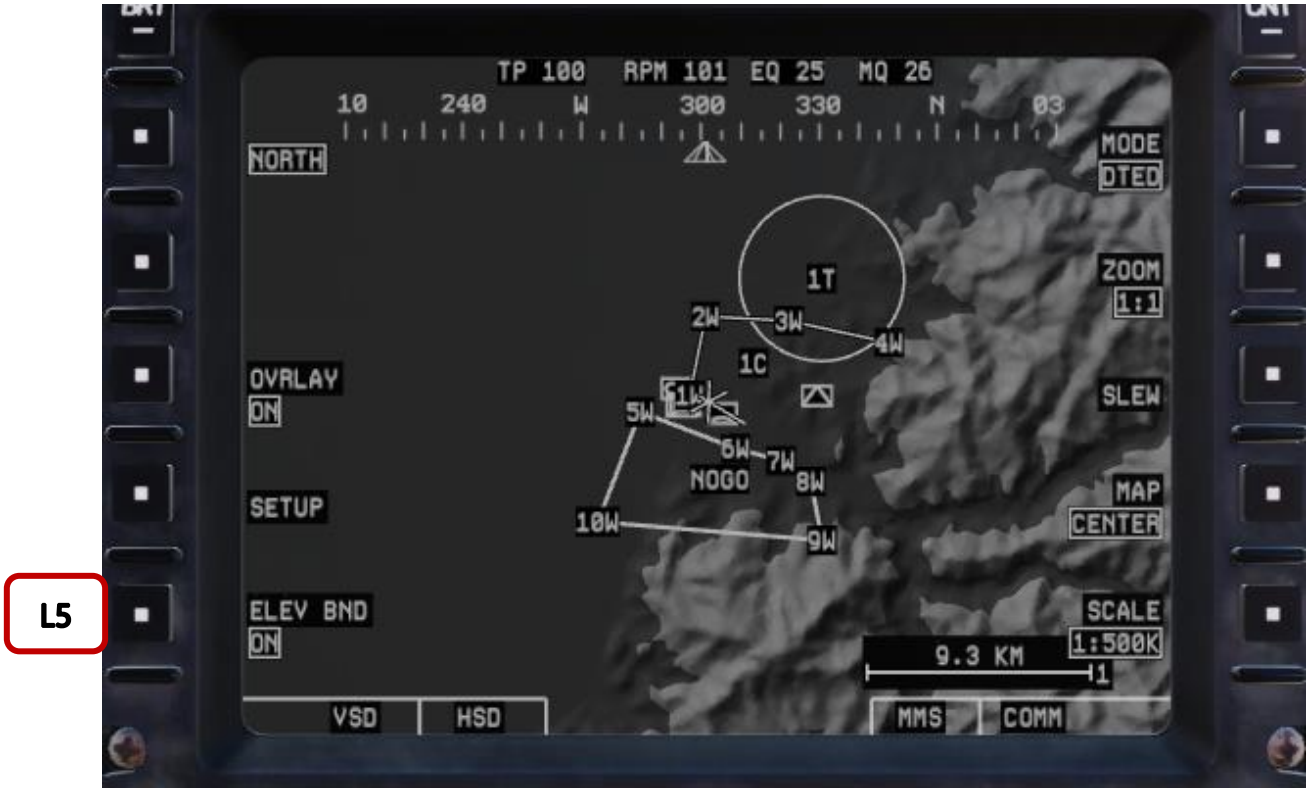


Figure 79 - RMS page (DTED display)

POLYCHOP SIMULATIONS

11.12.2.2 Slew Mode

Slew Mode allows repositioning of the map to any Waypoint, Target Point, Control Point or UTM position. While in Slew Mode, the function of certain bezels and symbology is changed. A box with the text SLEW is displayed in the top center of the screen while in this mode.

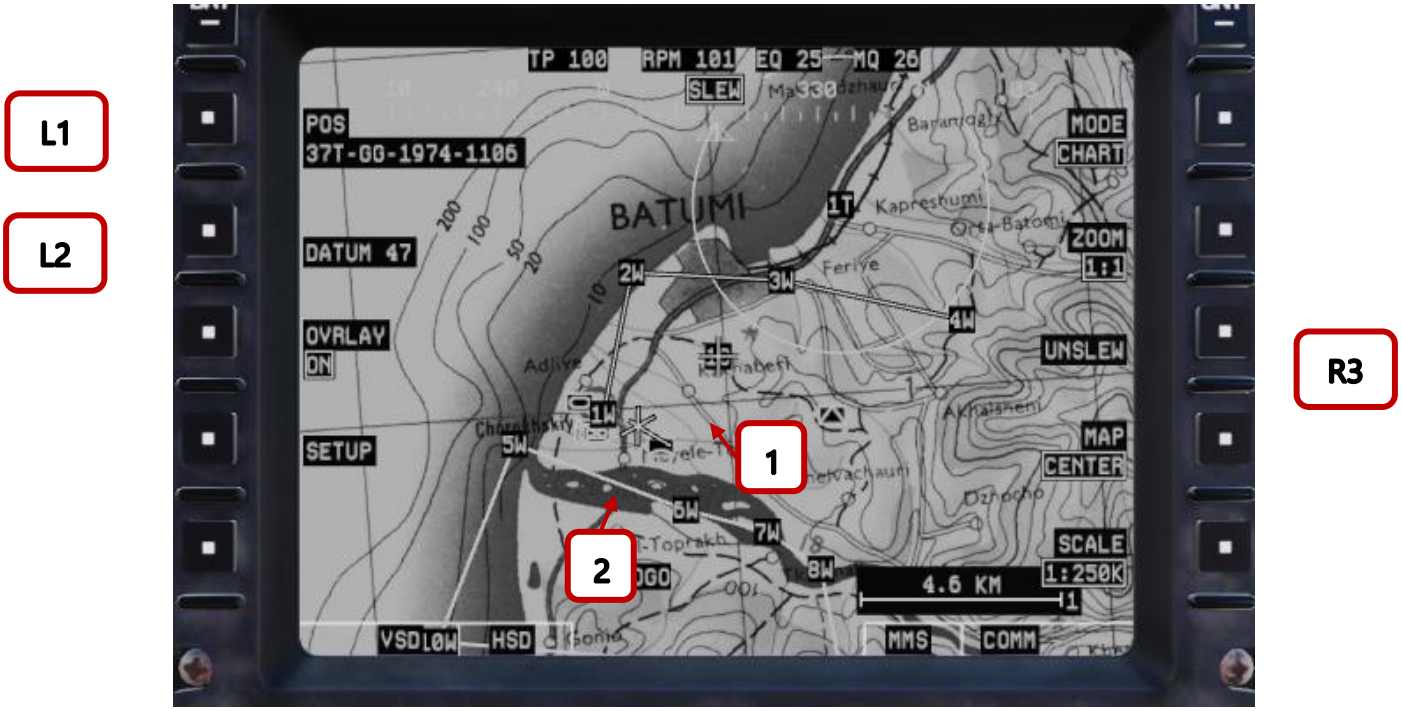


Figure 80 - RMS Slew mode

1. Center Cross
2. Aircraft Symbol

L1. *POS*. Selecting this bezel will activate data entry mode. A cursor will display to the right of the POS text, allowing input of either a Waypoint, Target Point or Control Point identifier. Alternatively, a UTM coordinate entry can be made. On entry of a valid position or navigation point identifier, the UTM coordinates will be displayed and the display map will center on the position.

L2. *DATUM*. This entry has no function

R3. *UNSLEW*. Pressing R3 will exit Slew mode and return the bezel functions to default

11.12.3 RMS Setup

The RMS Setup page is used to configure the Rotorcraft Map System and the symbology displayed in both RMS and HSD pages.



Figure 81 - RMS Setup page

L1. *ACTIVE ROUTE*. Selection toggles between LINE/IDS, LINE and NONE. Controls visibility of the Active Route. Selection is indicated by the coverage of the box. LINE/IDS will display both the Active Route line and navigation point IDs. LINE will display the Active Route line but hide the navigation point IDs. NONE will hide both the Active Route line and navigation point IDs

L2. *LINES*. Toggles between LINE/IDS, LINE and NONE. Controls visibility of all Battlefield Graphics Lines. Selection is indicated by the coverage of the box. LINE/IDS will display both the Battlefield Graphics

R1. *WAYPOINTS*. Controls visibility of all Waypoints

R2. *TARGETS*. Selection options of TARGET/RING, TARGET and NONE. Controls visibility of Target Points and Threat Rings. Selection is indicated by the coverage of the box. TARGET/RING will display both the Threat Ring and Target Point IDs. TARGET will display the Target Point IDs but hide the Threat Rings. NONE will hide both Threat Rings and Target Point IDs

R3. *DATAFRAME*. Not implemented

POLYCHOP SIMULATIONS

line and navigation point IDs. LINE will display the Battlefield Graphics line but hide the navigation point IDs. NONE will hide both Battlefield Graphics line and navigation point IDs

L3. *AREAS*. Toggles between LINE/NAME, LINE/IDS, LINE and NONE. Controls visibility of all Battlefield Graphics Areas. Selection is indicated by the coverage of the box. LINE/NAME will display the Battlefield Graphics Area line, name and navigation point IDs. LINE/IDS will display both the Battlefield Graphics Area line and navigation point IDs but hide the name. LINE will display the Battlefield Graphics Area line but hide the navigation point IDs and name. NONE will hide the Battlefield Graphics Area line, navigation point IDs and name

L4. *CONTROL POINTS*. Selects visibility of all Control Points

L5. *FEATURE SELECT*. Not implemented

R4. *ELEV BAND*. Sets the elevation of the display band on the DTED. Selecting the bezel will activate Data Entry Mode. Elevation band will be changed upon valid entry of an elevation between -1000 to 9999 meters or -3280 to 32767 feet

R5. *RMS*. Returns to the **Rotorcraft Map System** page.

11.12.4 RMS Feature Select

Not implemented.

POLYCHOP SIMULATIONS

11.12.5 NAV Setup

The NAV Setup page acts as an entry point for various navigation functions including the creation, modification and deletion of navigation points, routes and battlefield graphics.

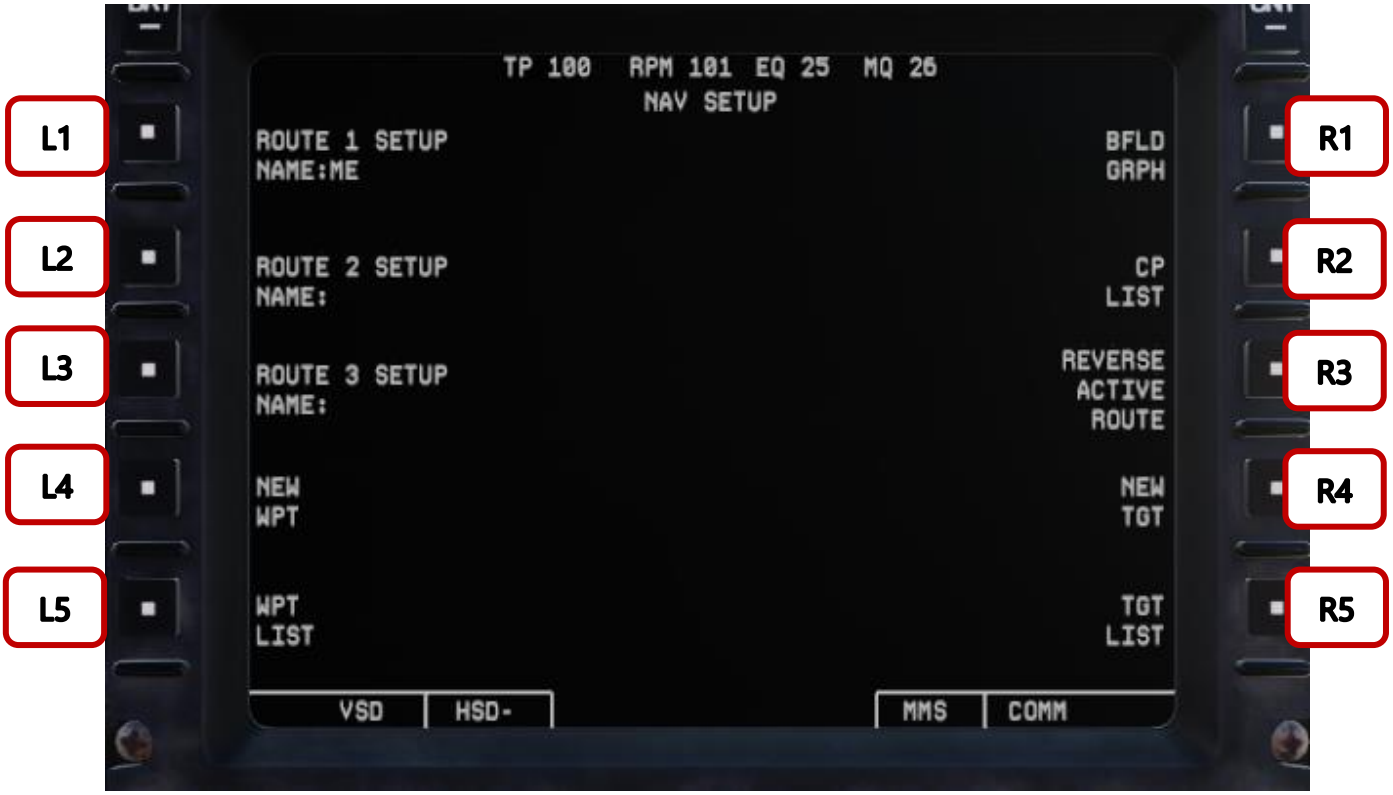


Figure 82 - NAV Setup page

L1. *ROUTE 1 SETUP*. Enters the **Route Setup** page for Route 1

L2. *ROUTE 2 SETUP*. Enters the **Route Setup** page for Route 2

L3. *ROUTE 3 SETUP*. Enters the **Route Setup** page for Route 3

L4. NEW WPT. Opens the **New Waypoint** page

L5. WPT LIST. Opens the **Waypoint List** page

R1. *BFLD GRPH*. Opens the **Battlefield Graphics** page

R2. *CP LIST*. Opens the **Control Point List** page

R3. *REVERSE ACTIVE ROUTE*. Pressing this bezel will reverse the order of the active route

R4. *NEW TGT*. Opens the **New Target Point** page

R5. *TGT LIST*. Opens the **Target Point List** page

POLYCHOP SIMULATIONS

11.12.6 Route Setup

The Route Setup page allows for the creation, modification and deletion of routes. The Route Setup page is identical for each of the three available route slots.

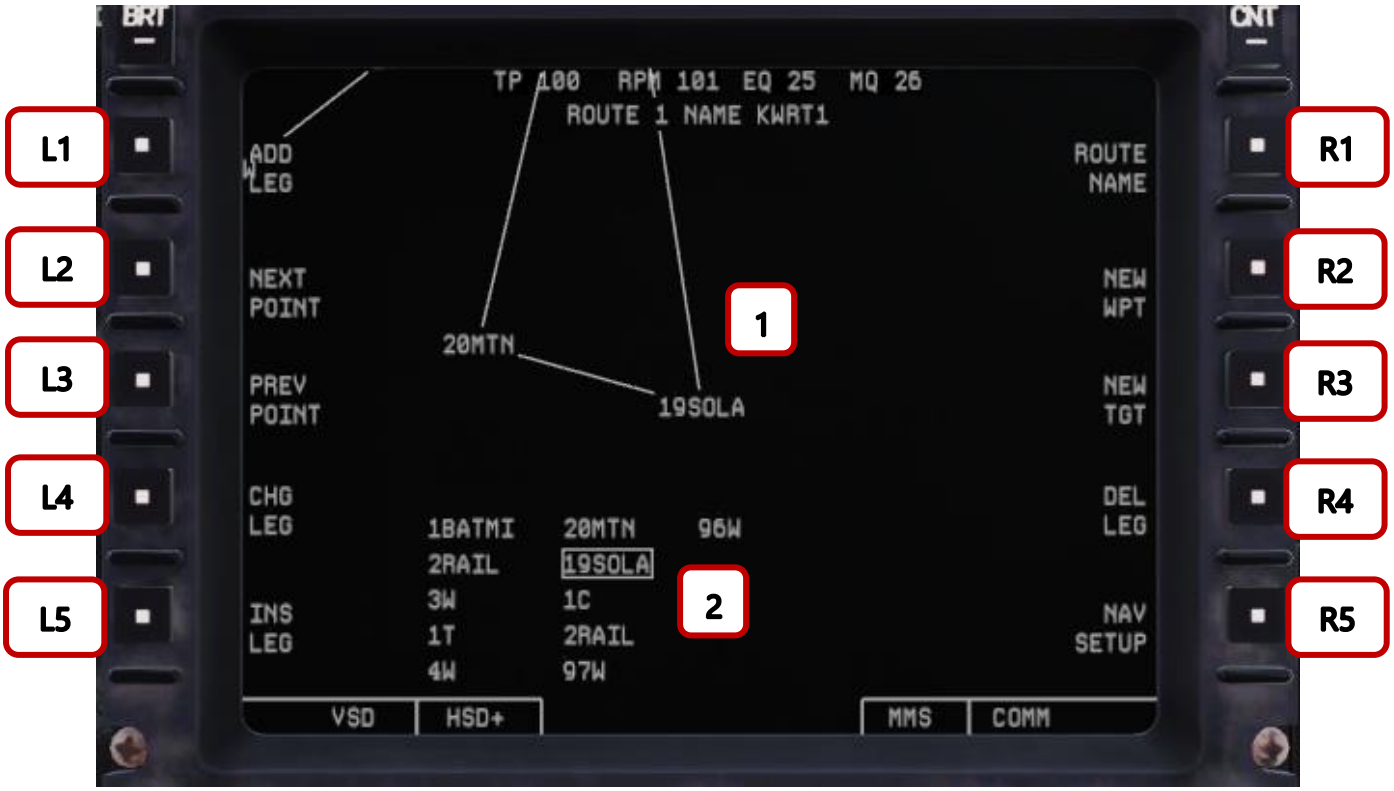


Figure 83 - Route Setup page

1. Route Map. The Route Map will remain centered on the selected Route Point
2. Route Leg List. The selected Point is boxed

INFO:

The scale and orientation of the Route Map in the Route Setup page is determined by the settings applied in the HSD and RMS pages

L1. *ADD LEG*. Pressing this bezel will activate the data entry mode and a cursor will appear after the last Leg in the Route Leg List. Entering a valid waypoint, control point or target point ID will add the point to the Route

R1. *ROUTE NAME*. Activates data entry mode to rename the route

R2. *NEW WPT*. Exits the Route Setup page and opens the **New Waypoint** page

R3. *NEW TGT*. Exits the Route Setup page and opens the **New Target Point** page

POLYCHOP SIMULATIONS

L2. *NEXT POINT*. Selects the next Route Point

L3. *PREV POINT*. Selects the previous Route Point

L4. *CHG LEG*. Enables data entry mode on the selected item. Entering a valid waypoint, control point or target point ID will update the Route

L5. *INS LEG*. Inserts a new Leg into the Route after the selected Route Point. Entering a valid waypoint, control point or target point ID will add the new point to the Route

R4. *DEL LEG*. Enables data entry mode. A cursor will appear under the bezel text. The user has three input options:

- Pressing ENTER without making any other input will delete the currently selected Route Point and update the Route
- Entering 'ALL' and pressing ENTER will clear the Route
- Entering a valid Waypoint, Target Point or Control Point name that is part of the route will delete that Route Point from the Route

R5. *NAV SETUP*. Exits the Route Setup page and opens the **NAV Setup** page

POLYCHOP SIMULATIONS

11.12.7 Waypoint List

The Waypoint List page displays up to 99 stored waypoints, as well as the present position of the aircraft and the DIRECT POINT if currently in use.

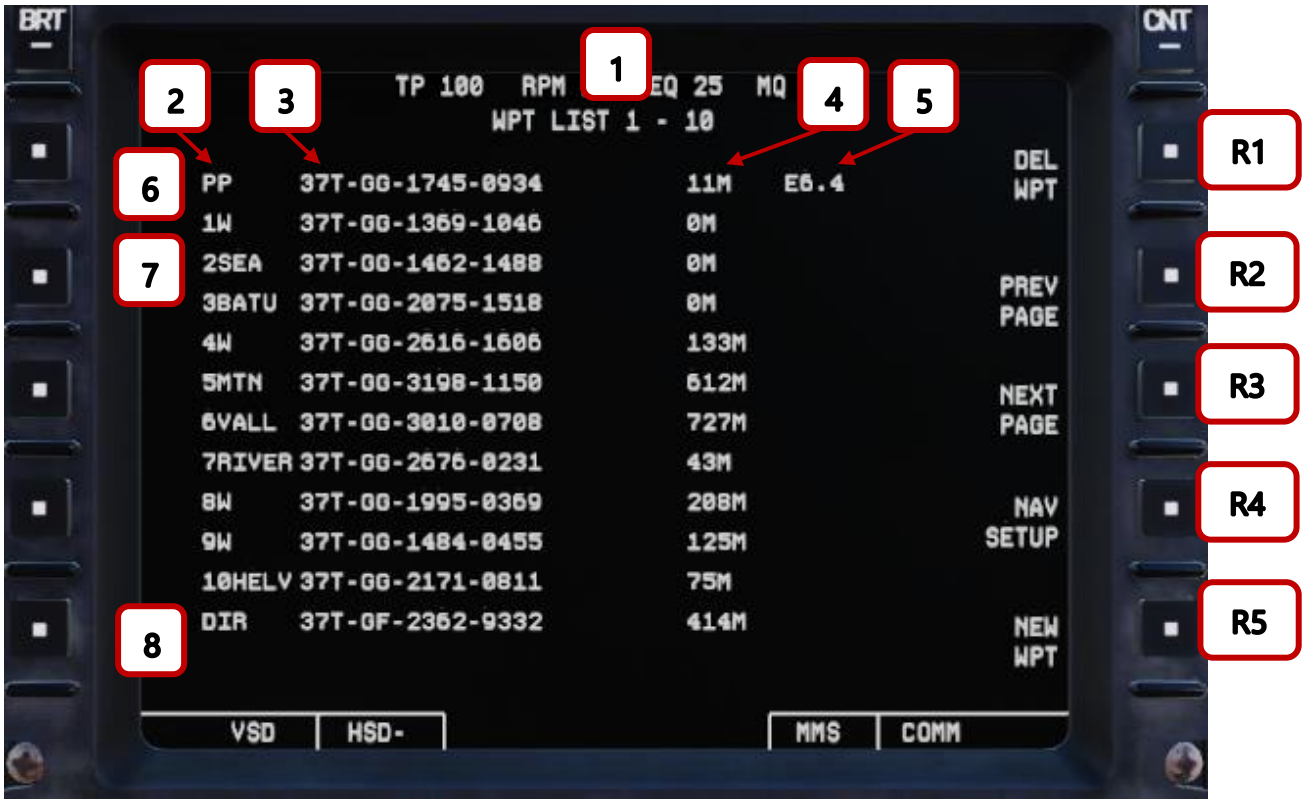


Figure 84 - Waypoint List page

1. Waypoint List Header, with currently shown waypoint minimum and maximum indices
2. Waypoint Name
3. Waypoint Position (UTM or LL)
4. Waypoint Elevation (M or F)
5. Magnetic Declination (Present Position only)
6. Present Position. Present Position is always shown as the first item in the list
7. List of up to ten Waypoints. Displayed Waypoints can be cycled using the PREV PAGE and NEXT PAGE bezels
8. *DIR POINT*. If DIR POINT is currently in use, the selected position and elevation will be displayed at the bottom of the list

R1. *DEL WPT.* Delete Waypoint
R2. *PREV PAGE*. Opens previous page of the Waypoint List
R3. *NEXT PAGE*. Opens next page of the Waypoint List
R4. *NAV SETUP*. Exits the Route Setup page and opens the **NAV Setup** page
R5. NEW WPT. Opens the **New Waypoint** page.

POLYCHOP SIMULATIONS

11.12.8 New Waypoint

The New Waypoint page is used to create and edit Waypoint data.

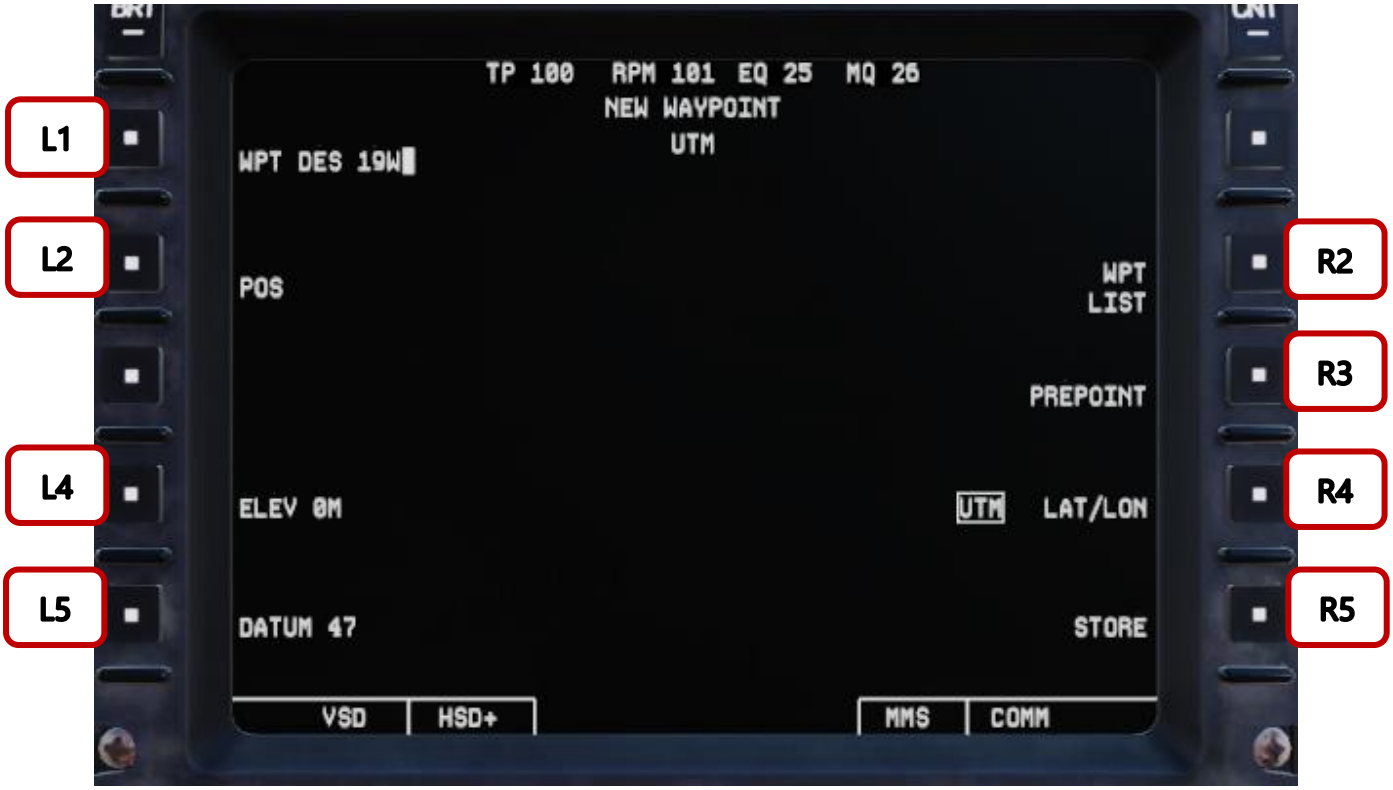


Figure 85 - New Waypoint page

L1. *WPT DES*. ##W

L2. *POS*. entry is user friendly and is automatically configured for data entry, depending on selected UTM/LL format

L4. *ELEV*. is automatically configured for entry

L5. *DATUM*. is automatically filled with current datum

R2. WPT LIST. Opens the **Waypoint List** page

R3. *PREPOINT*. used to add or remove a waypoint from the MMS prepoint

R4. *UTM LAT/LON*. Toggles coordinate mode between UTM and Lat/Long

R5. *STORE*. Save Waypoint

POLYCHOP SIMULATIONS

11.12.9 Control Point List

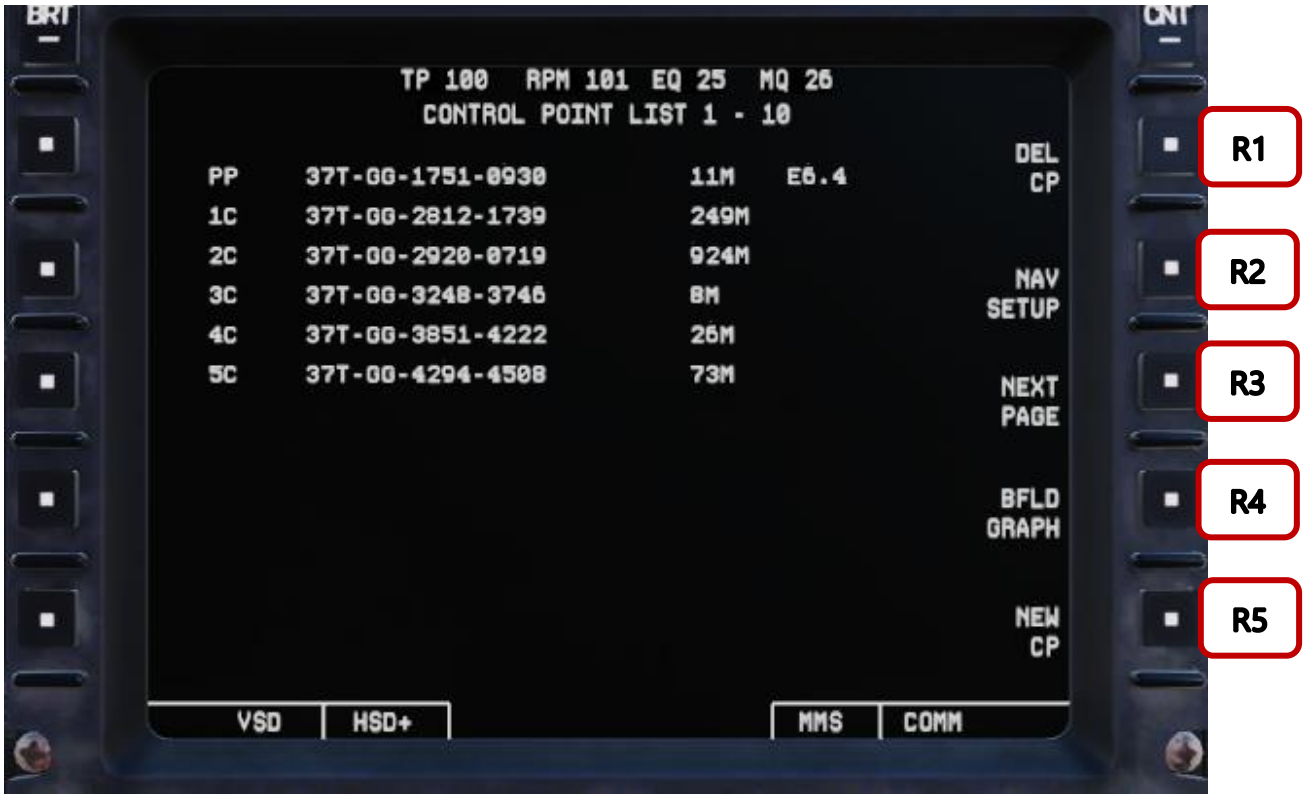


Figure 86 - Control Point List page

R1. *DEL CP*. Delete Control Point

R2. *NAV SETUP*. Exits the Route Setup page and opens the **NAV Setup** page

R3. *NEXT PAGE*. Opens next page of the control point List

R4. BFLD GRPH. Opens the **Battlefield Graphics** page

R5. *NEW CP*. Opens the **New Control Point** page

POLYCHOP SIMULATIONS

11.12.10 New Control Point

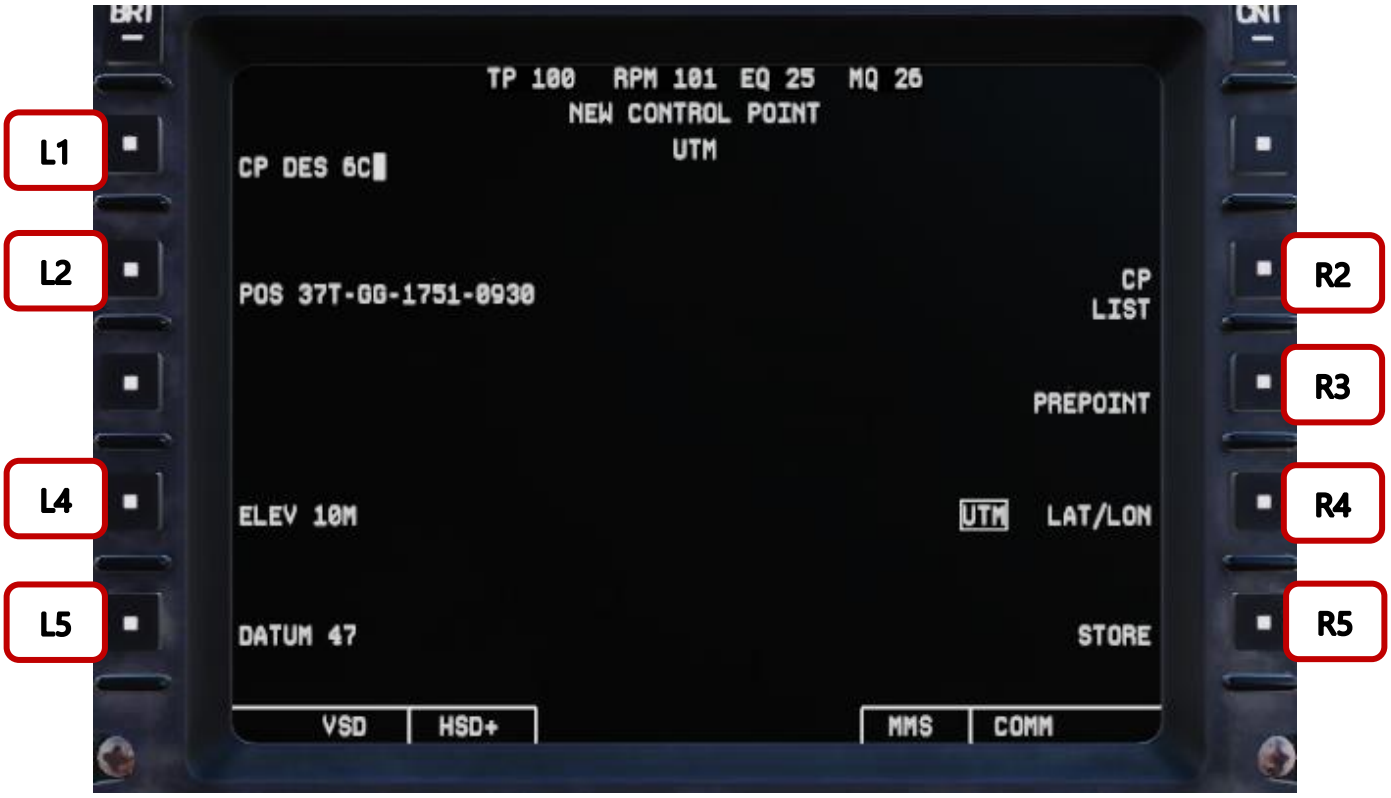


Figure 87 - New Control Point page

L1. *CP DES*. ##C

L2. *POS*. entry is user friendly and is automatically configured for data entry, depending on selected UTM/LL format

L4. *ELEV*. is automatically configured for entry

L5. *DATUM*.

R2. *CP LIST*. Opens the **Control Point List** page

R3. *PREPOINT*. used to add or remove a waypoint from the MMS prepoint

R4. *UTM LAT/LON*. Toggles coordinate mode between UTM and Lat/Long

R5. *STORE*. Save Control Point

POLYCHOP SIMULATIONS

11.12.11 Target Point List

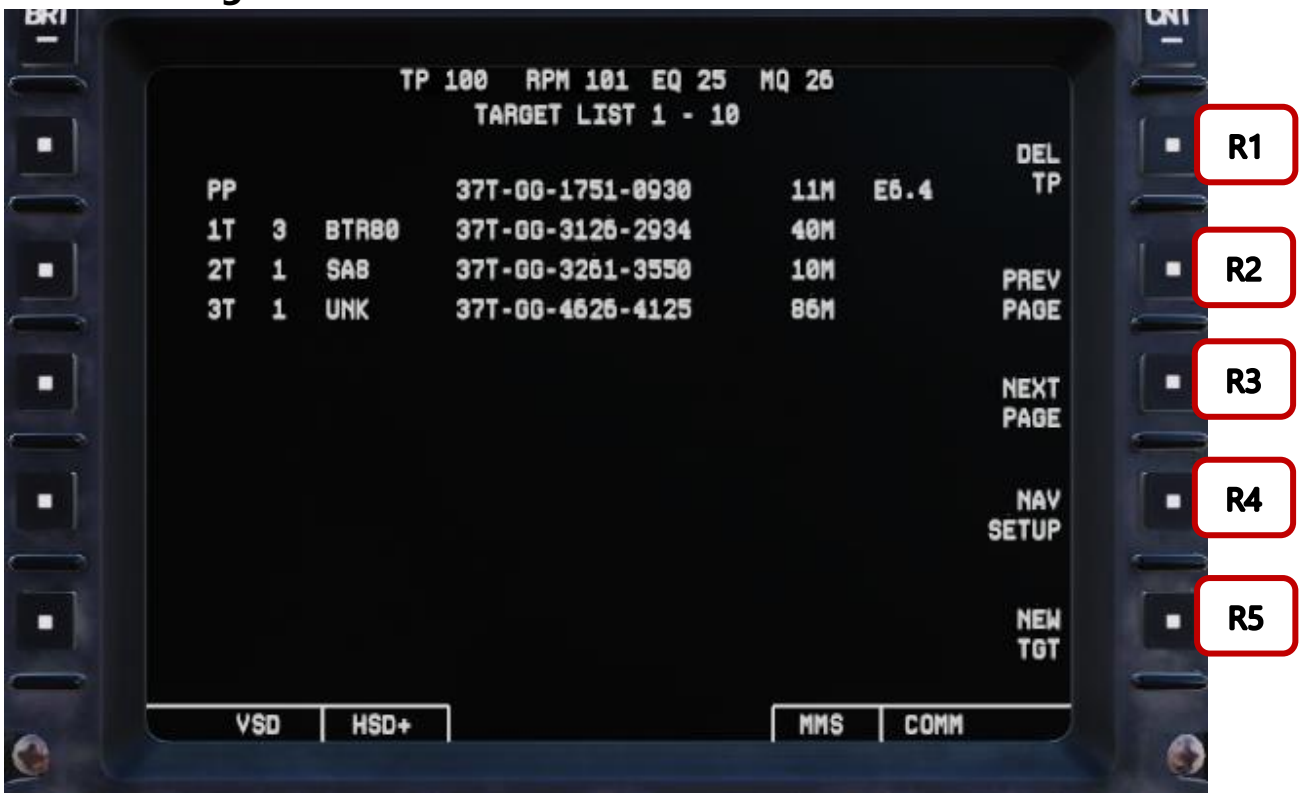


Figure 88 - Target Point List page

R1. *DEL TP*. Delete Target Point

R2. *PREV PAGE*. Opens previous page of the target point List

R3. *NEXT PAGE*. Opens next page of the target point List

R4. *NAV SETUP*. Exits the TP List page and opens the **NAV Setup** page

R5. *NEW TGT*. Opens the **New Target Point** page

POLYCHOP SIMULATIONS

11.12.12 New Target Point

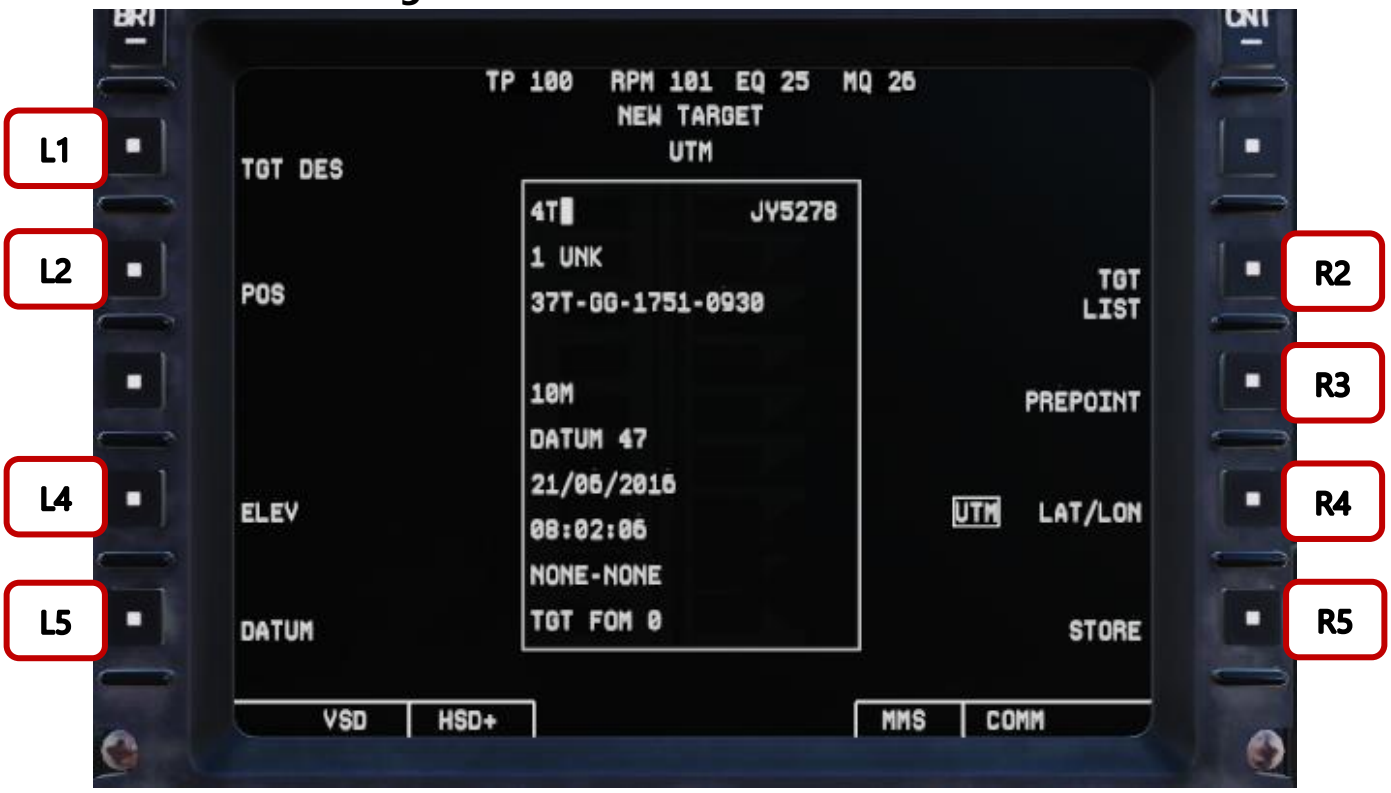


Figure 89 - New Target Point page

L1. *CP DES*. ##T

L2. *POS*. entry is user friendly and is automatically configured for data entry, depending on selected UTM/LL format

L4. *ELEV*. is automatically configured for entry

L5. *DATUM*. s boxed and the current DATUM is displayed followed by a cursor

R2. *TGT LIST*. Opens the **Target Point List** page

R3. *PREPOINT*. used to add or remove a waypoint from the MMS prepoint

R4. *UTM LAT/LON*. Toggles coordinate mode between UTM and Lat/Long

R5. *STORE*. Save Target Point

POLYCHOP SIMULATIONS

11.12.13 Direct Point

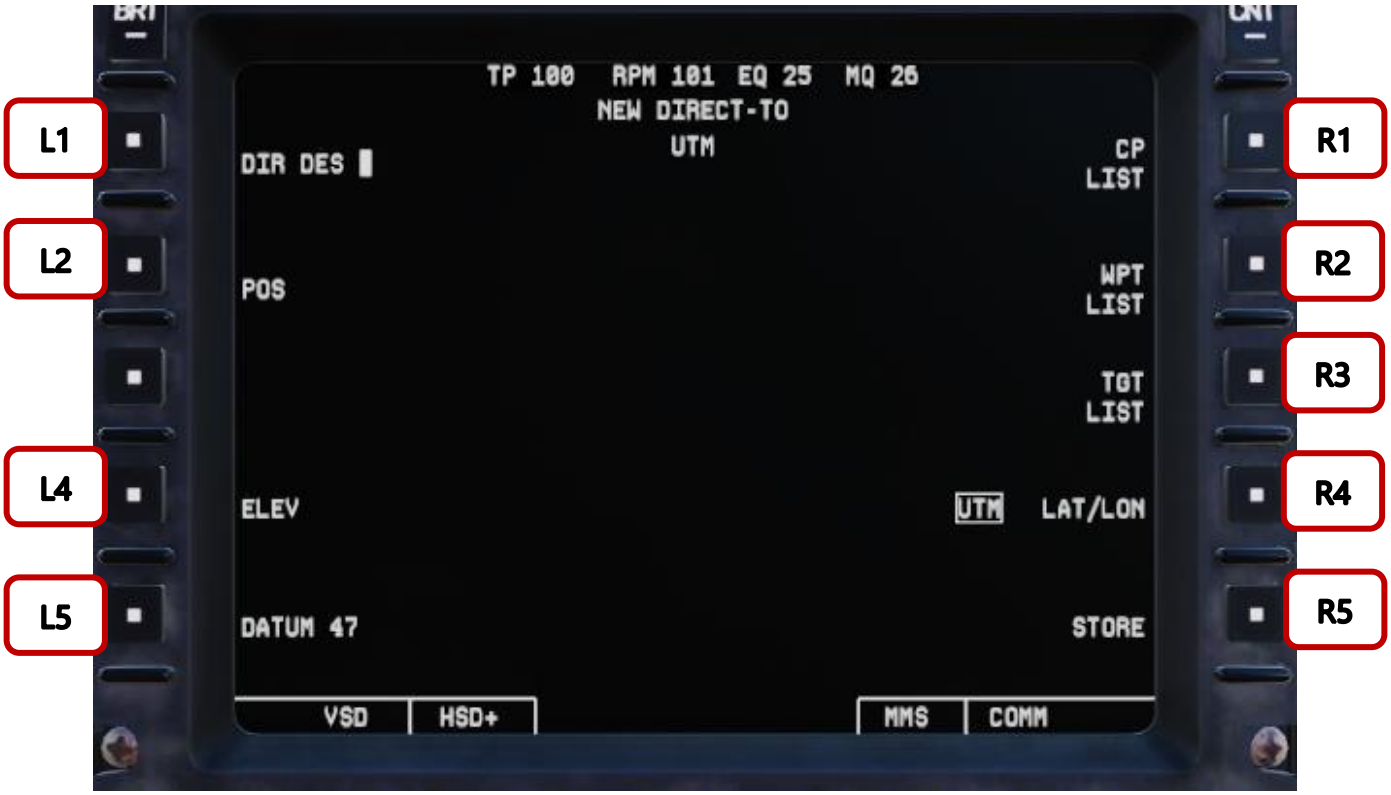


Figure 90 - Direct Point page

L1. *CP DES*. ##

L2. *POS*. entry is user friendly and is automatically configured for data entry, depending on selected UTM/LL format

L4. *ELEV*. is automatically configured for entry

L5. *DATUM*. is automatically configured with current datum and hot cursor

R1. *CP LIST*. Opens the **Control Point List** page

R2. WPT LIST. Opens the **Waypoint List** page

R3. *TGT LIST*. Opens the **Target Point List** page

R4. *UTM LAT/LON*. Toggles coordinate mode between UTM and Lat/Long

R5. *STORE*. Save Direct Point

POLYCHOP SIMULATIONS

11.12.14 Battlefield Graphics

Battlefield lines or areas are defined by waypoints, targets, or control points

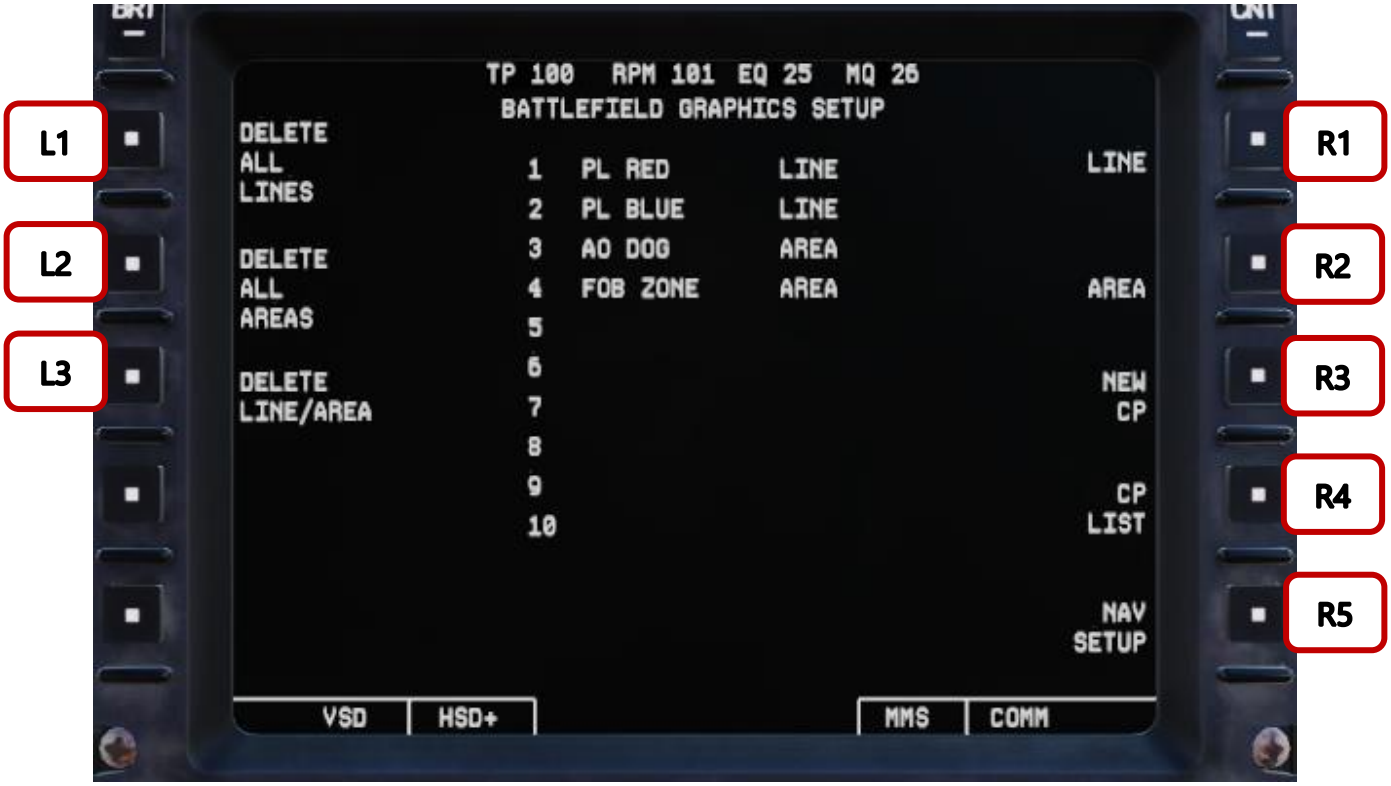


Figure 91 - Battlefield Graphics page

L1. *DELETE ALL LINES*. is used to delete all lines

L2. *DELETE ALL AREAS*. is used to delete all areas

L3. *DELETE LINE/AREA*. is used to delete a specific line or area

R1. *LINE*. Select a specific (existing) Line

R2. *AREA*. Select a specific (existing) Area

R3. *NEW CP*. Opens the **New Control Point** page

R4. *CP LIST*. Opens the **Control Point List** page

R5. *NAV SETUP*. Opens the **NAV Setup** page

INFO:

Battlefield Lines & Areas rely on the existence of waypoints/target points/control points to be created, if none exist you must first create these points as required.

POLYCHOP SIMULATIONS

11.12.14.1 Create New Line

Creates, modifies, and manages battlefield graphic lines

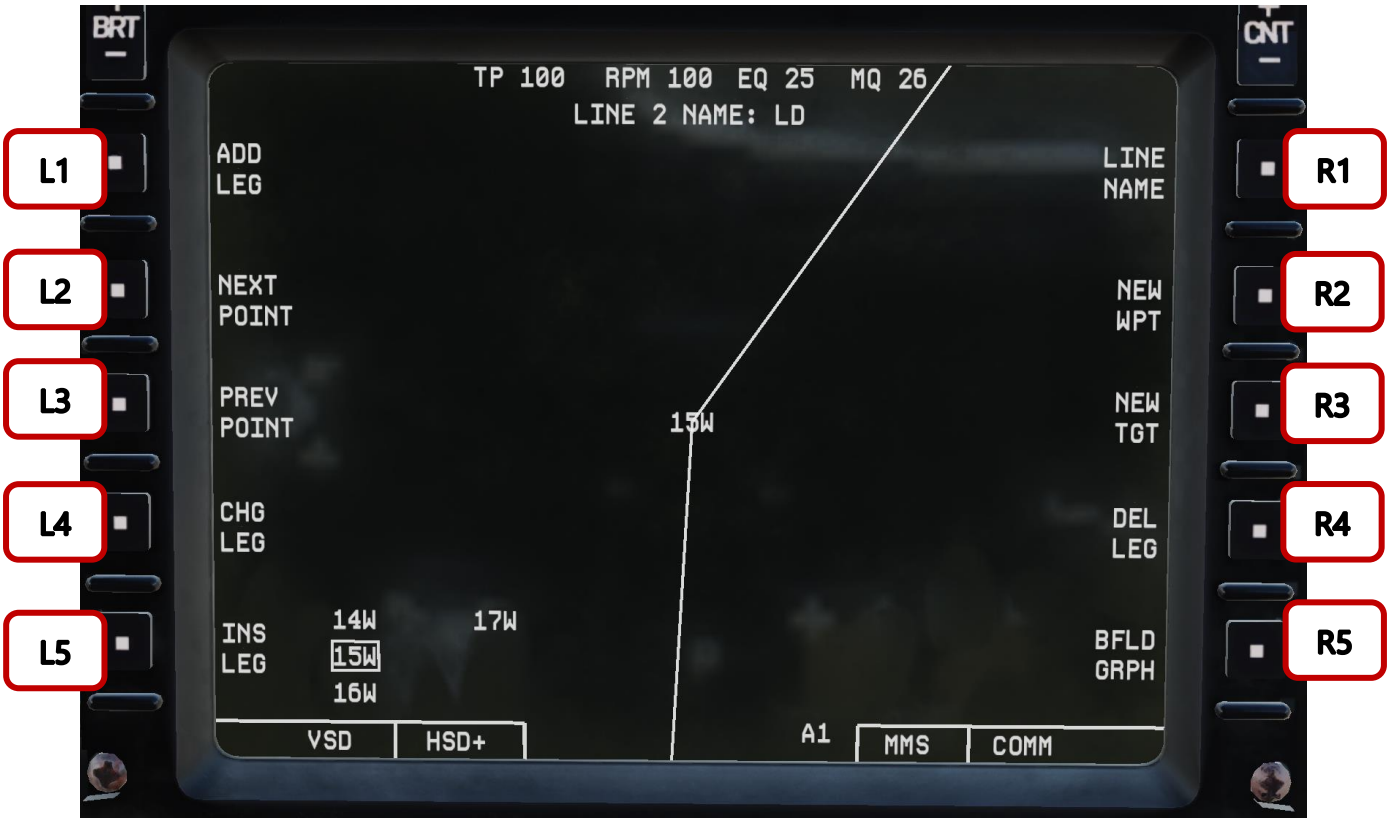


Figure 92 - Battlefield Graphic Line page

L1. *ADD LEG*. used to add a leg

L2. *NEXT POINT*. used to select the next point

L3. *START LINE*. used to start a line

L4. *CHG LEG*. used to change a leg of an existing line

L5. *INS LEG*. . used to insert a leg into an existing line

R1. *LINE NAME*. used to rename the line

R2. *NEW WPT*. Opens the **New Waypoint** page.

R3. *NEW TGT*. Opens the **New Target Point** page

R4. *DEL LEG*. used to delete a leg

R5. *BFLD GRPH*. Opens the **Battlefield Graphics** page

POLYCHOP SIMULATIONS

11.12.14.2 Create New Area

Creates, modifies and manages battlefield graphic areas.

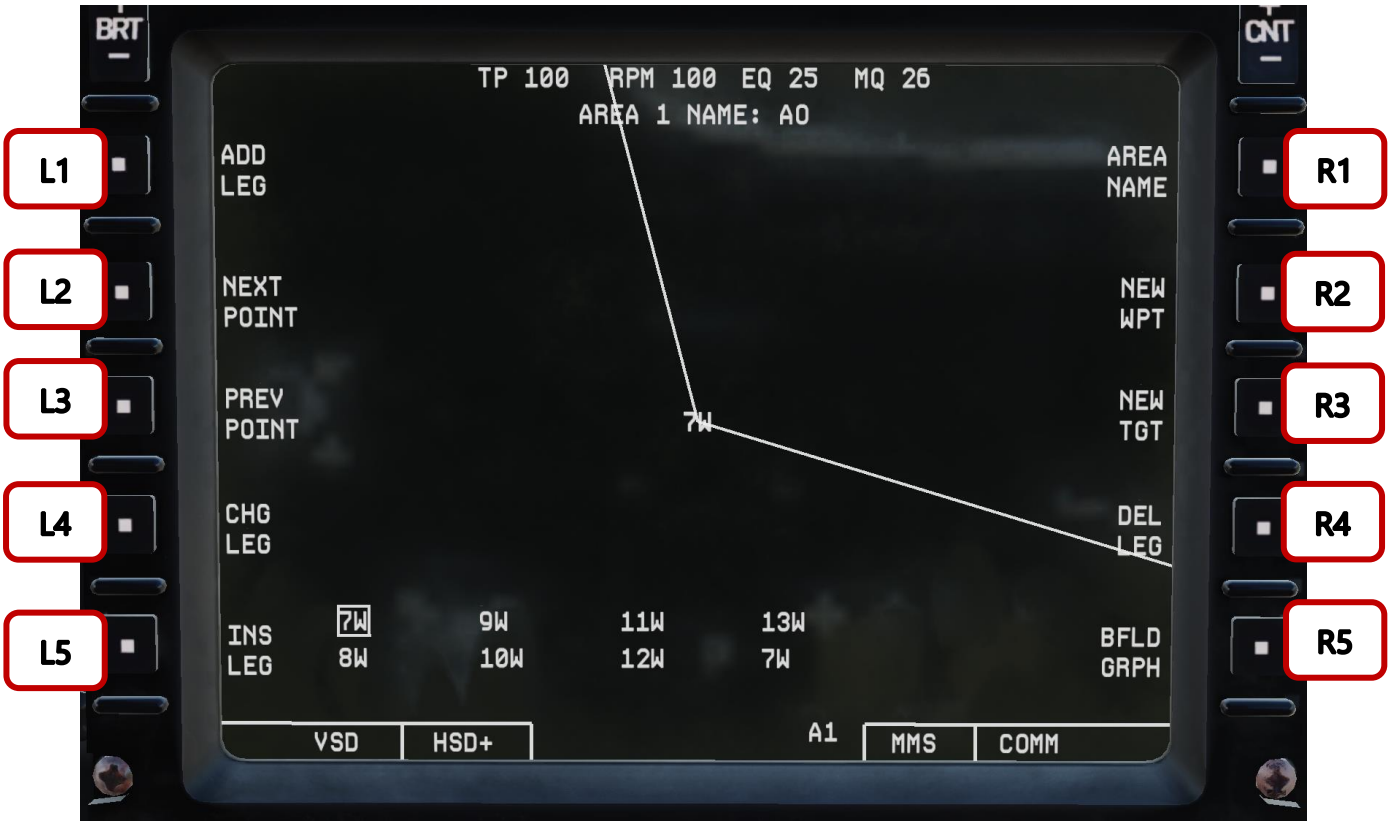


Figure 93 - Battlefield Graphic Area page

L1. *ADD LEG*. used to add a leg

L2. *NEXT POINT*. used to select the next point

L3. *START AREA*. used to start an area

L4. *CHG LEG*. used to modify a leg

L5. *INS LEG*. used to add a leg in an existing area

R1. *AREA NAME*. used to rename the AREA

R2. *NEW WPT*. Opens the **New Waypoint** page.

R3. *NEW TGT*. Opens the **New Target Point** page

R4. *DEL LEG*. . used to delete a leg

R5. *BFLD GRPH*. Opens the **Battlefield Graphics** page

POLYCHOP SIMULATIONS

11.13 COMM Pages

11.13.1 COMM

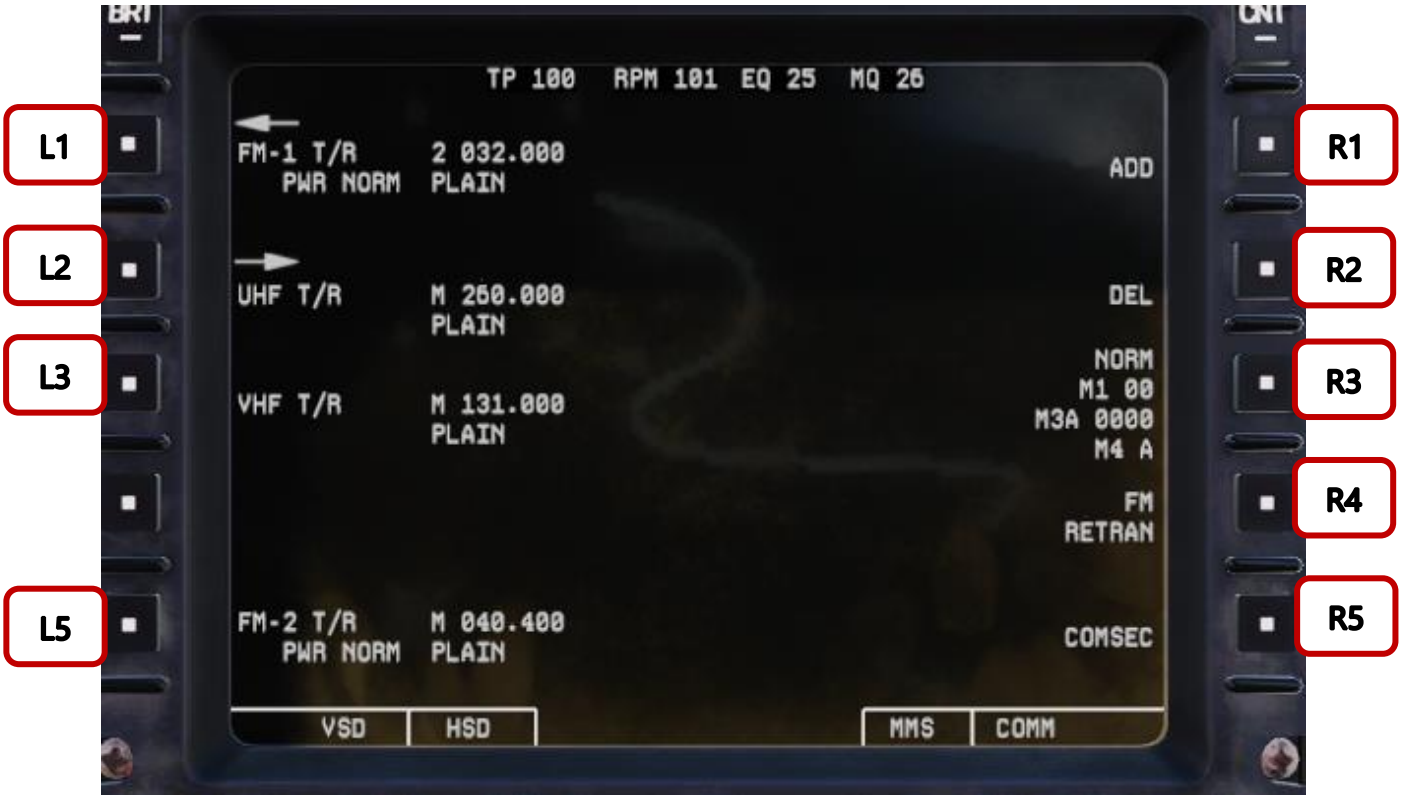


Figure 94 - COMM page

L1. *FM-1 T/R*. displays the frequency list for the corresponding FM-1 radio.

L2. *UHF T/R*. displays the frequency list for the corresponding UHF radio.

L3. *VHF T/R*. displays the frequency list for the corresponding VHF radio.

L5. *FM-2 T/R*. displays the frequency list for the corresponding FM-2 radio.

R1. *ADD*. Add frequency to frequency list

R2. *DEL*. Delete preset frequency

R3. *IFF*. (IFF) information display.

R4. *FM RETRAN*. Opens the FM Retrans page

R5. *COMSEC*. Opens the COMSEC page

POLYCHOP SIMULATIONS

11.13.2 IFF 1

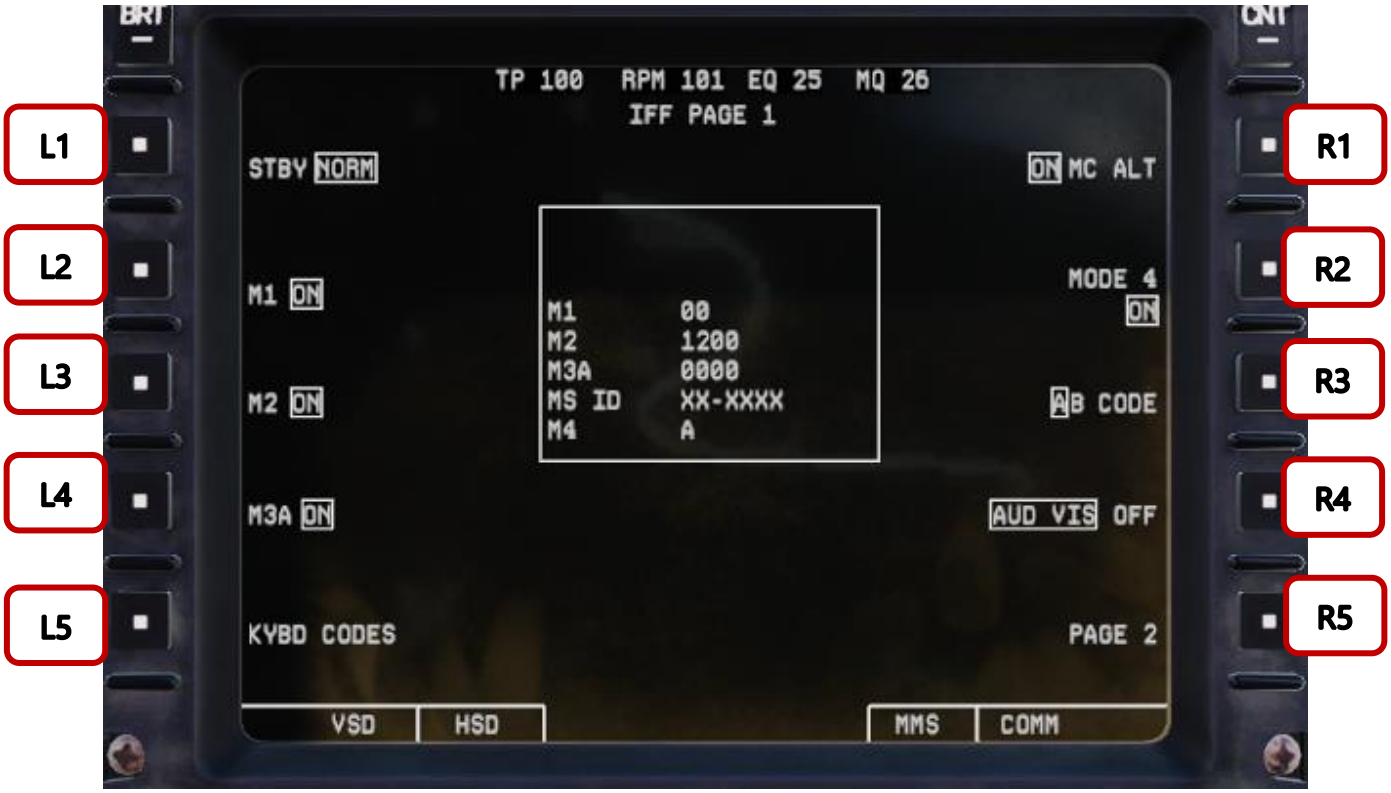


Figure 95 - IFF page 1

L1. *STBY or NORM.* The selected option is boxed.

L2. *M1*. ON-OFF switch for mode 1.

L3. *M2*. ON-OFF switch for mode 2.

L4. *M3A*. ON-OFF switch for M3A.

L5. *KYBD CODES*. Activates MFK to input changes to Mode 1, Mode 2, Mode 3A, and Mode S codes.

R1. *MC ALT* . Selects ON or OFF condition for altitude reporting to a Mode C interrogation.

R2. *MODE 4*. Selects Mode 4 ON or OFF condition.

R3. *CODE*. Selects Mode 4 Code A (today's code) or Code B (tomorrow's code). The selected option is boxed.

R4. *AV MODE.* Selects the method used to notify the crew when interrogation occurs.

R5. *PAGE 2.* Opens page 2 of the IFF menu

POLYCHOP SIMULATIONS

11.13.3 IFF 2

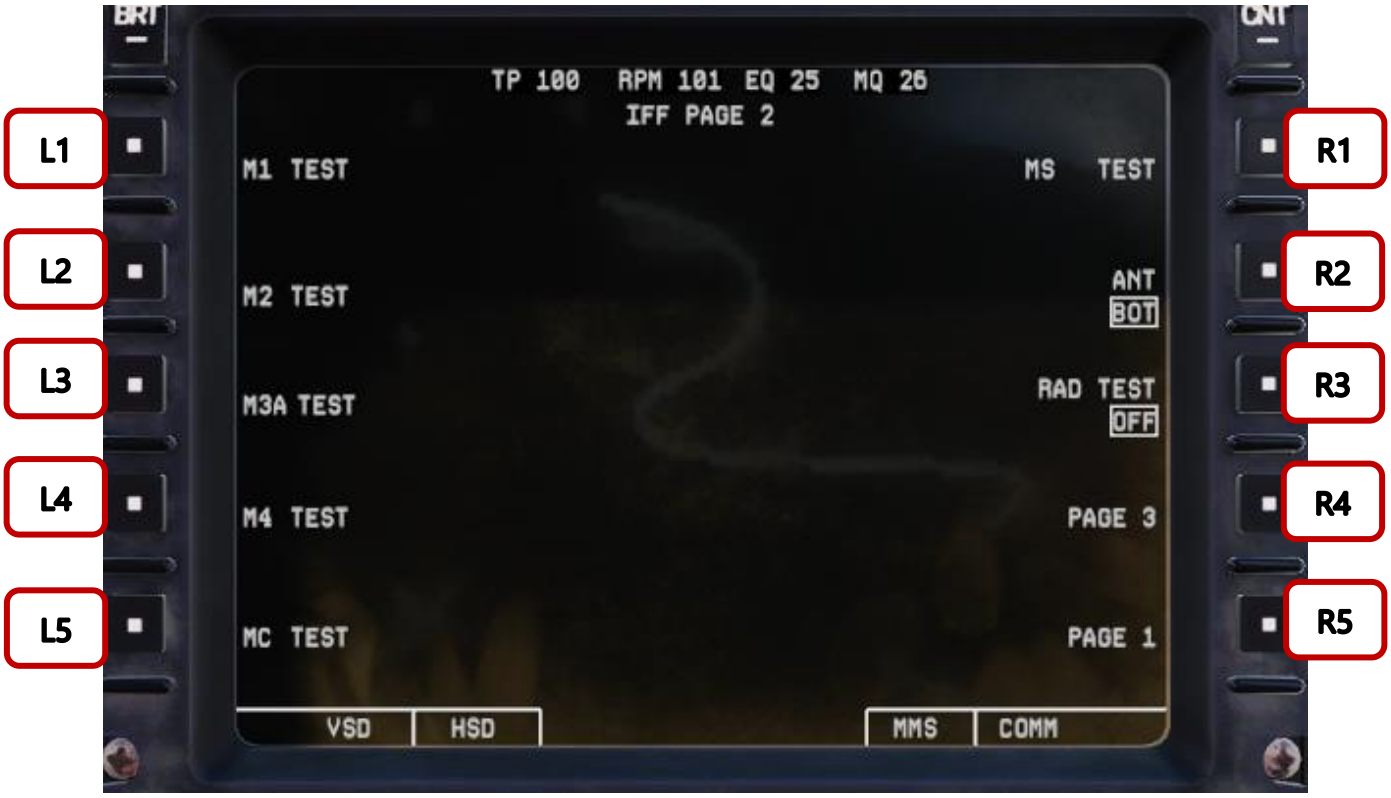


Figure 96 - IFF page 2

L1. *M1.* Test Status.

L2. *M2.* Test Status.

L3. *M3A.* Test Status.

L4. *M4.* Test Status.

L5. *MC.* Test Status.

R1. *MS.* Test Status.

R2. *ANT.* Selection (TOP | BOT | DIV)

R3. *RAD TEST.* ON or OFF condition.

R4. *PAGE 3.* Opens page 3 of the IFF menu.

R5. *PAGE 1.* Returns to page 1 of the IFF menu.

INFO:

GO/NO-GO appears after test.
GO/NO-GO remains on the page until a different page is selected.
All tests can be performed simultaneously.
When NO-GO is displayed, press key and run test again.

POLYCHOP SIMULATIONS

11.13.4 IFF 3

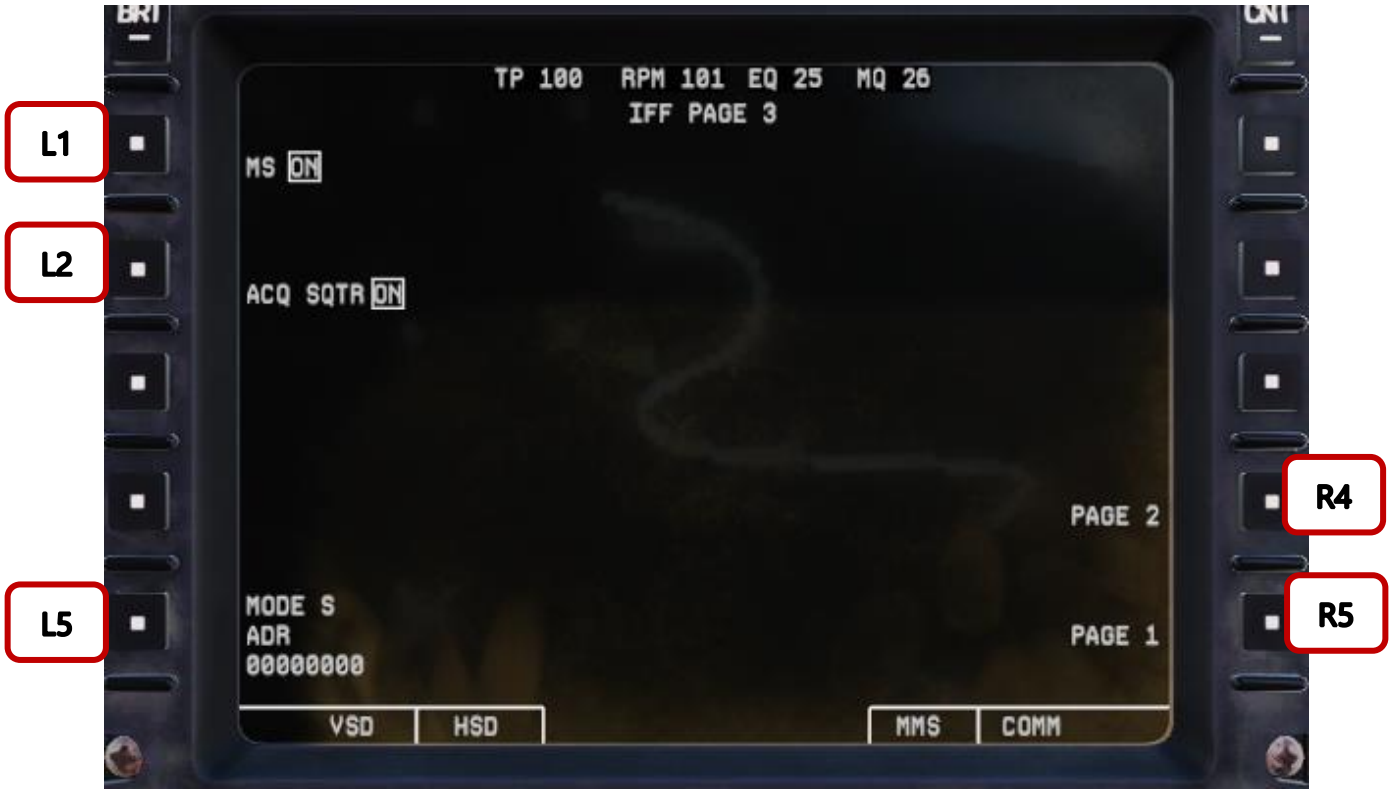


Figure 97 - IFF page 3

L1. *MS.* ON-OFF switch for Mode S.

L2. *ACQ SQTR*. ON-OFF switch for Acquisition Squitter.

L5. *MODE S ADR*. Displays and allows temporary changes to the Mode S Address.

R4. *PAGE 2.* Returns to page 2 of the IFF menu

R5. *PAGE 1.* Returns to page 1 of the IFF menu

11.14 Weapons Pages

11.14.1 Weapons

The Weapons page allows for configuration of the loaded weapons. This page is dynamic and will change based on the aircraft's payload. Configuration options for both left and right weapons are displayed, which may result in options for different weapons being displayed at the same time.

11.14.1.1 Gun



Figure 98 - Weapons page (Gun)

L3. *ROUNDS ENTER*. causes the existing value to blank and be replaced with a cursor, using the MFK, then enter a value between 0 and 500, then press ENTER key

R3. *GUN RETICLE.* Successive presses cycles the offset from zero, to minus two, to minus four, and back to zero. The selected option is boxed.

R5. *WEAPONS BIT/SET-UP PAGE.* Displays the WEAPONS BIT/SET-UP PAGE

POLYCHOP SIMULATIONS

11.14.1.2 Rockets

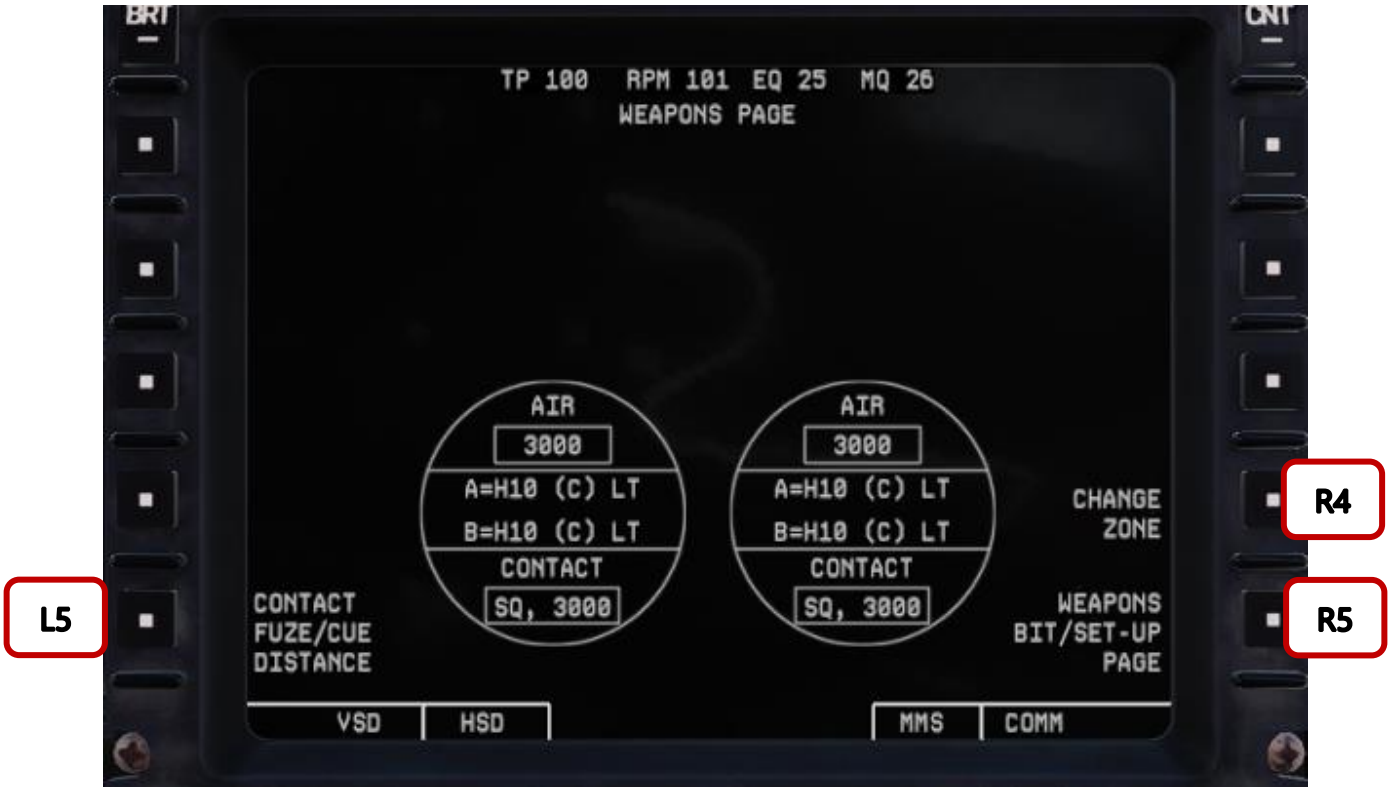


Figure 99 - Weapons page (Rockets)

L5. *CONTACT FUZE/CUE DISTANCE.* changes the rocket fuze distance text at L5 to blank and be replaced by the text CONTACT FUZE on the top line and S, B, 10-45 on the second line with a cursor below it.



R4. *CHANGE ZONE.* Activating this key changes zones from A to B to ALL.

R5. *WEAPONS BIT/SET-UP PAGE.* Displays the first WEAPONS BIT/SET-UP PAGE

POLYCHOP SIMULATIONS

11.14.1.3 Hellfire



Figure 100 - Weapons page (Hellfire)

L1. *TOGGLE MISSILE CODE.* exchanges the primary codes for the alternate codes, and vice versa.

L2. *ECCM.* activates ECCM (electronic counter-countermeasure) in a spun-up selected missile.

R1. *PRI/ALT CODES.* selected, the text is replaced by PRI, followed by a cursor, on the top line, and ALT beneath it. A letter (A through H, or a manual code) is selected on the MFK followed by ENTER. Next, the text at R1 displays ALT followed by a cursor. The ALT code is entered using the same procedure.

R2. *DEICE.* jettisons deice cover on the missiles.

R5. *WEAPONS BIT/SET-UP PAGE.* Displays the first WEAPONS BIT/SET-UP PAGE

POLYCHOP SIMULATIONS

11.14.1.4 ATAS



Figure 101 - Weapons page (ATAS)

R5. *WEAPONS BIT/SET-UP PAGE.* Displays the first WEAPONS BIT/SET-UP PAGE

POLYCHOP SIMULATIONS

11.14.2 Weapons BIT/Setup

The Weapons BIT/Setup pages are almost identical to the Weapons pages, with some small changes to bezel options.

11.14.2.1 Gun



Figure 102 - Weapons BIT/Setup page (Gun)

L3. *ROUNDS ENTER*. causes the existing value to blank and be replaced with a cursor, using the MFK, then enter a value between 0 and 500, then press ENTER key

R3. *GUN RETICLE.* Successive presses cycles the offset from zero, to minus two, to minus four, and back to zero. The selected option is boxed.

R5. *WEAPONS PAGE.* returns to the WEAPONS PAGE

POLYCHOP SIMULATIONS

11.14.2.2 Rockets

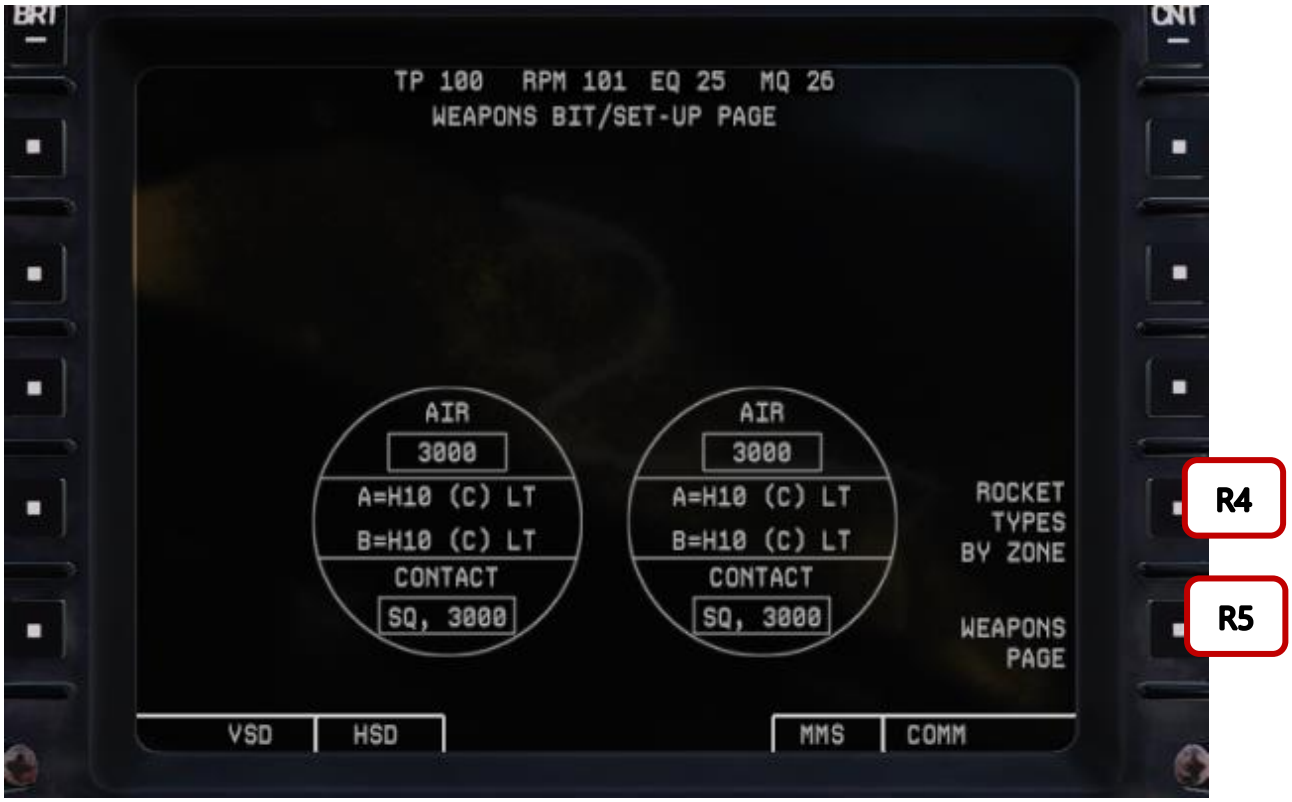


Figure 103 - Weapons BIT/Setup page (Rockets)

R4. *ROCKET TYPES BY ZONE*. display the second weapons BIT/set-up page (warhead types).

R5. *WEAPONS PAGE.* returns to the WEAPONS PAGE

POLYCHOP SIMULATIONS

11.14.2.3 Rocket Types



Figure 104 - Weapons BIT/Setup page (Rocket Types)

R4. *ZONE A, ZONE B.* identifies the active zone.

R5. *WEAPONS PAGE.* returns to the WEAPONS PAGE

11.14.2.4 Hellfire



Figure 105 - Weapons BIT/Setup page (Hellfire)

L1. *HELLFIRE BIT.* initiates Hellfire BIT.

R1. *MISSILES PER CODE*. selects the number of primary and alternate coded missiles.

R5. *WEAPONS PAGE.* returns to the WEAPONS PAGE

POLYCHOP SIMULATIONS

11.14.2.5 ATAS



Figure 106 - Weapons BIT/Setup page (ATAS)

L5. *ATAS BIT.* Perform ATAS BIT.

R5. *WEAPONS PAGE.* returns to the WEAPONS PAGE

INFO:

Text "ATAS BIT" remains boxed while BIT is in progress.

BIT failure for the ATAS launchers is an "X" overlaid on the missile.

POLYCHOP SIMULATIONS

11.14.3 Pilot Sparse Weapons Page (Gun)

The Pilot Sparse Weapons page is used for employment of the selected weapon. It contains a mixture of VSD and weapon symbology. For further information about the VSD symbology, see **VSD**.

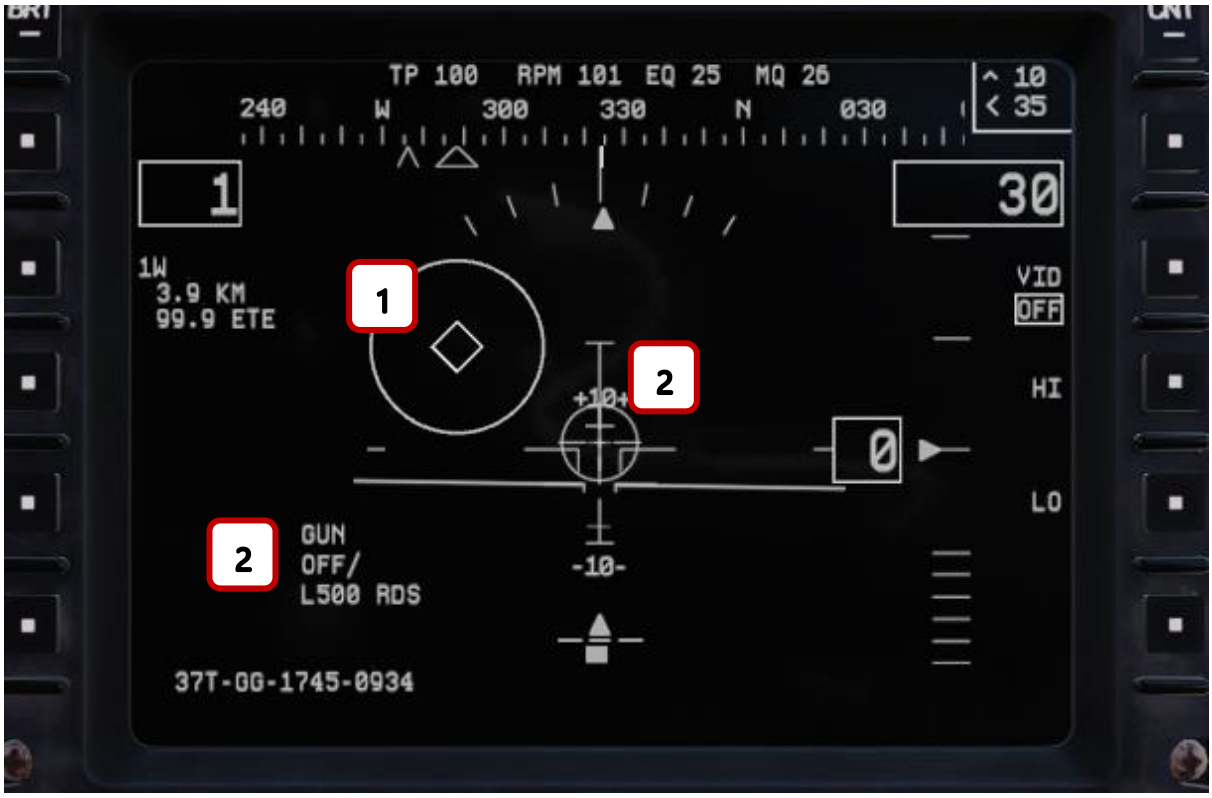


Figure 107 - Pilot Sparse Weapons page (Gun)

1. MMS Line of Sight indicator
2. Gun Reticule
3. Weapon Status indicator and Ammunition Counter

INFO:

The MMS Line of Sight indicator is simply a reference point for the MMS, and does not provide and ranging elevation or windage adjustment information for employing the gun.

POLYCHOP SIMULATIONS

11.14.4 Pilot Sparse Weapons Page (Rockets)

The Pilot Sparse Weapons page is used for employment of the selected weapon. It contains a mixture of VSD and weapon symbology. For further information about the VSD symbology, see **VSD**.

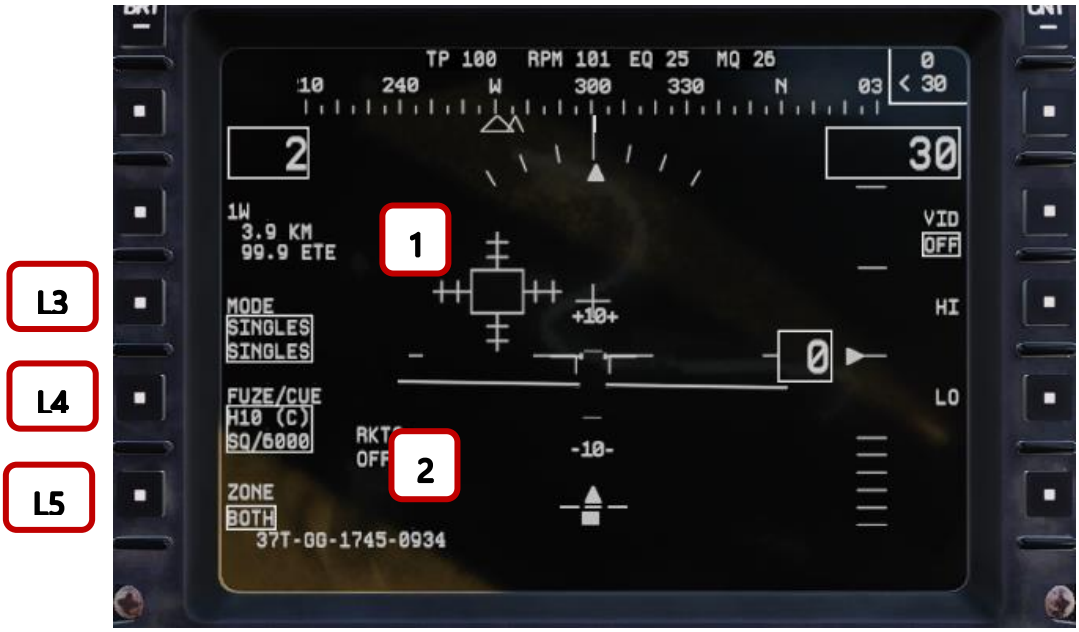


Figure 108 - Pilot Sparse Weapons page (Rockets)

1. Rocket Cue. The rocket cue azimuth is fed from the MMS line of sight, and the elevation is based on the Cue elevation source and type of rocket selected
2. Weapon Status indicator

INFO:

The Rocket Cue range tables are designed for employing rockets in a stationary hover. They do not factor in airspeed or windage and should not be considered an accurate method of targeting.

L3. *MODE.* Selects firing mode. Pressing L3 will cycle between SINGLES/SINGLES, SINGLES/PAIR, RIPPLE/SINGLES and RIPPLE/PAIR. If only a single rocket pod is installed, will cycle between SINGLES and RIPPLE.

L4. *FUZE/CUE.* Pressing this bezel will active data entry mode and allow configuration of fuze and cue range source. See **Weapons Page - Rockets** for details on use of this input

L5. *ZONE.* Selection of rocket pod zone to employ. Cycles between BOTH/A/B. Configuration will change based on zone selection. If BOTH is selected, the configuration of one A will be used

POLYCHOP SIMULATIONS

11.14.5 Pilot Sparse Weapons Page (Hellfire)

The Pilot Sparse Weapons page is used for employment of the selected weapon. It contains a mixture of VSD and weapon symbology. For further information about the VSD symbology, see **VSD**.

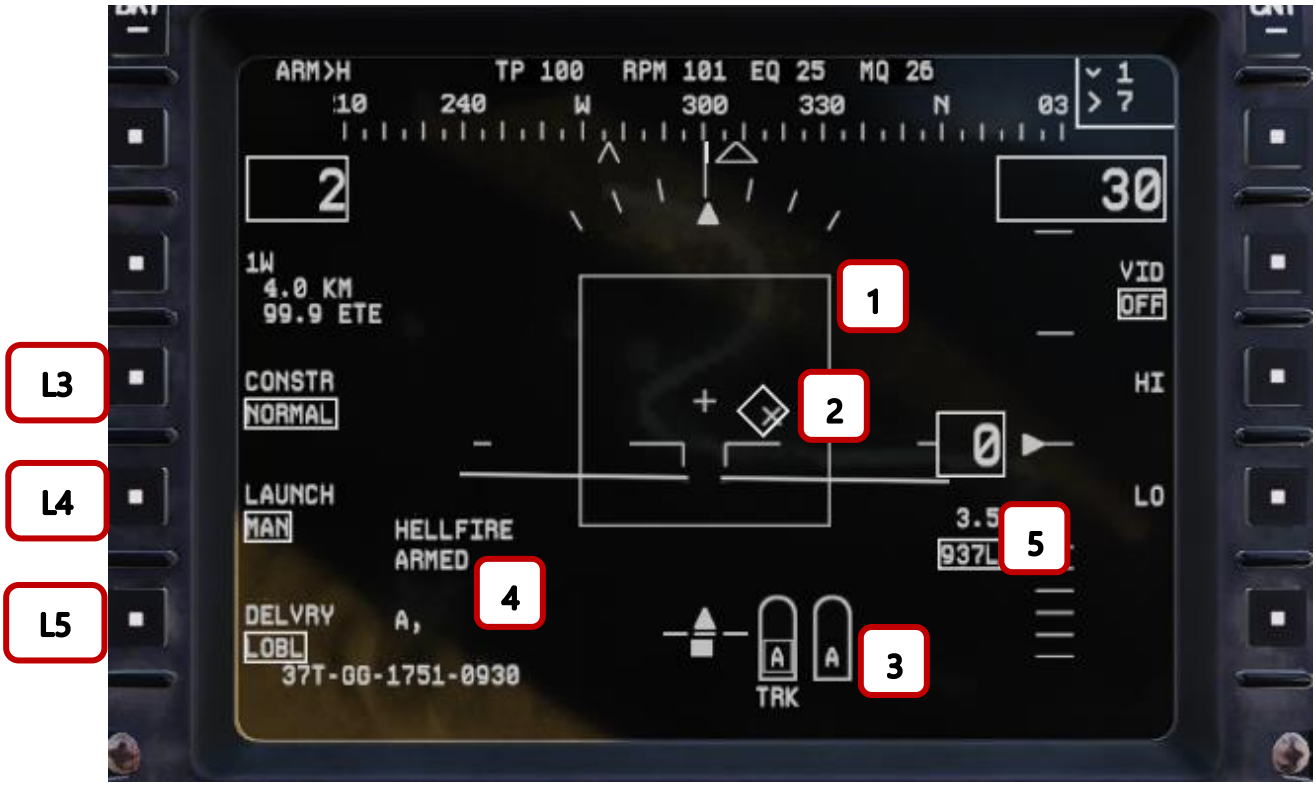


Figure 109 - Pilot Sparse Weapons page (Hellfire)

1. Hellfire constraints box (LOBL mode, within constraints)
2. MMS Line of Sight indicator (diamond) and Hellfire seeker cue (cross)
3. Missile selection and code status indicator
4. Missile status indicator
5. Range, range source and estimated missile flight time in seconds

L3. *CONSTR.* Hellfire Constraints mode. Toggles between NORMAL and OVRIDE (Override)

L4. *LAUNCH.* Launch Mode. Cycles from STBY (Standby), to MAN (Manual), NORM (Normal) and RIPL (Ripple)

L5. *DELVRY.* Hellfire Delivery method. Cycles between LOBL (Lock on Before Launch), DIR (Lock on After Launch, Direct), LO, (Lock on After Launch, Low), HI (Lock on After Launch, High)

POLYCHOP SIMULATIONS

11.14.6 Pilot Sparse Weapons Page (ATAS)

The Pilot Sparse Weapons page is used for employment of the selected weapon. It contains a mixture of VSD and weapon symbology. For further information about the VSD symbology, see **VSD**.

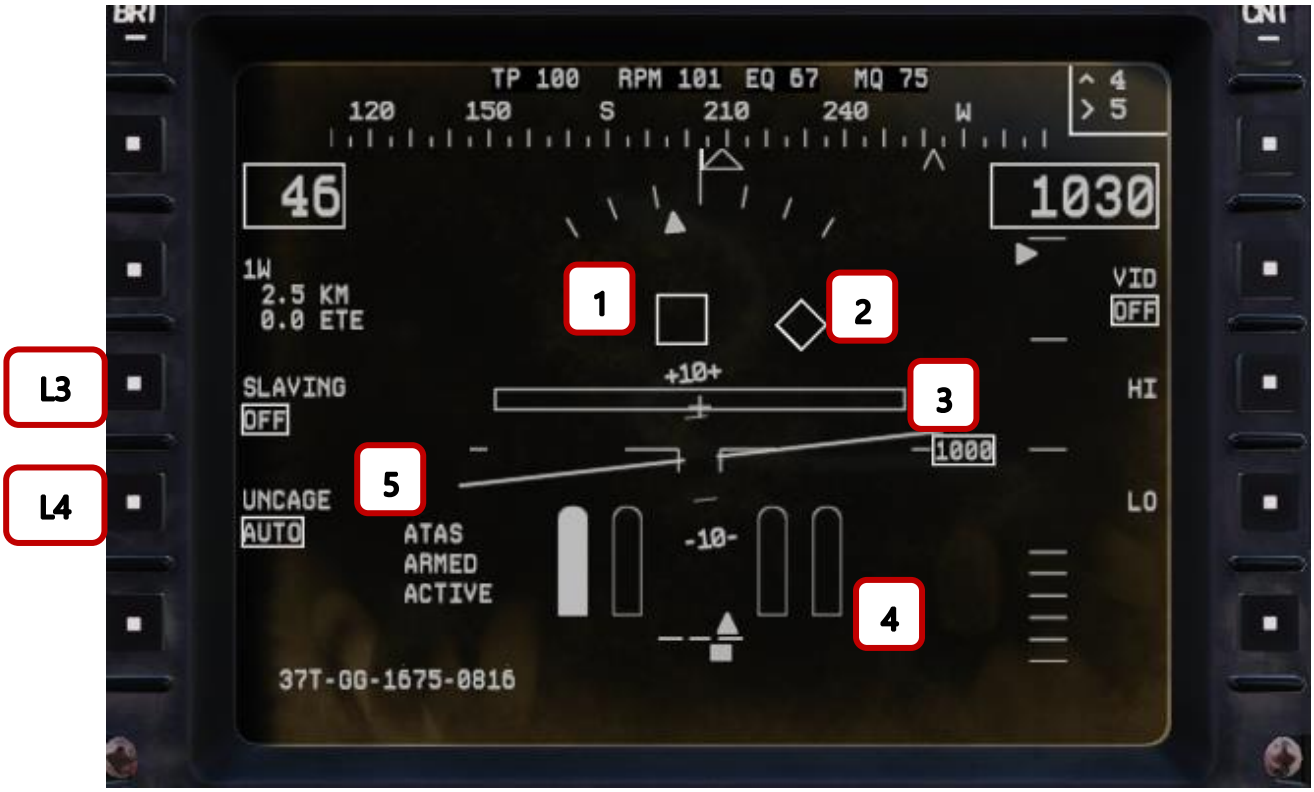


Figure 110 - Pilot Sparse Weapons page (ATAS)

1. ATAS Seeker cue (uncaged, tracking)
2. MMS Line of Sight indicator
3. Superelevation cue
4. Missile selection indicator
5. System status

L3. *SLAVING.* Toggles between OFF and ON

L4. *UNCAGE.* Toggles between AUTO and MAN (Manual)

POLYCHOP SIMULATIONS

11.14.7 Copilot Sparse Weapons Page (Gun)

The Copilot Sparse Weapons page is used for employment of the selected weapon. It contains weapon symbology overlaid onto VSD and MMS symbology. For further information about the VSD and MMS symbologies, see **VSD** and **MMS**.

For information on the Gun symbology, see **Pilot Sparse Weapons Page (Gun)**.



Figure 111 - Copilot Sparse Weapons page (Gun)

POLYCHOP SIMULATIONS

11.14.8 Copilot Sparse Weapons Page (Rockets)

The Copilot Sparse Weapons page is used for employment of the selected weapon. It contains weapon symbology overlaid onto VSD and MMS symbology. For further information about the VSD and MMS symbologies, see **VSD** and **MMS**.

For information on the Rockets symbology, see **Pilot Sparse Weapons Page (Rockets)**.



Figure 112 - Copilot Sparse Weapons page (Rockets)

POLYCHOP SIMULATIONS

11.14.9 Copilot Sparse Weapons Page (Hellfire)

The Copilot Sparse Weapons page is used for employment of the selected weapon. It contains weapon symbology overlaid onto VSD and MMS symbology. For further information about the VSD and MMS symbologies, see **VSD** and **MMS**.

For information on the Hellfire symbology, see **Pilot Sparse Weapons Page (Hellfire)**.



Figure 113 - Copilot Sparse Weapons Page (Hellfire)

POLYCHOP SIMULATIONS

11.14.10 Copilot Sparse Weapons Page (ATAS)

The Copilot Sparse Weapons page is used for employment of the selected weapon. It contains weapon symbology overlaid onto VSD and MMS symbology. For further information about the VSD and MMS symbologies, see **VSD** and **MMS**.

For information on the ATAS symbology, see **Pilot Sparse Weapons Page (ATAS)**.

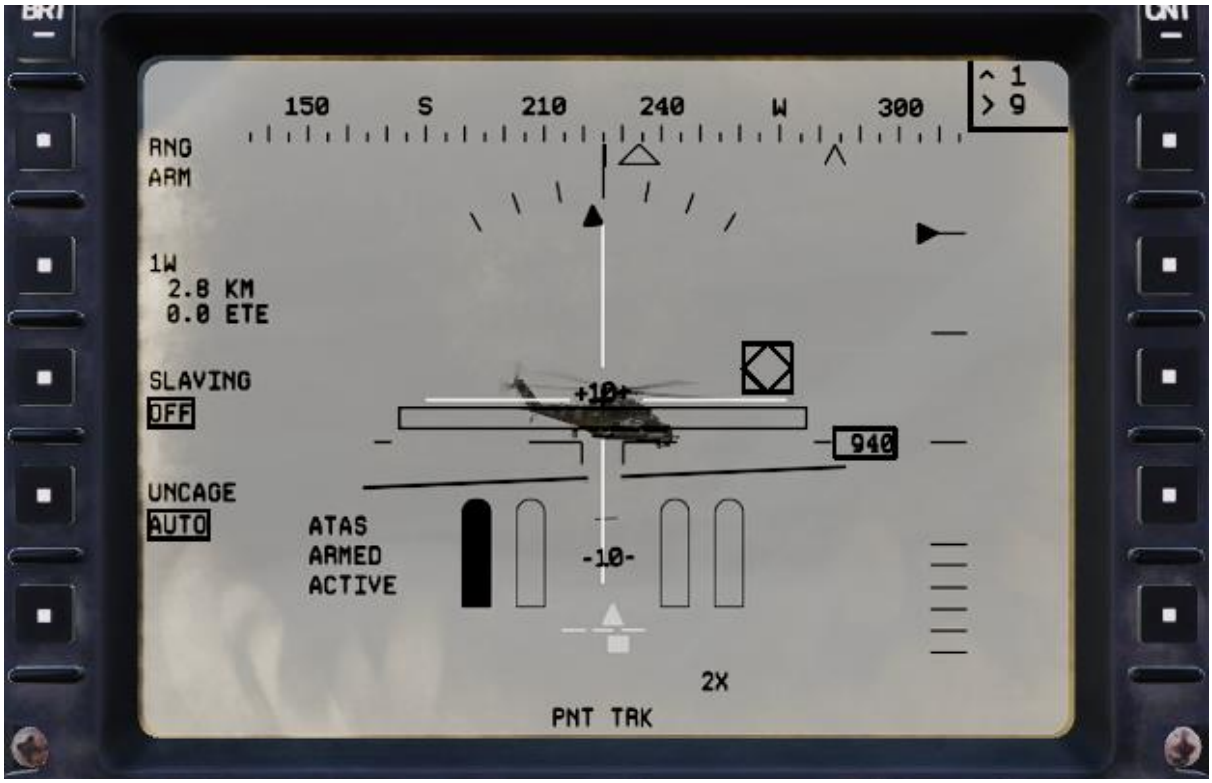


Figure 114 - Copilot Sparse Weapons page (ATAS)

POLYCHOP SIMULATIONS

11.15 Pages

11.15.1 ASE Setup

The ASE (Aircraft Survival Equipment) page is used to control the **AN/APR-39 Radar Detection Set**, the **AN/AVR-2 Laser Detection Set** and the **AN/ALQ-144 IRCM** if available.

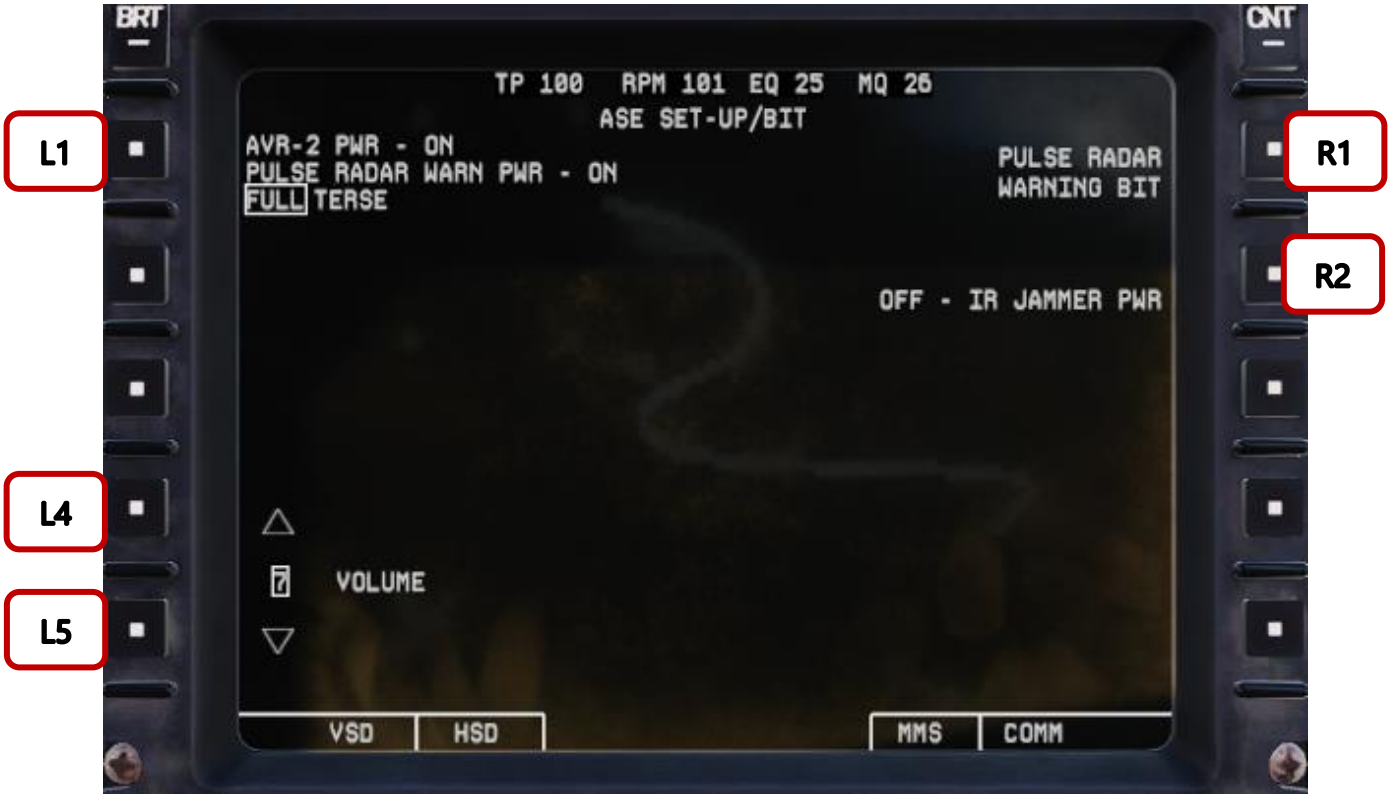


Figure 115 - ASE Setup page

L1. *FULL/TERSE.* Displays power status of the **AVR-2 Laser Detection Set** and the **AN/APR-39 Radar Detection Set**, and the selected mode of operation of the AN/APR-39. When set to FULL, the AN/APR-39 annunciator operates in Mode 1. When set to TERSE, the AN/APR-39 annunciator operates in Mode 2 L4/L5. VOLUME Indicates and sets the volume of the AN/APR-39 annunciator

R1. *PULSE RADAR WARNING BIT.* Runs the self-test of the AN/APR-39. Bezel text is boxed while test is running

R2. *IR JAMMER PWR.* Indicates and sets the power status of the **AN/ALQ-144 IRCM**

POLYCHOP SIMULATIONS

Armament Control Panel

12.1 Overview

The Armament Control Panel (ACP) is located on the Center Console and controls armament and jettisoning of stores.

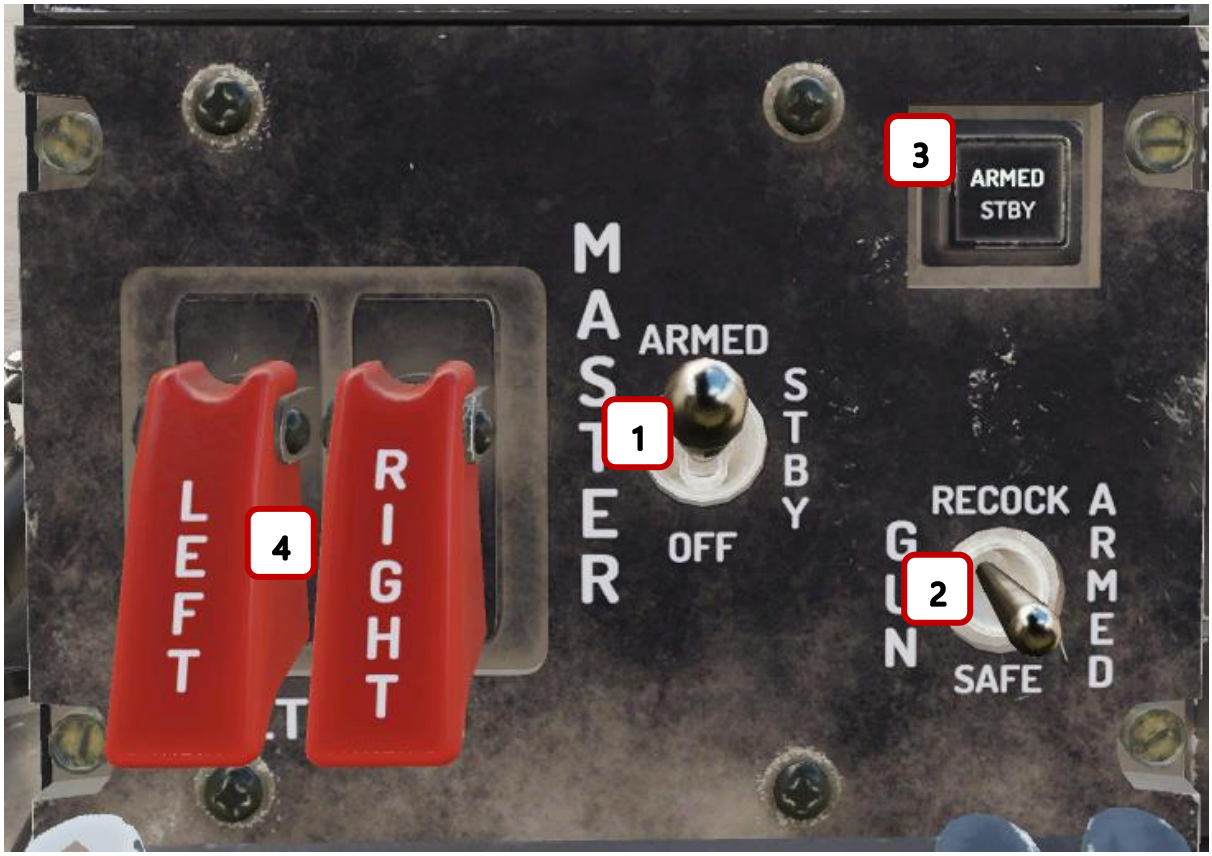


Figure 116 - Armament Control Panel

1. Master Arm Switch
2. Gun Arm Switch
3. Master Arm Status Indicator
4. Left and Right Stores Jettison Switches

INFO:

Power is only supplied to the weapons pylons when the Master Arm Switch is placed in STANDBY or ARMED positions.

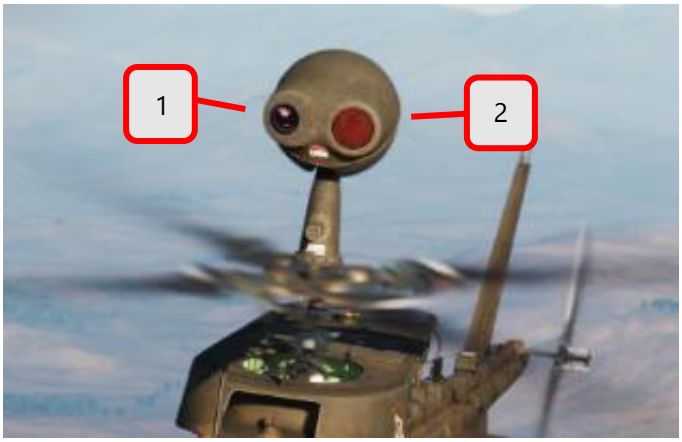
When rearming the gun or changing weapon loadout to a gun at an airfield or FARP, the weapon must be recocked before it can be fired.

POLYCHOP SIMULATIONS

Mast Mounted Sight

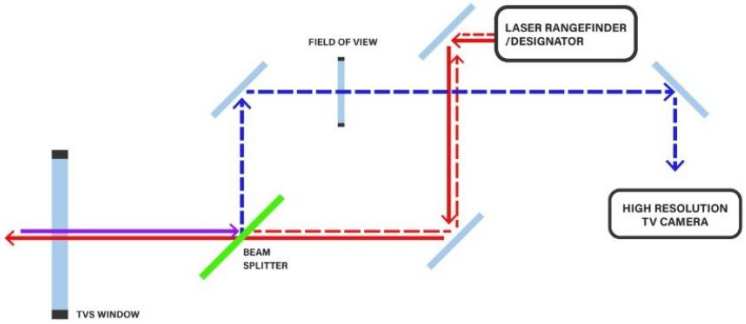
13.1 Overview

The Mast Mounted Sight (MMS) is a stabilized sensor suite mounted above the main rotor. It consists of a Television Sensor (TVS), a Thermal Imaging Sensor (TIS), and a Laser Rangefinder/Designator.

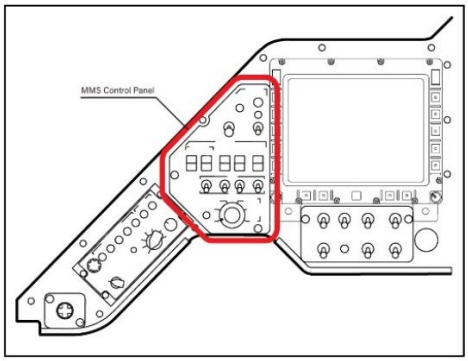


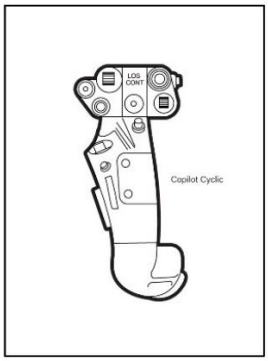
1. Tele Vision Sensor (TVS) Laser Range Finder Designator (LRF/D)
2. Thermal Imaging Sensor (TIS)

The TVS & LRF/D optics are able to share the same external lens by means of a beam splitter lens, whereas the TIS requires its own dedicated lens



The MMS is controlled from the Copilot Cyclic Grip and the MMS Control Panel situated next to the Copilot's MFD.





POLYCHOP SIMULATIONS

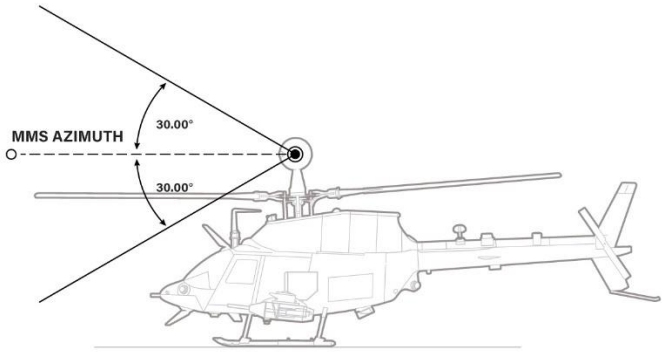
The MMS has a variety of functions and modes that allow it to search, detect, locate and track targets.

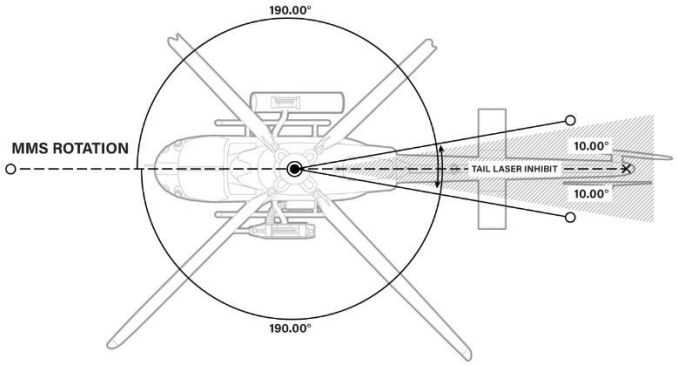
The MMS combined with the CDS and IDM allows for target designation and storage, handover and remote engagement.

The MMS can rotate
+/- 190° in azimuth

and

+/- 30° elevation
relative to the aircraft heading and pitch.





CAUTION:

Tail Laser inhibits are NOT reliable. Do NOT fire the laser within 10 degrees of the tail.

POLYCHOP SIMULATIONS

13.2 Mast Mounted Sight System Control Panel

The Mast Mounted Sight System Control Panel contains inputs for controlling the MMS modes, video displays and Laser Rangefinder/Designator.

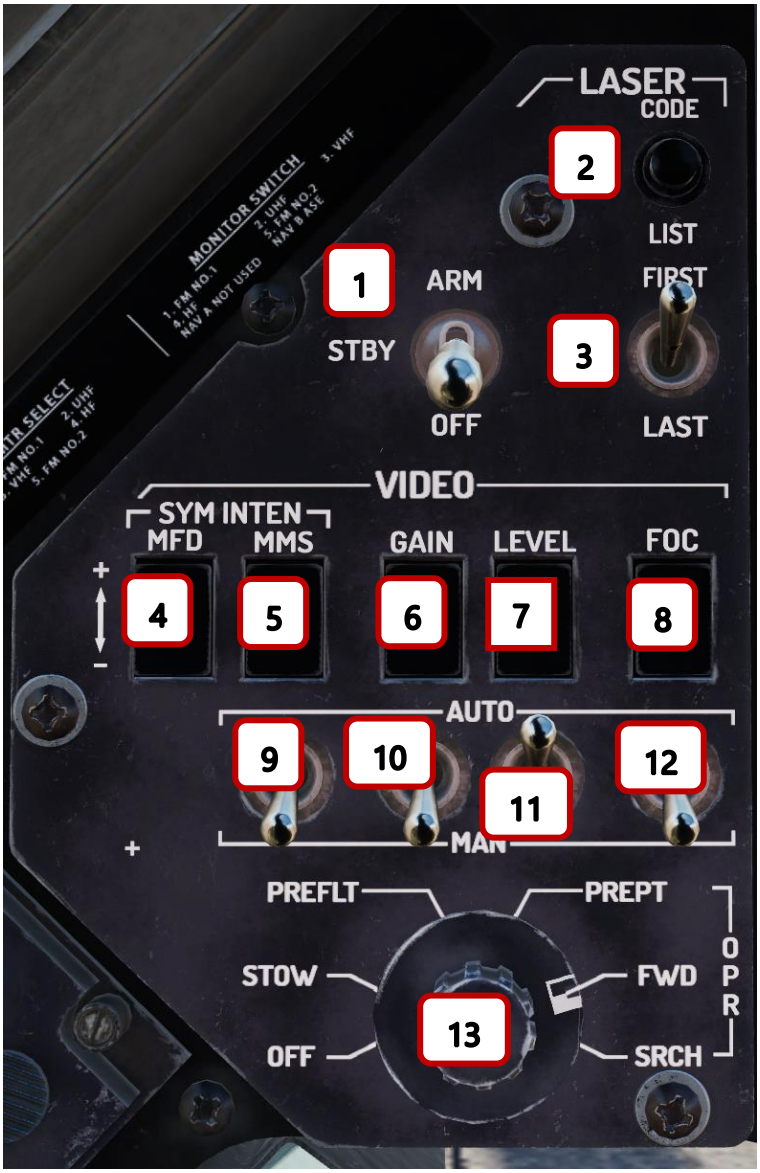


Figure 117 – Mast Mounted Sight System Control Panel

1. Laser Switch ARM/STBY/OFF.
2. Code List Switch. Displays the Code List page on the Copilot MFD
3. First/Last Switch. Not functional in DCS
4. MFD Symbol Intensity Switch.
5. MMS Symbol Intensity Switch.
6. Video Gain Switch.
7. Video Level Switch.
8. Video Focus Switch.
9. MMS Symbol Intensity Auto/Manual Switch. Not functional in DCS
10. Video Gain Auto/Manual Switch. Not functional in DCS
11. Video Level Auto/Manual Switch. Not functional in DCS
12. Video Focus Auto/Manual Switch. Not functional in DCS
13. MMS Mode Selector

POLYCHOP SIMULATIONS

13.3 Pages

13.3.1 MMS Page

Most MMS modes share common elements, these are detailed below. Specific symbology is detailed in the relevant MMS Mode chapter.

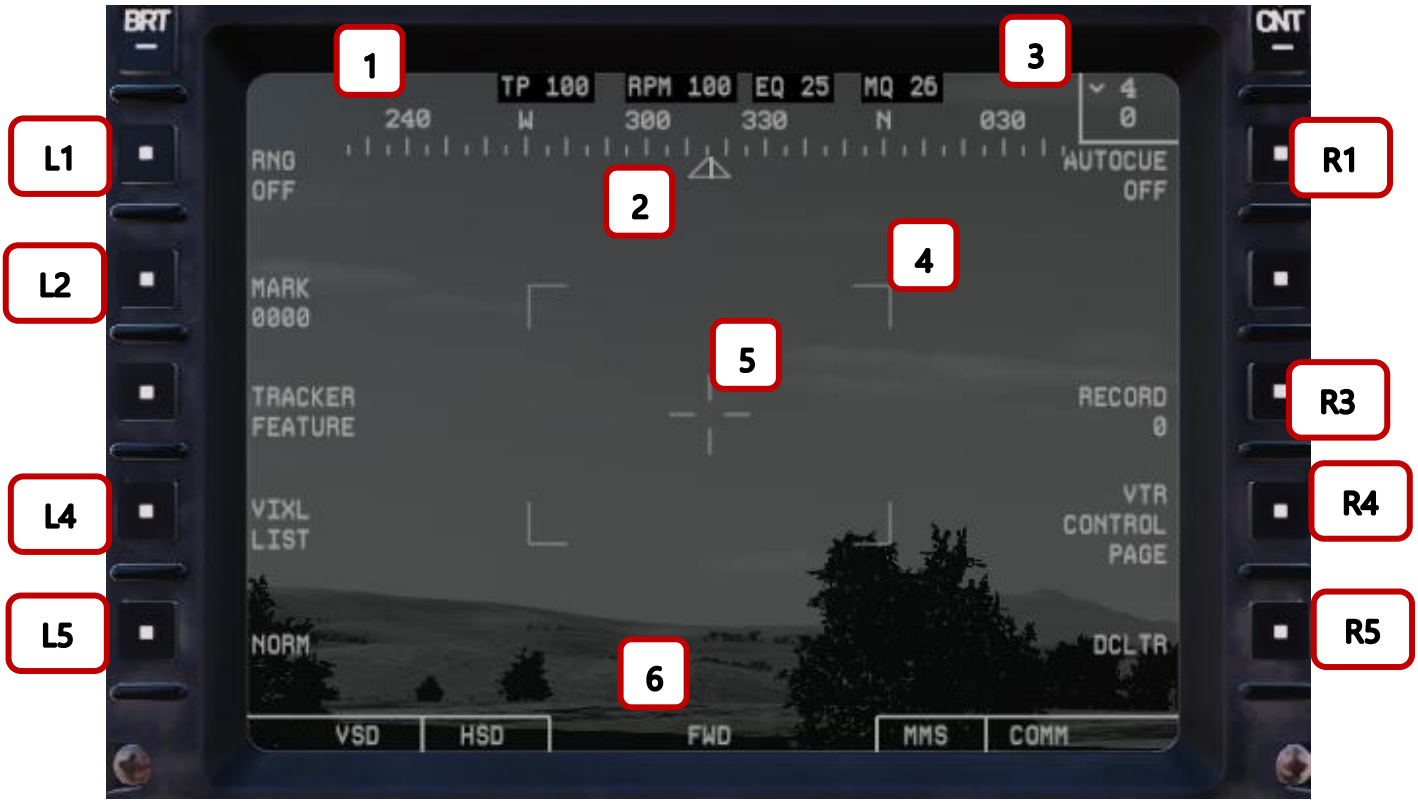


Figure 118 - General MMS page

1. Heading Tape
2. MMS Azimuth Caret and Lubber Line
3. MMS Azimuth and Elevation Display
4. Narrow Field of View Constraints
5. MMS Crosshair
6. MMS Advisory Display

L1. *RNG.* The R2 bezel displays the current status, selected code and last returned range of the Laser Rangefinder/Designator. Activation of the bezel will display the Laser Code List and allow selection of one of the eight laser code presets through successive presses of the bezel

L2. *MARK.* Not implemented

R1. *AUTOCUE.* Not implemented

R3. *RECORD.* Not implemented

R4. *VTR CONTROL PAGE.* Not implemented

POLYCHOP SIMULATIONS

L4. *VIXL LIST.* Not implemented

L5. *NORM.* Displays the current TV/TIS mode. When displaying feed from the TVS, displays either NORM or INV for normal or inverted display. When TIS is selected, displays WHOT or BHOT for white hot/black hot display. Pressing the bezel will cycle between the two modes for the current feed

R5. *DCLTR.* Activation of the R5 key will cycle the declutter mode of the MMS display. On the first press, all MFD symbology will be hidden. A second press will hide both MFD and MMS symbology. A third press will revert to displaying all symbology

13.3.1.1 Laser Code List

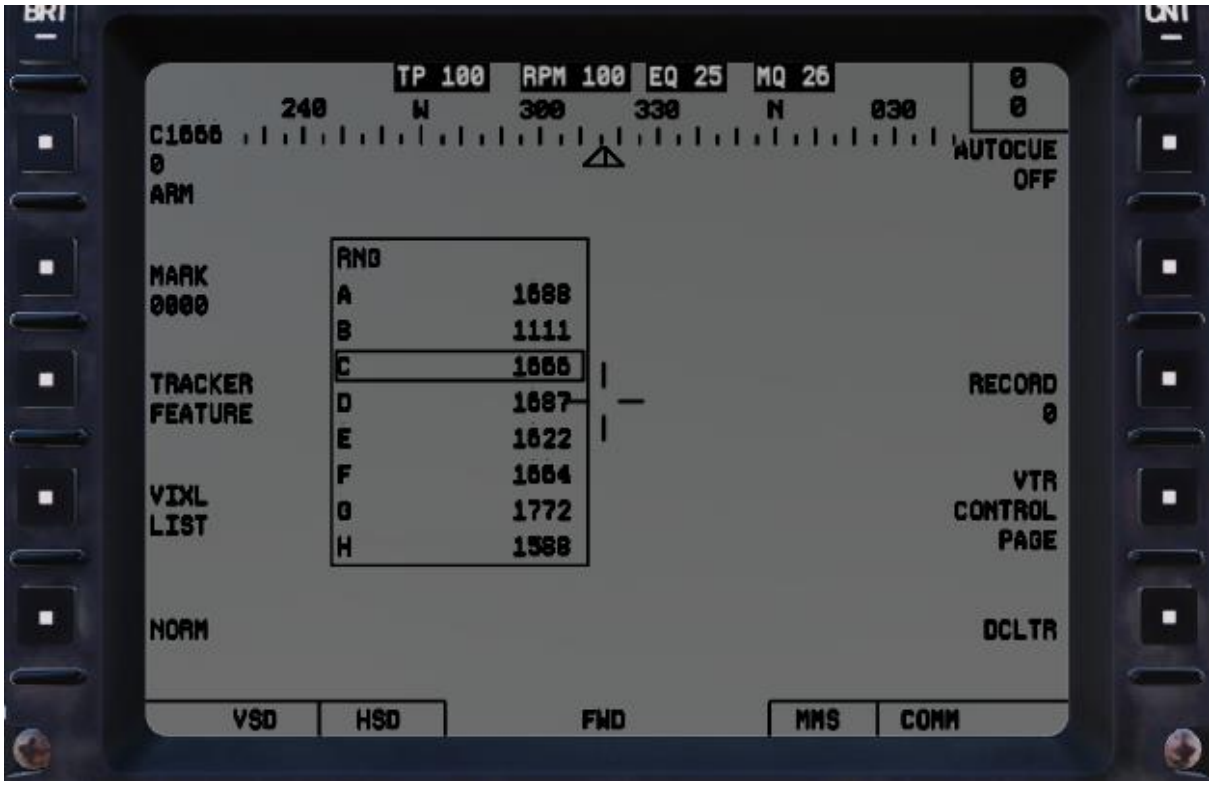


Figure 119 - MMS Laser Code List

POLYCHOP SIMULATIONS

13.3.2 Laser Codes Page

The Laser Codes Page is activated by pressing the Laser Code List button on the MMS Control Panel. This page allows the edit to set the eight preset laser codes.

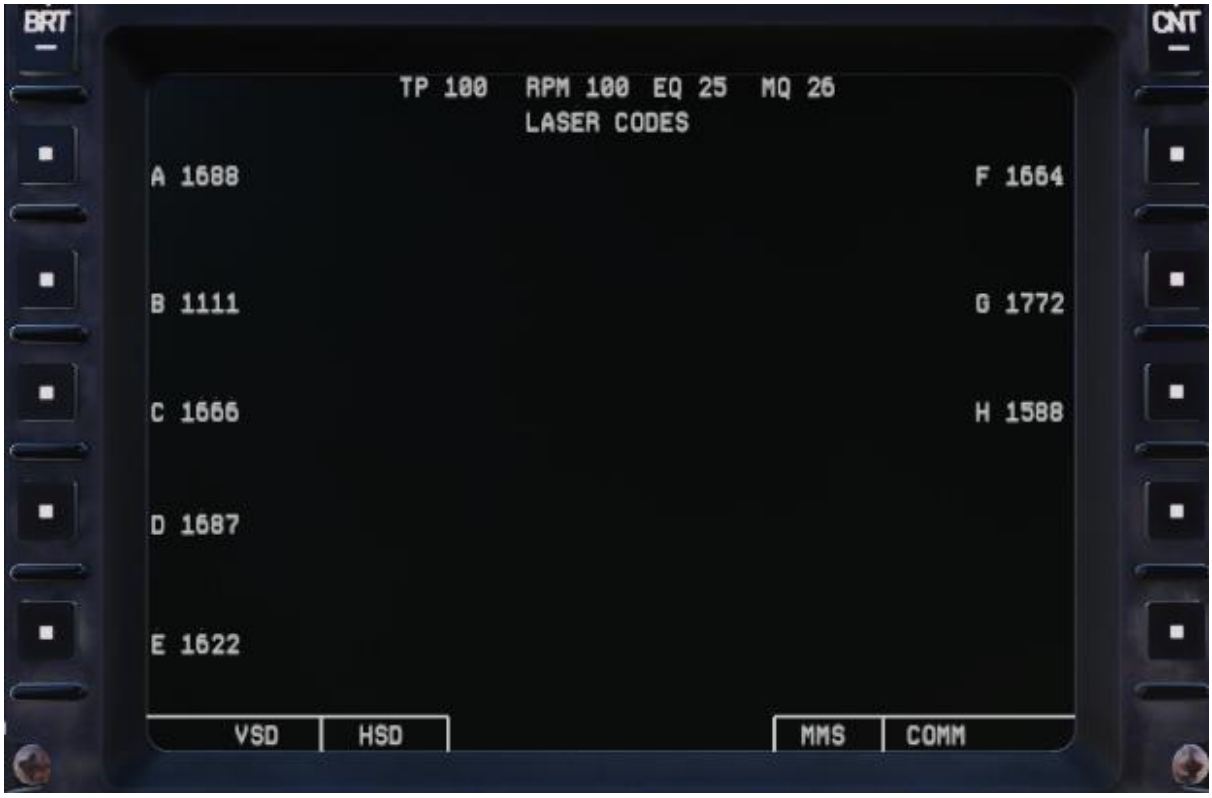


Figure 120 - Laser Code List page

Pressing a bezel will activate Data Entry Mode on the selected Laser Code, allowing entry of a new code preset.

POLYCHOP SIMULATIONS

13.4 Modes

13.4.1 Stow

When the STOW position is selected using the MMS Mode Selector, the MMS slews to the travel position and prepares for shutdown. The MMS may be shutdown once the advisory 'STORE COMPLETE' is displayed by moving the MMS Mode Selector to the OFF position.



Figure 121 - MMS Stow mode

POLYCHOP SIMULATIONS

13.4.2 Preflight

The Preflight mode is for performing setup and tests of the MMS system such as bore sighting the laser rangefinder/designator. These operations are not necessary in DCS World and so the Preflight mode is largely non-functional.

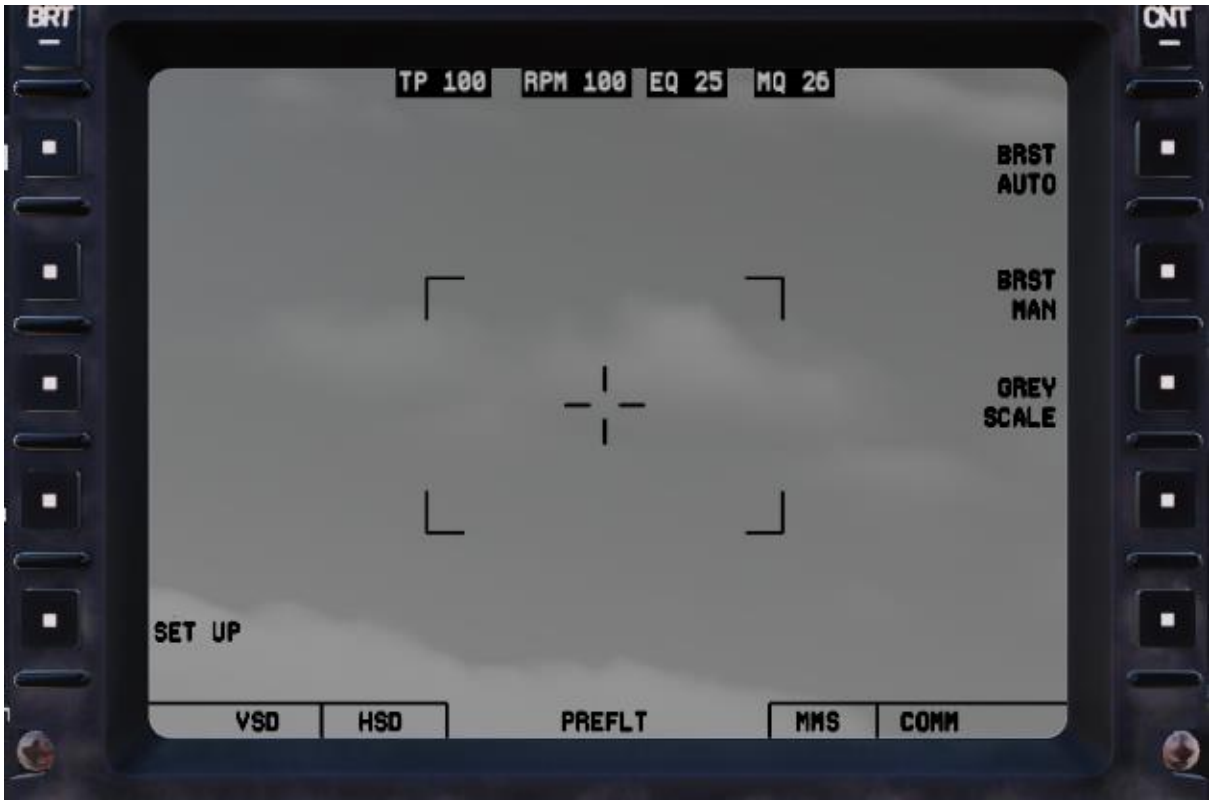


Figure 122 - MMS Preflight mode

POLYCHOP SIMULATIONS

13.4.3 Pre-Point

The Pre-Point mode is one of the three operational modes of the MMS. Pre-Point mode will slew the MMS to a selected Waypoint, Control Point or Target Point. The desired point can be entered into – and selected from – the Pre-Point list.



Figure 123 - MMS Pre-Point mode

7. Pre-Point Cursor. Indicates the position of the designated point. If the designated point is off-screen, indicates the relative direction to the point

R2. *PPT*. Selecting this bezel displays the Pre-Point dialog and activates data entry mode. Repeated selection of the bezel will cycle the selected list item. Upon entry of a valid Waypoint, Control Point or Target Point ID, the point will be entered into the selected list item. The Pre-Point list will close after 5 seconds of the last key activation, and Data Entry Mode will be deactivated.

POLYCHOP SIMULATIONS

13.4.4 Forward

The second of the three operational modes of the MMS, Forward mode will slave the MMS to the aircraft's heading. When in Forward mode, the elevation can be trimmed using the MMS Slew LOS input, however changing the MMS azimuth will break the MMS out of Forward mode and into Uncommanded mode.



Figure 124 - MMS Forward mode

POLYCHOP SIMULATIONS

13.4.5 Search

Search mode is the last of the three operational modes of the MMS. Search mode consists of two sub-modes: Sweep Search and Point Search. The sub-mode can be toggled using the L2 bezel.

13.4.5.1 Sweep Search

The Sweep Search sub-mode slews the MMS azimuth back and forth between defined search angles. The angles and speed of search can be adjusted using the bezel keys.



Figure 125 - MMS Sweep Search mode

L2. *SWEEP*. Indicates the selected Search sub-mode. Activation of the bezel toggles between Point Search and Sweep Search

L3. *SLOW*. Displays the selected sweep speed. Toggles between SLOW and FAST

L4. *SWEEP ANG*. Pressing the L4 bezel cycles between the available sweep angles

R2. *OSET*. Cycles Sweep Offset between 90° left, center and 90° right. Selected offset is boxed

POLYCHOP SIMULATIONS

13.4.5.2 Point Search

The Point Search sub-mode slews the MMS between entered Waypoints, Control Points and Target Points. The MMS will dwell on each point for a specified amount of time before slewing to the next point in the list. Up to four points can be entered.

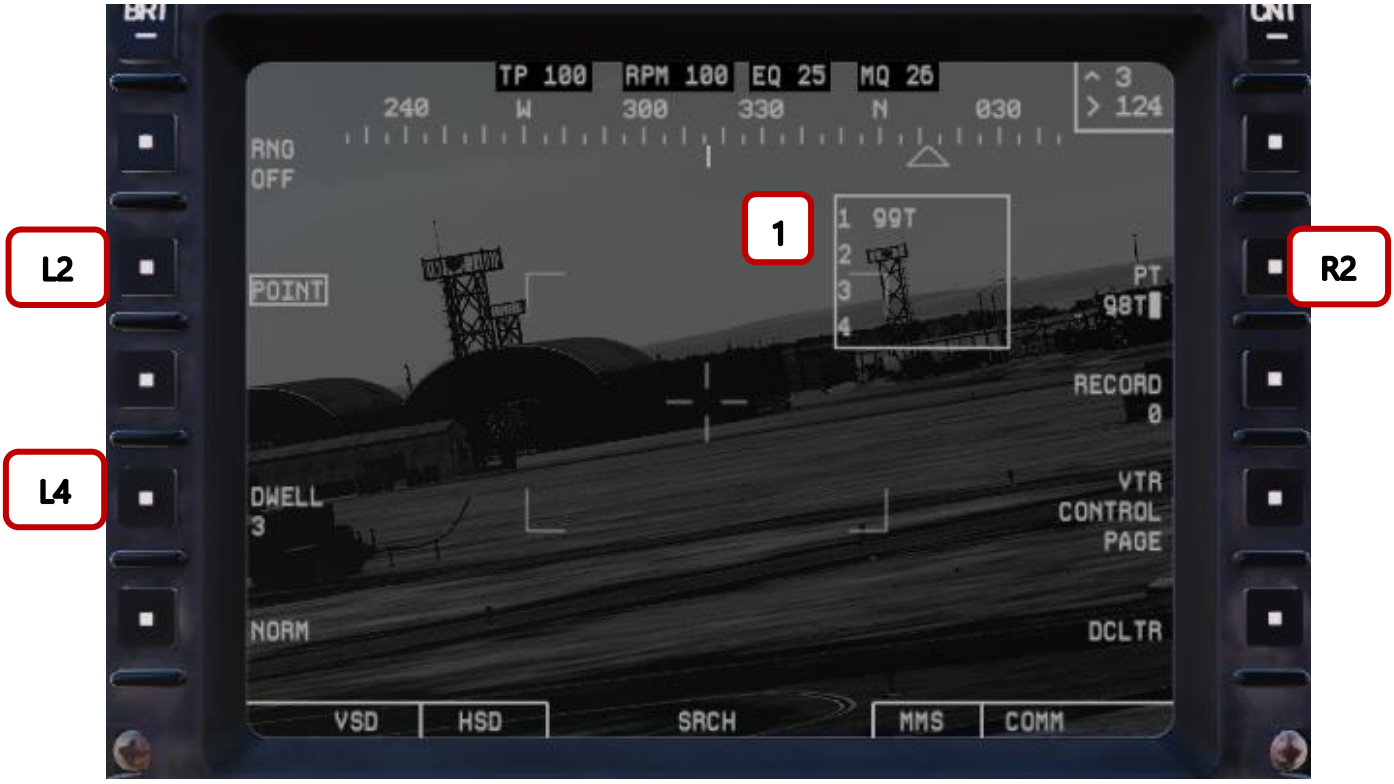


Figure 126 - MMS Point Search mode

1. Search Point List

L2. *POINT.* Indicates the selected Search sub-mode. Activation of the bezel toggles between Point Search and Sweep Search

L4. *DWELL.* Displays the current Dwell Time in seconds. Selection of this bezel key activates Data Entry Mode and allows input of a Dwell Time

R2. *PT.* Pressing R2 displays the Search Point List and activates Data Entry Mode. Successive presses cycle items in the list. Upon entry of a valid Waypoint, Control Point or Target Point ID the currently selected list item is designated with the entered point

POLYCHOP SIMULATIONS

13.4.6 Manual

The Manual mode of the MMS can be accessed from any of the MMS operational modes by toggling the Manual/Slave button on the Copilot Cyclic Grip.

Manual mode provides stabilized slew control over the MMS using the MMS LOS Slew input, and is also a required intermediary mode when switching to or between any of the three operational modes.



Figure 127 - MMS Manual mode

13.4.7 Uncommanded

The Uncommanded 'mode' can only be entered in certain circumstances by 'breaking out' of another mode or incorrect use of the MMS. The MMS is considered uncommanded when not in any other mode, and this is communicated to the user by the lack of mode advisory text in the MMS display.

When the MMS is uncommanded, stabilization is disabled and slew inputs are not smoothed. The MMS can be brought back to a commanded mode at any time using the Manual/Slave button on the Copilot Cyclic Grip.



Figure 128 - MMS Uncommanded mode

POLYCHOP SIMULATIONS

13.4.8 Point Track

The Point Track mode is a sub-mode of Manual mode that is accessed by pressing the Point Track button on the Copilot Cyclic Grip. Upon activation of the button, the MMS will attempt to locate and lock onto an area of high contrast within the MMS field of view. This is displayed in real time using the gate symbology that expands and, if a target is found, is displayed around the area of high contrast. When the track is completed, the MMS will slew to the target location and maintain a lock until the target is either obscured or otherwise lost from the field of view.

Point Track can also be accessed from Area Track and Offset Track modes by pressing the Point Track button.

Point Track can be exited at any time by pressing the Manual/Slave button to return to Manual mode.

Point Track is an entry point to the Acquire On The Move and Offset Acquire sub-modes.

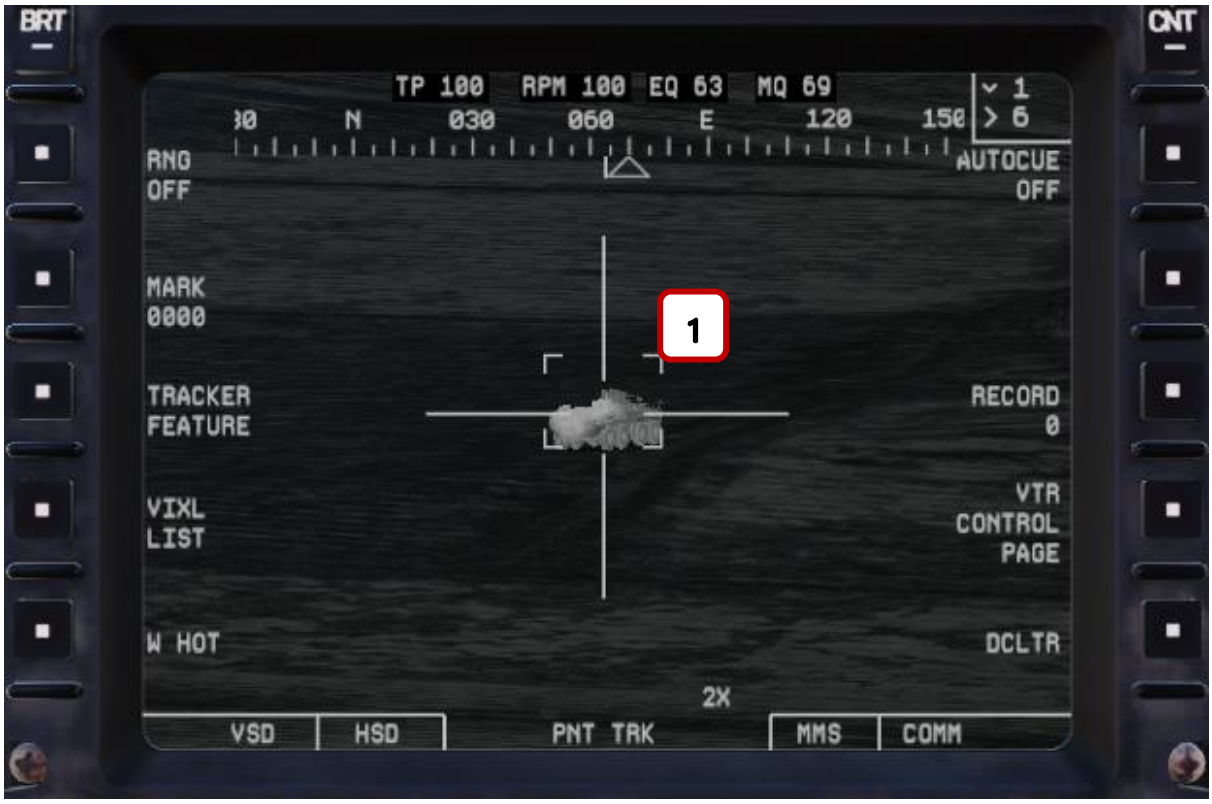


Figure 129 - MMS Point Track mode

1. Primary Gate

POLYCHOP SIMULATIONS

13.4.9 Area Track

The Area Track mode will lock the MMS to the current bearing and elevation. Area Track mode is a sub-mode of Manual mode and can be accessed by selecting the Area Track button on the Copilot Cyclic Grip.

Area Track can be exited at any time by pressing the Manual/Slave button to return to Manual mode.

Area Track is an entry point to the Acquire On The Move sub-mode.



Figure 130 - MMS Area Track mode

POLYCHOP SIMULATIONS

13.4.10 Acquire On The Move

Acquire On The Move (ACQ OM) is a sub-mode of Area Track or Point Track that allows for the designation and acquisition of up to six simultaneous contrast locks.

Acquire On The Move is entered from Area Track or Point Track by pressing the Area Track button. Upon activation, the crosshair will change and the Acquire On The Move diamond will be displayed. Using the MMS LOS Slew input, the user can slew the diamond cursor over targets where it will automatically establish a track upon release of the LOS Slew input. Selecting the Area Track button will allow for successive tracks.

The last selected targe will be displayed in the Transition Target Gate (thick box), denoting it as the Transition Target, while other targets are displayed in Secondary Target Gates (thin box).

Using the Target Select Switch on the Copilot Cyclic Grip, the user can select which target is the Primary Target.

Pressing the Point Track button will designate the Transition Target as the Primary Target, and the MMS will slew and establish a Point Track lock onto this target.

Pressing Manual/Slave will eliminate all tracked targets from the display.

In order to access the Offset Acquire sub-mode from Acquire On The Move, the Transition Target Gate must be eliminated. This can be done by cycling the Transition Target until no Transition Target Gate is displayed.

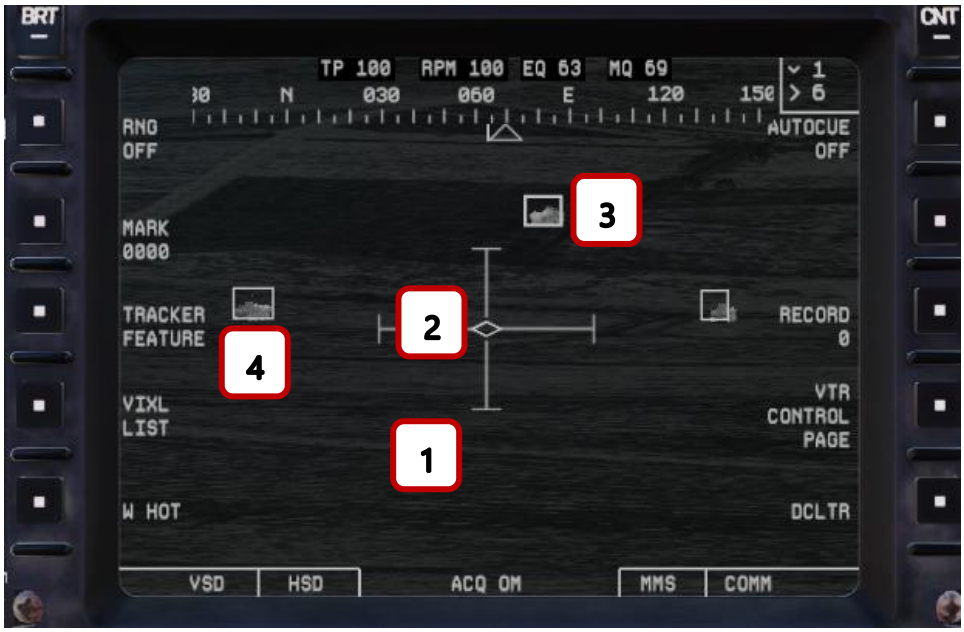


Figure 131 - MMS Acquire On The Move mode

1. Acquire On The Move reticule
2. Acquire On The Move diamond cursor
3. Transition Target Gate
4. Secondary Gate

POLYCHOP SIMULATIONS

13.4.11 Offset Acquire

The Offset Acquire sub-mode is accessed from Point Track and Acquire On The Move modes by selecting the Point Track button.

Entering Offset Acquire enables the Laser Hit Point reticule, which can be slewed using the MMS LOS Slew inputs. The Laser Hit Point reticule displays the location at which the Laser Rangefinder/Designator will be fired, allowing for designation or ranging of an offset location from the current MMS target.

The offset Laser Hit Point location is relative to any track and so will maintain a relative position to a moving target.

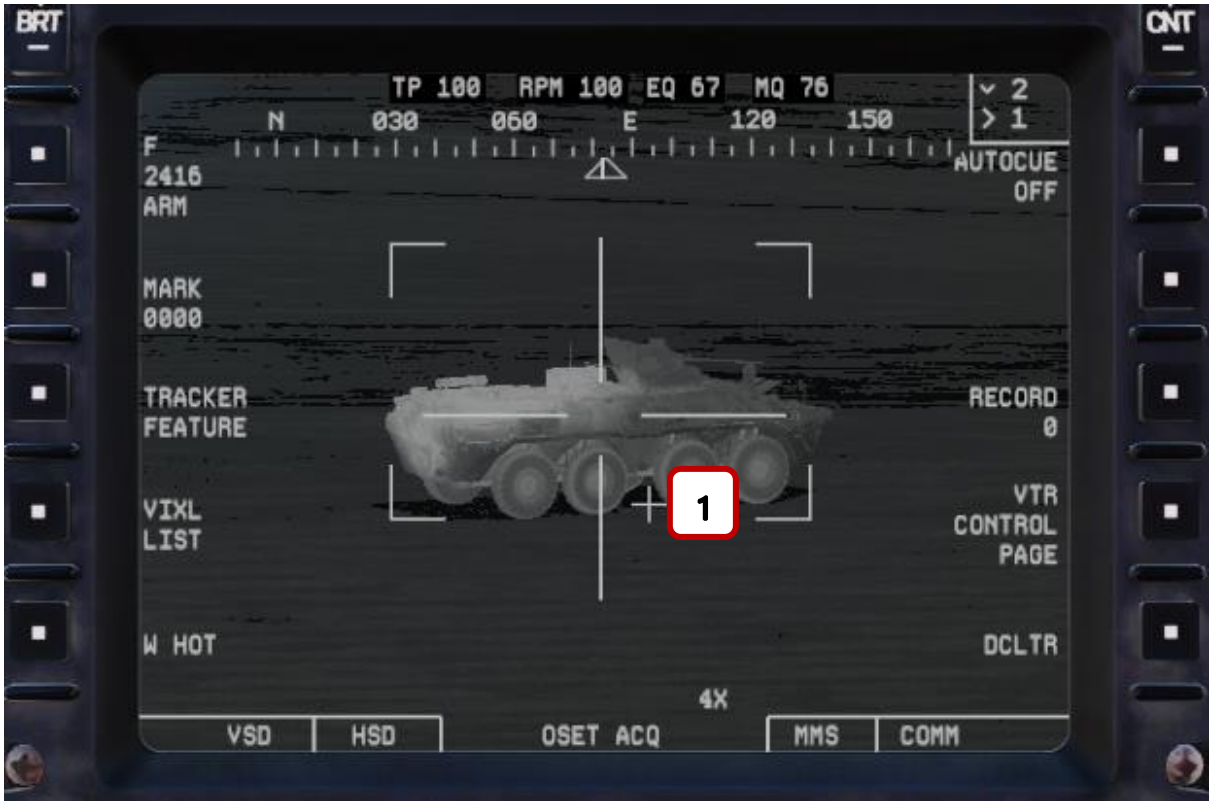


Figure 132 - MMS Offset Acquire mode

1. Laser Hit Point reticule

POLYCHOP SIMULATIONS

13.4.12 Offset Track

The Offset Track is a sub-mode of Offset Acquire which slews the MMS to the Laser Hit Point reticule.

Offset Track is entered by pressing the Point Track button while in Offset Acquire mode.

Pressing the Point Track button a further time will return to Point Track mode.

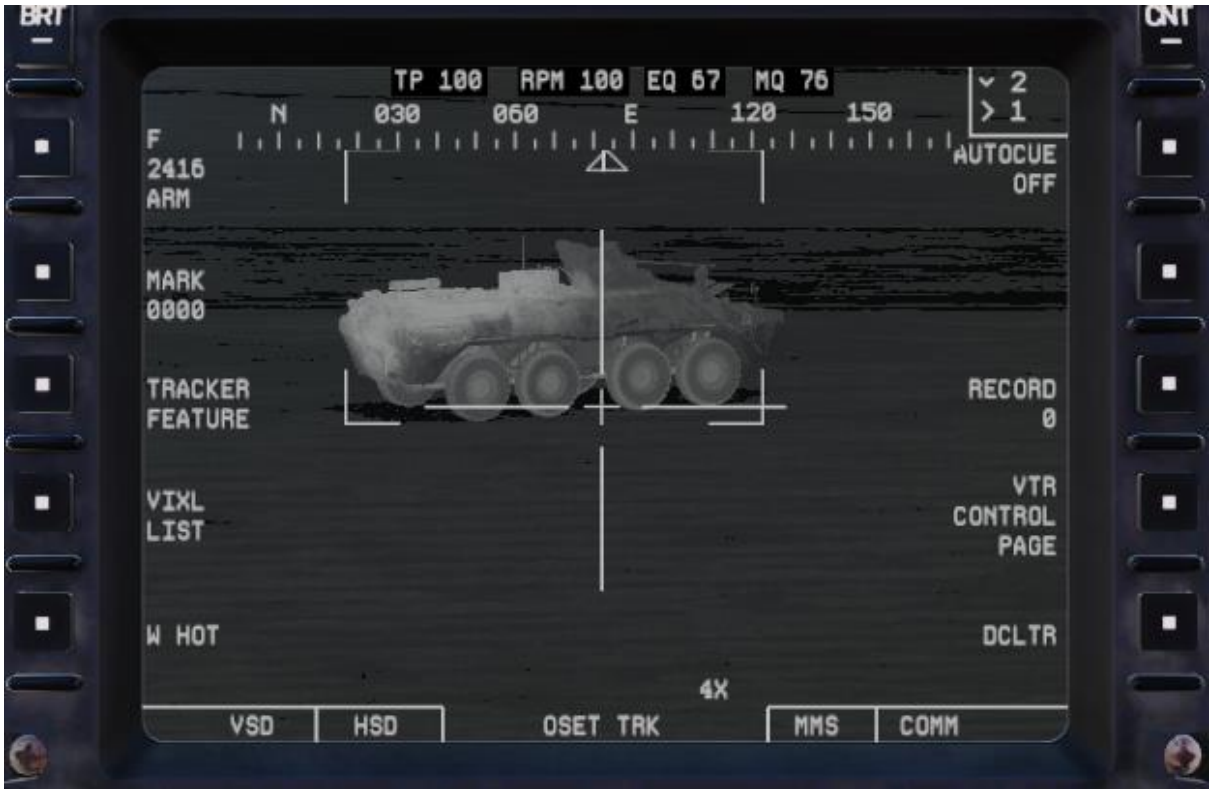


Figure 133 - MMS Offset Track mode

POLYCHOP SIMULATIONS

13.4.13 Target Locate Mode

Target Locate is an intermediate mode for storing targets as Target Points. Target Locate is activated by pressing the Target Designate switch on the Copilot Cyclic Grip.

When in Target Locate mode, the user can locate the target using the Laser Rangefinder/Designator. This will attempt to determine the position of the desired target. When the position has been determined, the symbology changes and the options to store the target are enabled.

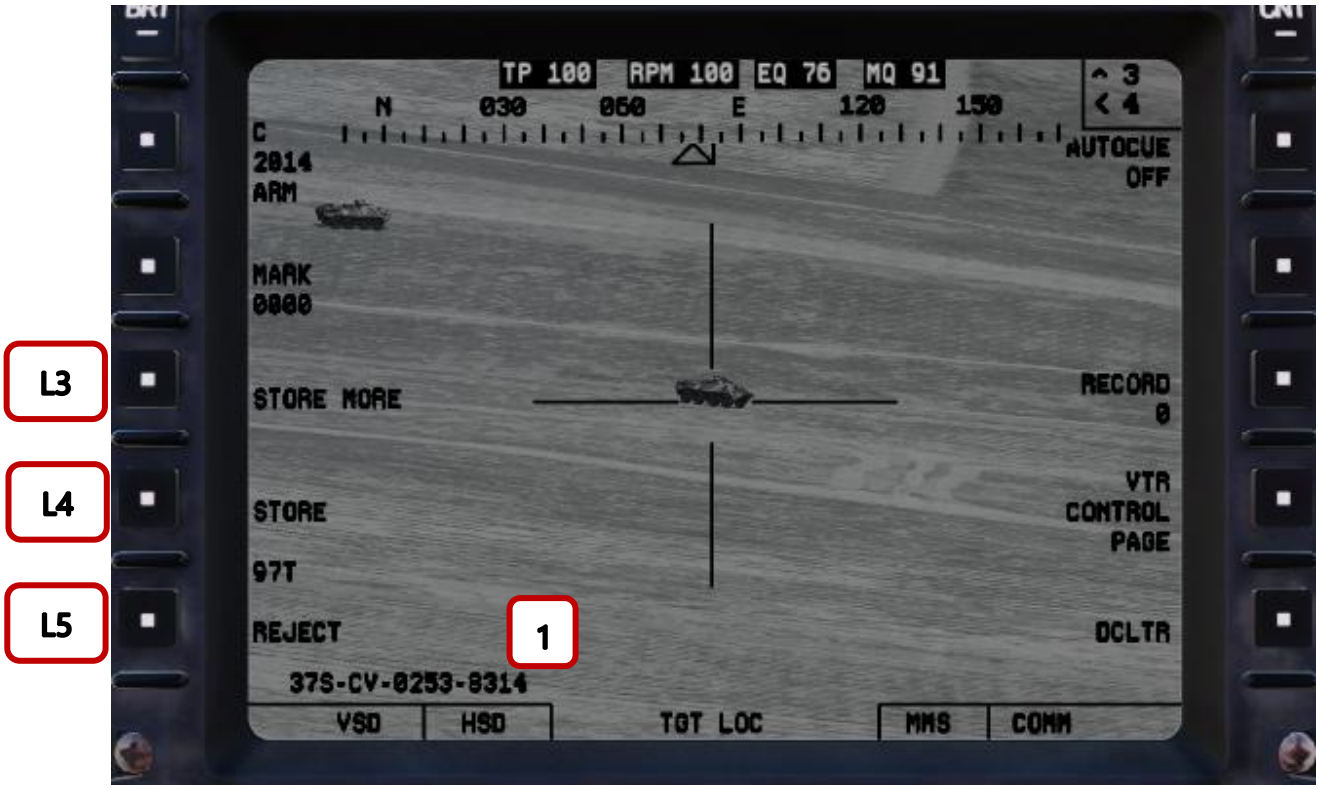


Figure 134 - MMS Target Locate mode

1. Target Position

L3. *STORE MORE*. Activation of the L3 bezel displays the Hands On Grip User Interface (HOG) menu system for target storage

L4. *STORE*. Pressing L4 stores the target at the displayed index (e.g. 97T). Target metadata reverts to defaults

L5. *REJECT*. Rejects the target and exits Target Locate mode

POLYCHOP SIMULATIONS

Radios

14.1 Overview

UHF-AM Radio Set AN/ARC-164 (UHF)	Two-way voice communication in frequency range of 225.000 to 399.975MHz.
VHF-AM Radio Set – AN/ARC-186 (VHF)	Two-way voice communication in frequency range of 116.000 to 151.975MHz
VHF-FM Radio Set – SINCGARS (FM-1 or 2) AN/ARC-201D	Two-way voice communication and transmission in frequency range of 30.000 to 87.975 MHz with frequency hopping, scan, and secure capability (FM-1 and FM-2).
AN/ARC-231 SATCOM RADIO	Not Implemented in DCS

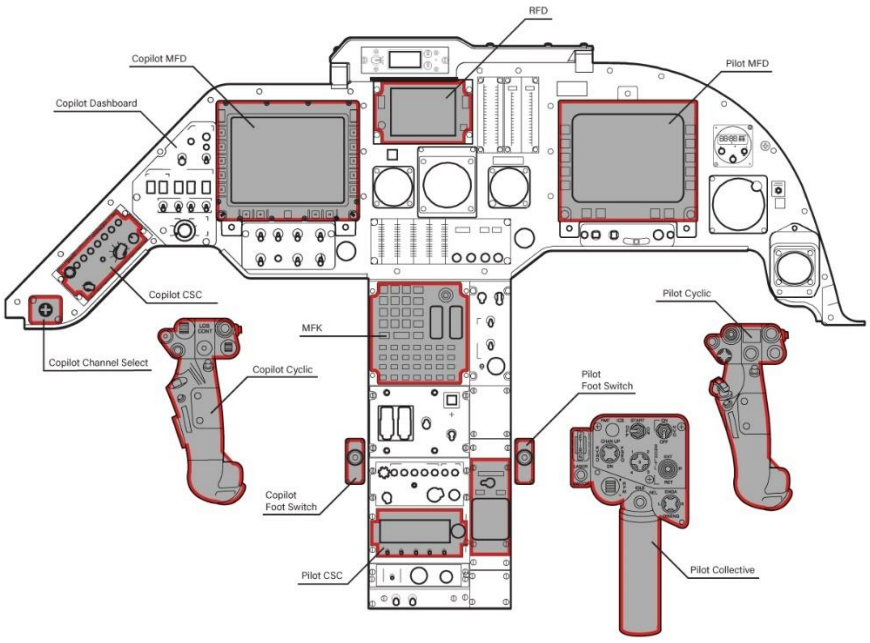


Figure 135 - Communications interface

Control of the radios is via a combination of panels;

- The RFD displays the current frequencies of each radio,
- MFDs & MFK are used to change frequencies,
- communications system control panel provides volume controls to each pilot,
- combined with channel select & radio select switches.

POLYCHOP SIMULATIONS

14.2 Communications System Control Panel

The communication system provides two way communications between the crewmembers and allows the crew to communicate outside the helicopter through four communication radios.

Radio selection and audio volume are controlled through the communication system control (CSC) panel.



Figure 136 - Pilot Communications System Control Panel

1. *Volume Control Knob.* Controls audio output level to the headset
2. *Radio Monitor Switches.* Controls audio output level for each radio. A pulled knob inhibits audio for the selected radio
3. *NAV Switches.* The NAV switches are not used in the OH-58D
4. *ICS Rotary Switch.* When operating in "NORM", you must key the ICS
5. *VOX Control.* Causes the intercom system to be voice activated for crew communications
6. *Transmit Selector Switch.* Allows pilot to remotely select a radio if the CSC transmit select switch is in remote (RMT) position.
7. *MIC Switch.* Allows pilot to select between MIC1 & MIC2.

POLYCHOP SIMULATIONS

14.3 Radio Select Switch

A five-position switch that is used to select a radio for transmit.

Radio selections are also made with the radio select switch located on the pilot collective control head. The channel and frequency to which each radio is tuned is displayed on the RFD Setting the desired frequencies and modes into the radios is accomplished using the MFK and MFD.



Figure 137 - Radio Select Switch

Pressing to the 1 (upper left) position selects FM-1	Pressing straight down on 5 (center position) selects FM-2	Pressing to the 2 (upper right) selects UHF
Pressing to the 3 (lower right) selects VHF		Pressing to the 4 (lower left) SATCOM, if installed (not implemented in DCS),

As a radio is selected, an arrow will appear beside the frequency of that radio on the MFD.

This arrow will point to the right or left indicating that the pilot or CPO respectively is selected to transmit on that radio. (see Figure 133 - **Remote Frequency Display**)

POLYCHOP SIMULATIONS

14.4 Channel Select Switch

CHAN select switch — Located on the Pilot collective head and CPO CSC panel

A four-position, center spring-loaded to neutral switch that works in conjunction with the MFD, MFK, and RFD.



Figure 138 - Copilot Channel Select Switch

Pressing the switch to the CHAN UP position changes the channel number and frequency of the radio currently selected for transmitting to the next higher channel and frequency stored on the frequency list.

Pressing the switch to the left, or KYBD (keyboard), position activates the keyboard if a manual frequency is currently selected on the RFD, to allow the entry of manual channels and frequencies for the radios selected for transmitting.

Pressing the switch to the DN (down) position changes it to the next lower channel and frequency stored.

INFO:

If the COMM page is being displayed, pressing the switch to the right or FREQ position, brings up the Frequency page with channel numbers and call signs for the selected radio on the MFD.

POLYCHOP SIMULATIONS

14.5 Remote Frequency Display

The Remote Frequency Display (RFD) acts as the primary display for the radios. The RFD displays the Pilot and Copilot (see Left & Right arrows indicated #4 & #5 below) selected radios, channels and modes.

The RFD is primarily a display and cannot make inputs to change channels, frequencies or radio selection.

These inputs are made through the MFD **COMM** pages or via the Pilot and Copilot Channel Select controls or Communication System Control panels.

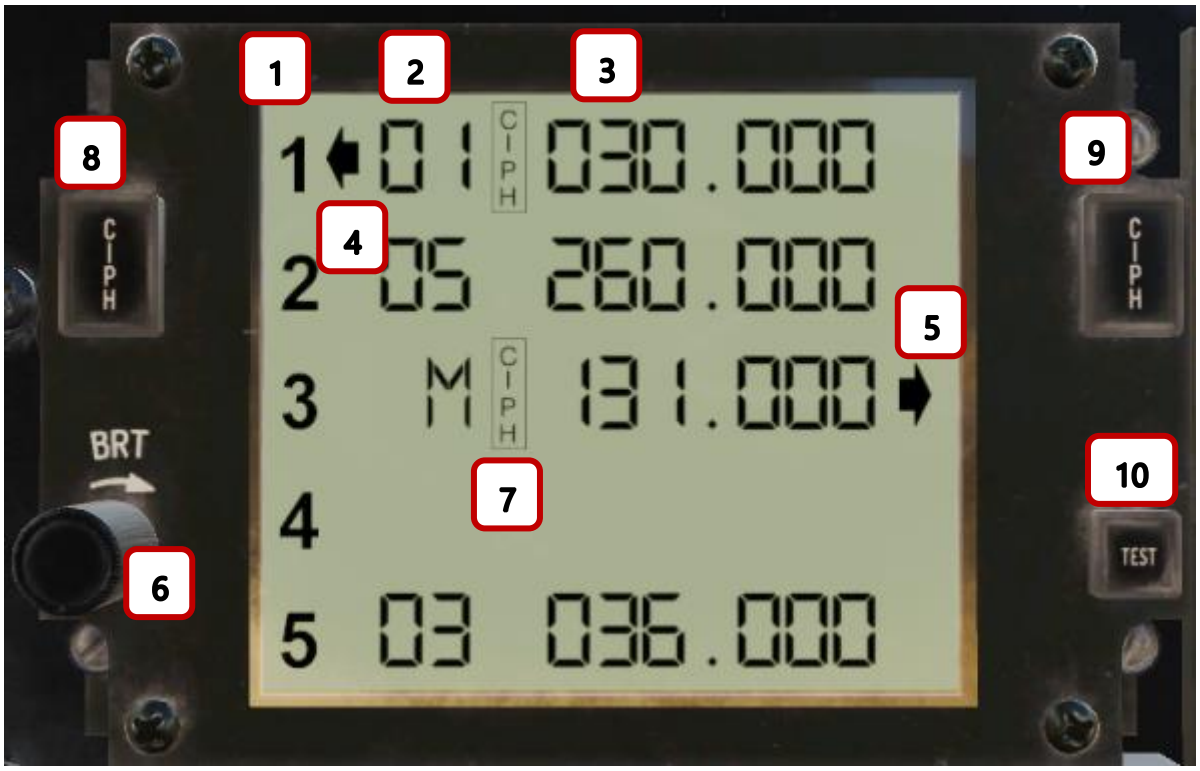


Figure 139 - Remote Frequency Display

1. List of available radios:
    1) FM1
    2) UHF
    3) VHF
    4) Not Used
    5) FM2
2. Channel Indicator
    - M: Manual frequency
    - 01-19: Selected channel
3. Selected frequency
4. Copilot selected radio indicator
5. Pilot selected radio indicator
6. Brightness control rheostat
7. CIPH mode indicator
8. Copilot CIPH mode selector
9. Pilot CIPH mode selector
10. Test button. Holding test button will display test pattern

POLYCHOP SIMULATIONS

Aircraft Survivability Equipment

15.1 Introduction

The Aircraft Survivability Equipment (ASE) suite is comprised of defensive systems to alert and protect the crew against threats.

The DCS OH-58D Kiowa Warrior module contains a range of systems designed to alert and protect against incoming radar, infra-red, laser and missile threats.

15.2 Flare Dispenser

A single flare dispenser is mounted to the bottom of the aircraft. The dispenser can carry up to 30 flares and can be manually dispensed using a switch on the Pilot's Collective Grip or controlled automatically by the **Common Missile Warning System.**

15.3 AN/APR-39 Radar Detection Set

The Radar Detection Set (RDS) consists of four sensors mounted on the exterior of the aircraft, an electronic control unit and the Radar Detector Indicator (RDI).

The RDI displays the positions and severity of radar threats, as well as alerting the crew via the use of synthesized voice messages.

The RDI also displays and alerts to laser threats from the **Laser Detection Set.**

The RDI volume and BIT test can be controlled via the MFD **ASE Setup** page.

1. Missile Alert lamps (not in use)
2. Brightness control
3. Night/Day brightness control switch

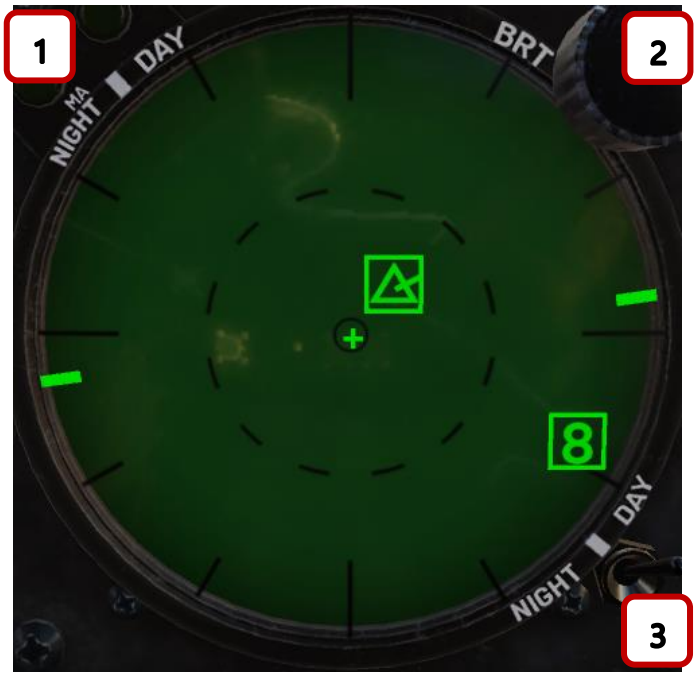


Figure 140 Figure 134 - Radar Detector Indicator

POLYCHOP SIMULATIONS

15.3.1 Radar Detector Indicator Symbology

The following list displays the Radar Indicator symbology and the associated name as spoken by the annunciator while in Mode 1.

Symbol	Threat Type	Annunciator Name
Tick	Unknown Emitter (searching)	“RADAR SEARCHING”
U	Unknown Emitter (tracking or acquisition)	“UNKNOWN”
2	SA-2	“SA SA 2”
3	SA-3	“SA SA 3”
5	SA-5	“SA SA 5”
6	SA-6	“SA SA 6”
8	SA-8	“SA SA 8”
10	SA-10	“SA SA 10”
11	SA-11	“SA SA 11”
FW	Fixed Wing aircraft	“FIXED WING”
RW	Rotary Wing aircraft	“HELICOPTER”
Gun	Air Defence Artillery	“GUN GUN”
ZU	ZU-23-4 Shilka	“ZU ZU 23”
Star/Asterisk	Laser Threat	“LASER”

POLYCHOP SIMULATIONS

15.3.2 Radar Detector Annunciator Messages

The AN/APR-39 contains a synthetic speech annunciator that relays information to the crew. The AN/APR-39 has two modes of operation. In Mode 1, full length messages are spoken. In Mode 2, terse messages are used. The mode of operation can be selected from the **ASE Page** on the MFD.

Message	Description
"APR-39 POWER UP"	Broadcast on receiving power. APR-39 will initiate start up test sequence.
"SELF TEST"	Beginning of the self-test.
"SET VOLUME"	Announced during the start-up self-test. APR-39 will count from one to twelve (Mode 1) or from five to one (Mode 2) to allow volume adjustment and annunciator tests
"APR-39 OPERATIONAL"	Self-test has completed successfully. Accompanied by a visual indicator of the radar and laser receiver status on the Radar Indicator.
"APR-39 FAILURE"	The self-test has failed. The Radar Indicator will display the status of the receivers.
"RADAR SEARCHING"	A radar has been detected in search mode. The Radar Indicator will display a tick in the direction of the threat.
"SA SA 6 TWELVE O'CLOCK TRACKING"	A radar threat is tracking the aircraft. The annunciator will speak the threat name and direction. The Radar Indicator will display the threat icon.
"SA SA 6 TWELVE O'CLOCK LAUNCH"	A radar threat has launched. The annunciator will speak the threat name and direction. The Radar Indicator will display a flashing threat icon.
"ALL LOCKS BROKEN"	Annunciated when all tracking radars have disappeared from the threat picture.
"THREAT DETECTION DEGRADED"	When more than 7 threats are simultaneously tracked, the detection effectiveness of the RWR is degraded. While degraded, the RWR will only track the 7 highest priority threats.
"THREAT DETECTION RESTORED"	Threat detection is no longer in degraded mode of operation.

POLYCHOP SIMULATIONS

15.4 AN/AVR-2 Laser Detection Set

The Laser Detection Set (LDS) consists of four detectors placed on the sides of the aircraft. The LDS is connected to the Radar Detector Set for power and display functions.

15.5 AN/ALQ-144 Infra-red Missile Countermeasures Device

The Infra-red Missile Countermeasures Device (IRCM) is a passive deflector system for countering infra-red guided threats. It has no method to alert the crew to threats.

15.5.1 IR Jammer Circuit Breaker Switch

INFO:
The IRCM is not equipped by default and must be installed using the **Mission Editor option**. Installing the IRCM removes the Common Missile Warning System.

The IR Jammer Circuit Breaker Switch is positioned to the right of the Overhead Console Panel above the Pilot's head. Placing the switch in the ON position provides power to the IRCM.



Figure 141 - IR Jammer Circuit Breaker Switch

POLYCHOP SIMULATIONS

15.6 AN/AAR-57 Common Missile Warning System

The Common Missile Warning System (CMWS) consists of five detectors placed around the airframe, an electronic control unit, a Control Indicator (CI) and CMWS Control Panel (CMWS CP).

The CMWS detects incoming missile threats and provides visual and audio alerts, and can automatically respond to threats by dispensing flares.

INFO:

Installing the IRCM removes the Common Missile Warning System.

15.6.1 CMWS Circuit Breaker Switch

The CMWS Circuit Breaker Switch is positioned to the right of the Overhead Console Panel above the Pilot's head. Placing the switch in the ON position provides power to the CMWS.



Figure 142 - CMWS Circuit Breaker Switch

POLYCHOP SIMULATIONS

15.6.2 CMWS Control Panel

The CMWS Control Panel is located at the rear left of the Center Console and contains controls for the CMWS.

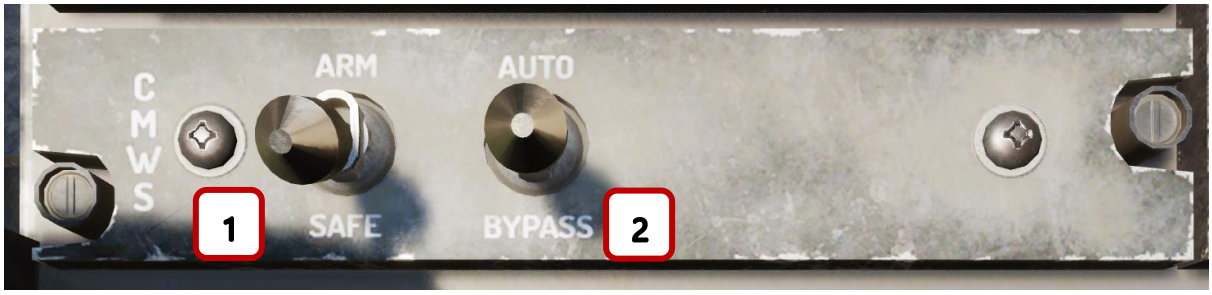


Figure 143 - CMWS Control Panel

1. ARM/SAFE Switch. Placing the switch into SAFE prevents dispensing of countermeasures
2. AUTO/BYPASS Switch. Placing the switch into AUTO allows the CMWS to automatically dispense countermeasures on detection of a threat. Placing the switch into BYPASS prevents automatic dispensing of countermeasures. When the CMWS is placed into BYPASS mode, the CMWS Control Indicator will display BYPASS

POLYCHOP SIMULATIONS

15.6.3 CMWS Control Indicator

The CMWS Control Indicator is a panel located on the center of the dashboard. It contains a status and threat display and various controls.



Figure 144 - CMWS Control Indicator

1. *OFF/ON/TEST* Knob. Placing to ON powers on the CMWS and initiates the start-up self-tests. Momentarily placing to TEST initiates BIT tests
2. *AUDIO* Dial. Controls annunciator volume
3. *LAMP* Dial. Controls display brightness
4. *STATUS* Display. Displays various CMWS status messages
5. *THREAT* Display. Threat quadrants will flash when a threat is detected from the relevant receiver
6. *DISPSENSING* Indicator. Illuminates when the CMWS is actively dispensing countermeasures
7. *READY* Indicator. Illuminates when the CMWS is ready to dispense countermeasures

INFO:

The CMWS includes a Weight On Ground safety switch that prevents use when on the ground.

POLYCHOP SIMULATIONS

Lighting

16.1 Exterior Lighting

16.1.1 Anti-Collision Lights

The Anti-Collision lights consist of Upper and Lower strobe lights.



Figure 145 - Anti-collision Light

The Anti-Collision lights are controlled by the Anti-Collision Lights Switch on the Overhead Console.



POLYCHOP SIMULATIONS

16.1.2 Position Lights

The Position Lights consist of red, green and white lights found on the tail section of the aircraft.



Figure 146 - Position Lights

The Position Lights have three modes – OFF, DIM and BRIGHT.



The modes can be selected using the three-way Position Lights Switch on the Overhead Console.

POLYCHOP SIMULATIONS

16.1.3 Searchlight

The searchlight is a controllable light, located in the underside of the nose section.

The light is controlled by two switches labelled SRCH LT on the pilot collective control head (3, 4, Figure 2-19). The searchlight can be turned on by the three-position toggle switch labelled ON, NVG, OFF.

Placing the switch in the ON position turns on the white searchlight. The center, or NVG position, turns on an NVG-compatible light, and the OFF position turns off both lights. The searchlight control switch, labelled EXT, RET, L, R, is center spring-loaded off.

The switch controls searchlight direction. The light receives power through the battery emergency bus and is protected by the SRCH LT CONTR and SRCH LT PWR circuit breakers.





POLYCHOP SIMULATIONS

16.1.4 Formation Lights

Six white infrared (IR) NVG-compatible formation lights are mounted on the aircraft. Two forward lights are mounted on the left and right top of the nose section.

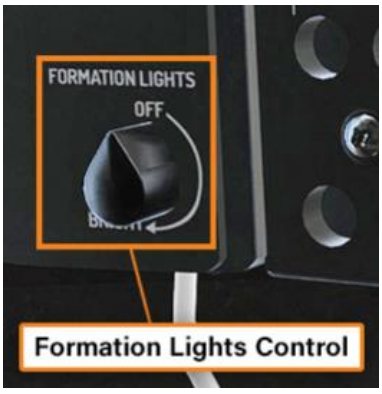
One is mounted in the lower forward area of the aft electronics bay door. One is mounted aft of the lower anticollision light and just above the tiedown mount. One is mounted on top of the tail cone structural support, and one is mounted on the right side of the fuselage in the avionics bay area.

The POS LT circuit breaker located on the aft overhead console panel, protects the circuit for the NVG formation lights.



Operation of the NVG formation lights is controlled by the FORMATION LIGHTS rotary switch located next to the overhead console panels.

When the knob is rotated just out of the OFF position, the lights are at minimum intensity, when the knob is rotated to the BRT position the lights are at maximum intensity



POLYCHOP SIMULATIONS

16.2 Interior Lighting

16.2.1 Instrument Lights





Figure 147 - Instrument Lighting

POLYCHOP SIMULATIONS

16.2.2 Console Lights



Figure 148 - Console Lighting

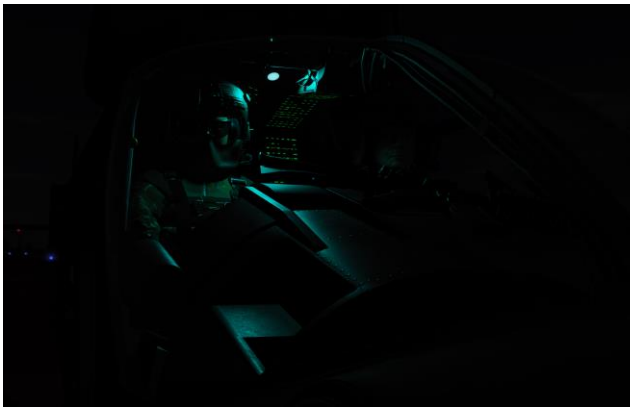
POLYCHOP SIMULATIONS

16.2.3 Flood Light

Brightness knob & Flood mode switch



Green Flood Light



White Flood Light



POLYCHOP SIMULATIONS

16.2.4 Flashlight



The utility light is located between the Pilot & copilot seats



POLYCHOP SIMULATIONS

Level 2 Manned/Unmanned System (L2MUMS)

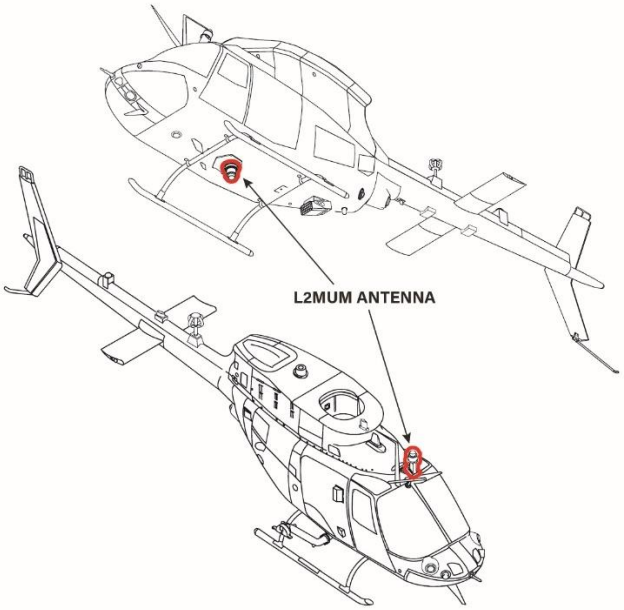
17.1 Overview

The Level 2 Manned/Unmanned System (L2MUMS) allows for the transmission of video and metadata between aircraft. It can be used to transmit and receive sensor data in real time between both unmanned and manned platforms.

While users can both receive data from another platform and transmit their own data at the same time, the receiver has no control over the platform that is transmitting data or the transmission itself.

The L2MUMS is a federated system that runs on a separate processor stored in the rear of the aircraft.

Two antennae mounted on the top and bottom of the aircraft allow line of sight communication to other platforms.



The Copilot's MFD displays data from the L2MUMS, and is used to make inputs to the system.

The Copilot can select whether the MFD displays and accepts input for either the IMCPU or L2MUMS by pressing the INIT bezel on the MFD.

INFO:

The L2MUMS is not accessible from the Pilot's MFD.

POLYCHOP SIMULATIONS

17.2 Pages

17.2.1 Platform Reception Selection

The Platform Reception Selection page is used to select and connect to incoming transmissions. There are 5 preset channels for Kiowa-to-Kiowa communication, while any available unmanned platforms are displayed below.

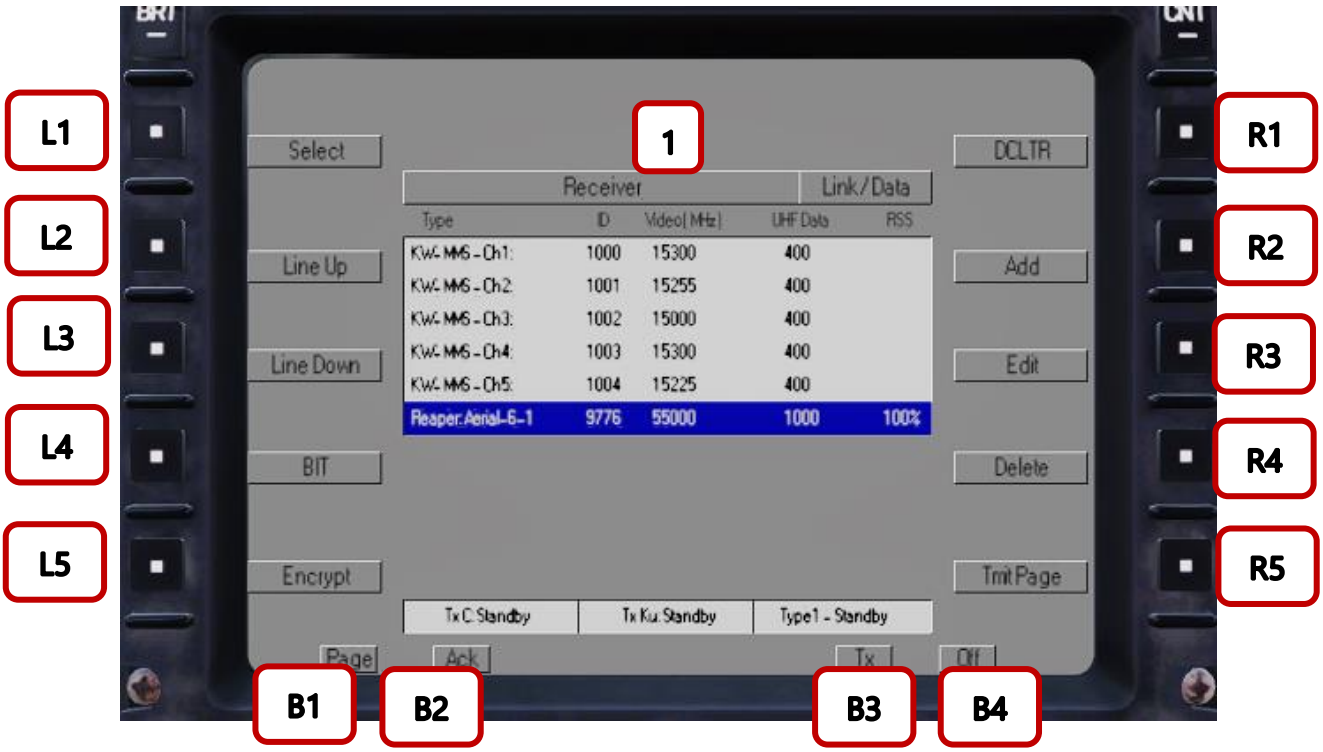


Figure 149 - Platform Reception Selection page

1. List of available channels. The selected channel is highlighted. Each channel has a Type Name, ID, Frequency, Bandwidth and Relative Signal Strength (RSS). RSS is only displayed when the channel is selected

L1. *Select*. Selects the highlighted channel for reception of data
L2. *Line Up*. Highlights the previous entry in the list
L3. *Line Down*. Highlights the next entry in the list
L4. *BIT*. Not implemented
L5. *Encrypt*. Not implemented

R1. *DCLTR*. Activating this bezel enables the declutter mode, hiding the bezel key text
R2. *Add*. Not implemented
R3. *Edit*. Not implemented
R4. *Delete*. Not implemented
R5. *Tmit Page*. Navigates to the **Platform Transmission Selection** page

B1. *Page*. Navigates to the **Video Display** page
B2. *Ack*. Acknowledges L2MUMS alerts
B3. *Tx*. Toggles and indicates transmission of sensor data
B4. *Off*. Not implemented

POLYCHOP SIMULATIONS

17.2.2 Platform Transmission Selection

The Platform Transmission Selection page is used to select a preset channel on which to transmit the video and sensor data from the player aircraft.

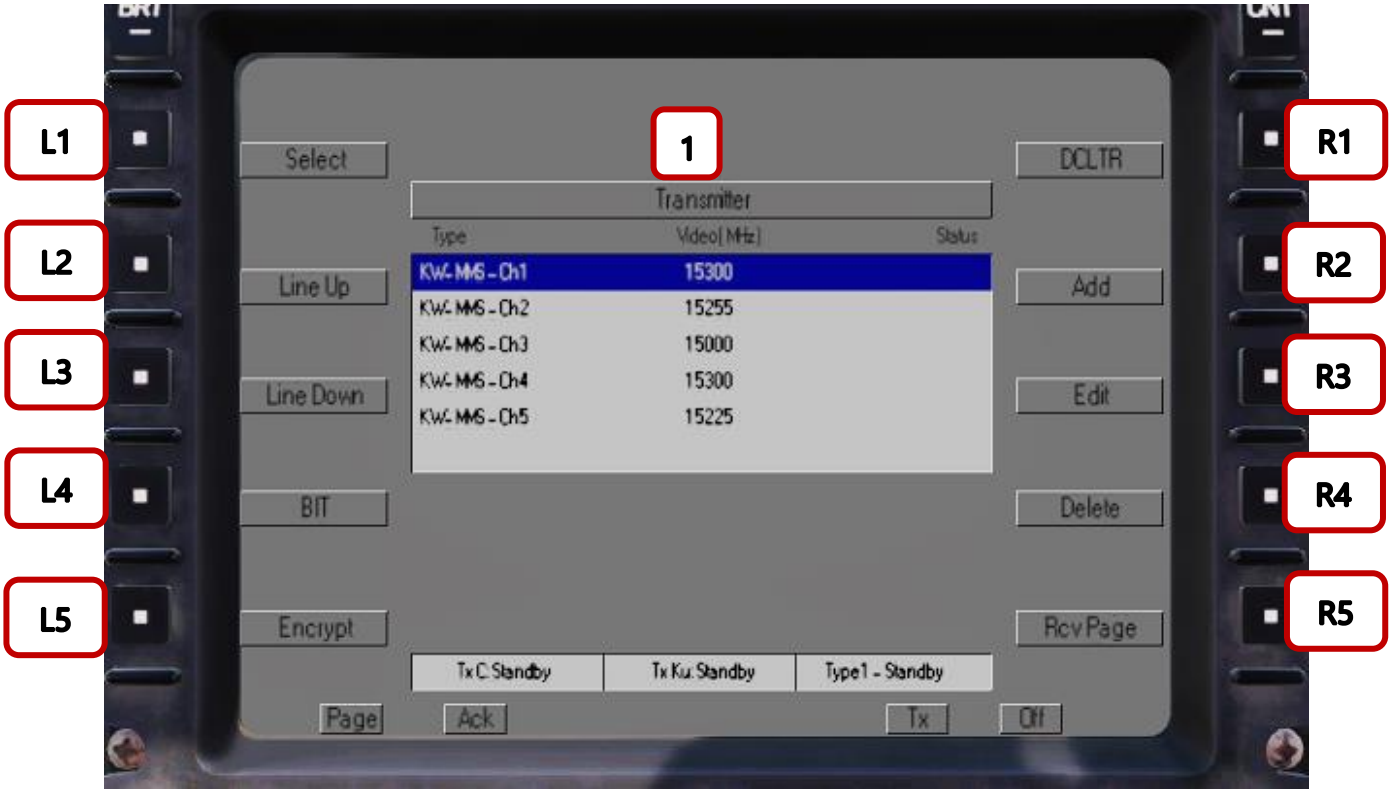


Figure 150 - Platform Transmission Selection page

1. List of available channels. The selected channel is highlighted. Each channel has a Type Name, Frequency and Status. Status is only displayed when the channel is selected

L1. *Select*. Selects the highlighted channel for reception of data
L2. *Line Up*. Highlights the previous entry in the list
L3. *Line Down*. Highlights the next entry in the list
L4. *BIT*. Not implemented
L5. *Encrypt*. Not implemented

R1. *DCLTR*. Activating this bezel enables the declutter mode, hiding the bezel key text
R2. *Add*. Not implemented
R3. *Edit*. Not implemented
R4. *Delete*. Not implemented
R5. *Rcv Page*. Navigates to the **Platform Reception Selection** page

B1. *Page*. Navigates to the **Video Display** page
B2. *Ack*. Acknowledges L2MUMS alerts
B3. *Tx*. Toggles and indicates transmission of sensor data
B4. *Off*. Not implemented

POLYCHOP SIMULATIONS

17.2.3 Video Display

The Video Display page displays the video feed from the connected platform, as well as target and platform metadata.

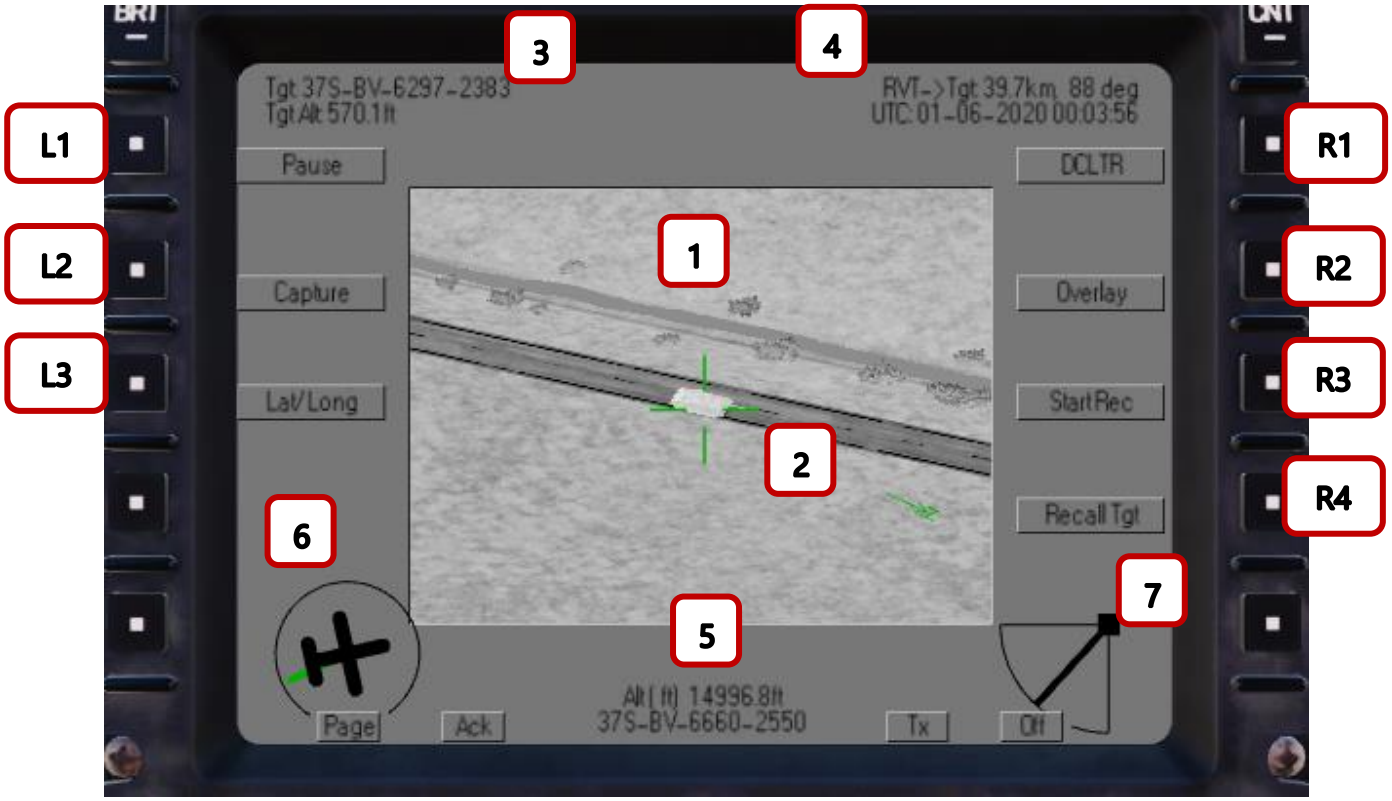


Figure 151 - Video Display page

1. Video Display
2. North Indicator and Stare Point Marker. Indicates relative bearing to north
3. Stare Point position and elevation
4. Relative bearing and distance from Receiving platform to Stare Point, and Current Time
5. Platform Position and Altitude
6. Platform Heading and Sensor Heading (green)
7. Sensor Elevation

L1. *Pause*. Not implemented

L2. *Capture*. Not implemented

L3. *Lat/Long or UTM*. Toggles metadata display between UTM and Lat/Long coordinate systems

R1. *DCLTR*. Activating this bezel enables the declutter mode, hiding the bezel key text

R2. *Overlay*. Successive presses of the Overlay bezel cycle the overlay mode:

- OFF (no overlays displayed)
- Level 1 (Stare Point Marker and North Indicator are displayed)
- Level 2 (Stare Point and Relative Position data in top left and top right are displayed along with all items from Level 1)
- Level 3 (Platform Position and Altitude at bottom are displayed along with all items from Level 2 and Level 1)
- Level 4 (Platform and Sensor Heading icons and Sensor Elevation icons are displayed along with all items from Level 3, Level 2 and Level 1)

R3. *Start Rec*. Not implemented

R4. *Recall Tgt*. Not implemented

B1. *Page*. Navigates to the **Map Display** page

B2. *Ack*. Acknowledges L2MUMS alerts

B3. *Tx*. Toggles and indicates transmission of sensor data

B4. *Off*. Not implemented

POLYCHOP SIMULATIONS

17.2.4 Map Display

The Map Display page shows the platform, sensor and target data overlayed on chart maps.



Figure 152 - Map Display page

1. Platform Position
2. Own Position
3. Platform Heading
4. Stare Point Location and Sensor Field of View

L1. *UAS Tgt*. Momentarily centers the map on the Stare Point

L2. *UAS Locn*. Momentarily centers the map on the Platform Position

L3. *UAS Scroll*. Centers and locks the map on the Platform Position. When the map is locked and the bezel text is highlighted blue, the other centering keys are disabled. A successive press of the L3 bezel disables the locking

R1. *DCLTR*. Activating this bezel enables the declutter mode, hiding the bezel key text

R2. *Zoom In*. Increases the map zoom. Available zoom levels are 50%, 75%, 100%, 200% and 400%. Current zoom level is displayed at the bottom right of the map

R3. *Zoom Out*. Decreases the map zoom. Available zoom levels are 50%, 75%, 100%, 200% and

POLYCHOP SIMULATIONS

L4. *Own Posn*. First press of this bezel momentarily centers the map on Own Position. A second press of this bezel locks the map on the Own Position. When the map is locked and the bezel text is highlighted blue, the other centering keys are disabled. A successive press of the L4 bezel disables the locking

400%. Current zoom level is displayed at the bottom right of the map

R4. *Scale In*. Increases the map scale. Available scales are 1:50000, 1:100000, 1:250000 and 1:500000

R5. Scale Out. Decreases the map scale. Available scales are 1:50000, 1:100000, 1:250000 and 1:500000

B1. *Page*. Navigates to the **Platform Reception Selection** page

B2. *Ack*. Acknowledges L2MUMS alerts

B3. *Tx*. Toggles and indicates transmission of sensor data

B4. *Off*. Not implemented

POLYCHOP SIMULATIONS



Section 3

PROCEDURES

Aviation

18.1 Startup

18.1.1 Before Engine Start

7. Set **Essential Bus Switch** to START
8. Set **FADEC Circuit Breaker Switch** to ON
9. Set **Ignition Circuit Breaker Switch** to ON
10. Turn **Ignition Keylock Switch** to ON position
11. Set **Battery 1 Switch** to ON
    a. Pilot MFD, RFD, VSI and MPD will switch on
    b. Acknowledge Warning Cautions Advisories (WCA's) as required by pressing the REC/ACK switch DOWN.
12. Verify **FADEC Auto/Manual Switch** (on dash) is in AUTO

18.1.2 Engine Start

- Advance **Throttle** to IDLE position
  - The Start Switch must be activated within 60 seconds of advancing the throttle or the engine will not start. To reset, set Throttle to OFF and toggle the FADEC Circuit Breaker Switch OFF, then ON. Then restart the procedure.
- Hold **Start Switch** for two seconds then release
  - Pilot MFD may flicker as voltage drops during start process
  - Acknowledge cautions as necessary
- Wait for NG to stabilize between 63-65%
- **DC Generator Switch** – ON
- **AC Generator Switch** – ON
- **Essential Bus Switch** to RUN
- **Fuel Boost Switch** - ON
- Particle Separator Circuit Breaker – IN
- Advance Throttle to OPEN
  - Acknowledge cautions as necessary
  - Wait for TGT and RPM to stabilize
- Set **MPD Selector Switch** to display NR/NP
- Increase NP/NR to 100% using the **RPM Trim Switch**
  - Increase until one amber chiclet is visible on the NR **VSI**

18.1.3

18.1.4 ENGINE RUNUP — CPO/PILOT.

1. PC DTS-V — Mission load as required.
2. NAV align — Initiate as required.
3. ASE — Switches on as required.
4. Avionics — Configure as required:
    a. Communication radios — Check; program frequencies, configure secure equipment, and configure frequency hopping equipment.
    b. Transponder — Program and check.
5. MMS startup checks — Complete as required.
6. Navigation systems — Configure as required.
    a. Waypoints, control points, and targets — Check/load.
    b. Routes — Check/construct.
    c. Battlefield graphics — Check/construct.
    d. HSD−, HSD+, and RMS — As required.
7. IDM — Initialize as required.
8. Weapons systems — Initialize and check as required.
9. ASE — Check as required.
10. Set CI PWR switch to ON.
11. Verify CI Status alphanumeric display and indicators briefly light.
12. Verify CI software version number briefly displays. Message COMM FAIL is displayed on CI at CMWS power cycle for approximately 30 seconds - this is normal.
13. Verify “A-A-R 57 Self Test” voice alert is heard at start of SBIT over ICS and CI displays BIT I/P.
14. (Ground only) Verify "SEQ LOAD" flashes "SEQ XXXX" for approximately two minutes. (Only if Sequencer OFP is being updated.)
15. After one second, verify four CI quadrant indicators simultaneously flash twice.
16. If SYS PASS message appears, system is mission capable.
17. L2MUM — Startup system as required.
    a. CPO LWCMFD INIT Button — Press.
    b. L2MUM Circuit Breaker Switch — L2MUM (ON).

POLYCHOP SIMULATIONS

18.2 Shutdown

18.2.1 ENGINE SHUTDOWN.

1. Flight controls — Cyclic centered, pedals neutral, collective down.
2. FORCE TRIM switch — FORCE TRIM.
3. Present position — Store or record as required.
4. MMS — OFF.
5. Throttle — Reduce to idle for two minutes.
6. BATT 1 — Check as follows:
   a. MPD SEL — Select RECT LD% – S GEN LD%.
   b. BATT 1 switch — OFF.
   c. MPD — Check S GEN LD% and note indication.
   d. BATT 1 switch — BATT 1. Check S GEN LD% increase should not exceed 10%.
7. AC GEN switch — OFF.
8. FUEL BOOST switch — OFF.
9. ASE — Off as follows:
   a. RADAR WARN switch — WARN (OFF).
10. IFF circuit breaker switch – OFF.
11. Standby attitude indicator — Caged.
12. SCAS PWR — OFF.
13. PC DTS-V — Mission store as desired.
14. FADEC MONITOR and ENGINE HISTORY pages — Check and record faults or values that exceed limitations.
15. Throttle — Closed and monitor TGT.
16. Overhead switches — Set; off except battery, required lights, and the following:
   a. ESNTL BUS switch — START when NG is less than 10%.
   b. COMPT BLWR switch — AUTO.
   c. FADEC switch – FADEC.
   e. PART SEP circuit breaker — OUT.
17. Battery and light switches — OFF, when main rotor blades stop turning.
18. Ignition keylock switch — OFF, remove key as required.

POLYCHOP SIMULATIONS

18.3 FADEC & Engine Management

18.3.1 Switching to FADEC Manual mode

18.3.2 MPD Bit Operation

1. Move **MPD BIT Switch** to BIT
2. Verify WRN indicator illuminates
3. On the right-side boxes “PN” followed by numbers displays
4. Wait for 5 seconds
5. On the right-side boxes “ERR” followed by a digit if any error happened

18.4 Warnings, Cautions, Advisories

18.4.1 Acknowledgement and Recall of Cautions and Advisories

Pressing the REC/ACK switch to the ACK position will delete the message from the screen.

When an advisory has been acknowledged, it will stay in memory as long as the advisory condition exists.

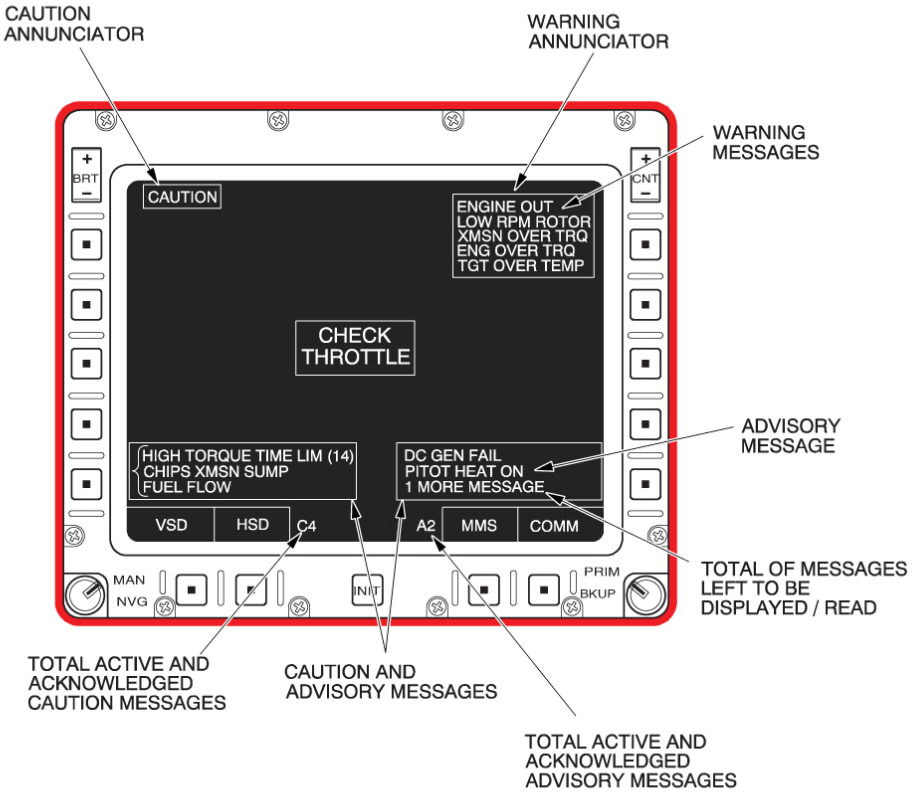
Pressing the switch to the REC position will return any stored caution or advisory messages to the display, up to a total of five cautions and advisories.



Figure 153 - REC/ACK switch

If more than five messages are active at one time, only five messages will be displayed plus a note of how many more messages are yet to be displayed.

POLYCHOP SIMULATIONS



The total number of active advisories is displayed to the right of the letter A at the bottom of the screen.

INFO:

When a warning condition occurs, the warning is displayed, accompanied by a warning audio. The FADEC FAIL warning consists of a high/low tone.

The LOW RPM ROTOR warning consists of a continuous whoop.

The audio signal for all other warnings is a continuous three “bongs” per second.

The audio signal can be silenced by moving the REC/ACK (recall/acknowledge) switch on the MFK to the ACK position

The Full list of **Warnings Cautions & Advisories** is included in the **Appendix**

INFO:

Aircraft Caution, Warning, and Advisory messages will not display on the CPO (left) LWCMFD while in the L2MUM mode.

POLYCHOP SIMULATIONS

Navigation

19.1 Overview

The navigation tools in the OH-58D Kiowa Warrior comprise of Waypoints, Control Points and Target Points which can be used as building blocks to create Routes and Battlefield Graphics. This data can also be exported and imported using the Data Transfer System.

19.2 Data Transfer System

19.2.1 Saving Mission Data

1. Select INIT button on the respective **MFD Auxiliary Control Panel** to display the **Initial Page.**
2. Select L4 and the Data Entry Mode will be activated.
3. Type the password into the MFK (Default is **"58D"**), and press Enter. This will display the **Data Loader** page.
4. Select R1, R2 or R3 to open the **Store Mission** page of the selected slot to display.
5. Select B4 to change the Mission Name, enabling the Data Entry.
6. Select the MFD bezels of the things you want to store.
7. Select B3 to save the selected mission slot.

INFO:

Missions are saved in a folder called "DCS_OH58D" at your Saved Games folder. You can find the missions by the name MissionN.json, in which N is the slot (1-3). If you are not familiar with json files restrain from editing them, because it could cause unexpected behaviour.

POLYCHOP SIMULATIONS

19.2.2 Loading Mission Data

1. Select INIT button on the respective **MFD Auxiliary Control Panel** to display the **Initial Page.**
2. Select L4 and the Data Entry Mode will be activated.
3. Type the password into the MFK (Default is **"58D"**), this will display the **Data Loader** page.
4. Select L1, L2 or L3 to open the **Load Mission** page of the selected slot to display. (There might be less than 3 options depending on the number of .json present in your folder)
5. Select the MFD bezels of the things you want to load.
6. Select B3 to load the selected mission slot.

INFO:

Missions are saved in a folder called "DCS_OH58D" at your Saved Games folder. You can find the missions by the name MissionN.json, in which N is the slot (1-3). If you are not familiar with json files restrain from editing them, because it could cause unexpected behavior.

POLYCHOP SIMULATIONS

19.3 Waypoints

19.3.1 Creating a Waypoint

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select L4, **New Waypoint** page
    - Data Entry Mode will be automatically activated
    - A cursor will appear at the L1 field
4. Enter a valid waypoint ID
5. Enter a valid UTM or LL position
    - When in LL mode, the Latitude is entered and L2 followed by the Longitude at L3
6. Enter a valid Elevation
7. Select R5 STORE to create the waypoint
    - Upon the creation of a valid waypoint, the page will autocomplete the next available waypoint slot in preparation for another waypoint to be created

INFO:
A waypoint ID must be in the following formats:

- #W
- ##W
- #ABCDE
- ##ABCD

POLYCHOP SIMULATIONS

19.3.2 Editing a Waypoint

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select L4, **New Waypoint** page
    - Data Entry Mode will be automatically activated
    - A cursor will appear at the L1 field
4. Enter an existing waypoint ID. If done correctly, the position and altitude data will correlate to the one on the list.
5. Enter a valid UTM or LL position
    - When in LL mode, the Latitude is entered and L2 followed by the Longitude at L3
6. Enter a valid Elevation
7. Select R5 STORE to update the waypoint
    - Upon the creation of a valid waypoint, the page will autocomplete the next available waypoint slot in preparation for another waypoint to be created

INFO:
A waypoint ID must be in the following formats:

- #W
- ##W
- #ABCDE
- ##ABCD

POLYCHOP SIMULATIONS

19.3.3 Deleting a Waypoint

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select L5, **Waypoint List** page
4. Select R1, DEL WP
    - Data Entry Mode is activated and a cursor appears next to the bezel
5. Enter a waypoint number between 0 and 99 to delete the waypoint at that index

INFO:

When attempting to delete any navigation point, the system first checks whether it is part of a Battlefield Graphic or Route.

If the system finds that the navigation point is part of a Route, the navigation point is not deleted, the Data Entry Mode is cancelled and the bezel text will display a flashing 'RTE #' indicating the Route using the navigation point.

If the system finds that the navigation point is part of a Battlefield Graphic, the navigation point is not deleted, the Data Entry Mode is cancelled and the bezel text will display a flashing 'BF ##' indicating the Battlefield Graphic using the navigation point.

POLYCHOP SIMULATIONS

19.3.4 Deleting all Waypoints

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select L5, **Waypoint List** page
4. Select R1, DEL WP
    - Data Entry Mode is activated and a cursor appears next to the bezel
5. Enter 'ALL'
    - Upon entry all waypoints will be checked and deleted

INFO:

When attempting to delete any navigation point, the system first checks whether it is part of a Battlefield Graphic or Route.

If the system finds that the navigation point is part of a Route, the navigation point is not deleted, the Data Entry Mode is cancelled and the bezel text will display a flashing 'RTE #' indicating the Route using the navigation point.

If the system finds that the navigation point is part of a Battlefield Graphic, the navigation point is not deleted, the Data Entry Mode is cancelled and the bezel text will display a flashing 'BF ##' indicating the Battlefield Graphic using the navigation point.

19.4 Control Points

19.4.1 Creating a Control Point

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select R2 to enter **Control Point List** page
4. Select R5, **New Control Point** page
    - Data Entry Mode will be automatically activated
    - A cursor will appear at the L1 field
5. control point ID
6. Enter a valid UTM or LL position
    - When in LL mode, the Latitude is entered and L2 followed by the Longitude at L3
7. Enter a valid Elevation
8. Select R5 STORE to create the waypoint
    - Upon the creation of a valid control point, the page will autocomplete the next available control point slot in preparation for another control point to be created

INFO:
A control point ID must be in the following formats:

- #C
- ##C
- #ABCDE
- ##ABCD

POLYCHOP SIMULATIONS

19.4.2 Editing a Control Point

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select R2 to enter **Control Point List** page
4. Select R5, **New Control Point** page
5. Enter a valid UTM or LL position
    - When in LL mode, the Latitude is entered and L2 followed by the Longitude at L3
6. Enter a valid Elevation
8. Select R5 STORE to update the waypoint
    - Upon the creation of a valid control point, the page will autocomplete the next available control point slot in preparation for another control point to be created

INFO:
A control point ID must be in the following formats:

- #C
- ##C
- #ABCDE
- ##ABCD

19.4.3 Deleting a Control Point

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select R2 to enter **Control Point List** page
4. Select R1, DEL CP
    - Data Entry Mode is activated and a cursor appears next to the bezel
5. Enter a control point number between 0 and 99 to delete the waypoint at that index

INFO:
When attempting to delete any navigation point, the system first checks whether it is part of a Battlefield Graphic or Route.
If the system finds that the navigation point is part of a Route, the navigation point is not deleted, the Data Entry Mode is cancelled and the bezel text will display a flashing 'RTE #' indicating the Route using the navigation point.
If the system finds that the navigation point is part of a Battlefield Graphic, the navigation point is not deleted, the Data Entry Mode is cancelled and the bezel text will display a flashing 'BF ##' indicating the Battlefield Graphic using the navigation point.

POLYCHOP SIMULATIONS

19.4.4 Deleting all Control Points

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select R2, **Control Point List** page
4. Select R1, DEL CP
    - Data Entry Mode is activated and a cursor appears next to the bezel
5. Enter 'ALL'

INFO:

When attempting to delete any navigation point, the system first checks whether it is part of a Battlefield Graphic or Route.

If the system finds that the navigation point is part of a Route, the navigation point is not deleted, the Data Entry Mode is cancelled and the bezel text will display a flashing 'RTE #' indicating the Route using the navigation point.

If the system finds that the navigation point is part of a Battlefield Graphic, the navigation point is not deleted, the Data Entry Mode is cancelled and the bezel text will display a flashing 'BF ##' indicating the Battlefield Graphic using the navigation point.

19.5 Target Points

19.5.1 Creating a Target Point

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select R4, **New Target Point** page
    - Data Entry Mode will be automatically activated
    - A cursor will appear at the L1 field
4. Select L2 to set entry mode on position line
5. Enter a valid UTM or LL position
    - When in LL mode, the Latitude is entered and L2 followed by the Longitude at L3
6. Enter a valid Elevation
7. Select R5 STORE to create the waypoint
    - Upon the creation of a valid target point, the page will autocomplete the next available target point slot in preparation for another target point to be created

19.5.2 Creating a Target Point using the MMS 'Store' function

INFO:

This procedure has to be performed in the left MFD, since the right side has a simplified MMS symbology.

1. Select B3 to activate the **MMS** in any mode
2. Set the Laser Arm Switch to armed
3. Press the Target Designate button
4. Press L1 to select any Laser Code (Not Range mode)
5. Hold down the Laser Fire Switch for 3 to 5 seconds
6. Press L4 to save the target point on the highest slot available

19.5.3 Creating a Target Point using the MMS 'Store More' function

INFO:

This procedure has to be performed in the left MFD, since the right side has a simplified MMS symbology.

1. Select B3 to activate the **MMS** in any mode
2. Set the Laser Arm Switch to armed
3. Press the Target Designate button
4. Press L1 to select any Laser Code (Not Range mode)
5. Hold down the Laser Fire Switch for 3 to 5 seconds
6. Press L3 to open the **HOG** menu

INFO:
To navigate through this menu you have to use the following controls:

- MMS FOV Narrow - Scroll Up
- MMS FOV Wide - Scroll Down
- MMS Point Track - Select
- MMS TV/TIS - Back

7. Fill with the MFK the index you want the target point saved
8. Fill the Target Quantity
9. Fill the Target Type
10. Fill the Artillery number
11. If you need to modify previous data select modify, else you can finish the process by selecting Store.

POLYCHOP SIMULATIONS

19.5.4 Editing a Target Point

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select R4, **New Target point** page
    - Data Entry Mode will be automatically activated
    - A cursor will appear at the L1 field
4. Enter an existing target point ID. If done correctly, the position and altitude data will correlate to the one on the list.
5. Enter a valid UTM or LL position
    - When in LL mode, the Latitude is entered and L2 followed by the Longitude at L3
6. Enter a valid Elevation
7. Select R5 STORE to update the target point
    - Upon the creation of a valid target point, the page will autocomplete the next available target point slot in preparation for another target point to be created

INFO:
A Target Point ID must be in the following formats:

- #T
- ##T

19.5.5 Deleting a Target Point

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select R2 to enter **Control Point List** page
4. Select R1, DEL TP
    - Data Entry Mode is activated and a cursor appears next to the bezel
5. Enter a control point number between 0 and 99 to delete the waypoint at that index

INFO:

When attempting to delete any navigation point, the system first checks whether it is part of a Battlefield Graphic or Route.

If the system finds that the navigation point is part of a Route, the navigation point is not deleted, the Data Entry Mode is cancelled and the bezel text will display a flashing 'RTE #' indicating the Route using the navigation point.

If the system finds that the navigation point is part of a Battlefield Graphic, the navigation point is not deleted, the Data Entry Mode is cancelled and the bezel text will display a flashing 'BF ##' indicating the Battlefield Graphic using the navigation point.

19.6 Direct To Mode

19.6.1 Creating a Direct To Point

1. Select B2 to activate the **HSD**
2. Select L1 to enter **DIRTO Point** page.
3. Select L1 again to enter the Id of the desired created point or press enter to continue with a custom point.
4. If it's a custom point enter a valid UTM or LL position
    - When in LL mode, the Latitude is entered and L2 followed by the Longitude at L3
5. If it's a custom point a valid Elevation
6. Select R5 STORE to update the waypoint

19.7 Routes

19.7.1 Creating a Route

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select L1, L2 or L3 depending on the slot you want to create
4. Select R1 to change the name
5. Select L1 and introduce a navigation point name
    a. Repeat with all the legs you need
6. To change an already created leg use CHG LEG on L4
7. To delete an already created leg use DEL LEG on R4
8. To create a leg between already created legs use INS LEG on L5
9. To move from one leg to other use L2 and L3
10. Press R5 when finished to go back to **NAV Setup** page

19.7.2 Editing a Route

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select L1, L2 or L3 depending on the slot you want to create
4. Select R1 to change the name
5. To change an already created leg use CHG LEG on L4
6. To delete an already created leg use DEL LEG on R4
7. To create a leg between already created legs use INS LEG on L5
8. To move from one leg to other use L2 and L3
9. Press R5 when finished to go back to **NAV Setup** page

19.7.3 Selecting a Route

1. Select B2 to activate the **HSD**
2. Continuously press R3 until the name of the desired route is on the screen.

POLYCHOP SIMULATIONS

19.8 Battlefield Graphics

19.8.1 Creating a Line Graphic

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select R1 to enter **BFLD GRPH** page
4. Select R1 to create a line
5. Fill the index you want to give the Line
6. Select R1 to change the name
7. Select L1 and introduce a navigation point name
    - Repeat with all the legs you need
8. To change an already created leg use CHG LEG on L4
9. To delete an already created leg use DEL LEG on R4
10. To create a leg between already created legs use INS LEG on L5
11. To move from one leg to other use L2 and L3

19.8.2 Editing a Line Graphic

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select R1 to enter **BFLD GRPH** page
4. Select R1 to select a line
5. Fill the index of the line you want to edit
6. Select R1 to change the name
7. Select L1 and introduce a navigation point name
    - Repeat with all the legs you need
8. To change an already created leg use CHG LEG on L4
9. To delete an already created leg use DEL LEG on R4
10. To create a leg between already created legs use INS LEG on L5
11. To move from one leg to other use L2 and L3

19.8.3 Deleting a Line Graphic

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select R1 to enter **BFLD GRPH** page
4. Select L3 to choose the drawing you want to delete
5. Fill the index

POLYCHOP SIMULATIONS

19.8.4 Creating an Area Graphic

1. Select B2 to activate the **HSD**
2. Select R2 to enter **NAV Setup** page
3. Select R1 to enter **BFLD GRPH** page
4. Select R2 to create a Area
5. Fill the index you want to give the Area
6. Select R1 to change the name
7. Select L1 and introduce a navigation point name
    - Repeat with all the legs you need
8. To change an already created leg use CHG LEG on L4
9. To delete an already created leg use DEL LEG on R4
10. To create a leg between already created legs use INS LEG on L5
11. To move from one leg to other use L2 and L3

19.8.5 Editing an Area Graphic

12. Select B2 to activate the **HSD**
13. Select R2 to enter **NAV Setup** page
14. Select R1 to enter **BFLD GRPH** page
15. Select R2 to select an Area
16. Fill the index of the Area you want to edit
17. Select R1 to change the name
18. Select L1 and introduce a navigation point name
    - Repeat with all the legs you need
19. To change an already created leg use CHG LEG on L4
20. To delete an already created leg use DEL LEG on R4
21. To create a leg between already created legs use INS LEG on L5
22. To move from one leg to other use L2 and L3

19.8.6 Deleting an Area Graphic

6. Select B2 to activate the **HSD**
7. Select R2 to enter **NAV Setup** page
8. Select R1 to enter **BFLD GRPH** page
9. Select L3 to choose the drawing you want to delete
10. Fill the index

POLYCHOP SIMULATIONS

Communication

20.1 Radios

20.1.1 Selecting a Radio

Radio selection and audio volume are controlled through the communication system control (CSC) panel (Figure 3-1). Radio selections are also made with the radio select switch located on the pilot collective control head (Figure 3-2).

20.1.2 Transmitting on a Radio

20.1.3 Adding or Editing a Channel

1. Select B4 to activate the **COMM** page.
2. Choose the radio you want to manipulate by selecting L1 to L5 (except L4)
3. Select R1 “ADD” to enable the Data entry at the bottom of the page
4. Fill the index of the radio
5. Fill the frequency
6. Fill the name or leave it empty and select the enter key

20.1.4 Deleting a Channel

1. Select B4 to activate the **COMM** page.
2. Choose the radio you want to manipulate by selecting L1 to L5 (except L4)
3. Select R2 “DEL” to enable the Data entry at the bottom of the page
4. Fill the index of the radio

20.1.5 Editing Frequencies from the Remote Frequency Display

1. Press the KYBD button from the pilot’s collective or the left knee of the copilot.
2. Fill the frequency on the active selected radio and channel
3. Press Enter to validate it

20.2 Improved Data Modem (IDM)

20.2.1 Overview

The Improved Data Modem is a subsystem of the CDS that handles data communication between aircraft. It is accessed using the Hands On Grip User Interface (HOG) and allows for messaging, target sharing, situational awareness on the battlefield and calling in fire support.

20.2.2 Special Controls

The HOG interface uses the Copilot Cyclic Grip controls for navigation and selection of menu items and has some specialized behaviors. The HOG interface is only accessible from the Copilot's MFD.

20.2.2.1 Frame Freeze/Target Select / HOG ON/OFF

The MMS Frame Freeze/Target Select button is used to toggle display of the HOG UI. This feature is not accessible when the MMS Frame Freeze/Target Select button has another use, such as when using the Acquisition On The Move sub-mode.

20.2.2.2 MMS FOV - Wide Toggle Up 2x > 4x / HOG Menu Item Scroll Up & MMS FOV - Narrow Toggle Down 2x > 4x / HOG Menu Item Scroll Down

The MMS FOV input serves to scroll selection up or down when the HOG UI is active.

20.2.2.3 MMS Point-Track On / HOG Menu Selection Confirm

The Point Track input is used to confirm selection of an item in the HOG UI. The outcome of selection depends on the menu item selected.

20.2.3 Changing Callsign ID

Each aircraft spawns with an ID based on the callsign they have been assigned in the mission editor. This can be changed to make it easier to distinguish different aircraft in mission.

1. Display the HOG UI by pressing Frame Freeze/Target Select / HOG ON/OFF on the Copilot Cyclic Grip
2. Navigate to 'OTHER FUNCS' using the MMS FOV input
3. Select 'OTHER FUNCS' by pressing Point Track
4. Enter the 'USER STAT' page using the MMS FOV input and Point Track inputs
5. Select the option under 'USERNAME'. The menu will automatically activate Data Entry Mode and a cursor will appear
6. Input your desired ID using the MFK and press Enter
7. Successful validation will update your ID and deactivate Data Entry Mode
8. Close the HOG UI by pressing Frame Freeze/Target Select / HOG ON/OFF on the Copilot Cyclic Grip

20.2.4 Sending a Position Report (POSREP)

A Position Report is an auto-generated message that includes your current position. Recipients can store the position as a Target Point for navigation.

1. Open the HOG UI
2. Select C2 MSGS
3. Select OWN POS
4. Select MOD HEADER
5. Select RECIPIENTS
6. Select the recipients for your POSREP message by highlighting and selecting the IDs available. Selected recipients will be marked with an asterisk.
7. Once the desired recipients have been selected, select BACK
8. Select BACK a second time to return to the SEND page
9. The first three recipients selected will now be displayed under the message subject header. Select CONFIRM SEND to send the message
10. The SEND STATUS page will be displayed and the message will be broadcast to the selected recipients. When the send process is complete, the top level page will be selected

20.2.5 Receiving a Message

When you receive a message from another player, an advisory will be displayed in the lower left of the MFD indicating the message type. The message will then be available in your inbox to open.

1. Open the HOG UI
2. Select INBOX
3. Select the received message from the list.

INFO:

Depending on the message type, different actions may occur

Fire Support requests will automatically display the relevant MMS page to view the mission data and accept the fire mission

C2 messages will display the received text and subject, with various options such as target storing and replying being available depending on the message contents

20.2.6 Sending a Free Text Message

Free text messages allow a custom input message to be sent.

1. Open the HOG UI
2. Select C2 MSGS
3. Select FREETXT
4. Select MAN ENTRY
5. Data Entry Mode will be activated on the first available line. There are seven free lines that can accept up to 16 characters using the MFK. Each line can be confirmed by pressing Enter or selecting the next line
6. When the message is complete, select SEND
7. Select MOD HEADER
8. Select RECIPIENTS
9. Select the recipients for your POSREP message by highlighting and selecting the IDs available. Selected recipients will be marked with an asterisk.
10. Once the desired recipients have been selected, select BACK
11. Select BACK a second time to return to the SEND page
12. The first three recipients selected will now be displayed under the message subject header. Select CONFIRM SEND to send the message
13. The SEND STATUS page will be displayed and the message will be broadcast to the selected recipients. When the send process is complete, the top level page will be selected

POLYCHOP SIMULATIONS

20.2.7 Sending a Predetermined Free Text Message

The Free Text category offers a number of alternative preset messages that can be used to issue or respond to orders and allow the inclusion and sharing of Target Points. These include the 'MOVE TO', 'HOLD AT', 'RDVZ AT', 'MOVING TO' and 'HOLDING AT' messages. The process for each type of message is the same.

1. Open the HOG UI
2. Select C2 MSGS
3. Select FREETXT
4. Select MOVE TO
5. Select CHANGE TGT
6. Data Entry Mode will be activated below the TARGET ID line. Entering a valid Target Point ID will add the target data to the message
7. Select BACK
8. Alternative: Select MAN ENTRY
9. Data Entry Mode will be activated below the MOVE TO line. Any data may be entered and will be displayed to the recipient (e.g. 'FARP') however a entering a valid UTM position will allow the recipient to store the position as a Target Point
10. Select BACK
11. When the message is complete, select SEND
12. Select MOD HEADER
13. Select RECIPIENTS
14. Select the recipients for your message by highlighting and selecting the IDs available. Selected recipients will be marked with an asterisk.
15. Once the desired recipients have been selected, select BACK
16. Select BACK a second time to return to the SEND page
17. The first three recipients selected will now be displayed under the message subject header. Select CONFIRM SEND to send the message
18. The SEND STATUS page will be displayed and the message will be broadcast to the selected recipients. When the send process is complete, the top level page will be selected

20.2.8 Sending a Spot Report (SPOTREP)

A Spot Report allows for sending of target data to other aircraft. A Target Point must have been stored in order for a Spot Report message to be sent.

1. Open the HOG UI
2. Select C2 MSGS
3. Select FREETXT
4. Select SPOT
5. The SPOT page will automatically populate with the data from the first available target starting from 99T down
6. To choose another Target Point, select the target ID at position 2
7. On the LOCATION page, use the MMS FOV inputs to cycle through the stored Target Points. The Target Point UTM position is displayed above. Selecting the Target Point using Point Track will return to the previous page
8. The target data can be modified by selecting the TYPE/AFFILIATION entry at position 4. This will present a series of pages that allow the user to select the target quantity, type, sub-type and affiliation
9. The target's activity status can be changed through the options presented by selecting the item at position 3
10. When the target data is complete, select SEND
11. Select MOD HEADER
12. Select RECIPIENTS
13. Select the recipients for your message by highlighting and selecting the IDs available. Selected recipients will be marked with an asterisk.
14. Once the desired recipients have been selected, select BACK
15. Select BACK a second time to return to the SEND page
16. The first three recipients selected will now be displayed under the message subject header. Select CONFIRM SEND to send the message
17. The SEND STATUS page will be displayed and the message will be broadcast to the selected recipients. When the send process is complete, the top level page will be selected

20.2.9 Storing a Target Point from a Received Message

Depending on the type of message received, a position may be available to be stored as a Target Point.

1. Open the HOG UI
2. Select INBOX
3. Select the received message from the list.
4. Select MSG ACTION
5. If the message contains a position that can be stored, the option STORE TGT will be displayed
6. Select STORE TGT
7. Select the position to store. Selected positions will be indicated by an asterisk
8. Select STORE
9. Data Entry Mode will be activated on the STORE AS item. Entering a valid Target Point ID will exit Data Entry Mode and return the user to the MSG ACTIONS page

POLYCHOP SIMULATIONS

Sensors

21.1 Mast Mounted Sensor (MMS)

21.1.1 Starting the MMS

1. Once power is applied to the aircraft, rotate the **MMS Mode Selector** to any of the operational modes (FWD/PREPT/SRCH) or PREFLT
2. For the operational modes, press the Manual/Slave button twice to slave the MMS to the selected mode

21.1.2 Shutting down the MMS

1. Rotate the **MMS Mode Selector** to STOW
2. The MMS will initiate shutdown procedures and slew to the stow point
3. When the advisory STORE COMPLETE is displayed, rotate the **MMS Mode Selector** to OFF

21.1.3 Controlling the MMS in Manual Mode

1. From any of the operational modes (Forward/Pre-Point/Search) or sub-modes (Area Track, Point Track, Acquire On The Move, Offset Acquire or Offset Track) or Uncommanded mode, press the Manual/Slave button on the Copilot Cyclic Grip
2. MMS Advisory will read 'MANUAL' and the Manual mode crosshair will be displayed
3. The MMS can be slewed using the MMS LOS Slew input and will be stabilized

21.1.4 Using the Laser Rangefinder

1. Set the **Laser Switch** to ARM
2. Slew the MMS to the desired target
3. Press and hold the Laser Fire button on the Copilot Cyclic Grip
4. Range to target will be displayed at L1

21.1.5 Using the Laser Designator

1. Set the **Laser Switch** to ARM
2. Slew the MMS to the desired target
3. If the text at L1 reads RNG, press L1 to display the Laser Code List
4. Successive presses of L1 will cycle selection of laser codes
5. Selecting a code will enable laser designation and display the code at L1
6. Press and hold the Laser Fire button on the Copilot Cyclic Grip
7. Range to target will be displayed at L1 and the target will be designated using the selected code

POLYCHOP SIMULATIONS

21.1.6 Slaving to Forward Mode

1. Rotate the **MMS Mode Selector** to FWD
2. Press Manual/Slave to enter Manual mode (if already in Manual mode, will directly proceed to Forward mode)
3. Press Manual/Slave a second time to enter Forward mode
4. MMS Advisory will read 'FWD'
5. MMS will automatically slew to face forward relative to aircraft direction
6. MMS LOS Slew inputs can be used to trim the elevation of the MMS, however excessive left/right input will break the MMS out of Forward mode

21.1.7 Pre-Point Mode

1. Rotate the **MMS Mode Selector** to PREPT
2. Press Manual/Slave to enter Manual mode (if already in Manual mode, will directly proceed to Pre-Point mode)
3. Press Manual/Slave a second time to enter Pre-Point mode
4. MMS Advisory will read 'PREPT'
5. If a Pre-Point has already been entered, MMS will automatically slew to the current Pre-Point
6. To enter a Pre-Point, select bezel R2 – PPT
7. Data Entry Mode is activated and the Pre-Point List is displayed
8. To select a different item in the list, press R2 until the desired item is selected
9. To enter or change a list entry, enter a valid Waypoint, Control Point or Target Point identifier and press Enter
10. The Pre-Point List will automatically close and Data Entry Mode will be canceled after 5 seconds of inactivity

21.1.8 Search Mode – Point

1. Rotate the **MMS Mode Selector** to SRCH
2. Press Manual/Slave to enter Manual mode (if already in Manual mode, will directly proceed to Search mode)
3. Press Manual/Slave a second time to enter Search mode
4. MMS Advisory will read 'SRCH'
5. If the text at L2 reads SWEEP, press L2 to switch to the Point Search mode
6. If search points have been entered, the MMS will automatically begin slewing to the next available target point in the Search Point List
7. To access and edit the Search Point List, press R2- PT to display the list and activate Data Entry Mode
8. To select a different item in the list, press R2 successively to highlight the desired item
9. Entering a valid Waypoint, Control Point or Target Point will add or edit the point to the selected list item
10. The amount of time the MMS lingers on a target point can be edited by pressing L4 – DWELL to activate Data Entry Mode and entering the number of seconds the MMS should dwell on target

21.1.9 Search Mode – Sweep

11. Rotate the **MMS Mode Selector** to SRCH
12. Press Manual/Slave to enter Manual mode (if already in Manual mode, will directly proceed to Search mode)
13. Press Manual/Slave a second time to enter Search mode
14. MMS Advisory will read 'SRCH'
15. If the text at L2 reads POINT, press L2 to switch to the Sweep Search mode
16. The MMS will begin sweeping according to the set sweep parameters
17. The sweep angle can be adjusted by pressing L4 to cycle between sweep angles
18. The sweep speed can be toggled between FAST and SLOW using L3
19. Pressing R2 will cycle between the sweep offsets

21.1.10 Area Track

1. Switch to Manual mode
2. Slew the MMS to the desired area
3. Press the Area Track button on the Copilot Cyclic Grip
4. MMS Advisory will read AREA TRK and the MMS LOS Slew inputs will no longer be functional
5. Area Track can be exited by toggling Manual/Slave

21.1.11 Point Track

1. Switch to Manual mode
2. Slew the MMS to the desired target
3. Press the Point Track button on the Copilot Cyclic Grip
4. The Acquisition Gate will expand until a contrast lock is achieved
5. If no contrast lock is achieved, adjust the MMS and press Point Track to initiate another attempt
6. When a lock is achieved, the Primary Gate will be displayed around the target and the MMS Advisory will display PNT TRK
7. The MMS LOS Slew inputs will no longer be functional
8. Point Track can be exited by toggling Manual/Slave

21.1.12 Acquire On The Move

1. Acquire On The Move mode must be entered from Area Track
2. Press the Area Track button to enter Acquire On The Move mode
3. The Acquire On The Move reticule and diamond Acquisition Cursor will be displayed, and the MMS Advisory will read ACQ OM
4. To designate a target, slew the Acquisition Cursor to the target using the MMS LOS Slew inputs
5. On release of the MMS LOS Slew inputs, the MMS will attempt to establish acquisition of the target
6. If acquisition is successful, the target will be displayed in the Target Transition Gate
7. To acquire further targets, press Area Track to display the Acquisition Cursor and follow the previous steps to establish further locks
8. When new targets are acquired, the Target Transition Gate moves to the new target and the previous targets are displayed with Secondary Gates
9. Up to six targets can be acquired
10. Pressing Target Select on the Copilot Cyclic Grip will cycle the Transition Target between the selected targets and off mode
11. Pressing Point Track when a Transition Target is selected will slew the MMS to the Transition Target
12. If no Transition Target is selected, pressing Point Track will enter Offset Acquire mode

21.1.13 Offset Acquire

1. Offset Acquire may be entered from Area Track, Point Track or Acquire On The Move modes
2. Press the Point Track button to enter Offset Acquire mode
3. The Laser Hit Point cursor is displayed and the MMS Advisory displays OSET ACQ
4. The MMS LOS Slew inputs will slew the Laser Hit Point cursor
5. Pressing Point Track a further time will enter Offset Track mode

21.1.14 Offset Track

1. Offset Track may be entered from Offset Acquire mode
2. Pressing the Point Track button will switch to Offset Track mode
3. The MMS will slew to the Laser Hit Point cursor and the MMS Advisory text will display OSET TRK
4. Pressing Point Track a further time will return to Point Track mode

POLYCHOP SIMULATIONS

Weapons

22.1 M3P Machine Gun

22.1.1 Firing

1) Set **Master Arm Switch** on the Armament Control Panel to ARMED.
2) If the gun has not already been cocked, or has been reloaded, set and hold **Gun Arm Switch** on the Armament Control Panel to RECOCK, then release to ARMED.
3) Action the gun by selecting the LEFT position on the **Weapon Select Switch** on the Pilot's Cyclic Grip.
4) If using the MMS to acquire the target, align the **MMS Line of Sight** indicator in the Spare Weapons page or the ODA to the aircraft reference symbol.
5) If using the Grease Pencil Marker or the PDU, align the crosshair onto the target
6) Action the **Weapon Fire Switch** on the Pilot's Cyclic Grip.
   a. First detent will fire up to a 1 second burst each time the input is pressed.
   b. Second detent will fire until the input is released.

INFO:

None of the sighting methods for the gun use any computed methods to calculate required elevation for aiming at a target. The MMS Line of Sight indicator as displayed on the PDU or MFDs simply shows relative azimuth and elevation of the MMS, and does not factor in range or any other factors for an accurate trajectory calculation.

POLYCHOP SIMULATIONS

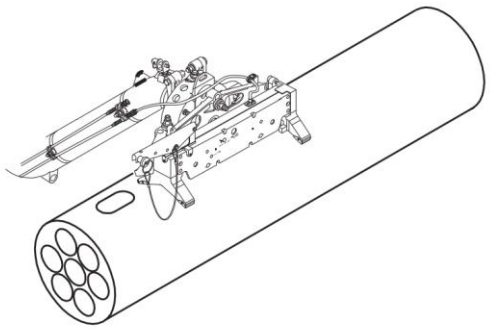
22.1.2 Infrared Zoom Laser Illuminator Designator

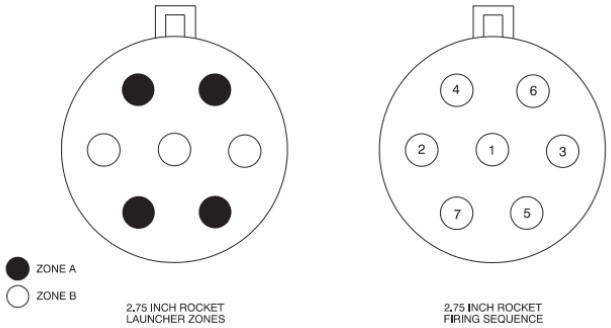
The Infrared Zoom Laser Illuminator Designator (IZLID) can be actioned at any time using the **IZLID Switch** on the Pilot Collective Grip.

INFO:

The IZLID is attached to the M3P and is only available when the M3P is installed.

22.2 Rockets





22.2.1 Rocket Cue Setup

The elevation of the Rocket Cue is determined by the selected type of rocket and the selected range. For APKWS only the elevation difference is also factored.

To achieve an accurate elevation the rocket type, desired range and fuze options must be programmed into the IMCPU.

22.2.1.1 Selecting Rocket Type

1. Enter the **Weapons Page** by actioning the **Weapon Select Switch** UP
2. Press R5 to enter the **Weapons BIT/Setup Page**
3. Press R4 to open the Rocket Types subpage
4. Press R4 again to activate Data Entry Mode for Zone A
5. Enter the corresponding number to the type of rocket installed in Zone A
6. Press Enter to validate entry and activate Data Entry Mode for Zone B
7. Enter the corresponding number to the type of rocket installed in Zone B
8. Press Enter to validate entry

22.2.1.2 Selecting Fuze and Range

1. Enter the **Weapons Page** by actioning the **Weapon Select Switch** UP
2. For rockets that use a contact fuze, press the bezel L5 to activate Data Entry Mode for Contact Fuze/Cue Distance
3. Enter the desired fuze setting (“S” for Superquick, “B” for Bunker or a value between 10 and 45 for Delay)
4. Press Enter to validate and activate Date Entry Mode for the range options
5. Enter the desired range or range source (“L” for Laser, “N” for current Navigation Point, or a value between 500 and 8000 meters
6. Press Enter to validate and exit Data Entry Mode
7. For rockets that use an airburst fuze, press the bezel L5 to activate Data Entry Mode for Airburst Fuze/Cue Distance
8. Enter the desired range or range source (“L” for Laser, “N” for current Navigation Point, or a value between 500 and 8000 meters/

POLYCHOP SIMULATIONS

9. Press Enter to validate and exit Data Entry Mode

INFO:

Bunker and Delay fuze settings for rockets are currently not available in DCS World. Contact fuze rockets will always detonate upon impact.

Airburst fuze distance/time settings for rockets are currently not available in DCS World. Illumination and Smoke rockets detonate at a fixed distance that cannot currently be set.

22.2.2 Unguided

1. Set **Master Arm Switch** on the Armament Control Panel to ARMED
2. Action the rockets by pressing in the direction of the desired pylon using the **Weapon Select Switch** on the Pilot's Cyclic Grip
3. If using the MMS to acquire the target, align the **Rocket Cue** in the Spare Weapons page or the ODA to the aircraft reference symbol
4. If using the Grease Pencil Marker or the PDU, align the crosshair onto the target
5. Action the **Weapon Fire Switch** on the Pilot's Cyclic Grip

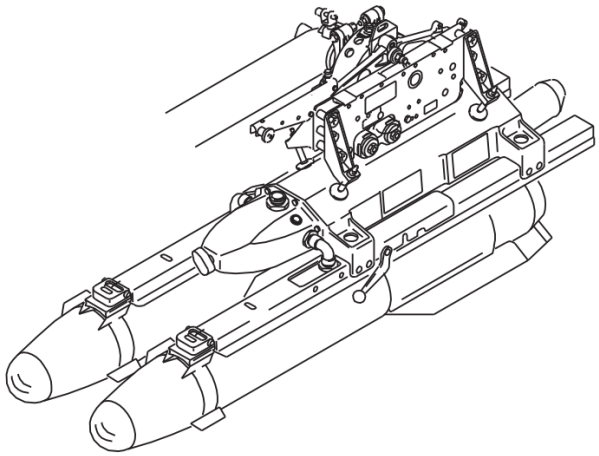
22.2.3 APKWS

1. Ensure Laser Code is set correctly for APKWS and Laser is ARMED and is in a designating mode
2. Set **Master Arm Switch** on the Armament Control Panel to ARMED
3. Action the rockets by pressing in the direction of the desired pylon using the **Weapon Select Switch** on the Pilot's Cyclic Grip
4. Hold down the Laser Fire Switch on the Copilot's Cyclic Grip until a range is returned, an elevation difference is displayed and the Rocket Cue updates
5. If using the MMS to acquire the target, align the **Rocket Cue** in the Spare Weapons page or the ODA to the aircraft reference symbol
6. If using the Grease Pencil Marker or the PDU, align the crosshair onto the target
7. Action the **Weapon Fire Switch** on the Pilot's Cyclic Grip

INFO:

The Rocket Cue elevation calculation is a guide, and will not provide high accuracy. The real-world system used data taken from firing at a 50' hover, and the DCS World OH-58D Kiowa Warrior uses this same data.

22.3 Hellfire



22.3.1 LOBL

1. Set **Master Arm Switch** on the Armament Control Panel to ARMED
2. Action the missiles by pressing in the direction of the desired pylon using the **Weapon Select Switch** on the Pilot's Cyclic Grip.

INFO:

If there is hellfires present in both pylons, any side of the Weapons Select Switch will work seamlessly.

3. Set the Launch Mode to MAN/NORM as required by pressing L4.
4. Set the Delivery Mode to LOBL by pressing L5 repeatedly until the L5 label shows LOBL.
5. Make sure below the hellfire symbol there is "RDY" as status.
6. Point with the nose of the aircraft to the general area the laser will be pointing
    a. If it's going to use the laser by your own MMS, acquire the target.
    b. Set the laser to ARMED.
    c. Hold down the Laser Fire Switch.
    d. Use the MMS cues to direct the aircraft to the laser spot.
7. Find the laser spot in your Hellfire Weapons Page, depicted as an "x", and place it inside the LOBL constrains markings.
8. When the line switches to solid and the status of the missile is "TRK" you are clear to shoot.
9. Action the **Weapon Fire Switch** on the Pilot's Cyclic Grip

22.3.2 LOAL

1. Set **Master Arm Switch** on the Armament Control Panel to ARMED

2. Action the missiles by pressing in the direction of the desired pylon using the **Weapon Select Switch** on the Pilot's Cyclic Grip.
3. Set the Launch Mode to MAN/NORM as required by pressing L4.
4. Set the Delivery Mode to LOAL by pressing L5 repeatedly until the L5 label shows HI, LO or DIR.
5. Make sure below the hellfire symbol there is "RDY" as status.

INFO:

If there is hellfires present in both pylons, any side of the Weapons Select Switch will work seamlessly.

22.3.2.1 DIR Point Reference

a. Press B2 to open HSD page.
b. Press L1 to open DIR Point Page.
c. Press L1 again to select an already created waypoint or, press L1 and then Enter to be able to write directly a custom position.
d. Press in the direction of the desired pylon using the **Weapon Select Switch** on the Pilot's Cyclic Grip to go back to your hellfire page.
e. Make sure you are in the correct Delivery and Launch modes.
f. Turn the aircraft into the DIR Point azimuth.
g. A box with DIR written inside will appear, as with the "x" from the laser in LOBL mode, put it inside the constrains box.
h. When the line switches to solid you are clear to shoot.
i. Action the **Weapon Fire Switch** on the Pilot's Cyclic Grip.

22.3.2.2 Pre-Point Reference

a. Press B3 on the left MFD to open MMS page.
b. Make sure you are in the "Prepoint" mode by looking at the MMS Rotator.
c. Press R2 to open de PPT pop up.
d. Write on the MFK the point you want to select as prepoint.
e. Make sure you are in the correct Delivery and Launch modes.
f. Turn the aircraft into the DIR Point azimuth.
g. A big cross will appear, as with the "x" from the laser in LOBL mode, put it inside the constrains box, and it will convert into a small plus sign "+".
h. When the line switches to solid you are clear to shoot.
i. Action the **Weapon Fire Switch** on the Pilot's Cyclic Grip.

22.3.3 Hellfire parameters

22.3.3.1 Launch modes

There are 3 modes in this aircraft. MAN (manual), NORM (normal) and RIPL (ripple).

The MAN mode allows the pilot to select manually the missile that it's going to be launched and changes it into the main code.

The NORM mode steps in the normal sequence the missiles must be launched.

The ripple mode is for rapid launched missiles with alternated launched codes.

22.3.3.2 Delivery modes

As already described in this manual, there is 2 delivery modes LOAL and LOBL, the LOBL mode or Lock on Before Launch requires the detection of the laser with the missile before the launch. Which gives the missile the ability to calculate the path without an extra consumption of fuel.

On the other side LOAL or Lock on After Launch, lets the user fire using a given point as a reference. This is useful in situations in which the aircraft can’t be exposed. This mode has 3 sub modes: HI, LO and DIR.

The first two will make the missile loft to gain altitude and energy that will make it fly further, the DIR mode will be used in engagements at shorter distances when lofting could make the missile not able to find the missile.

22.3.3.3 Constrain modes

Hellfire has two different constrain modes and those are NORMAL and ORIDE, for standard operations the normal mode will be the one used. The ORIDE or override mode lets the user fire without taking the constrains in consideration, that reduces the probability of a hit if not used correctly.

22.4 ATAS

22.4.1 Unslaved

1. Set **Master Arm Switch** on the Armament Control Panel to ARMED
2. Action the missiles by pressing in the direction of the desired pylon using the **Weapon Select Switch** on the Pilot's Cyclic Grip
3. Check that the bezel on L3 "SLAVE" is on OFF
4. Press **Missile Activate** button to start using the coolant
5. Move the nose of the aircraft to the general area of the objective
6. Press the **Weapon Fire First Detent** to make the seeker start sounding
7. On a high pitch sound, you are cleared to shoot by pressing the Second Detent of the **Weapon Fire Button.**

INFO:

Each pylon loaded with ATAS has 5 minutes of coolant approximately. Only activate the missiles when needed. In case of emptying the coolant the missile will lose some sensitiveness. The coolant will be resupplied on rearm.

22.4.2 Slaved

1. Set **Master Arm Switch** on the Armament Control Panel to ARMED
2. Action the missiles by pressing in the direction of the desired pylon using the **Weapon Select Switch** on the Pilot's Cyclic Grip
3. Check that the bezel on L3 "SLAVE" is on "ON"
4. On the right MFD, track your target with the MMS
5. Align your MMS with the ATAS box.
6. Press **Missile Activate** button to start using the coolant
7. Press the **Weapon Fire First Detent** to make the seeker start sounding
8. On a high pitch sound, you are cleared to shoot by pressing the Second Detent of the **Weapon Fire Button.**

INFO:

Each pylon loaded with ATAS has 5 minutes of coolant approximately. Only activate the missiles when needed. In case of emptying the coolant the missile will lose some sensitiveness. The coolant will be resupplied on rearm.

22.5 Copilot's Personal Weapon (M4)

OH-58D Kiowa Warrior crews were equipped with Personal Defense Weapons when deployed in combat. Due to the Kiowa's close quarters engagement style, there were times when these weapons were used offensively by the crew from the air in place of other onboard weapon systems.

We have brought this feature to the DCS World OH-58D Kiowa Warrior by allowing the Copilot to deploy their M4 Carbine as a handheld weapon.

As well as being controllable by mouse or joystick, the M4 can also be used with VR controllers, and we recommend this method for anyone using VR. Since the method of use is slightly different, we have documented both cases.

The method used in game is automatically decided by the detection of any powered VR hand controls.

22.5.1 Using VR Hand Controllers

1. With VR controllers switched on and in-game hands visible, as the Copilot reach and touch the latch securing the weapon.
2. The latch will open and the weapon will snap to the player's hands. If using only the right controller, the weapon will align to the position and rotation of the hand. If using both controllers, the weapon will be aimed using the position of the right hand and the relative direction to the left hand
3. The weapon can be fired using the trigger on the controller
4. The weapon will reload automatically – no input is needed

22.5.2 Using Mouse Input

1. As the Copilot, click on the latch securing the weapon.
2. The latch will open and the Copilot avatar will reach out and unsecure the weapon, then move into the ready position, aiming the weapon outside the aircraft
3. From this position the player can use the mouse to aim the weapon within the limits of the aircraft frame and seat harness restrictions using the M4 Mouse Left/Right and M4 Mouse Up/Down inputs (Default: Mouse X & Mouse Y)
4. The weapon can be fired using the M4 Mouse Fire input (Default: Mouse Button 1)

INFO:

You may need to toggle Clickable Mouse Cockpit Mode (Default: LAlt+C) in order to switch between control of the weapon and interacting with other cockpit elements.

The control bindings for the Copilot Weapon using Mouse Input mode can be found in the OH-58D Copilot Personal Weapon category

L2MUMS

23.1 Receiving from an Unmanned Platform

23.1.1 Prerequisites

In order to receive data from an Unmanned Platform you must be running a mission with a friendly MQ-9 Reaper aircraft within reception distance (dependent on relative altitude and terrain).

If there is no suitable target within range of the Reaper aircraft, then the camera will default to forward slaved position.

These steps assume that the aircraft is powered and the L2MUMS Circuit Breaker Switch is ON.

23.1.2 Steps

1. Select the INIT bezel on the Copilot's MFD to display the **Platform Reception Selection** page
2. If a suitable Unmanned Platform is available, it will appear on the list of available channels below the preset Kiowa channels
3. Press 'Line Down' on bezel L3 to highlight the desired channel
4. Press bezel L1 'Select' to select the highlighted channel for reception
5. If a platform is transmitting on the channel, the Relative Signal Strength will be displayed. The higher the Relative Signal Strength, the stronger the reception. If the Relative Signal Strength is zero, then no transmission is being received
6. Select bezel B1 'Page' to switch to the **Video Display**
7. Video feed and metadata from the platform should be present

POLYCHOP SIMULATIONS

23.2 Receiving from a Manned Platform

23.2.1 Prerequisites

The DCS OH-58D Kiowa Warrior can only receive data from player controlled Kiowa Warrior manned aircraft (this does not apply to unmanned aircraft).

In order to receive data from another player aircraft you must be in a multiplayer environment where another player is transmitting data from their OH-58D. You must also be within reception range and know the preset channel the transmitter is using.

23.2.2 Steps

1. Select the INIT bezel on the Copilot's MFD to display the **Platform Reception Selection** page
2. Press 'Line Down' on bezel L3 to highlight the desired channel
3. Press bezel L1 'Select' to select the highlighted channel for reception
4. If a platform is transmitting on the channel, the Relative Signal Strength will be displayed. The higher the Relative Signal Strength, the stronger the reception. If the Relative Signal Strength is zero, then no transmission is being received
5. Select bezel B1 'Page' to switch to the **Video Display**
6. Video feed and metadata from the platform should be present

POLYCHOP SIMULATIONS

23.3 Transmitting

23.3.1 Prerequisites

The DCS OH-58D Kiowa Warrior can only receive data from player controlled Kiowa Warrior manned aircraft (this does not apply to unmanned aircraft).

In order to transmit data to another player aircraft you must be in a multiplayer environment where another player is awaiting reception of data in their OH-58D. You must also be within reception range and know the preset channel the receiver is using.

23.3.2 Steps

1. Select the INIT bezel on the Copilot's MFD to display the **Platform Reception Selection** page.
2. Select 'Tmit Page' bezel to navigate to **Platform Transmission Selection.**
3. Press 'Line Down' on bezel L3 to highlight the desired channel.
4. Press bezel L1 'Select' to select the highlighted channel for transmission.
5. The Status will now read 'Standby'.
6. To start transmission, press bezel B3 'Tx'. If you are transmitting, the bezel text will remain highlighted and the Status will now read 'On'.
7. Transmission can be stopped at any time by pressing bezel B3.

POLYCHOP SIMULATIONS

Mission Editor

INFO:

An example mission with all Mission Editor features can be found in the 'Docs' folder of your OH-58D install.

24.1 Creating the Default Route

The 'default' route is created using the standard DCS Mission Editor waypoints. Waypoints are always read from the mission file and loaded into the aircraft before all other mission navigation points. They are automatically assigned to ROUTE 1.



Figure 154 - Creating the Default Route in the Mission Editor

1. In the Mission Editor, place an OH-58D(R) helicopter and set the SKILL option to 'Player' or 'Client'
2. On the lower right panel, select 'Add'
3. Click anywhere on the map to create a waypoint. Successive clicks will add further waypoints

When the mission is run and the aircraft started, the created route will appear in the **NAV Setup** page under Route 1 and will be named 'ME' (Mission Editor).

POLYCHOP SIMULATIONS

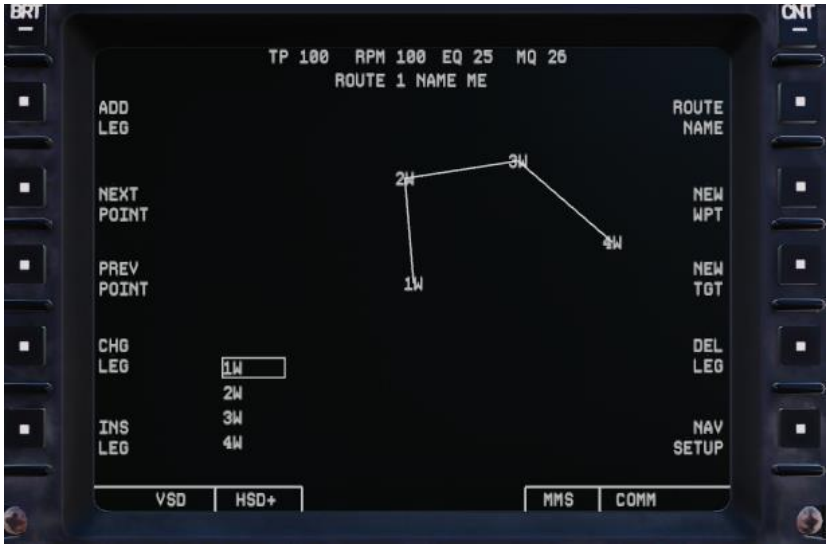


Figure 155 - Viewing the Default Route

POLYCHOP SIMULATIONS

24.2 Creating Non-Route Waypoints

It is possible to create further Waypoints in the Mission Editor that are not part of any Route or Battlefield Graphic (unless later assigned).

These Waypoints will be loaded after the Default Route.

1. In the Mission Editor, place an OH-58D(R) helicopter and set the SKILL option to 'Player' or 'Client'
2. On the lower right panel, open the 'Navigation Target Fix Points' tab
3. Click anywhere on the map to create a Navigation Target Fix Point.
4. In the Comment field, enter the desired name of the Waypoint. It is recommended to follow the standard naming convention of #W where # is a number between 1 and 99

When the mission is started, the aircraft will load the created Waypoints and attempt to assign them to the index based on the name supplied in the Comment field. If the ID matches an already existing waypoint (such as one created in the Default Route), it will override this waypoint and any Route or Battlefield Graphic will be affected by the change.

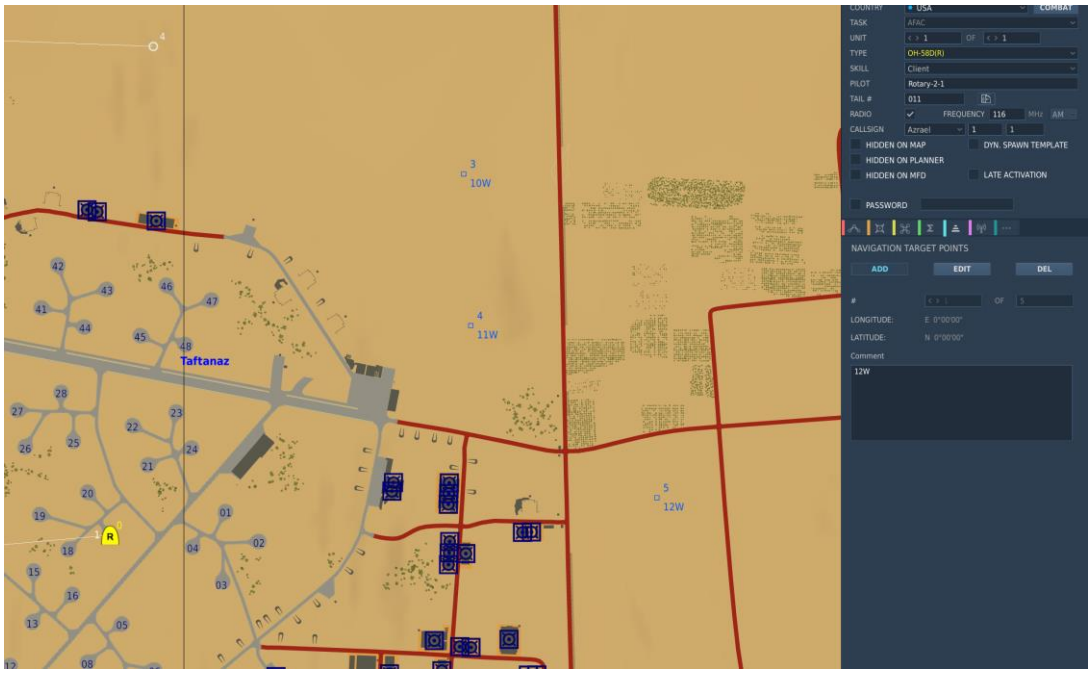


Figure 156 - Adding Non-Route Waypoints in the Mission Editor

INFO:

Waypoints that are not part of a Route or Battlefield Graphic will never be displayed in the HSD or RMS pages. They will be visible only in the Waypoint List, until assigned to a Graphic or Route

POLYCHOP SIMULATIONS

24.3 Creating Control Points

Control Points can be created in the Mission Editor using the same method as creating Non-Route Waypoints. Control Points will be loaded after the Default Route and Non-Route Waypoints.



Figure 157 - Creating Control Points in the Mission Editor

1. In the Mission Editor, place an OH-58D(R) helicopter and set the SKILL option to 'Player' or 'Client'
2. On the lower right panel, open the 'Navigation Target Fix Points' tab
3. Click anywhere on the map to create a Navigation Target Fix Point.
4. In the Comment field, enter the desired name of the Control Point. It is recommended to follow the standard naming convention of #C where # is a number between 1 and 20

The loaded Control Points will be visible on the **HSD**, **RMS** and **Control Point List** pages.

POLYCHOP SIMULATIONS

24.4 Creating Target Points

Target Points may be created in the Mission Editor using the same method as Waypoints and Control Points. Target Points allow for an extended naming convention to supply basic target data.

Target Points are loaded after the Default Route, Non-Route Waypoints and Control Points.

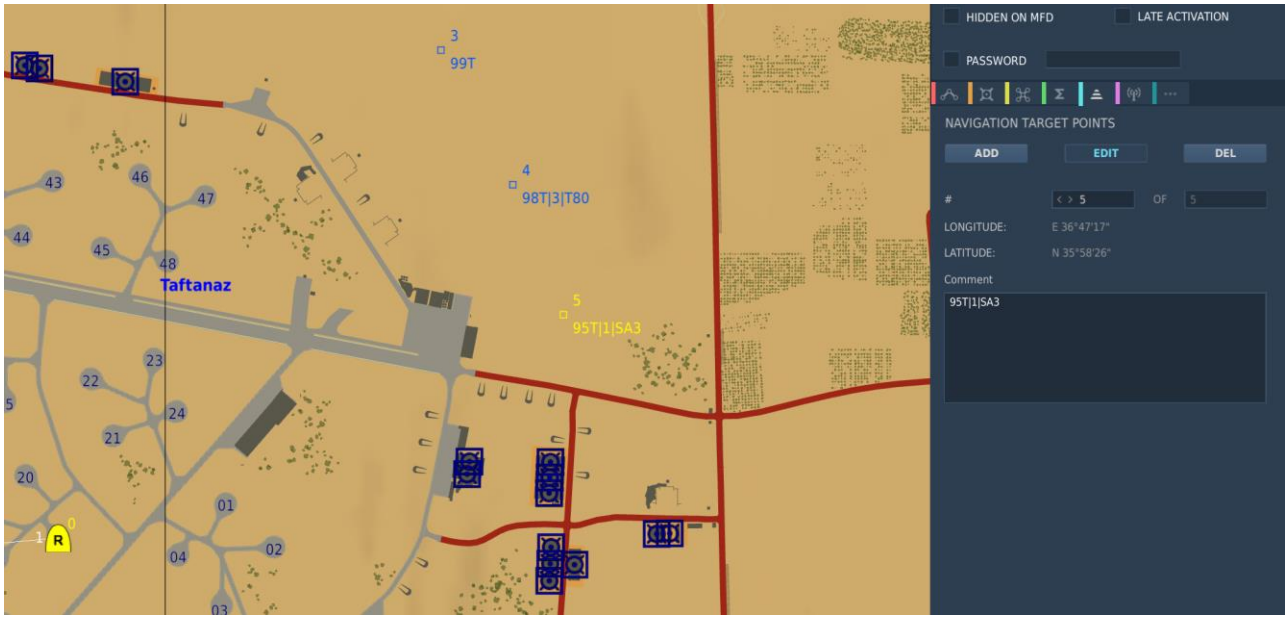


Figure 158 - Creating Target Points in the Mission Editor

1. In the Mission Editor, place an OH-58D(R) helicopter and set the SKILL option to 'Player' or 'Client'
2. On the lower right panel, open the 'Navigation Target Fix Points' tab
3. Click anywhere on the map to create a Navigation Target Fix Point.
4. In the Comment field, enter the desired name of the Target Point. It is required to follow the standard naming convention of #T where # is a number between 1 and 99, e.g. 98T
5. The Target Quantity can be entered by appending a vertical bar character '|' and a number, e.g. 98T|3
6. The Target Subtype can be entered by appending a further vertical bar character and Target Subtype entry, e.g. 98|3|T80

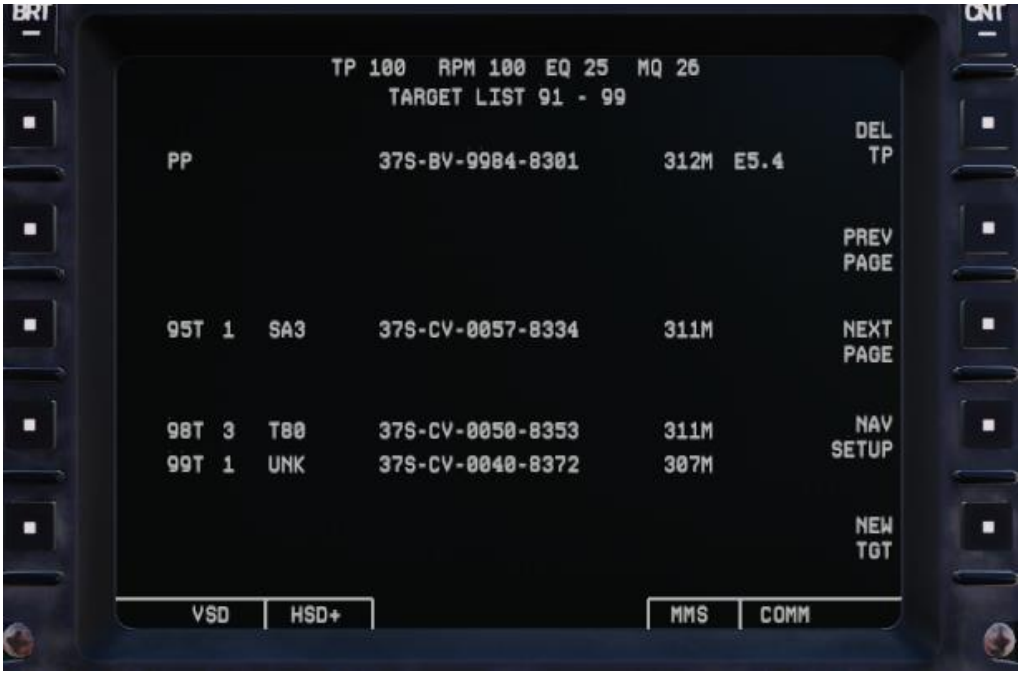


Figure 159 - Viewing the created Target Points

POLYCHOP SIMULATIONS

24.4.1 Target Subtypes

The following table contains the valid target strings that can be entered in the Mission Editor.

UNK
T54
T55
T62
T64
T72
T80
T90
TYPE 59
TYPE 69
TYPE 62
PT76
BRDM2
BRDM1
PT76
D30
BM21
2S1
2S3
ZU23
ZPU
ZPU2
ZPU4
ZSU23-4
ZSU57-2
SA2
SA4
SA6
SA8
SA9
SA11
SA13

SA7
SA14
SA16
SA18
S60
SA5
2S6
SA3
MIG17
MIG21
MIG23
MIG25
MIG27
MIG31
SU21
SU25
AN12
IL76
IL28
YAK28
AN2
MIG29
SU17
SU24
SU27
SU7
TU26
MIG19
MI2
MI4
MI8
MI17

MI24
MI28
MI31
MI26
KA50
MI6
BRDM
RKH
MT55
IMR
TMM
BTR60
BTR70
BTR80
BMP1
BMP2
BMP3
BMD
YW531
TYPE77
MTLB
OT64
BM11
BM21
BM14
BM16
BM24
BMD25
M1985
FROG5
FROG7
SCUD

24.5 Creating Routes with Drawings

In addition to the method provided with the Default Route, additional Routes may be created using the DCS Mission Editor's Draw tools.

1. Select the Draw mode on the left panel of the Mission Editor

2. On the right panel, select the Line tool



Figure 160 - Creating Routes using Drawings in the Mission Editor

3. Click on the map to place line points and draw the line. To complete the line, double click to place the last point
4. The line must be named using a specific format to be imported as a Route. In the Name box, enter 'KW-ROUTE#' where # is the number of the route you wish to create – 1,2 or 3.

5. On mission start, the line points will be loaded as Waypoints and the Route created.

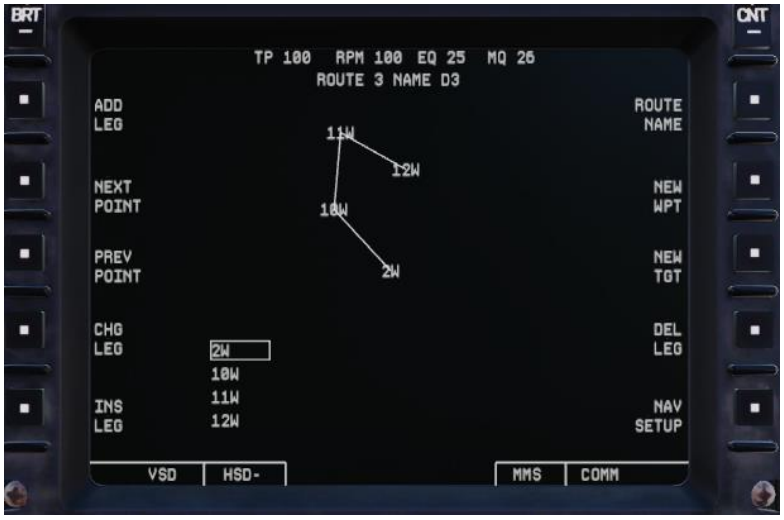


Figure 161 - Viewing one of the resultant Routes

INFO:

Setting the name to 'KW-ROUTE1' will override any created Default Route.

If any point of the drawn line overlaps an existing placed Waypoint, Control Point or Target Point, the existing point will be used in place of creating a new Waypoint.

24.6 Creating Battlefield Graphics with Drawings

Battlefield Graphics can be created using the Mission Editor drawing tools. This simplifies the creation of graphics and allows mission designers to communicate areas of interest to players.

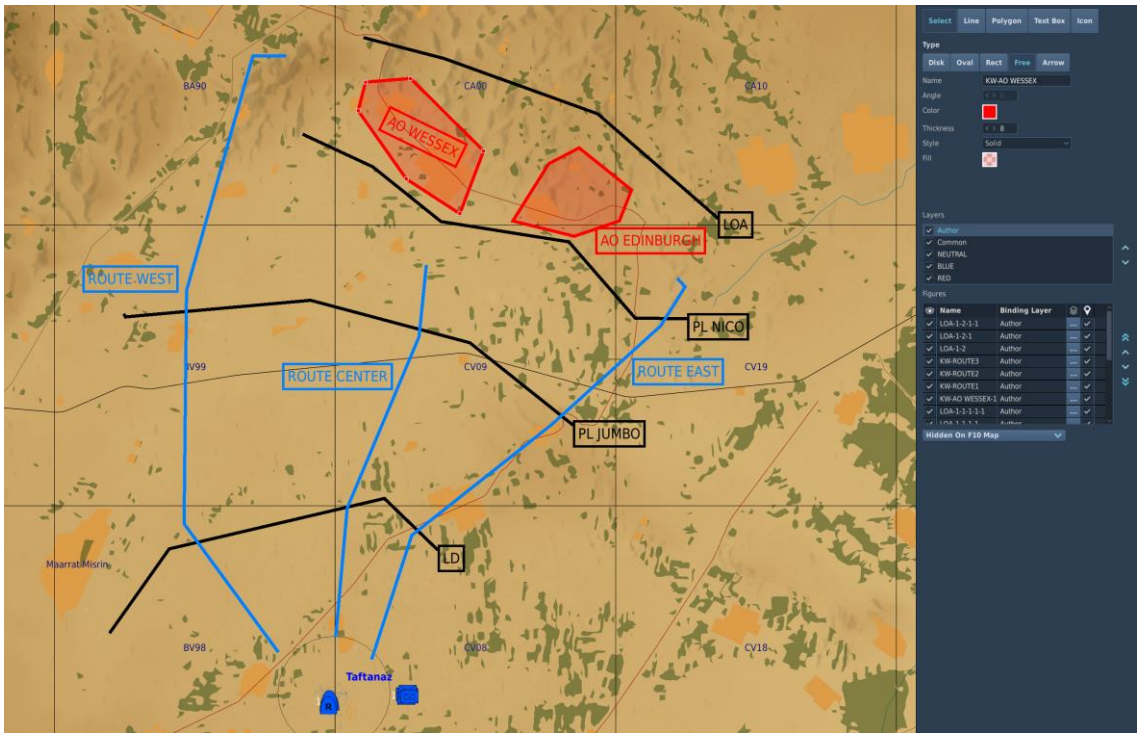


Figure 162 - Example mission using drawings to create Routes and Battlefield Graphics



Figure 163 - Viewing the created Battlefield Graphics and selected Route in the RMS page

24.6.1 Creating Line Drawings

5. Select the Draw mode on the left panel of the Mission Editor
6. On the right panel, select the Line tool
7. Click on the map to place line points and draw the line. To complete the line, double click to place the last point
8. The line must be named using a specific format to be imported as a Battlefield Graphic. In the Name box, enter 'KW-#' where # is the name of the line, e.g. KW-PL JUMBO.
9. On mission start, the line points will be loaded as a Battlefield Graphic Line

INFO:

If any point of the drawn line overlaps an existing placed Waypoint, Control Point or Target Point, the existing point will be used in place of creating a new Waypoint.

24.6.2 Creating Area Drawings

1. Select the Draw mode on the left panel of the Mission Editor
2. On the right panel, select the Polygon tool
3. Select Free under the Polygon Type
4. Click on the map to place line points and draw the shape. To complete the shape, double click to place the last point
5. The shape must be named using a specific format to be imported as a Battlefield Graphic. In the Name box, enter 'KW-#' where # is the name of the shape, e.g. KW-AO WESSEX.
6. On mission start, the shape points will be loaded as a Battlefield Graphic Area

INFO:

If any point of the drawn line overlaps an existing placed Waypoint, Control Point or Target Point, the existing point will be used in place of creating a new Waypoint.

APPENDIX

Glossary

ASE	Aircraft Survivability Equipment
CG	Centre of Gravity
CMWS	Common Missile Warning System
EGI	Embedded GPS / INS
EQ	Engine Torque
ETE	Estimated Time Enroute
FADEC	Full Authority Digital Electronic Control
FAT	Free Air Temperature
GPS	Global Positioning System
HSD	Horizontal Situation Display
HSI	Horizontal Situation Indicator
IGE	In Ground Effect
INS	Inertial Navigation System
IZLID	Infrared Zoom Laser Illuminator / Designator
LMC	Linear Motion Compensation
LRF/D	Laser Rangefinder / Designator
MFD	Multi-Function Display
MFK	Multi-Function Keyboard
MMS	Mast Mounted Sight
MPD	Multi-Parameter Display
MGRS	Military Grid Reference System
Ng	Percentage of gas producing turbine speed
Np	Percentage of power turbine speed
Nr	Percentage of main rotor speed
OGE	Out of Ground Effect
PP	Present Position
R/C	Rate of Climb
R/D	Rate of Descent
RMS	Rotorcraft Map System
RPM	Revolutions per minute
SCAS	Stability and Control Augmentation System
TAS	True Airspeed
TGT	Target or Turbine Gas Temperature
TIS	Thermal Imaging System
TRQ	Torque
VSD	Vertical Situation Display
VSI	Vertical Scales Instrument
WOG	Weight On Gear
WPT	Waypoint
XTK	Cross-track

Warnings Advisories & Cautions

26.1 WARNING messages

There are 11 warning messages that can be displayed, prioritized as follows:

1. ENGINE OUT
2. LOW RPM ROTOR
3. LOW FUEL PRES
4. FADEC FAIL
5. HIGH RPM
6. XMSN OVER TRQ
7. ENG OVER TRQ
8. TGT OVER TEMP
9. ENG OVER SPD
10. CHECK THROTTLE *
11. LOW ALTITUDE

26.2 CAUTION messages

The following is a complete list of cautions:

- AC GEN FAIL
- ADU FAIL
- BATT CHGR FAIL
- CHIPS ENG FREEWHEEL
- CHIPS ENG LOWER
- CHIPS ENG UPPER
- CHIPS T/R GEARBOX
- CHIPS XMSN SUMP
- CHIPS XMSN UPPER
- DC GEN FAIL
- EGI FAIL
- ENG TRQ TIME LIM [ ] *
- FADEC MANUAL
- FUEL BOOST FAIL
- FUEL FILTER BYP
- FUEL LOW
- HIGH OIL PRESS ENG
- HIGH OIL TEMP ENG
- HIGH OIL TEMP XMSN
- HIGH TEMP T/R GRBX
- HIGH TGT TIME LIM [ ] *
- IFF FAIL
- IFF MODE 4 FAIL
- INS FAIL
- INV FAIL
- LEFT MCPU FAIL
- LOW HYD PRESS
- LOW OIL PRESS ENG
- LOW OIL PRESS XMSN
- LOW OIL QUANTITY ENG
- MAST TRQ TIME LIM [ ] *
- MISSILE UNLATCHED
- NP TIME LIM [ ] *
- OIL BYP ENG
- P/R DISENG
- QE 30 MIN LIM [ ] *
- RECT FAIL
- RIGHT MCPU FAIL
- SCAS DISENG
- TGT START LIM [ ] *
- TGT 30 MIN LIM [ ] *
- WEAPONS FAIL
- YAW DISENG

26.3 ADVISORY messages

The following is a complete list of advisory messages:

- AIR MSN REJECTED
- AIR MSN X UPDATE
- AIR RQST MSN
- AIR SA
- ALARM (alarm name) *
- ARTY MSN X UPDATE
- ASE FAIL
- BINGO FUEL *
- BRIDGE MSG RCVED
- BYPASS (EBF Switch – segment light)
- CHANGE FS NET *
- CODE NOT ACCEPTED
- COMM FAIL — XX
- DANGER CLOSE
- DTS FAIL
- EGI BATT LOW *
- EN/BDA MSG RCVED
- ENG ANTI-ICE ON
- EXT PWR
- FADEC DEGRADE *
- FADEC MAINT *
- FILTER (EBF Switch – segment light)
- FIRE INHIBIT – EYESAFE
- FLIGHT CARD ERROR
- FM-1 CT FAIL
- FM-1 CUE
- FM-1 FAIL
- FM-1 HUB LOW
- FM-1 PT FAIL
- FM-2 CT FAIL
- FM-2 CUE
- FM-2 FAIL
- FM-2 HUB LOW
- FM-2 PT FAIL
- FREETXT MSG RCVED
- FRIENDLIES SA
- FUEL CHECK *
- GATEWAY LOST *
- GEOREF SA
- GPS DIVERGENT
- GPS FAIL *
- HF MIN RNG XXXX *
- HVR DEGRADED *
- HVR DRIFT *
- IDM FAIL
- IDM SHUTDOWN CMPLT
- IFF MISMATCH
- IFM FAIL
- IMAGE RECEIVED *
- INFO MSG RCVED
- INPROG MSG RCVED
- INVALID COMMAND
- LASER CODE MISMATCH
- LAUNCHER SAFED
- LEFT LAUNCHER FAIL
- LIMIT EXCEEDED **
- MAP CARD ERROR
- MAYDAY MSG RCVED
- MISSION CARD ERROR
- MMS FAIL
- MOPP MSG RCVED
- NAV INVALID
- NAV NOT ALIGNED
- NAV UPDT REQUIRED *
- NBC SA
- NET JOIN? — FS
- NO AUTO START *
- NO CODE
- NO VIXL NET *
- OBSERVED SA
- OBSTCL MSG RCVED
- OIL TEMP ENG *
- ONE YAW CHAN OFF *
- ORDERS MSG RCVED
- OVERLAY MSG RCVED
- PC-DTS FAIL
- PITOT HEAT ON
- POSREP MSG RCVED
- PSWD EXPIRES IN XX-YYYY *
- P(Y) CODE INVALID *
- REDCON MSG RCVED
- RHE FAIL
- RIGHT LAUNCHER FAIL
- RMS FAIL
- ROUTE MSG RCVED
- RS232 — FS
- SECURITY CHALLENGE *
- SITREP MSG RCVED
- SPOT MSG RCVED
- STRIKE MSG RCVED *
- STRIKE SA
- TARGET ID — XXT
- THREAT MSG RCVED *
- THREAT SA
- TI NET INOP
- TI NET SHUTDOWN
- TIMER (timer name) *
- UHF FAIL
- VDU FAIL
- VHF FAIL
- VIDEO CARD ERROR
- VIDEO CARD FULL
- WEDGE CONSTANT ZERO
- WOG MISMATCH
- WPN NOT ACTIONED
- WPN NOT ARMED